

**Verified Control Envelope Design:
Scaling for Complexity and Automation
v2.0**

Aditi Kabra

April 2026

1

School of Computer Science
Carnegie Mellon University
Pittsburgh, PA 15213

2

3

4

Thesis Committee:

5

André Platzer, Co-chair

6

Stefan Mitsch, Co-chair

7

Eunsuk Kang

8

Armando Solar-Lezama (MIT)

9

*Submitted in partial fulfillment of the requirements
for the degree of Doctor of Philosophy.*

10

11

Copyright © 2026 **Aditi Kabra**

12

March 27, 2026
DRAFT

*For Vandana Kabra,
who spent a large part of her life cultivating mine.*

14

15

Abstract

Many cyber-physical systems, such as trains, planes, and self-driving cars, are safety-critical but difficult to reason about. The task of designing controllers for such systems is complex (the subject of an entire field, control theory), made even more challenging by the need to ensure correctness over all of the infinitely many possible scenarios that the system may face. This thesis develops techniques that let computers *automatically* synthesize the conditions that define correct control solutions, with *mathematical guarantees* that these conditions are correct.

Symbolic control envelopes are our representation of the control conditions that characterize *sets of safe control solutions*. They are represented parametrically in symbols that can be instantiated with any real-valued input (e.g., for a train control envelope, train weight w). Control envelopes provide a path to designing complex controllers that still have mathematical correctness guarantees by allowing separation of concerns during controller design. A verified (i.e., mathematically correct) safe control envelope can first identify the set of control solutions that ensure the safety-critical control objectives, and then non-formal techniques, e.g., machine learning, can optimize within that envelope for secondary objectives.

This thesis starts by designing, as a target for automation, the first verified train control envelope that accounts for all the forces in a realistic train kinematics model, demonstrating the feasibility and potential benefits of applying verified control envelopes to a realistic industrial-scale system. The thesis then automates the process of designing such a control envelope by creating the first framework for *symbolic control envelope synthesis*. The framework takes as input the *shape* of a control system, which indicates what control and environment behaviors are physically possible and what the desired control behavior is, making the synthesis question well-defined. It then automatically identifies the *symbolic control conditions* indicating when a given control action is correct, which is shown to correspond to the non-deterministic control policies of players in *hybrid games* (games with both continuous and discrete dynamics).

This thesis tackles the hybrid game control envelope synthesis problem in its full generality, developing the theory to solve for *all of differential game logic* (two-player, zero-sum games). It introduces a specialized algorithm, CESAR, to solve for an important class of control problems (time-triggered control, where the controller repeatedly polls to provide control decisions with some maximum time latency). By strategically using AI along with verification, it provides a general framework that leverages large language models for sound, scalable synthesis.

	Acknowledgments	52
TODO.		53

Contents

54

1	Introduction	1	55
1.1	Introduction	1	56
1.2	Structure	3	57
1.3	Related Work	5	58
2	Background	9	59
2.1	Differential Dynamic Logic	9	60
2.2	Differential Game Logic	11	61
2.3	Semantics	12	62
2.3.1	Differential Dynamic Logic	13	63
2.3.2	Differential Game Logic	13	64
3	Verified Train Controllers	17	65
3.1	Introduction	17	66
3.2	Related Work	19	67
3.3	FRA Model of Train Kinematics	19	68
3.4	Mathematical Model Abstraction	20	69
3.5	Model Structure	21	70
3.5.1	Model Description	21	71
3.5.2	Stopping Distance: Conservative	23	72
3.6	Safe Efficiency Improvements	26	73
3.6.1	Bound on Gradient	26	74
3.6.2	Bound on Curve Resistance	26	75
3.6.3	Tight Stopping Distance Approximation	27	76
3.6.4	Effect of Air Pressure Brakes	29	77
3.6.5	Exploiting Resistance	31	78
3.7	Kinematic Train Model Proofs	32	79
3.8	Evaluation	33	80
3.8.1	Stopping Behavior in Troughs	35	81
3.8.2	Stopping Behavior on Crests	36	82
3.9	Conclusion	37	83

84	4	Automatic Verification	39
85	4.1	Introduction	39
86	4.2	Related Work	40
87	4.3	Automated Theorem Proving	41
88	4.3.1	Pipeline Description	41
89	4.3.2	Case Studies	42
90	4.3.3	Outcomes	42
91	4.3.4	Verification Impact by Example	42
92	4.4	Conclusion	45
93	5	Control Envelope Synthesis via Angelic Refinement	47
94	5.1	Introduction	47
95	5.2	Background	48
96	5.3	Approach	50
97	5.3.1	Problem Definition	50
98	5.3.2	An Optimal Solution	51
99	5.3.3	Controllable Invariants	51
100	5.3.4	One-Shot Fallback Refinement	52
101	5.3.5	Bounded Fallback Unrolling Refinement	54
102	5.3.6	Proving Optimality via the Dual Game	56
103	5.3.7	Implementing the Reduction Oracle	57
104	5.3.8	The CESAR Algorithm	58
105	5.4	Benchmarks and Evaluation	58
106	5.5	Related Work	60
107	5.6	Conclusion	61
108	6	Generalized Hybrid Game Control Envelope Synthesis	63
109	6.1	Introduction	63
110	6.2	Differential Game Logic Background	65
111	6.3	Overview	67
112	6.4	Prefixes and Suffixes	69
113	6.4.1	Game Suffix Construction	70
114	6.4.2	Game Trees and Plays	71
115	6.4.3	Game Prefix	72
116	6.5	Subvalue Maps	73
117	6.5.1	Subvalue Map Definition	73
118	6.5.2	Existential Subvalue Map Projection	75
119	6.5.3	Universal Subvalue Map Projection	76
120	6.6	Inductive Subvalue Maps	77
121	6.7	Synthesis Framework for Inductive Subvalue Map with Efficient Checks	80
122	6.7.1	Invariant Generation	82
123	6.8	Maximal Solution	83
124	6.8.1	Recovering Solution Optimality	84
125	6.9	Relationship to Runtime Monitoring	84

6.10	Evaluation	85	126
6.11	Related Work	87	127
6.12	Conclusion	88	128
7	Automatic Synthesis	89	129
7.1	Overview	89	130
7.1.1	Pipeline Description	90	131
7.1.2	Outcomes	91	132
7.2	Conclusion	92	133
8	Autoformalization	93	134
8.1	Overview	94	135
8.2	Methodology	96	136
8.3	Related Work	98	137
8.4	Evaluation	98	138
8.5	Discussion	100	139
8.6	Conclusion	101	140
9	Conclusion	103	141
9.1	Future Work	105	142
A	CESAR (Appendix)	107	143
A.1	Reduce Operation	107	144
A.2	Parachute: reduce of Non-solvable Dynamics	108	145
A.3	Benchmarks	109	146
A.3.1	ETCS Train	109	147
A.3.2	Sled	109	148
A.3.3	Parachute	110	149
A.3.4	Intersection	111	150
A.3.5	Curvebot	113	151
A.3.6	Corridor	114	152
A.3.7	Power Station	114	153
A.3.8	Coolant	115	154
A.4	Simplification Impact Example	118	155
B	Generalized Hybrid Game Synthesis (Appendix)	121	156
B.1	Naïve Ordering Counterexample	121	157
B.2	Inductive Subvalue Map Loop Condition Discussion	122	158
B.2.1	Counterexample for Projection Condition Weakening	122	159
B.2.2	Counterexample for Recursive Condition Weakening	122	160
B.3	Example of Subvalue Map Shortfall for Runtime Monitoring	123	161
B.4	Benchmarks	123	162
B.4.1	New Examples Discussion	123	163
B.4.2	Quadcopter	124	164

165	B.4.3	Full Benchmark Listings	125
166	B.4.4	Rewrite Heuristics	128
167	B.4.5	Refinements	129
168	B.5	Proofs	130
169	B.6	Additional Definitions	138
170	B.6.1	dGL Action Semantics	139
171	B.6.2	Labeled Game Trees	139
172	B.6.3	Label-free Game Trees	140
173	B.6.4	Demonic Subvalue Projection	141
174	B.6.5	Universal Projection of Demonic Subvalue Maps	141
175	B.6.6	Strategy Set Generation	141
176	B.6.7	Inductive Demonic Subvalue Maps	143
177	B.6.8	Game Prefix	143
178	B.6.9	Solving Function for Inductive Demonic Subvalue Maps	144
179	C	Automatic Verification and Synthesis (Appendix)	145
180	C.1	Details of Verification Experiments	145
181	C.2	Tactic for Model 6	146
182	C.3	Other Case Studies	148
183	C.4	Verification Prompts	153
184	C.5	Synthesis Prompts	192
185	D	Proofs	211
186		Bibliography	239

List of Figures

187

1.1	Chapter relationships	5	188
2.1	Example of a dL proof tree.	11	189
2.2	dGL axiomatization and derived axioms and rules	15	190
3.1	Relationship between train models in this chapter.	22	191
3.2	Comparison of start braking (dashed vertical line) and stopping points (solid vertical lines) in a trough from 0.5% downhill at start to −0.5% uphill at 7000ft, with end of movement authority (dotted vertical line) at 7910ft.	36	192 193 194
3.3	Initial conditions of proof (Corollary 3.1) not satisfied $\nexists$: an underpowered train on a 3% downhill slope needs air brakes to stay stopped; the stopping point is closer than the rolling distance during brake rampup.	37	195 196 197
3.4	Initial conditions of proof (Corollary 3.1) not satisfied $\nexists$: an underpowered train on a crest from 3% uphill to −3% downhill slows down uphill despite full tractive effort and regains speed on the downhill segment, but needs air brakes to stay stopped; in this scenario, the movement authority ends past the rolling distance during brake rampup.	38	198 199 200 201 202
4.1	Schematic representation of tradeoffs in formal methods applied to Hybrid Systems.	40	203 204
4.2	Automated Theorem Proving Pipeline. Appendix C.4 shows prompts.	41	205
4.3	dGL Proof Rules and Axioms relevant to the proof of Model 6. The rules are as follows: differential invariant (dI), differential cut (dC), loop (loop), and exists right (existsr).	44	206 207 208
5.1	Robot navigating a corridor (Model 8). A 2D robot must navigate safely within a corridor with a dead-end without crashing against a wall. The corridor extends infinitely on the bottom and on the right. The robot can choose between going left and going down with a constant speed V . The left diagram shows I^0 in gray. The right diagram shows I^1 under the additional assumption $VT < 2R$ (I^1 and I^0 are otherwise equivalent). A darker shade of gray is used for regions of I^1 where only one of the two available actions is safe according to G^1	55	209 210 211 212 213 214 215

216	6.1	An <i>inductive subvalue map</i> for a simple hybrid game and the associated policy.	
217		The subvalue formulas are numbered in the order in which they are derived by	
218		the algorithm presented in Section 6.7. We will revisit this example throughout	
219		the chapter, providing more explanations.	67
220	6.2	The game tree for subgame 6 of the game in Fig. 6.1, i.e., $6:(7:v := v + a)^*$	71
221	6.3	Benchmark summaries and results.	86
222	7.1	Example of envelope synthesis shown on Model 6	90
223	7.2	Synthesis Pipeline. Appendix C.5 shows prompts.	90
224	8.1	Autoformalization pipeline	97
225	8.2	Outcomes of autoformalization by GPT 4o, GPT-4.1 and o3, with zero-shot and	
226		multi-shot prompting.	99
227	8.3	Problem outcome by model.	100
228	A.1	Definition of reduce (exact elided). Notation $P\{e/x\}$ indicates P with unbound	
229		occurrences of x replaced by expression e . <code>odereduce</code> isolates the effect of	
230		reduce on ODEs. $\triangleright$ simplifies and quantifier eliminates P assuming A	108
231	B.1	Inductive subvalue maps S and S' . Locally at subgame α , S' seems more per-	
232		missive, but global analysis reveals S is more permissive.	121
233	B.2	Computed control envelope for quadcopter problem 23. Red regions are unsafe	
234		(obstacles or floor). Green regions are safe starting points per the computed	
235		control envelope. The x and y coordinates represent the quadcopter's position. . .	127
236	C.1	Formula ϕ_{init} for the Coolant case study.	149

List of Tables

237

2.1	Hybrid game operators for two-player hybrid systems	12	238
3.1	FRA train parameter instantiation example.	34	239
4.1	Verification: Adding a summarization step improves success rate compared to providing the agent the full history of tactic attempts and outcomes.	43	240 241
5.1	Summary of CESAR experimental results	60	242
7.1	Synthesis: ablation shows the benefit of the recovery segment of the pipeline (detailed proving and custom backtracking).	91	243 244
7.2	Synthesis experiments: Number of LLM calls, time to run, and dollar cost for each case study.	92	245 246
7.3	Synthesis experiments without recovery choice: Number of LLM calls and dollar cost for each case study. The costs for all other case studies exceeds the budget.	92	247 248
A.1	Benchmark listing.	109	249
A.2	Simplification impact on solution size.	119	250
B.1	Quadcopter benchmark suite generated parameters and solving outcomes. In the result column, ✓ indicates that a nonempty control envelope was computed, while ✗ indicates that algorithm timed out after 20 minutes, failing to compute an envelope.	126	251 252 253 254
B.2	Rewrite heuristics used in the evaluation problems.	128	255
C.1	Verification experiments: Number of LLM calls, time to run, and dollar cost for two sets of four parallel runs per case study.	145	256 257
C.2	Verification experiments without summary step: Number of LLM calls and dollar cost for two sets of four parallel runs per case study. The costs for all other case studies exceeds the budget.	146	258 259 260

Chapter 1

261

Introduction

262

Cyber-physical systems (CPS) are systems like trains, planes, and self-driving cars, where discrete software interacts with continuous physics. These systems are becoming an increasingly prominent part of our lives, and are often safety-critical, since software errors can have tragic or expensive consequences. For such systems, designing correct controllers is difficult, and requires careful reasoning about what the system must do *now* to stay safe *forever*, accounting precisely for environmental nondeterminism, interacting forces, and strategic control over an uncountably infinite state space. This is the subject of an entire field, control theory. Can we *automatically* identify the principles that will yield correct control for a given control system, with *mathematical guarantees* that our identification is correct? We seek not a single numeric control solution but symbolic expressions characterizing entire classes of solutions. This thesis provides a positive answer, guided by the following thesis statement.

263
264
265
266
267
268
269
270
271
272
273

Symbolic reasoning can automatically derive correct control constraints.

274

We encode the ability to reason *symbolically* about control systems such that computers become able to compute the expressions that characterize control solutions that are mathematically provably correct.

275
276
277

1.1 Introduction

278

Our representation of the constraints defining a *set of correct control solutions* is a *control envelope*. This thesis synthesizes control envelopes given the *shape* of the control system, i.e., what control decisions and environment dynamics are physically possible, and the desired control contract. Such information is necessary to make the synthesis question meaningful, and is generally already available to engineers in the form of client specifications. The specification of the control system may contain symbolic parameters representing any real number (e.g., train weight w while designing a train controller). The synthesized control envelope will be parametric in those symbols, making it adaptable. The generated control envelope is *verified*, that is, mathematically guaranteed to be correct for all of the (usually uncountably infinitely many) possible behaviors of the system.

279
280
281
282
283
284
285
286
287
288

Control envelopes are useful to synthesize controllers downstream. Suppose we start with a verified control envelope, representing a set of as many control solutions as possible that are all

289
290

291 guaranteed to satisfy the mission-critical or safety-critical control objectives. Then non-formal
292 methods, including machine learning, can be used to optimize within the control envelope for
293 secondary or evolving objectives¹. In this way control envelopes provide a way to disentangle
294 design of the safety-critical aspects of controllers, which deserve expensive verification, from
295 other concerns like performance and robustness which benefit from scalable non-formal opti-
296 mization techniques.

297 We address the control envelope synthesis problem for the very general class of all problems
298 where the shape of control can be expressed as a hybrid game (in differential game logic [120]).
299 A hybrid game is a game with both discrete and continuous dynamics. It is typically used to
300 model cyber-physical systems as a game played between a controller and an environment player.
301 The controller player chooses control actions (for example, for a train, whether to accelerate),
302 while the environment player resolves environmental non-determinism (such as how the curva-
303 ture of the track evolves, which affects train motion dynamics). A control envelope corresponds
304 to the nondeterministic policy of a controller player in a hybrid game against an adversarial en-
305 vironment player. The controller player wins if it maintains the desired control contract. Its
306 policy indicates which control actions it can take in a given state to continue to have a way to
307 win, thereby defining a space of safe control. Given a hybrid game specification defining what is
308 physically possible, we therefore synthesize the controller player’s control envelope. A special
309 property that our work considers (compared to controller synthesis) is that synthesized control
310 conditions should be as permissive as possible, allowing control actions unless they are genu-
311 inely unsafe. This way, there is as much room as possible for non-formal optimization within
312 the envelope, allowing for maximal performance on secondary objectives.

313 To solve the problem in its full generality, we develop key technical innovations like a new
314 abstraction to represent control envelopes in a form that composes at the level of imperative pro-
315 gram constructs (a *subvalue map*, or a symbolic analog to value tables from game theory [10]),
316 a hybrid games analog of Brzowski derivatives [25] to step through games syntactically, and
317 a precondition calculus that defines correct control envelopes that never let the controller get
318 stuck or lose the game. With these fundamental tools, it becomes possible to synthesize con-
319 trol envelopes through symbolic execution, except for subgames entailing unbounded sequences
320 of gameplay (loops). For these, our precondition calculus, like most [40], *checks* the defining
321 control conditions (loop invariants), which must be generated externally.

322 To *generate* such conditions about unbounded repetition games, we develop refinement and
323 rewrite procedures. *Refinements* rewrite games in a way that makes them easier to symbolically
324 execute, but harder for the controller player to play, which maintains soundness when the refined
325 game is used in synthesis. *Rewrites* also make games easier to symbolically execute, but do not
326 necessarily maintain soundness. Instead, any conditions they generate are checked retrospec-
327 tively to recover soundness. For a practically important class of problems (with *time-triggered*
328 *control*, the common controller design pattern where the controller repeatedly polls, chooses an
329 action, and then goes to sleep for some maximum time latency, and *action permanence*, a hybrid

¹One way to use the control envelopes in combination with a machine learning (ML) based optimization step is to *shield* an ML controller agent’s decisions. At training time, the agent is rewarded for staying within the envelope [132], and at runtime, fallbacks are enforced when it tries to deviate from the verified set of solutions [49, 97]. This allows behavior to be optimized by gradient descent within the safe solutions. Control envelopes can also statically verify a model [153].

analog to idempotence), we identify a systematic set of refinements that provide a procedure to solve control envelope synthesis.

For synthesis procedures when the engineer has a high-level idea for the control policy they want to include in the control envelope, we use large language models (LLMs) combined strategically with verification to soundly scale synthesis without having to manually identify new custom rewrite strategies. Synthesis uses *automated verification* as a subroutine and *guarantees* that the generated control envelope is correct. To make automated verification scale, we build a pipeline to automate theorem proving by having LLMs interact with a theorem prover. This allows scaling of the overall synthesis pipeline.

Finally, having addressed the challenge of synthesizing control conditions given the shape of the control system, we explore the problem of automatically discovering the shape of the control system itself. In practice, engineers often have informal specifications of the control system as natural language design documents. The task of transforming these into the formal input to control envelope synthesis is the problem of *autoformalization*. An experimental exploration of applying LLMs to autoformalization for hybrid games shows both challenges and promise.

Overall, this thesis studies the synthesis of control envelopes for cyber-physical systems. It enables computers to automatically design control envelopes with formal guarantees by recasting control-theoretic reasoning into *syntactic* formulations. This involves introducing a compositional representation and calculus for control envelopes, identifying systematic refinement and rewrite-based approaches to make synthesis effective across different problem classes, and developing frameworks that combine LLMs strategically with formal reasoning to push the boundaries of automatic verification and synthesis. Together, the techniques of this thesis make it possible to automatically derive symbolic constraints defining correct control. This lays the foundation for controllers that are both formally guaranteed to be correct and support high-performance, learning-based optimization.

1.2 Structure

The thesis is structured as follows.

Chapter 2 provides background on the logic that the thesis uses. We use hybrid *games* written in *differential game logic* (dGL) [120], and hybrid *systems* represented in a subset of dGL called *differential dynamic logic* (dL) [115]. dGL uses an imperative programming language-like syntax. It has a deductive proof system, meaning that we can prove statements about the games by applying syntactic, compositional proof rules. The choice of using a symbolic, deductive logic allows us to produce control envelopes with nice properties: they are parametric in input symbols, making them adaptable, they cleanly handle infinite state spaces, and reason precisely about continuous dynamics without discretization. This choice is crucial to our technical approach later in the thesis: it allows the use of classic programming language techniques like symbolic execution and is amenable to the use of LLMs because of proximity to code.

Chapter 3 establishes a challenging case study that will serve as a target for automation. An objective of our work is to be able to synthesize control envelopes complex enough to apply to real-world systems. Designing symbolic control envelopes with high fidelity to real-world dynamics even manually is so challenging that case studies from the literature still usually make

simplifications while modeling the environment and controller logically [91, 99, 100, 105, 126]. This leaves them a few steps away from being immediately applicable to real systems. So we first create, as a target for the synthesis tool, a verified control envelope that solves the practically important problem of train control [68] accounting for all the forces in a realistic train kinematics model [24]. The train control envelope defines what acceleration and braking actions are safe while ensuring that the train never goes past its end of motion authority, past which there is a risk of collision. This case study poses challenges common in other safety-critical embedded systems – complex dynamics with transcendental arithmetic, competing forces with subtle interaction, and effects whose exact magnitude is unknown at proof time.

Chapter 4 uses large language models to automate the process of proving theorems in dGL. This reduces the effort required to verify control envelopes, such as those from the train case study. Moreover, scalable theorem proving becomes a key component of scalable synthesis later in the thesis, because in order to scale, the synthesis procedure heuristically guesses control conditions and relies on verification as an oracle to check correctness. Next, the thesis shifts to the problem of synthesizing control conditions.

Chapter 5 solves control envelope synthesis for an interesting class of problems (with *time-triggered control*, the common controller design pattern where the controller repeatedly polls, chooses an action, and then goes to sleep for some maximum time latency, and *action permanence*, a hybrid analog to idempotence). To do so, it introduces the CESAR (Control Envelope Synthesis via Angelic Refinements) algorithm [69]. We implicitly characterize an optimal solution to control envelope synthesis using hybrid games, and then solve these games to obtain an explicit solution via symbolic execution aided by *game refinement* (transforming the game to be easier to execute symbolically while still preserving some properties).

Chapter 6 generalizes synthesis to all the problems where the shape of control can be characterized by a hybrid game. Given a hybrid game specification defining what is physically possible and what control properties are desired, we synthesize the controller player’s control envelope (nondeterministic policy). To solve the problem in its full generality, this chapter introduces the concept of a *subvalue map*, a symbolic analog to value tables [10], to represent control envelopes such that their verification and synthesis composes at the level of imperative program constructs. The chapter provides a precondition calculus to verify/synthesize such control envelopes. This precondition calculus again makes it possible to use symbolic execution for synthesis, but still runs into subgames that are too complex to symbolically execute. Because of the extreme generality of the problem we tackle in this chapter, unlike in CESAR, no single set of systematic refinements is sufficient to simplify all possible subgames that we can encounter. So instead we use template-based *rewrite heuristics* that help hypothesize solutions when symbolic execution alone fails. The hypothesized solutions are checked to recover correctness.

Chapter 7 increases scalability by leveraging large language models. Instead of heuristics that need to be encoded case-by-case, we use large language models to guess control conditions, with retrospective verification per our developed theory to maintain soundness. The resulting scalability lets us synthesize a control envelope for the train case study. Finally, *Chapter 8* performs an initial exploration of using LLMs to obtain the game specification itself from informal descriptions. While the use of this step *can lose correctness because of the lack of formal guarantees on the correctness of the specification*, it can still be useful to streamline the control envelope design process if used in combination with manual review and checking the generated

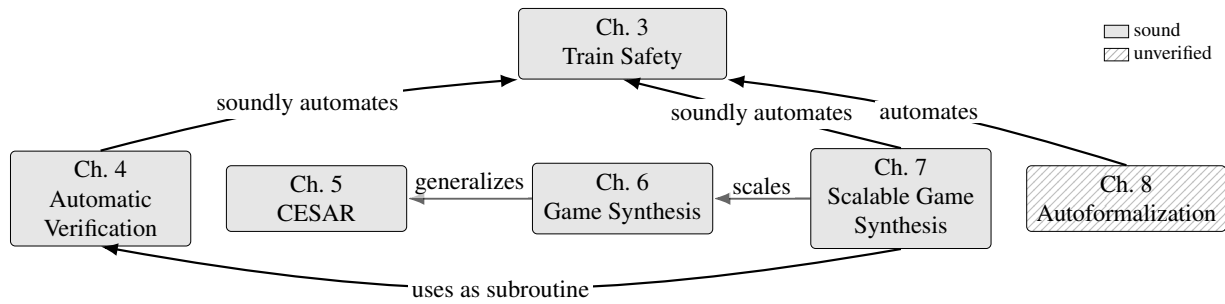


Figure 1.1: Chapter relationships

specification against real-world data when available.

The chapters are organized to be read in order. Fig. 1.1 shows a visualization of the relationship between them.

1.3 Related Work

We situate symbolic control envelopes within the larger scientific context. Individual chapters will have more specific and detailed related work sections pertinent to the specific contributions of each chapter. This thesis is based on some recently published works [67, 69, 71, 73]. Briefly, our work shows the following distinguishing features: 1. it is about permissive control *envelopes*, as opposed to deterministic controllers, 2. it uses *symbolic* reasoning (in contrast with numerical computation) to reason about infinitely many solutions at once, and 3. it reasons compositionally for rich *imperative program-like* control specifications.

Hybrid controllers. Instead of *controllers*, control *envelopes* strive to represent not one but as many safe control solutions of a system as possible, typically infinitely many. For our cyber-physical systems that deal in real-valued variables, this is often the difference between one controller and uncountably infinitely many controllers. From a synthesis perspective, maximally permissive control *envelope* synthesis involves an extra universal quantifier compared to controller synthesis. Thus, controller synthesis techniques [12, 90, 149], for which popular approaches use temporal logic [5, 12, 165], games [107, 154], and CEGIS-like guidance by counterexamples [1, 36, 134, 144], do not immediately apply. Control *envelope* synthesis does not, for example, fit the CEGIS quantifier alternation pattern $\exists\forall$. Controller synthesis approaches with a *compositional logic approach* include refinement [16, 104], Hoare logic [94], *Differential Floyd-Hoare logic* [60], and constructive proof [20]. They soundly construct *deterministic* policies, and do not provide a direct representation of nondeterministic policies (control envelopes) from which to derive *sets* of permissible control actions.

Numeric safety shields computed by numerical methods [2, 47, 80] share the objective of characterizing the space of safe control. Our approach is *symbolic*, with the usual (dis)advantages compared to numerical methods. Numerical safety shields can handle dynamical systems that are hard to analyze symbolically, but typically do not support unbounded state spaces, and can lose

rigorous formal safety guarantees when they discretize continuous systems. They do not produce shields that are *parametric* in the model’s parameters, except by increasing dimensionality which does not scale well. Our choice of a symbolic logic approach, that traditionally requires reasoning through a few manual steps of applying deductive proof rules, also provides a simple path to automation by large language models, which can choose which proof rules to apply and benefit from soundness checking by existing interactive theorem provers.

Runtime monitoring can also be performed using a *symbolic* approach [49, 97, 111, 139]. These runtime monitoring systems *use* control envelopes to monitor the safety of the system at runtime, resulting in *symbolic* shields. They raise an alarm upon deviation from the envelope, and can enforce safe fallback control. They bridge the gap between having a verified control envelope and enforcing it during actual system runs. We *synthesize* the control envelopes in the first place, providing the input to such monitoring systems. For example, in Chapter 3, our verified control envelope is applied by runtime monitoring system ModelPlex [97] to maintain train safety in our experiments.

Game theory and reinforcement learning [140, 146, 157] have the concept of nondeterministic player/agent policies which are similar in spirit to control envelopes. But these works operate at a different level of abstraction than our imperative program-like hybrid games. Also, our work is symbolic rather than numeric, and creates envelopes parametric in the symbolic parameters of the input game. Finally, our control envelope reasoning is syntactically formalized, suitable for computerized verification and formally justified synthesis.

Barrier functions and safe set approaches [4, 6, 33, 35, 96, 131, 145] also identify regions of safe and unsafe control. However, they typically model control by simple mathematical transitions in state space. We again solve for the higher level of abstraction of dL/dGL, with program structures such as loops, as well as the adversarial dynamics of games. Our envelopes indicate which actions, within the complex program structure, are safe, as opposed to just identifying safe regions in state space.

Hybrid automata provide an alternative formalism for modeling hybrid systems and games, in contrast with the logics used in this thesis which define game dynamics by imperative program-like constructs. However, we target specifications in a different formalism, at a different level of abstraction where the game has imperative programming constructs like loops and branches, and is closer to code. This different formalism, with different composition patterns and proof techniques, requires a different technical approach to verification and synthesis. Beyond the difference in the level of abstraction of supported input specifications, automata based formalisms typically use model checking to establish correctness, instead of symbolic proofs, leading to different strengths and weaknesses. When problems related to control envelope synthesis are treated in those frameworks [13, 41, 62, 155, 167], different tradeoffs become necessary, such as remaining within linear dynamics and disallowing model specifications in terms of symbolic parameters to make efficient model checking possible. Our approach lets us use program analysis techniques like loop invariants and their continuous analog, differential invariants, which have

traditionally been harder to automate but in return make it possible to solve classes of problems 482
that are difficult to treat in model checking frameworks. For example, we get access to a proof 483
calculus that is complete for reasoning symbolically about the very general class of dynamics 484
whose solutions are Noetherian functions [129]. This allows our reasoning to be symbolic and 485
parametric, even for highly complex dynamics. In a future where symbolic reasoning becomes 486
easier because of AI-based automation, this would permit powerful generality without sacrificing 487
scalability. 488

Chapter 2

489

Background

490

This chapter recalls the logics used throughout this thesis: differential dynamic logic (dL) [116] for reasoning about hybrid systems, and its extension, differential game logic (dGL) [120], for reasoning about hybrid games. The *compositional, symbolic* representation of hybrid systems that these logics provide will be crucial to our approach in the rest of the thesis. The chapter starts with high-level, intuitive explanations of the logics, followed by formal details for the interested reader.

491
492
493
494
495
496

2.1 Differential Dynamic Logic

497

Differential dynamic logic (dL) is a logic with a deductive proof system for hybrid systems [115, 121]. We provide an introduction to the logic in this chapter, but a more comprehensive treatment can be found in the literature [123].

498
499
500

Differential dynamic logic extends first-order logic with the notion of *hybrid programs*. A hybrid program runs according to a binary relation between states, mapping start states to end states that a program could reach. The program constructs include assignment, for example, $x := e_1$, which instantaneously assigns expression e_1 to variable x . In the special case of nondeterministic assignment, $x := *$, the transition relation accounts for any possible real value being assigned to x .

501
502
503
504
505
506

The test operator, as in $?Q$, aborts the current run if formula Q is false. The continuous evolution operator, $\{x' = f(x) \& Q\}$ follows the ordinary differential equation (ODE) $x' = f(x)$ for some nondeterministic amount of time, with evolution domain constraint Q being true throughout the evolution. Sequential composition, $\alpha; \beta$, runs program α followed by program β , for example, a discrete train controller α followed by the train's ODE β . The nondeterministic choice operator $\alpha \cup \beta$ runs either program α or β , for example, either accelerate the train with α or brake with β . The loop operator α^* runs hybrid program α any nondeterministically chosen $n \geq 0$ times. It is useful, e.g., for running a train control loop indefinitely. Eq. (2.1) summarizes the syntax of hybrid programs.

507
508
509
510
511
512
513
514
515

$$\alpha, \beta \equiv x := e \mid \alpha; \beta \mid ?Q \mid \alpha^* \mid \alpha \cup \beta \mid x := * \mid \{x' = f(x) \& Q\} \quad (2.1)$$

516

To express safety properties about hybrid programs, we use the box modality $[\alpha]Q$, which is

517

true in any state from which all runs of hybrid program α end in states in which the formula Q is true. For liveness properties, we use the diamond modality $\langle \alpha \rangle Q$, which is true in any state from which there is at least one run of hybrid program α that ends in a state in which Q is true. Besides these modalities, **dL** formulas are generated by the usual constructs of first-order logic, as summarized by Eq. (2.2), where $\sim \in \{<, \leq, =, \geq, >\}$ and θ_1, θ_2 are arithmetic expressions in $+$, $-$, $\cdot$, $/$ over the reals, x is a variable, α is a hybrid program.

$$\phi := \theta_1 \sim \theta_2 \mid \neg \phi \mid \phi \wedge \psi \mid \phi \vee \psi \mid \phi \rightarrow \psi \mid \forall x \phi \mid \exists x \phi \mid [\alpha] \phi \mid \langle \alpha \rangle \phi \quad (2.2)$$

For proofs of **dL** formulas, we use **dL** inference rules [117, 122]. To provide a flavor of how these rules work, we discuss a few here: *loop* (loop), *differential invariant* (dI), and *differential cut* (dC). A *sequent* $\Gamma \vdash \Delta$ with a finite list of antecedent formulas Γ and succedent formulas Δ is short for $\bigwedge_{\phi \in \Gamma} \phi \rightarrow \bigvee_{\psi \in \Delta} \psi$. An *inference rule* has the form

$$\frac{\text{premise 1} \quad \cdots \quad \text{premise n}}{\text{conclusion}},$$

where the premises and conclusion are sequents. The rule can be read as saying that if all premises are valid, then the conclusion is valid. We next show examples of **dL** inference rules.

$$\begin{aligned} \text{(loop)} \quad & \frac{\Gamma \vdash J \quad J \vdash [\alpha]J \quad J \vdash P}{\Gamma \vdash [\alpha^*]P} \\ \text{(dI)} \quad & \frac{\Gamma \vdash e_1 \leq e_2 \quad Q \vdash [x' := f(x)]e'_1 \leq e'_2}{\Gamma \vdash [x' = f(x) \ \& \ Q]e_1 \leq e_2} \\ \text{(dC)} \quad & \frac{\Gamma \vdash [x' = f(x) \ \& \ Q \wedge C]P \quad \Gamma \vdash [x' = f(x) \ \& \ Q]C}{\Gamma \vdash [x' = f(x) \ \& \ Q]P} \end{aligned}$$

As usual, the loop rule uses an invariant J that holds initially, inductively after each step, and implies the post condition that we seek to prove. A differential invariant preserves properties along the flow of a differential equation: if $e_1 \leq e_2$ initially and e_1 grows slower than e_2 , so $e'_1 \leq e'_2$ (where e'_1 and e'_2 are evaluated after substituting in the assignment $x' := f(x)$ from the differential equation), then it remains true that $e_1 \leq e_2$. A more general rule form and its explanation can be found in the literature [121, 122]. The idea behind a differential cut is that if formula C holds true at the end of every possible run of differential equation $x' = f(x)$, then C must hold true throughout its evolution. Differential cuts can be used to accumulate knowledge about a differential equation.

Fig. 2.1 shows an example of a **dL** proof. It proves that if initially position x is nonnegative and velocity v is nonnegative, then x remains nonnegative after following the ODE $x' = v$ for any duration of time. The theorem proved is at the bottom, $x \geq 0 \wedge v \geq 0 \vdash [x' = v]x \geq 0$. Applying proof rules creates a proof tree upwards. The proof uses the differential invariant rule (dI) to show that $x \geq 0$ throughout the evolution without actually having to solve the ODE by arguing instead that initially, $x \geq 0$, and based on its derivative x' , it never decreases. This ability to establish properties of ODEs without solving them is a key advantage when dealing with more complex systems where closed-form solutions may not be available or lie outside the decidable fragment of arithmetic. The proof also uses the differential cut rule (dC) to transform the conclusion into the right form to apply dI. Upon reaching a sequent such as $x \geq 0 \vdash x \geq 0$

that is free of modalities, it is possible to use a solver such as Z3 [39] to discharge the proof obligation (denoted by rule $\mathbb{R}$).

$$\begin{array}{c}
 \begin{array}{c}
 \mathbb{R} \frac{*}{x \geq 0 \vdash x \geq 0} \quad \mathbb{R} \frac{v \geq 0 \vdash v \geq 0}{v \geq 0 \vdash [x' := v]x' \geq 0} \\
 \text{dl} \frac{x \geq 0 \wedge v \geq 0 \vdash [x' = v \ \& \ v \geq 0]x \geq 0}{x \geq 0 \wedge v \geq 0 \vdash [x' = v]x \geq 0} \quad \text{truncated for brevity} \\
 \text{dc} \frac{x \geq 0 \wedge v \geq 0 \vdash [x' = v]x \geq 0}{x \geq 0 \wedge v \geq 0 \vdash [x' = v]x \geq 0}
 \end{array}
 \end{array}$$

Figure 2.1: Example of a dL proof tree.

2.2 Differential Game Logic

Differential game logic (dGL) is a logic for two-player hybrid games that has a relatively complete axiomatization. A detailed explanation can be found elsewhere [120, 123], but we provide an overview. A dGL game is played by two adversarial players, canonically called Angel and Demon. It extends dL with two versions of each program construct where there is nondeterminism: one where Angel gets to choose how the nondeterminism is resolved, and one where Demon gets to choose how it is resolved.

An example of a game is $(v := 1 \cap v := -1); \{x' = v\}$. In this game, first Demon chooses between setting velocity v to 1, or to -1. Then, Angel evolves position x as $x' = v$ for a duration of her choice. Differential game logic uses modalities to set win conditions for the players. For example, in the formula $[(v := 1 \cap v := -1); \{x' = v\}]x \neq 0$, Demon wins the game when $x \neq 0$ at the end of the game and Angel wins otherwise. The overall formula represents the set of states from which Demon can win the game, which is $x \neq 0$ because when $x < 0$, Demon has the *winning strategy* to pick $v := -1$, so no matter how long Angel evolves $x' = v$, x remains negative. Likewise, when $x > 0$, Demon can pick $v := 1$. However, when $x = 0$, Angel has a winning strategy: to evolve $x' = v$ for zero time, so that x remains zero regardless of Demon's choice.

Table 2.1 summarizes dGL's program notation. As before, assignment $x := \theta$ instantly changes the value of variable x to the value of θ . Challenge $? \psi$ continues the game if ψ is satisfied in the current state, otherwise Angel loses immediately. In continuous evolution $x' = \theta \ \& \ \psi$ Angel follows the differential equation $x' = \theta$ for some duration of her choice, but loses immediately on violating ψ at any time. Sequential game $\alpha; \beta$ first plays α and when it terminates without a player having lost, continues with β . Choice $\alpha \cup \beta$ lets Angel choose whether to play α or β . For repetition α^* , Angel repeats α some number of times, choosing to continue or terminate after each round. The dual game α^d switches the roles of players. For example, in the game $? \psi^d$, Demon passes the challenge if the current state satisfies ψ , and otherwise loses immediately. Notice that in games restricted to the structures listed above but without α^d , all choices are resolved by Angel alone with no adversary, and hybrid games coincide with hybrid systems in differential dynamic logic (dL) [123]¹.

¹The dL fragment still does have special properties like completeness results and additional proof rules than dGL [123, 128].

In order to express properties about hybrid games, differential game logic formulas refer to the existence of winning strategies for objectives of the games (e.g., a controller has a winning strategy to achieve collision avoidance despite an adversarial environment). The set of dGL formulas is generated by the same grammar as dL, shown in Eq. (2.2), except that α is a hybrid game instead of a hybrid program.

Comparisons of arithmetic expressions, Boolean connectives, and quantifiers over the reals are as usual. The modal formula $\langle \alpha \rangle \phi$ expresses that player Angel has a winning strategy to reach a state satisfying ϕ in hybrid game α . Modal formula $[\alpha] \phi$ expresses the same for Demon. The fragment without modalities is first-order real arithmetic. Its fragment without quantifiers is called *propositional arithmetic* $\mathcal{P}_{\mathbb{R}}$. $[\alpha \cup \beta] \phi$ expresses that Demon has a winning strategy when Angel chooses between α and β to achieve ϕ , while $[\alpha \cap \beta] \phi$ expresses that Demon has a winning strategy to achieve ϕ when Demon chooses whether to play game α or β . Correspondingly, $\langle \alpha \cap \beta \rangle \phi$ expresses that Angel has a winning strategy to achieve ϕ when Demon has a choice between α and β . A formula ϕ is *valid*, written $\models \phi$, iff it is true in every state ω . States are functions assigning a real number to each variable. For instance, $\phi \rightarrow [\alpha] \psi$ is valid iff, from all initial states satisfying ϕ , Demon has a winning strategy in game α to achieve ψ .

2.3 Semantics

This section recalls the semantics of dL [116] and dGL [120]. It is not essential to understanding the thesis at a high level, but is a quick reference for readers interested in the formal details. We list the semantics after introducing the necessary notation.

ω denotes a state, which is a mapping from variables to real numbers. $\mathcal{S}$ denotes the set of all states. A state maps each variable to a real number. $\omega[x]$ denotes the value of variable x in state ω . X^c denotes the complement of set X . ω_r denotes the state that is the same as ω except that variable x maps to value r .

Table 2.1: Hybrid game operators for two-player hybrid systems

Game	Effect
$x := \theta$	assign value of term θ to variable x
$?\psi$	Angel passes challenge if formula ψ holds in current state, else loses immediately
$(x'_1 = \theta_1, \dots, x'_n = \theta_n \ \& \ \psi)$	Angel evolves x_i along differential equation system $x'_i = \theta_i$ for choice of duration ≥ 0 , loses immediately when violating ψ
$\alpha; \beta$	sequential game, first play hybrid game α , then hybrid game β
$\alpha \cup \beta$	Angel chooses to follow either hybrid game α or β
α^*	Angel repeats hybrid game α , choosing to stop or go after each α
α^d	dual game switches player roles between Angel and Demon
$\alpha \cap \beta$	demonic choice $(\alpha^d \cup \beta^d)^d$ gives choice between α and β to Demon
$\alpha^\times$	demonic repetition $((\alpha^d)^*)^d$ gives control of repetition to Demon

2.3.1 Differential Dynamic Logic

Definition 2.1 (dL formula semantics). The *semantics of a dL formula* ϕ is the subset $\llbracket \phi \rrbracket \subseteq \mathcal{S}$ of states in which ϕ is true. It is defined inductively as follows

1. $\llbracket \theta_1 \sim \theta_2 \rrbracket = \{\omega \in \mathcal{S} : \omega \llbracket \theta_1 \rrbracket \sim \omega \llbracket \theta_2 \rrbracket\}$ where $\sim \in \{<, \leq, =, \geq, >\}$
2. $\llbracket \neg \phi \rrbracket = (\llbracket \phi \rrbracket)^c$
3. $\llbracket \phi \wedge \psi \rrbracket = \llbracket \phi \rrbracket \cap \llbracket \psi \rrbracket$
4. $\llbracket \phi \vee \psi \rrbracket = \llbracket \phi \rrbracket \cup \llbracket \psi \rrbracket$
5. $\llbracket \phi \rightarrow \psi \rrbracket = (\llbracket \phi \rrbracket)^c \cup \llbracket \psi \rrbracket$
6. $\llbracket \exists x \phi \rrbracket = \{\omega \in \mathcal{S} : \omega_x^r \in \llbracket \phi \rrbracket \text{ for some } r \in \mathbb{R}\}$
7. $\llbracket \forall x \phi \rrbracket = \{\omega \in \mathcal{S} : \omega_x^r \in \llbracket \phi \rrbracket \text{ for all } r \in \mathbb{R}\}$
8. $\llbracket p(\theta_1, \dots, \theta_k) \rrbracket = \{\omega \in \mathcal{S} : (\omega \llbracket \theta_1 \rrbracket, \dots, \omega \llbracket \theta_k \rrbracket) \in (p)\}$
9. $\llbracket \langle \alpha \rangle \phi \rrbracket = \{\omega : \nu \in \llbracket \phi \rrbracket \text{ for some state } \nu \text{ with } (\omega, \nu) \in \llbracket \alpha \rrbracket\}$
where $\llbracket \alpha \rrbracket$ is the semantics of hybrid program α in Def. 2.2.
10. $\llbracket [\alpha] \phi \rrbracket = \{\omega : \nu \in \llbracket \phi \rrbracket \text{ for all states } \nu \text{ such that } (\omega, \nu) \in \llbracket \alpha \rrbracket\}$
where $\llbracket \alpha \rrbracket$ is the semantics of hybrid program α in Def. 2.2.

Definition 2.2 (Semantics of hybrid programs). A hybrid program α is interpreted as a binary relation on states $\llbracket \alpha \rrbracket \subseteq \mathcal{S} \times \mathcal{S}$, mapping start states to end states that a program could reach, defined inductively as follows [116]:

1. $\llbracket x := \theta \rrbracket = \{(\omega, \nu) : \nu = \omega_x^{\llbracket \theta \rrbracket}\}$
2. $\llbracket x := * \rrbracket = \{(\omega, \nu) : \nu = \omega_x^r \text{ for some } r \in \mathbb{R}\}$
3. $\llbracket ?Q \rrbracket = \{(\omega, \omega) : \omega \in \llbracket Q \rrbracket\}$
4. $\llbracket \{x' = f(x) \ \& \ Q\} \rrbracket = \{(\omega, \nu) : \varphi(0) = \omega \text{ except at } x' \text{ and } \varphi(r) = \nu \text{ for a solution } \varphi : [0, r] \rightarrow \mathcal{S} \text{ of any duration } r \text{ satisfying } \varphi \models x' = f(x) \wedge Q\}$
5. $\llbracket \alpha \cup \beta \rrbracket = \llbracket \alpha \rrbracket \cup \llbracket \beta \rrbracket$
6. $\llbracket \alpha; \beta \rrbracket = \llbracket \alpha \rrbracket \circ \llbracket \beta \rrbracket = \{(\omega, \nu) : (\omega, \mu) \in \llbracket \alpha \rrbracket \text{ and } (\mu, \nu) \in \llbracket \beta \rrbracket\}$
7. $\llbracket \alpha^* \rrbracket = \llbracket \alpha \rrbracket^* = \bigcup_{n \in \mathbb{N}} \llbracket \alpha \rrbracket^n$, where $\alpha^{n+1} = \alpha^n ; \alpha$ and $\alpha^0 \equiv \text{?true}$.

2.3.2 Differential Game Logic

Def. 2.3 defines the semantics of dGL formulas. It uses $\varsigma_\alpha(\llbracket \phi \rrbracket)$ to denote *Angel's winning region function* (applied for game α and Angel's winning condition $\llbracket \phi \rrbracket$) which will be defined later in Def. 2.4. Similarly, $\delta_\alpha(\llbracket \phi \rrbracket)$ shows *Demon's winning region function* which will be defined later in Def. 2.4.

Definition 2.3 (dGL semantics). The *semantics of a dGL formula* ϕ is the subset $\llbracket \phi \rrbracket \subseteq \mathcal{S}$ of states in which ϕ is true. It is defined inductively as follows

1. $\llbracket \theta_1 \sim \theta_2 \rrbracket = \{\omega \in \mathcal{S} : \omega \llbracket \theta_1 \rrbracket \sim \omega \llbracket \theta_2 \rrbracket\}$ where $\sim \in \{<, \leq, =, \geq, >\}$
2. $\llbracket \neg \phi \rrbracket = (\llbracket \phi \rrbracket)^c$
3. $\llbracket \phi \wedge \psi \rrbracket = \llbracket \phi \rrbracket \cap \llbracket \psi \rrbracket$
4. $\llbracket \phi \vee \psi \rrbracket = \llbracket \phi \rrbracket \cup \llbracket \psi \rrbracket$
5. $\llbracket \phi \rightarrow \psi \rrbracket = (\llbracket \phi \rrbracket)^c \cup \llbracket \psi \rrbracket$
6. $\llbracket \exists x \phi \rrbracket = \{\omega \in \mathcal{S} : \omega_x^r \in \llbracket \phi \rrbracket \text{ for some } r \in \mathbb{R}\}$

- 646 7. $\llbracket \forall x \phi \rrbracket = \{\omega \in \mathcal{S} : \omega_x^r \in \llbracket \phi \rrbracket \text{ for all } r \in \mathbb{R}\}$
- 647 8. $\llbracket p(\theta_1, \dots, \theta_k) \rrbracket = \{\omega \in \mathcal{S} : (\omega \llbracket \theta_1 \rrbracket, \dots, \omega \llbracket \theta_k \rrbracket) \in (p)\}$
- 648 9. $\llbracket \langle \alpha \rangle \phi \rrbracket = \varsigma_\alpha(\llbracket \phi \rrbracket)$
- 649 10. $\llbracket [\alpha] \phi \rrbracket = \delta_\alpha(\llbracket \phi \rrbracket)$

650 A dGL formula ϕ is *valid*, written $\models \phi$, iff it is true in all states, i.e. $\llbracket \phi \rrbracket = \mathcal{S}$.

651 **Definition 2.4** (Semantics of hybrid games). The *semantics of a hybrid game* α is a function
 652 $\varsigma_\alpha(\cdot)$ that, for each set of Angel's winning states $X \subseteq \mathcal{S}$, gives the *winning region*, i.e. the set
 653 of states $\varsigma_\alpha(X)$ from which Angel has a winning strategy to achieve X in α (whatever strategy
 654 Demon chooses). It is defined inductively as follows

- 655 1. $\varsigma_{x=\theta}(X) = \{\omega \in \mathcal{S} : \omega_x^{\omega \llbracket \theta \rrbracket} \in X\}$
- 656 2. $\varsigma_{x'=f(x) \& Q}(X) = \{\varphi(0) \in \mathcal{S} : \varphi(r) \in X \text{ for some } r \in \mathbb{R}_{\geq 0} \text{ and (differentiable)}$
 657 $\varphi : [0, r] \rightarrow \mathcal{S} \text{ such that } \varphi(\zeta) \in \llbracket Q \rrbracket \text{ and } \frac{d\varphi(t)(x)}{dt}(\zeta) = \varphi(\zeta) \llbracket f(x) \rrbracket \text{ for all } 0 \leq \zeta \leq r\}$
- 658 3. $\varsigma_{?Q}(X) = \llbracket Q \rrbracket \cap X$
- 659 4. $\varsigma_{\alpha \cup \beta}(X) = \varsigma_\alpha(X) \cup \varsigma_\beta(X)$
- 660 5. $\varsigma_{\alpha; \beta}(X) = \varsigma_\alpha(\varsigma_\beta(X))$
- 661 6. $\varsigma_{\alpha^*}(X) = \bigcap \{Z \subseteq \mathcal{S} : X \cup \varsigma_\alpha(Z) \subseteq Z\}$
- 662 7. $\varsigma_{\alpha^d}(X) = (\varsigma_\alpha(X^c))^c$

663 The *winning region* of Demon, i.e. the set of states $\delta_\alpha(X)$ from which Demon has a winning
 664 strategy to achieve X in α (whatever strategy Angel chooses) is defined inductively as follows

- 665 1. $\delta_{x=\theta}(X) = \{\omega \in \mathcal{S} : \omega_x^{\omega \llbracket \theta \rrbracket} \in X\}$
- 666 2. $\delta_{x'=f(x) \& Q}(X) = \{\varphi(0) \in \mathcal{S} : \varphi(r) \in X \text{ for all } r \in \mathbb{R}_{\geq 0} \text{ and (differentiable)}$
 667 $\varphi : [0, r] \rightarrow \mathcal{S} \text{ such that } \varphi(\zeta) \in \llbracket Q \rrbracket \text{ and } \frac{d\varphi(t)(x)}{dt}(\zeta) = \varphi(\zeta) \llbracket \theta \rrbracket \text{ for all } 0 \leq \zeta \leq r\}$
- 668 3. $\delta_{?Q}(X) = (\llbracket Q \rrbracket)^c \cup X$
- 669 4. $\delta_{\alpha \cup \beta}(X) = \delta_\alpha(X) \cap \delta_\beta(X)$
- 670 5. $\delta_{\alpha; \beta}(X) = \delta_\alpha(\delta_\beta(X))$
- 671 6. $\delta_{\alpha^*}(X) = \bigcup \{Z \subseteq \mathcal{S} : Z \subseteq X \cap \delta_\alpha(Z)\}$
- 672 7. $\delta_{\alpha^d}(X) = (\delta_\alpha(X^c))^c$

673 The dGL axioms and proof rules [120] used in this thesis are summarized in Fig. 2.2.

$$\begin{aligned}
([\cdot]) \quad & [\alpha]P \leftrightarrow \neg \langle \alpha \rangle \neg P \\
(\langle := \rangle) \quad & \langle x := e \rangle p(x) \leftrightarrow p(e) \\
(\langle ' \rangle) \quad & \langle x' = f(x) \rangle p(x) \leftrightarrow \exists t \geq 0 \langle x := y(t) \rangle p(x) \quad (y'(t) = f(y)) \\
(\langle ? \rangle) \quad & \langle ?Q \rangle P \leftrightarrow Q \wedge P \\
(\langle \cup \rangle) \quad & \langle \alpha \cup \beta \rangle P \leftrightarrow \langle \alpha \rangle P \vee \langle \beta \rangle P \\
(\langle ; \rangle) \quad & \langle \alpha; \beta \rangle P \leftrightarrow \langle \alpha \rangle \langle \beta \rangle P \\
(\langle * \rangle) \quad & \langle \alpha^* \rangle P \leftrightarrow P \vee \langle \alpha \rangle \langle \alpha^* \rangle P \\
(\langle^d \rangle) \quad & \langle \alpha^d \rangle P \leftrightarrow \neg \langle \alpha \rangle \neg P \\
(\langle \cap \rangle) \quad & \langle \alpha \cap \beta \rangle P \leftrightarrow \langle \alpha \rangle P \wedge \langle \beta \rangle P \\
(\langle^\times \rangle) \quad & \langle \alpha^\times \rangle P \leftrightarrow P \wedge \langle \alpha \rangle \langle \alpha^\times \rangle P \\
([:=]) \quad & [x := e]p(x) \leftrightarrow p(e) \\
([']) \quad & [x' = f(x)]p(x) \leftrightarrow \forall t \geq 0 [x := y(t)]p(x) \quad (y'(t) = f(y)) \\
([?]) \quad & [?Q]P \leftrightarrow (Q \rightarrow P) \\
([\cup]) \quad & [\alpha \cup \beta]P \leftrightarrow [\alpha]P \wedge [\beta]P \\
([;]) \quad & [\alpha; \beta]P \leftrightarrow [\alpha][\beta]P \\
([*]) \quad & [\alpha^*]P \leftrightarrow P \wedge [\alpha][\alpha^*]P \\
([^d]) \quad & [\alpha^d]P \leftrightarrow \neg[\alpha]\neg P \\
([\cap]) \quad & [\alpha \cap \beta]P \leftrightarrow [\alpha]P \vee [\beta]P \\
([\times]) \quad & [\alpha^\times]P \leftrightarrow P \vee [\alpha][\alpha^\times]P \\
\\
(\text{loop}) \quad & \frac{\Gamma \vdash J, \Delta \quad J \vdash [\alpha]J \quad J \vdash P}{\Gamma \vdash [\alpha^*]P, \Delta} \quad (\text{ind}) \quad \frac{P \rightarrow [\alpha]P}{P \rightarrow [\alpha^*]P} \\
(\text{M}) \quad & \frac{P \rightarrow Q}{\langle \alpha \rangle P \rightarrow \langle \alpha \rangle Q} \quad (\text{FP}) \quad \frac{P \vee \langle \alpha \rangle Q \rightarrow Q}{\langle \alpha^* \rangle P \rightarrow Q} \\
(\text{M}[\cdot]) \quad & \frac{P \rightarrow Q}{[\alpha]P \rightarrow [\alpha]Q} \quad (\text{FP}^\times) \quad \frac{P \vee [\alpha]Q \rightarrow Q}{[\alpha^\times]P \rightarrow Q}
\end{aligned}$$

Figure 2.2: dGL axiomatization and derived axioms and rules

Chapter 3

674

Verified Train Controllers

675

An objective of this thesis is to be able to synthesize control envelopes complex enough to provide controllers for real-world systems. However, designing and verifying symbolic control envelopes with high fidelity to real-world dynamics is so challenging that most existing work makes simplifications in their physical *model* of the world (e.g., ignoring resistance in train motion), or handles only a subset of situations (e.g. only handling single-lane highway driving for autonomous car control). To demonstrate a symbolic control envelope that is complex enough to apply to the real world, we create one for *train control* that uses a detailed physical model of train motion¹. Simulated experiments demonstrate the utility of this train control envelope: in induced controllers, train safety is maintained, and by careful formal reasoning, inefficient, forced early braking is limited. This chapter is based on published work [68]. Subsequently, the goal of this thesis becomes making synthesis possible for control envelopes such as this one.

676
677
678
679
680
681
682
683
684
685
686

3.1 Introduction

687

This chapter uses control envelopes as a tool to design verified train controllers. This is a practically important, representative problem with challenges common in other safety-critical embedded systems: complex dynamics with transcendental arithmetic, competing forces with subtle interaction, and effects whose exact magnitude is unknown at proof time. Train controllers decide when to enforce braking to prevent movement authority² violation and collisions. They must account for all the competing influences that govern train motion. Uphill slopes decrease velocity, for example, which decreases resistance, which permits a more rapid increase in velocity, slope and curve effect, all while the train's brake force builds gradually until saturation as air pressure propagates along brake pipes. These complex interactions make it hard to design a safe and efficient train controller, and even harder to ensure it is always safe. This chapter designs and verifies train control envelopes (sets of controllers) for a realistic freight train kinematics model [23, 24], henceforth called the Federal Railroad Administration model (FRA model) after its appearance in influential technical reports published by the Federal Railroad Administration.

688
689
690
691
692
693
694
695
696
697
698
699
700

¹Train control envelopes are practically important and already widely deployed. They are also called *train protection systems*, or in the context of the specific control objective of safe braking, *positive train control* systems.

²The movement authority of a train is the region of the track where it can safely be per the scheduling system.

In handling these dynamics, this chapter also identifies generalizable verified control envelope design techniques.

Existing studies of formally verified train motion [100, 126, 168] do not account for at least two effects amongst track grade, track curvature, resistance, and air brake propagation time, rendering their results inapplicable to most real-world scenarios. We surmount the challenges of verification against the full dynamics of the FRA model, in which these effects interact subtly with each other. Our verification results are significant, because these parameters influence the motion of the train in safety-critical and/or performance-critical ways. Neglecting track slope profile and the gradual propagation of air pressure braking, in particular, can render otherwise verifiably safe train controllers unsafe, since their influence may diminish the train’s ability to decelerate, causing collisions. Our verification is valid for the realistic FRA model [23, 24].

Before verifying control envelope safety, we first design the control envelopes, balancing efficiency with provable safety. Conservative control envelopes are mathematically more simplistic, and easier to design and verify, but make railway operations inefficient, violating performance objectives. We start by presenting a conservative safe control envelope and then demonstrate how to iteratively make it more efficient by exploiting characteristics of the physical train dynamics for better but safe control.

Train control envelopes are assessed relative to a (changing) destination stopping point—called *end of movement authority*: overshoot of the end of movement authority is a safety violation, because that risks collision with other trains; efficiency is measured in terms of end of movement authority undershoot. We prove absence of end of movement authority overshoot when using our control envelopes in the FRA model by verification and demonstrate efficiency by simulation.

To keep our proofs as general and widely applicable as possible, we leverage nondeterministic controllers (control *envelopes*) and a paradigm of mathematical abstraction. Each control envelope is set-valued such that *all* of its control choices are simultaneously proved safe under all circumstances in the FRA model. The safety of these control envelopes implies the safety of *all* of their specializations [92], giving railroads significant freedom in how to adapt the verified control envelopes for their purposes. Control envelope verification follows a two-stage process: we first prove mathematical models of abstract train control motion, and then obtain proofs of the actual physical models of train control by *uniform substitution* [121] to replace the abstract function symbols of the mathematical models with physical terms specific to the FRA model or even specific railroads.

Crucially, our approach uses three different types of models: (i) the high-fidelity physics model describing the kinematic motion of trains along the track, (ii) our generalized mathematical abstractions of the physics model, and (iii) the simplified but *computable* approximations of motion models used by the respective train control envelopes. Our verification results prove that the safety of (i) derives from the safety of (ii) and that all control decisions following (iii) are safe in (ii).

Our proof is written in differential dynamic logic [115, 121], and performed using the hybrid systems theorem prover KeYmaera X [51]. We compare the efficiency with concrete control algorithms [23] for a number of train consists (arrangement of locomotives and cars) and scenarios [23]. Our verified control envelopes are imposed on unverified controllers via runtime enforcement checks using ModelPlex [98]. We illustrate their behavior in simulation. The proved

models are permissive and only interfere in train control operation when acting otherwise risks movement authority violation.

Contributions: The primary contribution of this chapter is the formal verification results justifying the safety of train control in the FRA model. The secondary contribution is the design of new, safe train control envelopes. Technical contributions are the ideas of the formal proofs, and of substituting a specialized ground physics model into control envelopes designed with an abstract mathematical model. Finally, our simulations demonstrate the impact of verified train control envelopes in different terrain. Proofs are available at <https://doi.org/10.1184/R1/19542610> with ideas sketched inline.

3.2 Related Work

Due to their significance as safety-critical transportation systems, there have been many efforts to verify the safety of train control systems [46]. One approach is extensive simulation of train braking models [43, 152]. However, simulation can only show safety in a limited number of cases and is less appropriate when free acceleration is interspersed with braking. Similar limitations apply to test-based safety assurance of train control [27]. Our work uses differential dynamic logic (dL), a logic with a deductive proof system for hybrid systems [115, 121]. We write mathematical proofs that guarantee safety over the infinite state space in a model of physical motion.

In the realm of formal verification, there have been many studies of railway systems [14, 32, 45]. Discrete aspects of train control have been verified at industrial scale [44, 55]. Many studies [21, 31, 76, 77, 158] focus on scheduling trains to avoid route intersections. Train communication systems have been formally verified [45, 159]. Such studies are complementary to our work, which focuses on the motion of the train as it interacts with the environment. Results on the correctness of the motion of trains permit correct interaction with scheduling.

Studies of train motion have verified the European train control system with moving blocks [126], and the Chinese train control system [168], while ignoring the effect of resistance, air brakes, track grade and curvature. The FRA model has been verified while ignoring grade and curve [100]. Our work differs by accounting for *all* forces in the FRA model, creating controllers designed and verified against realistic physics.

3.3 FRA Model of Train Kinematics

This section introduces the FRA model, which provides the forces acting on a train [24, Eq. (1)]. After the net force on the train has been identified, Newton’s second law, using train mass, determines the acceleration that the train experiences, which in turn determines change in velocity and change in position for train control design. The forces are

$$\sum F = -F_G - F_C - F_B + F_L - F_R = m_T a \quad (3.1)$$

where F_G denotes force due to track grade, F_C resistance due to track curvature, F_B force from the brakes, F_L force from the locomotive engine (tractive effort), F_R resistive forces, and m_T ,

781 train mass. Newton's second law, $\sum F = m_T a$, determines train acceleration a . Resistive force
 782 F_R follows the modified Davis equation [24, Eq. (18)]:

$$783 \quad F_R = A_R w + B_R n + C_R w v + D_R v^2 \quad ,$$

784 where A_R, B_R, C_R, D_R are experimentally determined positive constants whose numerical val-
 785 ues, given a choice of units, can be found in other sources [23, 37, 61]. Further, n is the number
 786 of axles, and w is the weight of the train.

787 Grade and curve forces depend on the train position p on the track. Grade force [24, Eq. (19)]
 788 is proportional to train weight w and average track grade $grade(p)$ underneath the train:

$$789 \quad F_G = A_G w \cdot grade(p) \quad .$$

790 Similarly, curve force [24, Eq. (21)] is a function of average track curvature $curve(p)$ along the
 791 train and train weight w :

$$792 \quad F_C = A_C w \cdot curve(p) \quad .$$

793 where A_C and A_G are positive multiplicative constants. Braking force can be modeled as the
 794 minimum of two linear functions to capture the effect of air pressure brake force buildup [24,
 795 Fig. 17] and stabilization. Let F_{B0} be the force acting immediately on brake application, f_p , the
 796 slope with which brake force increases, t , the elapsed time, and $F_{B\max}$, the force after air pressure
 797 in the air pressure brakes saturates.

$$798 \quad F_B = \min(F_{B0} + f_p t, F_{B\max}) \quad .$$

799 Brake enforcement and train protection algorithms [23, 24] approximately solve a differential
 800 equation derived from Eq. (3.1) to estimate the velocity and position of the train at future times.

801 3.4 Mathematical Model Abstraction

802 In order to maximize generality of the embedded software design, minimize verification ef-
 803 fort, and simplify future proof maintenance, we present a mathematical abstraction of the FRA
 804 model [23, 24]. Concrete verified control envelopes and their safety proofs for the fully ex-
 805 panded model can be obtained automatically from the verified abstract model using uniform
 806 substitution [121].

807 Our abstract train kinematic model in (3.2) is an ODE in time. The rate of change of position
 808 is velocity, and the rate of change of velocity is acceleration. The variables and constants in-
 809 volved, along with their signs, when relevant, are (i) Train position p , (ii) velocity v , (iii) velocity
 810 and position-independent component of acceleration a_l ranging from immediate braking ability
 811 $-b_{\max} < 0$ to maximum train engine acceleration $a_{\max} > 0$, (iv) acceleration due to air brakes
 812 a_a in range $a_{b\max} < 0$ to 0, (v) rate of change m_b of air brake acceleration, which is $m_p < 0$
 813 when brakes are ramping up and 0 otherwise, (vi) map a_s from position to acceleration due to
 814 grade, (vii) map a_c from position to acceleration due to curvature, and (viii) velocity-dependent
 815 resistance a_r . In the chosen sign convention, resistive acceleration is negative.

$$816 \quad p' = v, v' = a_l + a_a + a_s(p) + a_r(v) + a_c(p), a'_b = m_b \quad (3.2)$$

with $a_l \in [-b_{\max}, a_{\max}]$, $a_a = \max(a_b, a_{b\max})$, $m_b \in \{0, m_p\}$ 817

The Davis equation resistance $a_r(v) = -\frac{C_R w v + D_R v^2}{m_T}$ has the shape $a_r = a_1 v + a_2 v^2$ when 818
 $a_1 = -C_r g$ with gravity g summarizes the linear coefficient of velocity, and $a_2 = -\frac{D_R}{m_T}$ summa- 819
 rizes the quadratic coefficient. Grade and curvature are represented by unspecified but bounded 820
 functions a_s and a_c that map train positions to a numeric value for acceleration due to slope and 821
 average curvature, respectively. The quantity a_l summarizes locomotive tractive effort ($a_l \geq 0$) 822
 and train deceleration ($a_l < 0$) as commanded by the train controller, with adjustment for the 823
 velocity-independent resistance. 824

We later instantiate the proved abstract kinematic train model by **dl**'s uniform substitution 825
 [121] to easily get proofs for specific physical train models such as the FRA model. Similarly, 826
 proofs for specific train configurations result from also substituting values for coefficients, or 827
 even for a specific train state when additionally substituting speed and position. 828

3.5 Model Structure 829

We develop a conservative train control envelope in **dl** based on the abstract train kinematic 830
 model Eq. (3.2) of Section 3.3, presenting it modularly and introducing conceptually important 831
 model components and functions along the way. We address the challenge of representing track 832
 grade and curve, which are unknown at proof time, using unspecified maps. In order to reason 833
 about them, we bound the maps with assumptions quantifying over all arguments. Our solution 834
 permits us to capture the full Eq. (3.2) without conservatively neglecting $a_s(p)$ and $a_c(p)$ when 835
 reasoning about it during verification. It generalizes to other embedded software that must reason 836
 about unspecified, bounded functions, such as noise or potential fields (e.g. electro-magnetic or 837
 gravitational effect). 838

We prove the control envelope safe: relative to Eq. (3.2), the control envelope will provably 839
 never permit the train's position to exceed the end of movement authority e , though it might be 840
 inefficient, braking unnecessarily early. Later, by revising modular components and functions to 841
 be more arithmetically sophisticated, Section 3.6 will retain provable safety but make the control 842
 envelope more efficient. Fig. 3.1 shows the relationship between the resulting control envelope 843
 models. 844

3.5.1 Model Description 845

Train control operates in a time control loop. The control has a latency of time $T > 0$ —a 846
 controller has to wait at most this long before being able to change the throttle position. In 847
 practice, reaction time T is typically on the order of 1 second, but train controllers often keep 848
 decisions in effect for a 10 second period without revising them [37]. Every control cycle, the 849
 control envelope computes an overapproximation $\text{stopDist}(p, v, a_b)$ of the *stopping distance*, 850
 the distance that the train will travel before stopping if it were to accelerate during the current 851
 control cycle, but then brake continuously starting at the next control cycle, until it comes to 852
 a halt. If the distance left to the end of movement authority exceeds $\text{stopDist}(p, v, a_b)$, the 853
 control envelope continues free driving (with any acceleration or deceleration choice within the 854

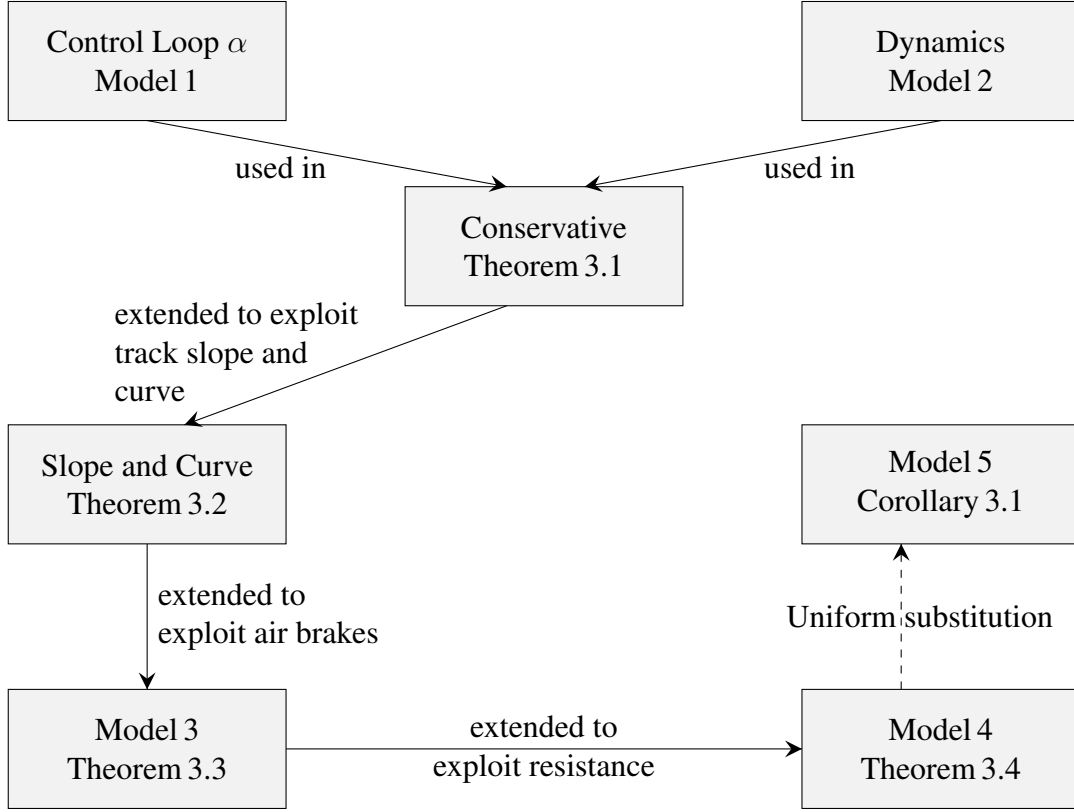


Figure 3.1: Relationship between train models in this chapter.

physical limits of the train), otherwise it brakes. This control cycle, $\alpha(\text{stopDist})$, is parametric in stopping distance stopDist , and expressed as a hybrid program in Model 1.

Model 1 Train control $\alpha(\text{stopDist})$.

Reset Timer	1	$t := 0;$
Free Driving	2	$(\ ?(e - p > \text{stopDist}(p, v, a_b)) ;$
	3	$a_l := *; \ ? - b_{\max} \leq a_l < a_{\max}; a_b := 0; m_b := 0)$
Braking	4	$\cup a_l := -b_{\max}; m_b := m_p)$

Train Dynamics Model 2 describes the physical motion of the train according to the abstract mathematical model of Eq. (3.2) augmented with a clock $t' = 1$ to switch back to control after at most time T . We leave all constants symbolic and prove safety for all values, so that individual railroads have the flexibility to instantiate them with the values that apply to their system and inherit our safety results. This generality in control envelope design comes at the cost of higher proof complexity, but compared to the alternative where we would initialize these constants with

a conservative value, this allows for more efficient, tailored controllers and control envelopes adapted to the specific rail operation.

Model 2 Train dynamics.

$$\text{Dyn.} \left| \begin{array}{l} 1 \quad p' = v, a'_b = m_b, t' = 1, \\ 2 \quad v' = a_l + \max(a_b, a_{b\max}) + a_s(p) + a_r(v) + a_c(p) \ \& \ t \leq T \wedge v \geq 0 \end{array} \right.$$

Stopping Distance In order to decide between free driving and braking, the control envelope computes the upper bound $\text{stopDist}(p, v, a_b)$ on the distance covered over one time period of acceleration and subsequently braking to a stop.

Thus, our models provide two distinct specifications of the distance that the train will take to stop. The first, indirect specification is through the differential equation in Model 2 that implicitly describes the physical motion of the train. The second, approximate specification is stopDist in Line 2 of Model 1, an explicit arithmetic expression that the control envelope can evaluate efficiently to make decisions at runtime.

Efficiency concerns demand that $\text{stopDist}(p, v, a_b)$ be as tight as possible; if the bound is too large, the control envelope would enforce braking unnecessarily. But verifiable safety requires $\text{stopDist}(p, v, a_b)$ to provably be an upper bound on the distance that the train covers (as determined by the dynamics). The tightest possible bound is the exact solution of the differential equation. However, even ignoring the effect of air brakes, the differential equation requires trigonometric solutions. Transcendental function arithmetic is undecidable [137]. To ensure mathematical provability³, we develop polynomial approximations, which is a delicate design task because automated decision procedures for polynomial real arithmetic validity are computationally expensive [38, 161]. This constrains the complexity of the polynomial approximations that can be used as upper bounds. We therefore strike a balance between conflicting concerns: striving for efficiency while satisfying mathematical provability.

To illustrate this approach, we start with a simple conservative expression for stopDist . This expression is similar in complexity to previous work [100, 126], but is now proved safe for the full model including slope, curve friction, air brake propagation, and aerodynamic drag. We later improve on this approximation, focusing on one contributing factor at a time.

3.5.2 Stopping Distance: Conservative

This section constructs a first, conservative control envelope by instantiating control loop α with an expression for stopDist , and proving it safe. Referring back to the train dynamics in Eq. (3.2), we first need an upper bound for v . Integrating this bound via $p' = v$ computes a stopping distance upper bound.

³In KeYmaera X, even when manually simplifying differential and modal expressions, arithmetic subgoals are outsourced to arithmetic decision procedures, which are subject to limitations in proving real arithmetic.

The first impediment to obtaining a provable upper bound for v is that grade and curvature maps a_s and a_c are arbitrary functions, constrained only by upper and lower bounds, and bounded gradients. At runtime, the train knows their exact values as the control envelope is instantiated with maps for the railroad it runs on. However, these maps are unknown at proof time. And yet, the proof has to show safety of the train control ahead of time for *all* possible track maps in order to justify safety of the train control envelope. In order to obtain a provable upper bound on stopping distance, the proof is therefore based on the limited information that we do have about the maps: upper bounds on the potential values of a_s and a_c .

A naïve upper bound on a_s is the value of acceleration that the train experiences when it is on the steepest permissible downward slope, m_s . Our proof will show that the distance required to stop for any permissible grade map cannot exceed the distance computed with the steepest downward slope. It first shows that the true acceleration is bounded above by an acceleration that uses the highest permissible value of grade acceleration, then that actual velocity cannot exceed the velocity computed using the worst-case acceleration, and consequently, that traveled distance cannot exceed the stopping distance computed using the worst-case estimate of velocity.

Accounting for grade force is important. On a downward hill, for example, a train with a control envelope that ignores grade would roll forward even at the end of its movement authority which may cause accidents. In contrast, we can safely ignore curve resistance when approximating `stopDist`, since resistances shorten stopping distance (upper bound 0). These simplifications result in differential equation $v'_2 = a_l + m_s + a_r(v)$, since $\max(a_b, a_{b\max}) = 0$ while the train is accelerating, where v_2 is the upper bound on v that we integrate to compute `stopDist`. However, the solution $v_2(t)$ is still transcendental:

$$\left(\sqrt{4(a_l + m_s)a_2 - a_1^2} \right) \cdot \frac{\tan \left(t \frac{\sqrt{4(a_l + m_s)a_2 - a_1^2}}{2} + \tan^{-1} \left(\frac{a_1 + 2a_2v_0}{\sqrt{4(a_l + m_s)a_2 - a_1^2}} \right) \right) - a_1}{2a_2}$$

The culprits are the linear and quadratic terms in velocity from the Davis equation. With another simplification of 0 as an upper bound for a_r (resistance always works against the train's motion), we derive a polynomial expression for `stopDist`:

$$\begin{aligned} v'_3 = a_l + m_s &\Rightarrow v_3(t) = v_0 + (a_l + m_s) \cdot t \\ p'_3 = v &\Rightarrow p_3(t) = p_0 + (v_0 + \frac{a_l + m_s}{2} \cdot t) \cdot t \end{aligned} \quad (3.3)$$

where v_0 and p_0 are the initial values of speed and position. The solutions provide a conservative stopping distance bound.

$$\text{stopDist}_b(p, v, a_b) = \underbrace{vT + \frac{a_{\max} + m_s}{2} \cdot T^2}_{\text{distance while accelerating}} + \underbrace{\frac{(v + (a_{\max} + m_s)T)^2}{2(b_{\max} - m_s)}}_{\text{stopping from increased speed}} \quad (3.4)$$

The conservative stopping distance `stopDistb` ignores its arguments p and a_b , but later refinements of `stopDist` functions also depend on p and a_b , which is why they are passed in.

The first two terms are the distance covered by the train in one control cycle of acceleration, while the third term is the distance that the train needs to stop should it start braking right after,

assuming the worst-case value of 0 for a_a . This conservative distance is similar to what has been used in the literature [126], but has been adjusted to account for grade force with its worst-case accelerating or decelerating effects. Substituting Eq. (3.4) into control cycle α results in the dL hybrid program of the conservative control envelope: $\alpha(\text{stopDist}_b)$.

A physically important quantity is *braking distance*, the distance that the train will travel before coming to a stop if it were to start braking right now. We derive an upper bound, $\text{brakeDist}_b(v, a_b)$, which will be crucial to our proofs and the initial assumptions, and is improved upon later.

$$\text{brakeDist}_b(v, a_b) = \frac{v^2}{2(b_{\max} - m_s)} \quad (3.5)$$

The last term of Eq. (3.4) is $\text{brakeDist}_b(v_3(T), 0)$ with v_3 according to Eq. (3.3) when $a_l = a_{\max}$.

Initial assumptions $\text{init}(\text{brakeDist})$ parametrized by brakeDist , and assumptions on air brakes initAirbrake are required to prove the conservative control envelope safe. Assumptions about unspecified functions are represented by universal quantification over their input. This representation permits derivation of a formula about the unspecified function at any point of the train's evolution by substituting the quantified input with current values.

$$\begin{aligned} \text{init}(\text{brakeDist}) = & a_{\max} > 0 \wedge b_{\max} > 0 \wedge a_1 < 0 \wedge a_2 < 0 \wedge e - p > \text{brakeDist}(v, 0) \\ & \wedge b_{\max} - m_s > 0 \wedge v \geq 0 \wedge m_s \geq 0 \wedge T > 0 \\ & \wedge \forall x (|a_s(x)| \leq m_s) \wedge \forall x (a_c(x) \leq 0) \\ \text{initAirbrake} = & m_p < 0 \wedge a_{b\max} < 0 \wedge m_b = 0 \wedge a_b = 0 \end{aligned} \quad (3.6)$$

Theorem 3.1 presents the dL formula representing the safety of the conservative control envelope.

Theorem 3.1. *The conservative braking control envelope guarantees that the train always remains within the end of the movement authority. The dL formula below is provable, where α is the control loop from Model 1 parameterized with Eq. (3.4) for stopDist .*

$$\text{init}(\text{brakeDist}_b) \wedge \text{initAirbrake} \rightarrow [(\alpha(\text{stopDist}_b); \text{Model 2})^*] e - p > 0$$

Proof. The proof has been done in the theorem prover KeYmaera X, but we present its central ideas here. We use loop invariant $e - p \geq \text{brakeDist}_b(v, 0) \wedge a_b \leq 0 \wedge v \geq 0$ and split into cases for free driving and braking corresponding to the nondeterministic choice in Lines 2–4 of Model 1. On braking, the invariant is maintained because the derivative of the distance that the train will take to come to a stop does not exceed the derivative of the distance to the end of movement authority, i.e., $(\text{stopDist}_b)' \leq (e - p)'$, by dI. On free driving for a control period T , we first restate that the train maintains a distance to the end of movement authority of at least stopDist adjusted for time t since the last control decision, i.e., $v(T - t) + \frac{a_{\max} + m_s}{2}(T - t)^2 + \frac{(v + (a_{\max} + m_s)(T - t))^2}{2(b_{\max} - m_s)}$ by dC and dI. The required inequality relation between the derivatives, $-v \geq -v + v'(T - t) - (a_{\max} + m_s)(T - t) + \frac{(v + (a_{\max} + m_s)(T - t))(v' - (a_{\max} + m_s))}{(b_{\max} - m_s)}$, holds because $v' - (a_{\max} + m_s) \leq 0$. $\square$

3.6 Safe Efficiency Improvements

We improve on the overapproximation for stopping distance in order to make our control envelope more efficient. The FRA model presents two challenges common in embedded controllers: it uses functions whose exact values are unknown at proof time (slope and curve maps), and has many interacting forces. Our techniques address these problems with two general principles: using quantified worst case bounds on unknown functions, and separation of dependencies. The first technique relies on the observation that the track changes are gradual and predictable (the rate of change of unknown functions is bounded). It drastically improves bounds on the effect of grade and curve over one time period of acceleration, after resolving *circular dependencies* between the variables of motion. The second technique improves the estimate `brakeDist` by accounting for air brake dynamics. It demonstrates a handling of triple integration, using *mode-splitting* to deal with the non-analytical change of behavior when brakes saturate. The third technique uses *Taylor polynomials* to capture the effect of resistance, which would otherwise lead to transcendental arithmetic. The track environment discussion in Section 3.6.1 and Section 3.6.2 will become assumptions of our models, while the calculations in Section 3.6.3 and Section 3.6.4 are machine-checked to be correct approximations as part of the KeYmaera X proofs.

3.6.1 Bound on Gradient

The train control envelope knows the current slope $a_s(p)$ and vertical curves of the track, which determine transitions from one track grade to another. This knowledge results in a bound $h_{\max}$ on the difference in grade per unit length [61, p.616–619]:

$$\left| \frac{\partial a_s(x)}{\partial x} \right| \leq h_{\max} \quad \Rightarrow \quad |a'_s(p)| \leq v h_{\max}$$

The second inequality follows from the first using the chain rule and $p' = v$. After time T , a_s could have increased by no more than $u h_{\max} T$, where u is some upper bound on v over the course of T time, which we derive later in this section.

3.6.2 Bound on Curve Resistance

Similar to bounding the gradient change, we compute an upper bound on the rate of change of curve resistance as a function of velocity using track geometry. Curve resistance depends on average curve $curve(p)$.⁴ Assume that the tightest permissible curvature for the railroad corresponds to radius r . The greatest change in average curvature occurs when a train goes from a track with the greatest permissible curvature to a straight track (or vice versa). Over a small period of time dt , the portion of the train transitioning from greatest curvature to 0 curvature is

⁴In practice, degree of curvature is the angle subtended by a 100ft arc of the track, 100ft chord of the track, or 100.7ft arc in the US. In Europe, radius is generally used [61]. In our rate of change of curvature derivation, we will use the mathematical idealization for average degree of curvature: $curve(p) = \int_{p-l}^p \frac{1}{lr(x)} dx$ where l is the length of the train and $r(x)$ is the radius of curvature at track point x . This approximately relates to the US definition by a multiplicative factor, with a small error introduced by the granularity of the 100ft measuring arc. This multiplicative factor can be rolled into the multiplicative coefficient c_{eff} .

dv , where v is velocity. So the rate of change of $\text{curve}(p)$ (taken in radians) with respect to time is $-\frac{v}{lr}$ where l is the length of the train. For a given train, a_c relates to curve with some constant multiplicative factor q . We use $a'_c = q \frac{v}{lr} = c_{cft} v$ with the constant factor $c_{cft} = \frac{q}{lr}$.

With this bound on the maximum rate of change of a_c , we now estimate the upper bound on curve resistance over time T , where p_0 is initial train position, to be

$$a_c(p_0) + uc_{cft}T$$

As before, u is an upper bound on velocity for duration T .

3.6.3 Tight Stopping Distance Approximation

The upper bounds $\bar{a}_s$ on gradient from Section 3.6.1 and $\bar{a}_c$ on curve resistance from Section 3.6.2 are summarized as:

$$a_s(p) \leq \bar{a}_s(p_0) = \min(m_s, a_s(p_0) + uh_{\max}T)$$

$$a_c(p) \leq \bar{a}_c(p_0) = \min(0, a_c(p_0) + uc_{cft}T)$$

This enables us to improve our estimation of stopping distance:

$$v'_4 = a_l + \bar{a}_s(p_0) + \bar{a}_c(p_0) \Rightarrow v_4(t) = v_0 + (a_l + \bar{a}_s(p_0) + \bar{a}_c(p_0))T \quad (3.7)$$

Upper bound v_4 is tighter than v_3 of Eq. (3.3) and thus integrates to an improved stopping distance estimate. It depends (transitively through $\bar{a}_s$ and $\bar{a}_c$) on the unknown upper bound u on velocity, which we still need to estimate provably correctly.

Circular Dependencies: The upper bound on velocity, u , is undefined in expression (3.7) above. We cannot use the bound v_4 for u , since v_4 itself is phrased in terms of u . The problem is a circular dependency between a_s and v : the bound on slope acceleration a_s depends on speed v , while the upper bound on speed v , in turn, depends on slope acceleration a_s ; likewise for a_c . Physically, this is because if the train is moving faster, we know less about the nature of the track—its curve and slope—after the passage of some time, as the train is farther from its previous position on the track. However, we need information about the grade curve in order to better estimate the velocity that the train is traveling at. In order to cut through these circular dependencies, we use the conservative estimations of these quantities from (3.3) as a base case to bootstrap incrementally finer computations, as presented below.

We first use the initial upper bounds m_s for a_s and 0 for a_c to get a conservative bound $v(t) \geq v_0 + (a_{\max} + m_s)t$, so that we can set $u = v_0 + (a_{\max} + m_s)T$. Since $(a_{\max} + m_s)$ is a positive upper bound on the train's acceleration, velocity could have increased no more than $(a_{\max} + m_s)T$. Hence u is indeed an upper bound on v through the T time interval. Substituting this u refines the gradient and curve resistance bounds.

$$\begin{aligned} \bar{a}_s(p_0) &= \min(m_s, a_s(p_0) + \overbrace{(v_0 + (a_{\max} + m_s)T)}^u h_{\max}T) \\ \bar{a}_c(p_0) &= \min(0, a_c(p_0) + \underbrace{(v_0 + (a_{\max} + m_s)T)}_u c_{cft}T) \end{aligned}$$

These expressions give the chosen definitions of $\bar{a}_s$ and $\bar{a}_c$ by replacing placeholder velocity bound u .

We could in principle further improve this upper bound on speed by using v_4 to obtain an even better bound on a_s and a_c , which could in turn yield an improved bound on v . However, extra levels of extrapolation increase proof cost and computation time when the control envelope is run. Each extra intermediate bound requires a constant number of extra proof steps, but provides diminishing efficiency gains in return. Intuitively, proof length is asymptotically linear in number of iterations because under optimal proof rule application ordering, each iteration induces one extra application of rule dC to introduce the intermediate bound into the proof tree, and rule dI to justify this intermediate bound.

The `stopDist` expression below uses v_4 with $u = v_0 + (a_{\max} + m_s)T$ to estimate stopping distance, which is sufficiently tight to make useful control decisions (see Section 3.8).

$$\text{stopDist}_s(p, v, a_b) = vT + \left(\frac{a_{\max} + \bar{a}_s(p) + \bar{a}_c(p)}{2} \right) T^2 - \frac{\left(v + (a_{\max} + \bar{a}_s(p) + \bar{a}_c(p))T \right)^2}{2(b_{\max} - m_s)} \quad (3.8)$$

We need further initial assumptions to prove the improved *slope-exploiting control envelope* $\alpha(\text{stopDist}_s)$ safe. These assumptions represent the result of the track environment discussion of Section 3.6.1 and 3.6.2 used for the computer-checked proof.

$$\begin{aligned} \text{init}_s(\text{brakeDist}) \equiv & \text{init}(\text{brakeDist}) \wedge a_{\max} - m_s + m_c > 0 \wedge h_{\max} \geq 0 \\ & \wedge m_c \leq 0 \wedge c_{\text{cft}} \geq 0 \wedge \forall x (a_c(x) \geq m_c) \wedge \forall x' \forall x (|a'_c| \leq x' c_{\text{cft}}) \wedge \forall x' \forall x (|a'_s| \leq x' h_{\max}) \end{aligned} \quad (3.9)$$

The technique of this section applies to *time-triggered controllers* (where a control loop runs with some known maximum latency and sensors measure current state every cycle) for physical systems with functions affecting the environment that are unknown except for bounds on their rate of change. The future value of the functions can be bounded in terms of their worst-case rate of change. Furthermore, these bounds can be used to compute bounds on other variables in the system, just as here a bound on velocity was used to bound slope and curve effect, which was again used to obtain a better bound on velocity. The situation arises frequently in practice: examples of unknown functions are a potential field, or a noise or error effect, which may have circular dependence with position.

Theorem 3.2. *The slope-estimating control envelope guarantees that the train stays within its movement authority. The dL formula below is provable, where α is the control loop from Model 1 parameterized with Eq. (3.8) for `stopDist`.*

$$\text{init}_s(\text{brakeDist}_b) \wedge \text{initAirbrake} \rightarrow [(\alpha(\text{stopDist}_s); \text{Model 2})^*] e - p > 0 \quad (3.10)$$

Proof. By proof in KeYmaera X. The proof builds on the ideas from Theorem 3.1. We apply the loop rule with the same loop invariant as Theorem 3.1. If the train brakes, differential invariant rule dI again shows that the loop invariant holds throughout differential equation evolution.

If the train chooses to accelerate, then as before, we restate that the train maintains at least a distance of `stopDists` adjusted for time t since the last control decision. Unlike before, $\bar{a}_s$ instead of m_s accounts for worst-case a_s , and $\bar{a}_c$ instead of 0 accounts for worst-case a_c . Again,

dI proves that this adjusted inequality remains true. In order to prove the required inequality on the derivatives, we first use differential cut rule dC to show $\bar{a}_s(p_0) \geq a_s(p)$ throughout the control cycle, and $\bar{a}_c(p_0) \geq a_c(p)$. There are two branches for each cut corresponding to how the min in $\bar{a}_s$ and $\bar{a}_c$ resolve. For example, using initial position and velocity p_0 and v_0 , for $\bar{a}_s$, we need to show that $a_s(p) \leq m_s$, and $a_s(p) \leq a_s(p_0) + uh_{\max}T$. While the former follows from the quantified assumption on $a_s(p)$, to prove the latter, we again adjust it for elapsed time t , to argue that $a_s(p) \leq a_s(p_0) + (v_0 + uh_{\max}(T - t))$, proved using dI. The required derivative inequality follows from instantiating the quantified assumption bounding the rate of change of $(a_s(p))' \leq uh_{\max}$ with the current position, and showing that u is an upper bound on v in the control loop. The argument for $\bar{a}_c$ is analogous. $\square$

3.6.4 Effect of Air Pressure Brakes

The term `brakeDistb` conservatively neglects the *significant* effects of air brakes to avoid reasoning about their time dependence. This section derives a tighter `brakeDista` that accounts completely for air brakes. It specifies a control envelope that simultaneously benefits from the slope and curve estimation of the previous section, and from air brake dynamics. The central insight required to prove the improved control envelope safe is how to compose reasoning about time-dependent air brake propagation and velocity-dependent slope and curve estimations from the previous section. We first show that `brakeDista`, the component of `stopDist` affected by air brakes, is the desired upper bound on distance to brake throughout the control loop. Then, holding `brakeDista` constant, we perform the differential reasoning on slope and curve estimation described in the previous section. The two results together permit an overall proof of safety of the air brake-exploiting control envelope.

To derive the improved `brakeDista`, we first compute some intermediate functions from air brake dynamics. In Eq. (3.2) during brake rampup, with slope relaxed pessimistically to m_s , and curve and resistance to 0, $\max(a_b, a_{b\max})$ evaluates to a_b , and m_b to m_p . The solution for v in the resulting differential equation $v' = b_{\max} - m_s + a_b$, $a'_b = m_p$ is quadratic in t :

$$v = v_0 - (b_{\max} - m_s + a_b)t + \frac{1}{2}m_pt^2 \quad (3.11)$$

Function t_b below computes the time the train takes to achieve full braking by subtracting current brake buildup a_b from maximal air braking $a_{b\max}$, and dividing by the rate of increase in air brake force m_p . If the train comes to a stop before air brake saturation, it instead evaluates to the time until the train stops, as computed by solving Eq. (3.11) for $v = 0$.

$$t_b(v, a_b) = \min\left((a_{b\max} - a_b)/m_p, \frac{(b_{\max} - m_s + a_b) - |(b_{\max} - m_s + a_b)^2 - 2m_pv|}{m_p}\right) \quad (3.12)$$

The distance that the train travels before either stopping or reaching maximum air brake effect is $\int_0^{t_b(v, a_b)} p dt = vt_b(v, a_b) + \frac{1}{2}(b_{\max} - m_s + a_b)t_b(v, a_b)^2 + \frac{1}{6}(m_p)t_b(v, a_b)^3$. The velocity of the train after this period of buildup, by Eq. (3.11), is $v_f = v - (b_{\max} - m_s + a_b)t_b(v, a_b) + \frac{1}{2}m_pt_b(v, a_b)^2$. So after the brakes finished ramping up, the distance traveled until the train comes to a halt is $\frac{v_f^2}{2(b_{\max} - m_s - a_{b\max})}$, using Newton's third equation of motion. If the train stops before finishing brake

1099 rampup, v_f evaluates to zero, as required. Adding the upper bounds on distance traveled before
 1100 and after brake rampup results in $\text{brakeDist}_a(v, a_b)$ in Eq. (3.13), an upper bound on braking
 1101 distance that accounts for the effect of air brakes. While this derivation for control envelope
 1102 design is manual, its result will be verified by computer-checked proof in Theorem 3.3.

$$\begin{aligned}
 \text{brakeDist}_a(v, a_b) &= vt_b(v, a_b) - \frac{1}{2}(b_{\max} - m_s + a_b)t_b(v, a_b)^2 \\
 &\quad + \frac{1}{6}(m_p)t_b(v, a_b)^3 + \frac{(v - (b_{\max} - m_s + a_b)t_b(v, a_b) + \frac{1}{2}m_pt_b(v, a_b)^2)^2}{2(b_{\max} - m_s - a_{b\max})} \\
 \text{stopDist}_a(p, v, a_b) &= vT + \left(\frac{a_{\max} + \bar{a}_s(p) + \bar{a}_c(p)}{2} \right) T^2 \\
 &\quad + \text{brakeDist}_a\left(v + (a_{\max} + \bar{a}_s(p) + \bar{a}_c(p))T, 0\right)
 \end{aligned} \tag{3.13}$$

1104 In order to prove the invariant that after a control cycle of braking, $e - p > \text{brakeDist}_a(v, a_a)$,
 1105 we must reason about the three dynamically distinct cases: (i) when $\max(a_b, a_{b\max})$ is $a_{b\max}$;
 1106 (ii) when it is a_b , and $t_b(v, a_b)$ evaluates to $(a_{b\max} - a_b)/m_p$; and (iii) when it is a_b but $t_b(v, a_b)$
 1107 evaluates to $\frac{(b_{\max} - m_s + a_b) - |(b_{\max} - m_s + a_b)^2 - 2m_pv|}{m_p}$. We split the model dynamics into these three evo-
 1108 lution domains, branching between the three possibilities in a loop to transition between modes
 1109 freely (Model 3). This does not affect the semantics of evolution: per train dynamics, mode tran-
 1110 sition happens at most once (from (ii) to (i) or (ii) to (iii)). Unrolling the loop to two iterations,
 1111 one per mode, suffices to model train behavior. This split formulation simplifies syntactic proofs.

Model 3 Mode-split train dynamics.

$$\begin{aligned}
 \text{Dyn.} \quad & \begin{cases} 1 & \{ \text{Model 2} \ \& \ a_{b\max} \geq a_b \} \\ 2 & \cup \{ \text{Model 2} \ \& \ a_{b\max} \leq a_b \wedge (a_{b\max} - a_b)/m_p \geq t_p \} \\ 3 & \cup \{ \text{Model 2} \ \& \ a_{b\max} \leq a_b \wedge (a_{b\max} - a_b)/m_p \leq t_p \}^* \end{cases} \\
 \text{where } t_p &= \frac{(b_{\max} - m_s + a_b) - |(b_{\max} - m_s + a_b)^2 - 2m_pv|}{m_p}
 \end{aligned}$$

1112 **Theorem 3.3.** *The air-brake-exploiting control envelope guarantees that the train stays within*
 1113 *its movement authority. The dL formula below is provable, where α is the control loop from*
 1114 *Model 1 parameterized with Eq. (3.13) for stopDist .*

$$1115 \quad \text{init}_s(\text{brakeDist}_a) \wedge \text{initAirbrake} \rightarrow [(\alpha(\text{stopDist}_a); \text{Model 3})^*] e - p > 0$$

1116 *Proof.* By proof in KeYmaera X. The high level idea is to use an outer loop invariant $e - p \geq$
 1117 $\text{brakeDist}(v, a_a) \wedge a_b \leq 0 \wedge v \geq 0$ and again split into free driving and braking cases.
 1118 On braking, we show that the outer loop invariant is maintained in each of the three dynamics
 1119 modes using an inner loop invariant consisting of 4 formulas, the most important of which is
 1120 $e - p > \text{brakeDist}(v, \max(a_b, a_{b\max}))$.

On free driving, we first show that $\text{brakeDist}_{a0} = \text{brakeDist}_a((v + (\bar{a}_{\max} + \bar{a}_s(p) + \bar{a}_c(p))T), 0)$ is truly an upper bound on all reachable $\text{brakeDist}(v, \max(a_b, a_{b\max}))$ values in the control cycle. Then, holding brakeDist_{a0} constant, we follow a proof similar to Theorem 3.2 to show that, in every mode, the increase in p does not exceed the decrease in distance buffer $vT + \left(\frac{a_{\max} + \bar{a}_s(p) + \bar{a}_c(p)}{2}\right)T^2$. The free driving inner loop invariant consists of 14 formulas, most of which state that various upper bound expressions (such as on velocity, grade and curve) remain upper bounds over the course of the loop. By monotonicity, then, $e - p > \text{distance buffer} + \text{brakeDist}_{a0} \geq \text{brakeDist}(v, \max(a_b, a_{b\max}))$. $\square$

3.6.5 Exploiting Resistance

Exactly accounting for the quadratic dependence of resistance on velocity, as discussed in Section 3.5.2, leads to an undecidable, transcendental exact solution for stopping distance. The control envelope must instead use an approximation. Since polynomial arithmetic is decidable, Taylor polynomials are a natural way to obtain decidable approximations. This section applies Taylor approximation to the FRA model, identifying techniques generalizable to verified control for other embedded systems with transcendental dynamics.

The Davis equation implies $v' \geq (a_{\max} + \bar{a}_s(p) + \bar{a}_c(p)) + a_1v + a_2v^2$, where slope and curve bounds $\bar{a}_s$ and $\bar{a}_c$ are from Section 3.6.3. This is a lemma for the dL proof of Theorem 3.4 justified as follows: consider two identical trains on tracks t_1 and t_2 , starting with the same velocity. We want to bound the velocity v_1 of the train on t_1 . Suppose t_2 is the track with worst case track and grade, and that a train on t_2 (the “ghost train”, that we have constructed for the sake of our argument) always accelerates so that $v'_2 = (a_{\max} + \bar{a}_s(p) + \bar{a}_c(p)) + a_1v + a_2v^2$. On the other hand, on track t_1 , the real train that we require a proof about only obeys the restriction $|a_s| < m_s$. If $v_2 - v_1$ is to become negative, it must cross the boundary where its value is 0. However, whenever $v_1 = v_2$, necessarily, $v'_2 > v'_1$. This ghost train argument serves a purpose similar to the “circular dependencies” argument of Section 3.6.3: reasoning about mutually influencing factors one at a time. The ghost train permits us to represent and reason about a transcendental bound on velocity, v_2 , derived using slope and curve estimates $\bar{a}_s$ and $\bar{a}_c$.

Given that $v' \geq (a_{\max} + \bar{a}_s(p) + \bar{a}_c(p)) + a_1v + a_2v^2$, the first-order Taylor polynomial of this expression for velocity is $v_0 + \left((a_{\max} + \bar{a}_s(p) + \bar{a}_c(p)) + a_1v_0 + a_2v_0^2\right)t$. Using this approximation at time T , with $a_l = a_{\max}$, as an upper bound for velocity after a time period of acceleration, we compute Eq. (3.14) for stopping distance that leverages resistance. While this derivation is manual, its result will be verified by computer-checked proof in Theorem 3.4.

$$\begin{aligned} \bar{v}' &= (a_{\max} + \bar{a}_s(p) + \bar{a}_c(p)) + a_1v + a_2v^2 \\ \text{stopDist}_t(v) &= vT + \left(\frac{a_{\max} + \bar{a}_s(p) + \bar{a}_c(p)}{2}\right)T^2 + \text{brakeDist}_a\left(v + (\bar{v}')T, 0\right) \end{aligned} \quad (3.14)$$

Unlike previous stopping distance estimates, this expression is *not* always an upper bound. It uses resistance for the original velocity v_0 , which is only a conservative bound when resistance is low enough to permit acceleration. This condition is captured by predicate vbound in Eq. (3.15).

$$\text{vbound}(v) \equiv (a_{\max} + \bar{a}_s(p) + \bar{a}_c(p)) + a_1v + a_2v^2 \geq 0 \quad (3.15)$$

1158 For the Taylor approximation control envelope (Model 4), we define the stopDist_{tp} predicate
 1159 (3.16), that unlike previous expressions for stopping distance, returns a truth value. It uses pre-
 1160 vious definitions stopDist_t from Eq. (3.14), vbound (3.15) to determine when stopDist_t
 1161 is applicable, and stopDist_s from Eq. (3.8) is used as a fallback.

$$\begin{aligned} \text{stopDist}_{tp}(p, v, a_b) \equiv & e - p > \text{stopDist}_s(p, v, a_b) \vee \\ & (\text{vbound}(v) \wedge e - p > \text{stopDist}_t(v)) \end{aligned} \quad (3.16)$$

1163 Higher order Taylor polynomials permit analogous reasoning. Theorem 3.4 expresses that the
Taylor polynomial control envelope in Model 4 is safe.

Model 4 Taylor polynomial control envelope and dynamics.

Reset Timer	1	$t := 0;$
Free driving	2	$((\text{?stopDist}_{tp}(p, v, 0); a_l := *; \text{?} - b_{\max} \leq a_l < a_{\max}; a_b := 0; m_b := 0)$
Braking	3	$\cup a_l := -b_{\max}; m_b := m_p);$
Dynamics	4	Model 3

1164
 1165 **Theorem 3.4.** *The Taylor polynomial control envelope guarantees that the train stays within its*
 1166 *movement authority. The dL formula below is provable.*

$$1167 \quad \text{init}_s(\text{brakeDist}_a) \wedge \text{initAirbrake} \rightarrow [(\text{Model 4})^*] e - p > 0$$

1168 *Proof.* By proof in KeYmaera X. We start by showing that the loop invariant from Theorem 3.3
 1169 is maintained (using dL rule loop). The case where the train is braking proceeds similar to
 1170 Theorem 3.3. When the train is accelerating, we need to show that the control envelope has
 1171 insisted on a sufficient distance margin (stopping distance), so that even after a time period, the
 1172 train has enough space to stop. As the Taylor polynomial computes the stopping distance, we
 1173 must prove that it actually is an upper bound. We use a monotonicity argument by introducing
 1174 an auxiliary variable that represents a “ghost” train, perpetually traveling down worst possible
 1175 slope $\bar{a}_s$ and curve $\bar{a}_c$. This isolates slope and curve from the effect of resistance, breaking
 1176 interdependence. We derive the Taylor polynomial result on the ghost train, show that it goes no
 1177 slower than the real train, and that consequently the Taylor polynomial result must hold for the
 1178 real train. Other elements of the proof remain similar to Theorem 3.1. $\square$

1179 3.7 Kinematic Train Model Proofs

1180 A proof for the FRA model [24] derives from the proof for our abstract mathematical models,
 1181 e.g. Theorem 3.4, by uniform substitution [121], which replaces abstract function symbols with
 1182 specific terms using the correspondence in Section 3.4. Model 5 lists the model resulting from
 1183 substitution.

Model 5 Train control envelope in kinematic motion model.

Reset Timer	1	$t := 0;$
Free Driving	2	$(\text{?stopDist}_{tp}(p, v, 0); a_l := *; \text{?} -b_{\max} \leq a_l < a_{\max}; a_b := 0; m_b := 0$
Braking	3	$\cup a_l := -b_{\max}; m_b := m_p);$
	4	$(\{\text{dyn} \ \& \ a_{b\max} \geq a_b\}$
Mode Split	5	$\cup \{\text{dyn} \ \& \ a_{b\max} \leq a_b \wedge (a_{b\max} - a_b)/m_p \geq t_p\}$
	6	$\cup \{\text{dyn} \ \& \ a_{b\max} \leq a_b \wedge (a_{b\max} - a_b)/m_p \leq t_p\})^*$

$$\text{with dyn} \equiv \{p' = v, a'_b = m_b, t' = 1, v' = a_l + \max(a_b, a_{b\max}) - \frac{A_G w \cdot \text{grade}(p)}{m_T} \\ - \frac{A_C w \cdot \text{curve}(p)}{m_T} - \frac{C_R w v + D_R v^2}{m_T} \ \& \ t \leq T \wedge v \geq 0\}$$

Corollary 3.1 (Kinematic train model is safe). *The train control envelope for the FRA model never overshoots the end of movement authority, i.e., the following formula is provable with* 1184
 $b_{\max} = \frac{F_B}{m_T} + \frac{A_R w + B_R n}{m_T}$ and $a_{\max} = \frac{F_L}{m_T} - \frac{A_R w + B_R n}{m_T}$. 1185
1186

$$\text{init}_s(\text{brakeDist}_a) \wedge \text{initAirbrake} \rightarrow [(\text{Model 5})^*] e - p > 0 . \quad 1187$$

Proof. By uniform substitution from Theorem 3.4, using the substitutions σ below: 1188

$$\sigma = \begin{cases} a_{\max} \mapsto \frac{F_L}{m_T} - \frac{A_R w + B_R n}{m_T} & b_{\max} \mapsto \frac{F_B}{m_T} + \frac{A_R w + B_R n}{m_T} \\ a_s(p) \mapsto -\frac{A_G w \cdot \text{grade}(p)}{m_T} & a_c(p) \mapsto -\frac{A_C w \cdot \text{curve}(p)}{m_T} \\ a_r(v) \mapsto -\frac{C_R w v + D_R v^2}{m_T} & \square \end{cases} \quad 1189$$

3.8 Evaluation

For validation, we use ModelPlex [98] to derive a controller monitor from Model 5 that measures 1191
the safety margin in decisions of previous brake enforcement envelopes [23, 24] and our verified 1192
control envelopes. That way, we measure if, and how well, our verified train control envelopes 1193
and existing control envelopes agree in order to assess the safety of those existing systems and 1194
the efficiency of our model. Existing brake enforcement envelopes brake to a full stop once 1195
engaged. 1196

The ModelPlex monitor computes a robustness measure indicating how close a decision is 1197
to losing the safety proof. When the robustness measure is positive, the decision is guaranteed 1198
to remain provably safe so that the system enjoys the safety proof of the verified model. When 1199
it is negative, emergency brakes should be applied for safety reasons. The ModelPlex controller 1200
monitor follows the structure of the verified model when it computes robustness. For example, a 1201
monitor for Lines 2–?? of Model 5 describes their effect with the formula $\text{stopDist}_{tp}(p, v, 0) \wedge$ 1202

Table 3.1: FRA train parameter instantiation example.

Param.	Value	Unit	Description
A_R	0.6	lbf/ton	(constant) weight coefficient
B_R	20	lbf/axle	(constant) train-size coefficient
C_R	0.01	lbf/ton-mph	(linear) speed coefficient
D_R	0.07 0.294	lbf/mph ²	(quadratic) aerodynamic coefficient (for loaded cars)
A_G	20	lbf/ton	(constant) weight coefficient

$-b_{\max} \leq a_l < a_{\max}$ which translates to the robustness measure

$$\min \left(\max \left(\min(e - p - \text{stopDist}_t(v), (a_{\max} + \bar{a}_s(p) + \bar{a}_c(p)) + a_1 v + a_2 v^2), \right. \right. \\ \left. \left. e - p - \text{stopDist}_a(p, v, 0) \right), a_l + b_{\max}, a_{\max} - a_l \right) .$$

The most important elements of the full Model 5 monitor are:

- in free driving (when stopDist_{tp} is satisfied) it combines remaining position margin (the larger of Taylor margin $e - p - \text{stopDist}_t$ when $\text{vbound}(v)$, or fallback margin $e - p - \text{stopDist}_a$) with acceleration choice robustness ($\min(a_l + b_{\max}, a_{\max} - a_l)$ from control decision $a := *; ? - b_{\max} \leq a_l < a_{\max}$) and speed robustness (v from evolution domain $\dots \& v \geq 0$);
- during braking, which is always allowed, it measures speed robustness (v per evolution domain constraint).

Because ModelPlex’s robustness measure combines multiple quantities of incompatible units, there is no direct interpretation of its magnitude, but only of its sign.

For validation, we implement the train model of Eq. (3.2) in Python by numerical integration, instantiating the model parameters per FRA model [23, 24]. These parameter values are estimated from train test runs and standards [23] and require careful consideration of their units (Table 3.1).

Our evaluation compares start braking and stopping points of trains, highlighting braking performance in terms of overshoot (safety risk) and undershoot (performance objective of a maximum undershoot of 1000ft [24]) of the end of movement authority. We follow [23, p. 47] and implement the baseline control envelopes using numeric forward Euler integration to simulate the model in order to determine the stopping distance. Our verified control envelopes neither use numeric integration nor include the dynamic model, but instead decide based on the stopping distance overapproximation stopDist_{tp} of Eq. (3.16).

The most interesting train behavior arises from the subtle interplay between air pressure propagation, aerodynamic/roll resistance, and acceleration/deceleration due to slope. It peaks on crests that change gradient from uphill to downhill and in troughs that change gradient from downhill to uphill. When calculating stopping distance, numerical integration in the baseline enforcement algorithms discretizes train speed and position to calculate forces, which overestimates

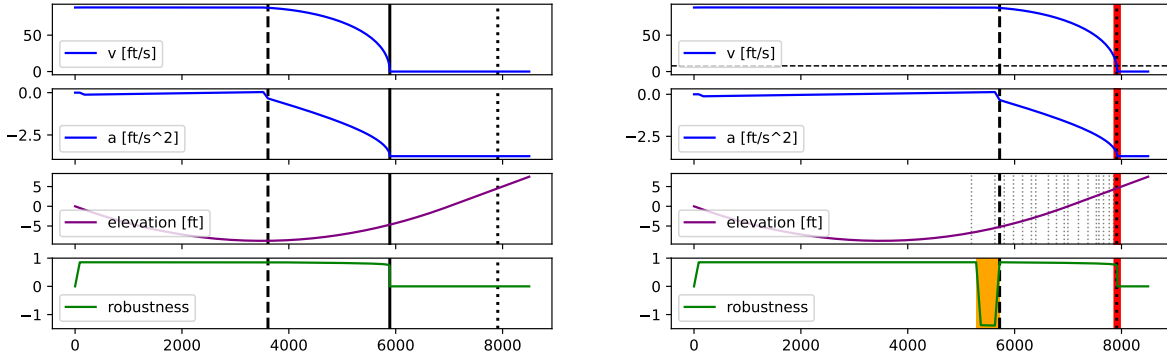
resistance while simultaneously underestimating available brake force. Acceleration/deceleration due to slope is even more subtle as it depends on the position of the train along the slope (e.g., on a crest changing from uphill to downhill, deceleration on the uphill segment is overestimated until the train passes the top, afterwards acceleration is underestimated). These effects do *not* balance out and thus make numerical integration errors unreliable and hard to predict. Moreover, changing the integration step size shifts how distance estimates are biased towards undershoot or overshoot (e.g., in typical configurations, brake rampup is the dominating influence on stopping distance, and so larger integration step sizes bias towards undershoot). As a result, for any given configuration of numerical integration in enforcement algorithms, we can construct scenarios where the numerical integration underestimates stopping distance and train enforcement exhibits unsafe behavior. Our formal models and proofs design *provably correct* stopping distance overapproximations instead of using numerical integration and are, therefore, not subject to these intricate safety tradeoffs.

3.8.1 Stopping Behavior in Troughs

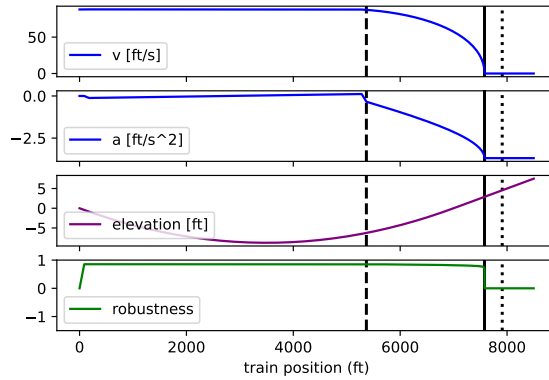
Our first validation in Fig. 3.2 follows [24, Fig. 10] with a train configuration of a medium (75 cars), loaded (car weight 286klbf), mixed freight train traveling at initial speed 60mph in a trough, with train starting position at 0ft. The trough is configured with uniform rate of change from 0.5% downhill to -0.5% uphill between positions 0ft and 7000ft. Locomotive tractive effort and locomotive braking compensates for speed loss and gain due to gradient. Air brakes engage only for the full braking maneuver when attempting to not exceed the movement authority. We configure our monitors and controllers to match the trough maximum uphill/downhill gradient of 0.5% for slope estimation. A sweep of movement authority endpoints in the range 4000ft–8000ft with 10ft steps identifies challenging configurations. The base enforcement algorithm [24] underestimates stopping distance for 19 (of the 400) endpoints. Fig. 3.2 compares the base enforcement algorithm to our verified control envelope on one of these challenging points at 7910ft, past the maximum uphill slope of the trough. The base enforcement algorithm in Fig. 3.2a uses numeric integration to determine the stopping distance, and adds a generous constant fudge factor plus speed-dependent safety offset that results in the train initiating braking at 3609ft. The maneuver finishes at 5889ft, stopping 2021ft short of the end of movement authority. In Fig. 3.2c, our verified control envelope does *not* use any such offsets. It initiates braking much later at position 5368ft and stops at 7578ft (significantly closer to the movement authority endpoint, limiting undershoot to 332ft).

Fig. 3.2b, removes the safety offset from the base enforcement algorithm in an experiment illustrating the subtle interplay between forces and their safety consequences. Even though the uphill segment of the trough helps reduce stopping distance, numeric integration overestimates this effect and initiates braking too late, which our monitor detects at 5368ft. Should the train ignore the warning, it overshoots by 9ft (highlighted in red⁵) with a remaining speed of $7.8 \frac{\text{ft}}{\text{s}}$ when passing the desired stopping point. Initiating fallback control $a_l = -b_{\max}$ at the monitor violation would have kept the train from overshooting the movement authority.

⁵The controller monitor subsequently no longer flags a violation, because the base enforcement algorithm then applies the correct decision of maximum braking (agreeing with the model). But it does so too late, as the earlier monitor warning was ignored.



(a) Base enforcement algorithm [24] initiates braking at 3609ft and stops at position 5889ft with an undershoot of 2021ft, which exceeds the 1000ft maximum undershoot performance objective [24]. (b) Base enforcement algorithm [24] without offset overshoots by 9ft. Light-gray dotted lines mark other challenging movement authority endpoints in the range 4000ft–8000ft that would also be violated.



(c) Verified control envelope (Model 5) limits undershoot to 332ft.

Figure 3.2: Comparison of start braking (dashed vertical line) and stopping points (solid vertical lines) in a trough from 0.5% downhill at start to -0.5% uphill at 7000ft, with end of movement authority (dotted vertical line) at 7910ft.

3.8.2 Stopping Behavior on Crests

An assumption in the brake enforcement algorithms and thus an initial condition in our proof is that train locomotives are not underpowered: their tractive effort is enough to overcome maximum uphill slope and stay stopped on the maximum downhill slope. Fig. 3.3 illustrates how an underpowered, initially stopped train starts rolling downhill until the air brakes build enough deceleration to stop the train. This configuration, violating initial conditions, is unsafe to start in the first place.

Underpowered locomotives are especially challenging on a crest where (full) tractive effort is needed to limit the speed loss on the uphill slope and regain desired speed on the downhill segment, but air brakes are needed to stay stopped. Fig. 3.4 compares the behavior of the base enforcement algorithms and our verified algorithm with underpowered locomotives on a crest with movement authority endpoint at 7920ft. In this experiment, we configure our monitors and

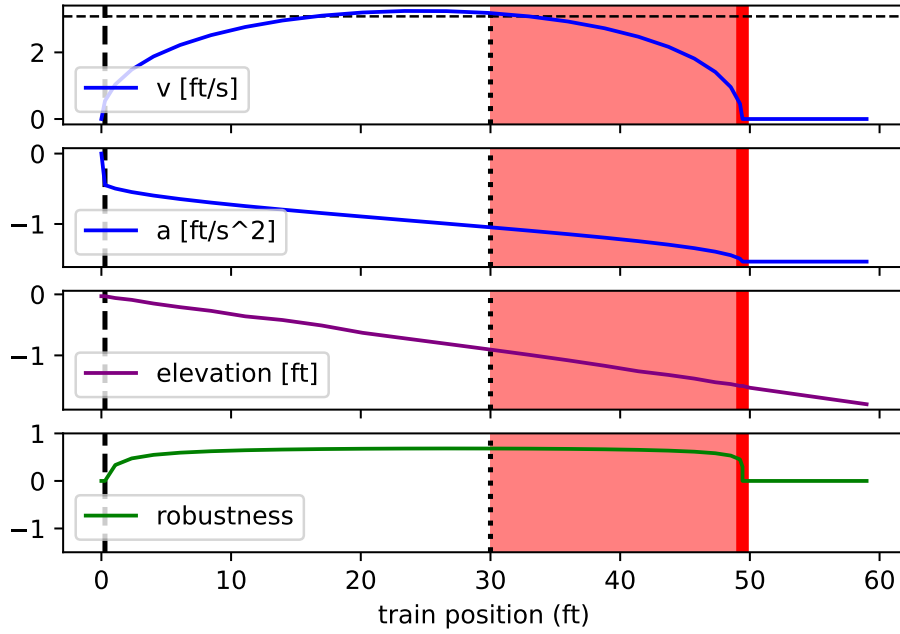


Figure 3.3: Initial conditions of proof (Corollary 3.1) not satisfied $\nexists$: an underpowered train on a 3% downhill slope needs air brakes to stay stopped; the stopping point is closer than the rolling distance during brake rampup.

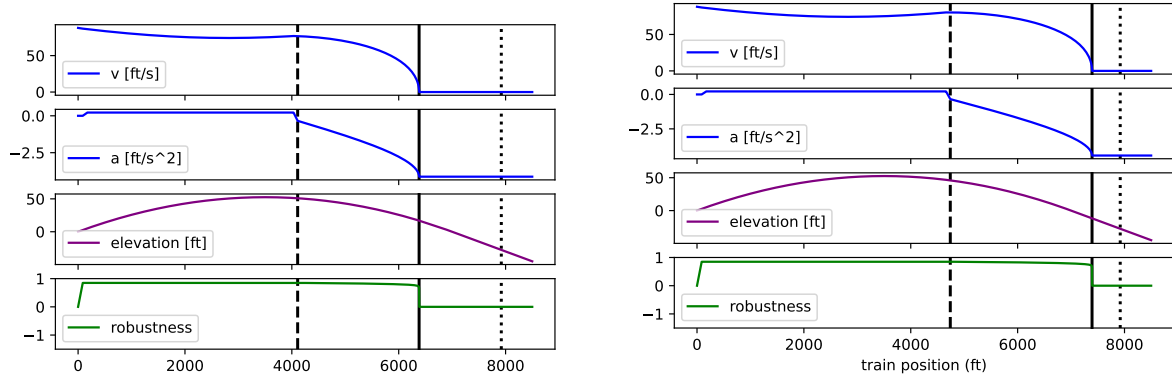
controllers for a maximum uphill/downhill gradient of 3% for slope estimation.

Numeric integration underestimates stopping distance in several configurations, whereas our verified control envelope correctly identifies the need to engage air brakes in time while simultaneously avoiding the inefficiencies of fudge factors.

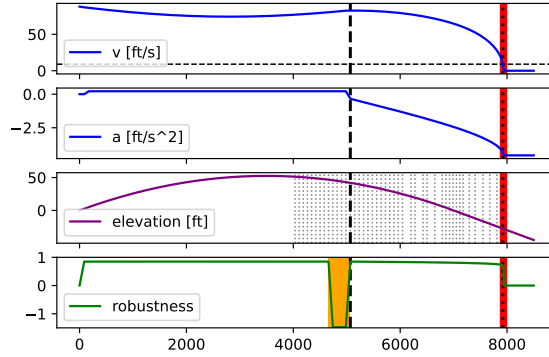
3.9 Conclusion

We created formally verified train control envelopes that account for the FRA model with all its competing influences of track grade, curve resistance, air brakes and Davis resistance. Control envelope efficiency is improved using techniques that generalize to resolve challenges in safety-critical embedded software design. Validation in simulation shows significant improvement in undershoot over conservative control envelopes that use safety offsets, and improved safety compared to control envelopes without safety offsets. VeriPhy[18], a verified pipeline that automatically converts verified dL models to verified executables can bridge the gap between the real arithmetic verification of the chapter and floating points used by software.

This chapter demonstrates that control envelopes can be designed to account for the complexity of the real world. It shows both the benefits and the challenges of doing so. The thesis next develops automation for the design process of control envelopes such as those discussed in this chapter.



(a) Base enforcement algorithm [24] undershoots by 1539ft. (b) Verified control envelope (Model 5) limits undershoot to 528ft.



(c) Base enforcement algorithm [24] without offset overshoots by 14ft ζ . Light-gray dotted lines mark other movement authority endpoints in the range 4000ft–8000ft that the algorithm would also overshoot.

Figure 3.4: Initial conditions of proof (Corollary 3.1) not satisfied ζ : an underpowered train on a crest from 3% uphill to -3% downhill slows down uphill despite full tractive effort and regains speed on the downhill segment, but needs air brakes to stay stopped; in this scenario, the movement authority ends past the rolling distance during brake rampup.

Chapter 4

1300

Automatic Verification

1301

In the previous chapter, we developed train control envelopes and then verified them. The verification process was time consuming and complex, and a significant contributor to the difficulty of creating the verified control envelope. In this chapter, we explore the use of large language models (LLMs) in combination with theorem prover KeYmaera X to *automate* the verification of control envelopes. This will be a crucial building block for scalable control envelope synthesis later in this thesis.

4.1 Introduction

1308

Extensive research has gone into developing formal methods for hybrid systems and games, with different approaches making different tradeoffs. Some tools gain impressive scalability by limiting the systems that they handle to mathematically simpler dynamics such as piecewise constant [26] or affine [82, 135]. Other tools very effectively handle nonlinear dynamics over short time horizons [3, 17] using numerical techniques, but lose precision of prediction over longer time horizons, struggling when time is unbounded. The general trend is that scalability is gained at the cost of expressiveness and precision. On the most expressive end are deductive, symbolic techniques in high-level logics [51, 119, 160, 166]. They accept nonlinear dynamics interacting with discrete control in rich, program-like structures, permit reasoning in terms of *symbolic parameters* that make theorems adaptable and re-usable, and cleanly handle infinite state space as well as unbounded domains including unbounded time horizons, all with precise reasoning free of discretization or approximation. However, they have one big disadvantage: despite efforts towards automation [58, 143], they generally require effort-intensive interactive theorem proving by an expert, which becomes a bottleneck for the complexity of systems handled and broader adoption¹. In particular, this thesis uses high level symbolic logics (dL, dGL) to express and reason about control envelopes, and therefore must contend with the same bottleneck.

LLMs have recently shown promise in automating (and therefore scaling) interactive theorem

¹Hybrid systems are fundamentally undecidable, and reasoning about real arithmetic is prohibitively expensive: for example polynomial real arithmetic is doubly exponential [38]. Still, using the deductive proving system of differential dynamic logic (dL), for example, it is possible to prove most theorems since dL is complete for algebraic invariants [127].

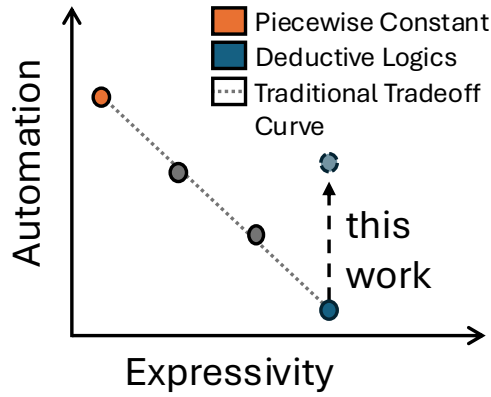


Figure 4.1: Schematic representation of tradeoffs in formal methods applied to Hybrid Systems.

proving for mathematics and (purely discrete) software [130, 133, 156]. This chapter explores the same premise for the data-scarce, mathematically complex domain of hybrid systems and games. We identify five challenging case studies in verification and create LLM-assisted pipelines to prove them. The implication is a change in the tradeoff landscape of formal methods for hybrid systems, as the application of LLMs brings unprecedented automation to symbolic, deductive verification, illustrated schematically in Fig. 4.1.

We create a pipeline that lets the LLM interact with the interactive theorem prover KeYmaera X over multiple rounds, analyzing the problem, and summarizing and fixing its own mistakes. This verification engine is already useful for manual control envelope design, since it automates the checking step after the human hypothesizes a control envelope. But additionally, in later chapters, it will be used as a subroutine while automatically *synthesizing* control envelopes. The key contribution of this chapter is an *automatic theorem proving system* that scales symbolic logic based verification for hybrid systems and games, with the pipeline design justified by ablation experiments on challenging case studies. An implementation is available online at <https://figshare.com/s/7f39a6756a7efa2e460e>.

4.2 Related Work

Hybrid games, and in particular the tactic language of KeYmaera X, are significantly more data-scarce than *LLM-based verification* in mathematics [130, 156] and software [84, 87, 133], with few proof and synthesis examples available online. To our knowledge, the work that this chapter is based on [74] is the first exploration of LLM-based verification and synthesis for hybrid systems and games using a deductive, high-level symbolic logic like differential game logic [120].

Theorem provers for hybrid systems traditionally focus on a mix of interactive proof techniques and proof heuristics for automation [102]. HHL Prover [160] and HHLPy [141] use proof scripts with annotated invariants for loops and differential equations [102]. IsaVODEs [164] uses mostly scripted proofs with automation provided by Isabelle [112]. Plaidypvs [142] embeds differential dynamic logic in PVS. KeYmaera X [51] selects proof tactics based on the syntactic shape of proof obligations; proof heuristics provide a manually selected priority between tactic alternatives, attempted in order with timeouts. These proof heuristics achieve automation in hy-

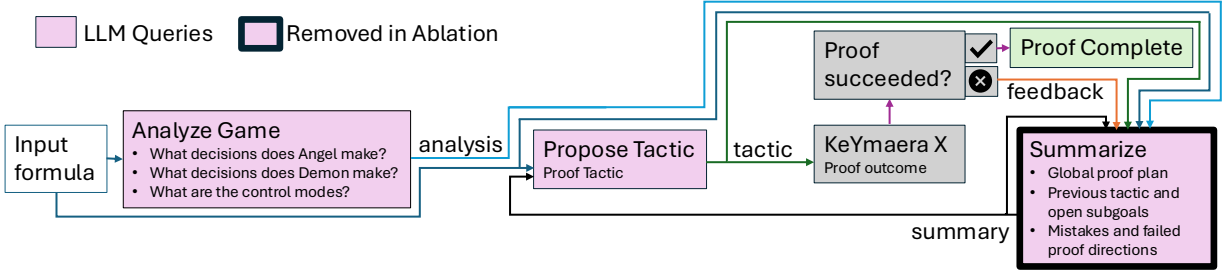


Figure 4.2: Automated Theorem Proving Pipeline. Appendix C.4 shows prompts.

brid systems, nonlinear continuous systems, and hybrid games, outperforming even proof scripts of other hybrid systems theorem provers [101, 102]. This chapter introduces a reasoning pipeline to push automation beyond the reach of KeYmaera X heuristics.

4.3 Automated Theorem Proving

We design an automated theorem proving (ATP) framework that uses LLMs to prove hybrid game/systems theorems (including those representing control envelopes) in KeYmaera X. A strength of our framework is its generality: we design the pipeline and prompts to focus on generic information about all parts of dGL, and the pipeline works across all five case studies without any case-study-specific engineering.

4.3.1 Pipeline Description

Our automated theorem proving system follows the pipeline in Fig. 4.2. A natural design is a conversational agent that iteratively proposes tactics that KeYmaera X checks. With access to the full conversation history, it corrects mistakes based on KeYmaera X output. This system fails to find proofs for three of our case studies. Therefore, to condense long contexts with limited signal, we add a summarization-and-reflection step. To make the LLM aware of the high-level meaning of the theorem, we add an initial problem analysis step. The analysis appears in subsequent prompts, having a small positive effect on robustness.

In the *Analyze Game* step, given information about the syntax of dGL, the LLM produces an analysis of the input formula that contains four parts: what actions are controlled by which player, what modes (regions of different qualitative behavior) the game has, and comments about the overall gameplay pattern. Then the proving loop begins. This consists of three steps repeated until the proof is complete. First, the LLM is prompted to propose a tactic. It is given a guide on the Bellerophon tactic language [52] that KeYmaera X supports, the formula to prove, and a summary of past proof attempts, if available. The second step is to run the proposed tactic in KeYmaera X. If the tactic completes the proof, the pipeline ends successfully. Otherwise, KeYmaera X produces some output indicating where the proof failed. This output proceeds to the third step, *Summarize*. In this step, the LLM uses the summary from the previous round (if available), and the proposed tactic and outcome of the present round to come up with a summary of the new proof state, currently known information, and suggestions/speculations about what

to do next. This summary feeds back into the next tactic proposal step, continuing the loop. Appendix C.4 lists the detailed prompts for each query. The pipeline is implemented using the Delphyne framework [85] for oracular programming with LLMs.

4.3.2 Case Studies

Five case studies are carefully developed to pose *challenging verification and synthesis problems*. The first is about maintaining safe populations under the *non-linear* predator-prey *Lotka-Volterra* model given the ability to introduce additional predators and prey at unpredictable intervals of *unbounded duration*. The second case study is about verifying the ability of a *train* control system to maintain safety over train dynamics with *Davis resistance (nonlinear) and time-dependent airbrakes*, similar to the previous chapter but without track slope and curve. This case study poses structural complexity with a *nested inner loop* over *different modes* of air brake operation, besides taking a step towards automatically verifying the train control system of Chapter 3. The third is maintaining safe temperatures during a *chemical reaction*. This system again requires reasoning about complex dynamics and different, arbitrarily alternating control modes over unbounded iterations. The solution to the dynamics lies *outside the decidable (polynomial) fragment of real arithmetic*, requiring non-trivial *deductive reasoning about differential equations via invariants*. The fourth is about the ability of a *coolant* system to meet heat absorption quotas in a nuclear power plant while limiting coolant discharge, with a *complex, timed, reach-avoid objective*, different discrete modes, and dynamics that behave very differently depending on initial conditions of variables. Anecdotally, it took a human expert three days to manually prove this case study. The fifth is about choosing the right initial conditions for a *Van der Pol* oscillator to maintain a safety property, which again requires reasoning about *nonlinear dynamics* over unbounded iterations, and therefore, effectively, *unbounded time*. The *chaotic* nature of the Van der Pol oscillator dynamics makes the *choice of initial conditions* delicate. Appendix C.3 presents these case studies in detail.

4.3.3 Outcomes

Table 4.1 summarizes the outcomes of running the tool on the case studies. A successful outcome means that the tool was able to find a proof within eight runs and a total LLM call budget of twenty four dollars. Appendix C.1 reports details. None of the case studies can be solved by KeYmaera X’s existing automation alone [143]. An ablation experiment demonstrates the significance of the summarization step. Instead of an LLM-generated summary, the tactic proposal step is given a list of all previously proposed tactics and their outcomes, solving only two of the five case studies. With the summarization step, all five case studies are solved. All experiments in this paper use OpenAI GPT-5 for LLM calls.

4.3.4 Verification Impact by Example

This section illustrates how the combination of a symbolic logic with the right level of expressivity and LLMs to automate reasoning resolves challenges beyond the reach of existing automatic techniques. For simplicity of explanation, we focus on one case study in particular, but the

Model 6 Lotka-Volterra Population Control.

assume	1	$x \leq \frac{g}{d} \wedge y \leq \frac{a}{b} \wedge \frac{g}{d} \geq x_{mn} \wedge \frac{a}{b} \geq y_{mn} \wedge x > 0 \wedge y > 0$
	2	$\wedge a > 0 \wedge b > 0 \wedge d > 0 \wedge g > 0 \wedge x \geq x_{mn} \wedge y \geq y_{mn} \rightarrow$
ctrl	3	$((x_add := *; ?x_add \geq 0; y_add := *; ?y_add \geq 0;$
	4	$x := x + x_add; y := y + y_add;$
plant	5	$\{x' = a \cdot x - b \cdot x \cdot y, y' = d \cdot x \cdot y - g \cdot y\}^d$
safe	6	$)^\times)(x \geq x_{mn} \wedge y \geq y_{mn})$

style of challenges and their resolutions generalize to all case studies and most dGL theorems including those about control envelopes.

Case Study: Lotka-Volterra. We introduce the first case study as a running example. A forest department must maintain predator population y and prey population x above minimum thresholds y_{mn} and x_{mn} respectively, for arbitrarily long. Because of unpredictable funding cycles, they can introduce additional prey and predators at unpredictable intervals, that could last arbitrarily long. This case study is inspired by an ARCH-COMP benchmark [54], but is modified by the addition of discrete control choices and adversarial game play.

Model 6 Explanation. Model 6 shows the dGL formula modeling this problem via a *game*, with two players, canonically called Angel (representing the forest department) and Demon (representing uncertainties). Angel chooses the number of animals to introduce (Line 3 and 4)², while Demon adversarially chooses the length of the interval till the next opportunity to introduce animals (Line 5), as well as how many iterations the game runs for (Line 6), forcing Angel to maintain safe populations for arbitrarily long³. Angel wins if in the end populations remain above the minimum thresholds (Line 6). The overall modal formula of form $\langle \alpha \rangle \phi$ is true in states where Angel has a strategy to play in α so that regardless of how Demon plays in α , she will win (i.e., postcondition ϕ holds). Model 6 asks us to prove that, under the assumptions of Lines 1 and 2, Angel can always win.

This case study is symbolic, with different behavior depending on the values of the parameters a, b, d, g , and variables x and y . Our reasoning must account for all uncountably infinite

²In the game $x_add := *$, Angel freely chooses any real number to assign to x_add . Then, $?x_add \geq 0$ asserts that x_add must be non-negative.

³In the game $\{x' = f(x)\}^d$, the state evolves per the flow of the differential equation $x' = f(x)$ for a duration chosen by Demon. In the game $(\alpha)^\times$, Demon chooses how many times to repeat the game α .

Traditional Proof Automation	Agent with Full History	Agent with Summarization
0/5	2/5	5/5

Table 4.1: Verification: Adding a summarization step improves success rate compared to providing the agent the full history of tactic attempts and outcomes.

possible values that these symbols can take. To verify this formula, we will necessarily have to reason about the *nonlinear dynamics of the Lotka-Volterra equation over unbounded time horizons* (Line 5). While only a few automatic solvers can reason about nonlinear dynamics, even these cannot reason about reachability over unbounded time horizons⁴. Then we must consider the discrete and adversarial dynamics: Angel chooses *some* positive real number of animals to introduce (Line 3) and we must verify whether *any* of her uncountably infinite choices will work over the nonlinear, unbounded-time dynamics. Additionally, we must consider what happens if this entire process repeats *arbitrarily many times* in a loop (Line 6). This explosion in possible behaviors is hard for traditional automatic techniques to deal with.

Yet a symbolic proof is possible by translating a high-level physical understanding of the problem to a clever choice of deductive proof rules. Looking closely at the qualitative behavior of the Lotka-Volterra dynamics reveals a way for Angel to win. The dynamics have an *equilibrium point* at $(x, y) = (g/d, a/b)$, which is above the minimum populations threshold per assumptions $a/b \geq y_{mn}$ and $g/d \geq x_{mn}$ (Model 6, Line 1). If Angel manages to set the populations to this equilibrium point at the start of each iteration, then she will satisfy her winning condition at the end of the iteration, regardless of how long Demon runs the dynamics. Since Angel can introduce any non-negative number of animals, all she needs at the start of each iteration is that predator and prey populations are no greater than the equilibrium point. Finally, many iterations of this maneuver string together correctly over loop iterations because at the end of each iteration, populations are at the equilibrium point, satisfying the requirement that populations are no greater than the equilibrium point.

$$\begin{array}{l}
\text{(loop)} \quad \frac{\Gamma \vdash J, \Delta \quad J \vdash [\alpha]J \quad J \vdash P}{\Gamma \vdash [\alpha^*]P, \Delta} \qquad (\exists R) \quad \frac{\Gamma \vdash p(e), \Delta}{\Gamma \vdash \exists x p(x), \Delta} \quad (\text{arbitrary term } e) \\
\\
\text{(dC)} \quad \frac{\Gamma \vdash [x' = f(x) \& Q]C, \Delta \quad \Gamma \vdash [x' = f(x) \& (Q \wedge C)]P, \Delta}{\Gamma \vdash [x' = f(x) \& Q]P, \Delta} \\
\\
\text{(dI)} \quad \frac{Q \vdash [x' := f(x)](F)'}{F \vdash [x' = f(x) \& Q]F}
\end{array}$$

Figure 4.3: dGL Proof Rules and Axioms relevant to the proof of Model 6. The rules are as follows: differential invariant (dI), differential cut (dC), loop (loop), and exists right (existsr).

To translate this high-level physical reasoning to a formal proof, the right choice of dGL proof rules⁵ is critical, and traditionally provided by a human expert who understands what proof rule is applicable when. To show the first step in our reasoning above, that $(x, y) = (g/d, a/b)$ is an equilibrium point of the dynamics, the rule most commonly used to prove invariants of differential equations (the *differential invariant* rule) does not immediately apply. Instead, a quick proof

⁴For instance, at ARCH-COMP 2025, where benchmark 5 involved Lotka-Volterra dynamics, the aim was only to solve for 3.64 time units.

⁵Fig. 4.3 lists all the rules referenced in this discussion; the intent is not to dwell on their individual details, but to highlight that the proof requires a sequence of complex rule applications that resist straightforward (traditional) automation.

is possible using the more rarely used *differential radical invariant* [57] rule (Eq. (4.1)). It allows us to conclude that polynomial equation $h_j = 0$ is an invariant of the dynamics if we can show that its Lie derivatives $\mathcal{L}_d(h_j), \mathcal{L}_d^2(h_j), \dots, \mathcal{L}_d^N(h_j)$ are zero, where N is the finite derivative order at which the chain of ideals $\langle h \rangle \subseteq \langle h, \mathcal{L}_p(h) \rangle \subseteq \dots$ reaches a fixed point.

$$(dRI) \quad \frac{\Gamma, Q \vdash \bigwedge_i^N \bigwedge_j \mathcal{L}_p^i(h_j) = 0}{\Gamma \vdash [x' = p \ \& \ Q] \bigwedge_j h_j = 0} \quad (4.1)$$

Eq. (4.2) shows how this applies to our case. Read from bottom to top, the differential radical invariant rule (dRI) applies to produce a sequent that is propositional in the decidable fragment of real arithmetic, which an automatic prover like Z3 [39] can discharge, establishing an equilibrium point of these nonlinear dynamics.

$$\frac{*}{\frac{x - g/d = 0 \wedge y - a/b = 0 \vdash d(ax - bxy) = 0 \wedge b(dxy - gy) = 0}{dRI \frac{x - g/d = 0 \wedge y - a/b = 0 \vdash [x' = ax - bxy, y' = dxy - gy](x - g/d = 0 \wedge y - a/b = 0)}} \quad (4.2)$$

This reasoning step must be carefully combined with the other reasoning steps to complete the proof. In this example we also need to apply rules such as the *loop* rule with the correct invariant $J \equiv x \geq x_{mn} \wedge y \geq y_{mn} \wedge x \leq g/d \wedge y \leq a/b \wedge g/d \geq x_{mn} \wedge a/b \geq y_{mn} \wedge a > 0 \wedge b > 0 \wedge d > 0 \wedge g > 0$ to demonstrate the inductiveness of Angel’s strategy over many iterations, and the *exists right* rule with a correct expression for Angel’s choice of `x_add` and `y_add`. Identifying and applying these rules requires both an understanding of the high-level solution strategy of the problem as well as dGL’s proof rules, leading to difficulties in automation using traditional techniques which lack the former and need careful, case-by-case encoding of the latter.

Proofs can be expressed succinctly as *tactic* scripts, which are sequences of proof rules with their relevant parameters that when applied in order, complete the proof. Our pipeline was able to automatically generate a 49 line tactic script that completed the proof for this case study, listed in Appendix C.2.

The verification pipeline is not yet able to automatically verify the full, most complex controller from Chapter 3. For this problem, the manually written tactic exceeded a thousand lines, making the proof so long that the pipeline “forgets” previous mistakes and is unable to complete the proof within the budget of our experiments. Future improvements to the pipeline, such as better summarization techniques, may help it scale to this proof as well. Nevertheless, the system is already useful in verification since it can still close the simpler lemmas that arise in the course of completing the larger proof. Further, the system is also useful for *synthesis*, which uses verification as a subroutine after breaking down the synthesis problem into smaller pieces. Because of how synthesis decomposes the problem, theorems to prove are generally shorter than the control envelopes, allowing this verification pipeline to help *synthesize* the full control envelope over the course of many smaller verification problems, which will be shown in Chapter 7.

4.4 Conclusion

This chapter created an automated theorem proving system that uses LLMs to prove theorems in KeYmaera X, and demonstrated its effectiveness on five challenging case studies. It already

1503 has practical utility for manual control envelope design, but becomes even more important in
1504 Chapter 7 as a subroutine for automatic control envelope synthesis.

Chapter 5

1505

Control Envelope Synthesis via Angelic Refinement

1506

1507

This chapter presents an approach for synthesizing provably correct control envelopes for hybrid systems. The algorithm fills in the blanks of a hybrid system’s sketch specifying the desired shape of the control envelope, the possible control actions, and the system’s differential equations. In order to maximize the flexibility of the control envelope, the synthesized conditions saying which control action can be chosen when should be as permissive as possible while establishing a desired safety condition from the available assumptions, which are augmented if needed. An implicit, optimal solution to this synthesis problem is characterized using hybrid systems game theory, from which explicit solutions can be derived via symbolic execution and sound, systematic game refinements. Optimality can be recovered in the face of approximation via a dual game characterization. The resulting algorithm, *Control Envelope Synthesis via Angelic Refinements (CESAR)*, is demonstrated in a range of safe control envelope synthesis examples with different control challenges. This chapter solves the synthesis problem for *time-triggered* control, and will be generalized by the next chapter.

1508
1509
1510
1511
1512
1513
1514
1515
1516
1517
1518
1519
1520

5.1 Introduction

1521

Control envelope design is challenging. Engineers are good at specifying the *shape* of a model and listing the possible control actions by translating client specifications, which is crucial for the fidelity of the resulting model. But identifying the exact control conditions required for safety in such a model is much harder, requiring design insights and creativity, and is the main point of the deep area of control theory. Most initial system designs are incorrect and need to be fixed before verification succeeds. Fully rigorous justification of the safety of the control conditions requires full verification of the resulting controller in the hybrid systems model. This chapter presents a synthesis technique that addresses this hard problem by filling in the holes of a hybrid systems model to identify a correct-by-construction control envelope that is as permissive as possible.

1522
1523
1524
1525
1526
1527
1528
1529
1530

Our approach is called *Control Envelope Synthesis via Angelic Refinements (CESAR)*. The idea is to implicitly characterize the optimal safe control envelope via hybrid games yielding maximally permissive safe solutions in differential game logic [123]. To derive explicit solutions

1531
1532
1533

Model 7 The train ETCS model (slightly modified from [126]). Framed formulas are initially blank and are automatically synthesized by our tool as indicated.

assum	1	$A > 0 \wedge B > 0 \wedge T > 0 \wedge v \geq 0 \wedge$
ctrlable	2	$\boxed{e - p > v^2/2B} \rightarrow \{ \{$
ctrl	3	$(\quad (? \boxed{e - p > vT + AT^2/2 + (v + AT)^2/2B} ; a := A)$
	4	$\cup (? \boxed{\text{true}} ; a := -B) \quad) ;$
plant	5	$(t := 0 ; \{p' = v, v' = a, t' = 1 \ \& \ t \leq T \wedge v \geq 0\})$
safe	6	$\}^*](e - p > 0)$

used for controller monitoring at runtime, we successively refine the games while preserving safety and, if possible, optimality. Our experiments demonstrate that CESAR solves hybrid systems synthesis challenges requiring different control insights.

Contributions. The primary contributions of this chapter behind CESAR are:

- optimal hybrid systems control envelope synthesis via hybrid games.
- differential game logic formulas identifying optimal safe control envelopes.
- refinement techniques for safe control envelope approximation, including *bounded fixpoint unrollings* via a recurrence, which exploits *action permanence* (a hybrid analogue to idempotence).
- a primal/dual game counterpart optimality criterion.

5.2 Background

Control Safety Envelopes by Example. In order to separate safety-critical aspects from other system goals during control design, we abstractly describe the safe choices of a controller with safe control envelopes that deliberately underspecify when and how to exactly execute certain actions. They focus on describing in which regions it is safe to take actions. For example, Model 7 designs a train control envelope [126] that must stop the train by the *end of movement authority* e located somewhere ahead, as assigned by the train network scheduler. Past e , there may be obstacles or other trains. The train’s control choices are to accelerate or brake as it moves along the track. The goal of CESAR is to synthesize the framed formulas in the model, that are initially blank.

Line 6 describes the *safety property* that is to be enforced at all times: the train driving at position p with velocity v must not go past position e . Line 1 lists *modeling assumptions*: the train is capable of both acceleration ($A > 0$) and deceleration ($B > 0$), the controller latency is positive ($T > 0$) and the train cannot move backwards as a product of braking (this last fact is also reflected by having $v \geq 0$ as a domain constraint for the plant on Line 5). These assumptions are fundamentally about the physics of the problem being considered. In contrast, Line 2 features a *controllability assumption* that can be derived from careful analysis. Here, this synthesized assumption says that the train cannot start so close to e that it won’t stop in time even if it starts

braking immediately. Line 3 and Line 4 describe a train controller with two actions: accelerating (1562
 $a := A$) and braking ($a := -B$). Each action is guarded by a synthesized formula, called an (1563
action guard that indicates when it is safe to use. Angel has control over which action runs, (1564
and adversarially plays with the objective of violating safety conditions. But Angel’s options (1565
are limited to only safe ones because of the synthesized action guards, ensuring that Demon still (1566
wins and the overall formula is valid. In this case, braking is always safe whereas acceleration (1567
can only be allowed when the distance to end position e is sufficiently large. Finally, the plant on (1568
Line 5 uses differential equations to describe the train’s kinematics. A timer variable t is used to (1569
ensure that no two consecutive runs of the controller are separated by more than time T . Thus, (1570
this controller is *time-triggered*. (1571

Overview of CESAR. CESAR first identifies the optimal solution for the blank of Line 2. In- (1572
tuitively, this blank should identify a *controllable invariant*, which denotes a set of states where (1573
a controller with choice between acceleration and braking has some strategy (to be enforced by (1574
the conditions of Line 3 and Line 4) that guarantees safe control forever. Such states can be char- (1575
acterized by the following dGL formula where Demon, as a proxy for the controller, decides (1576
whether to accelerate or brake: $[((a := A \cap a := -B); \text{plant})^*] \text{safe}$ where *plant* and *safe* (1577
are from Model 7. When this formula is true, Demon, who decides when to brake to maintain (1578
the safety contract, has a winning strategy that the controller can mimic. When it is false, De- (1579
mon, a perfect player striving to maintain safety, has no winning strategy, so a controller has no (1580
guaranteed way to stay safe either. (1581

This dGL formula provides an *implicit* characterization of the optimal controllable invariant (1582
from which we derive an explicit formula in $\mathcal{P}_{\mathbb{R}}$ to fill the blank with using symbolic execution. (1583
Symbolic execution solves a game following the axioms of dGL to produce an equivalent $\mathcal{P}_{\mathbb{R}}$ (1584
formula (Section 5.3.7). However, our dGL formula contains a loop, for which symbolic execu- (1585
tion will not terminate in finite time. To reason about the loop, we *refine* the game, modifying (1586
it so that it is easier to symbolically execute, but still at least as hard for Demon to win so that (1587
the controllable invariant that it generates remains sound. In this example, the required game (1588
transformation first restricts Demon’s options to braking. Then, it eliminates the loop using the (1589
observation that the repeated hybrid iterations $(a := -B; \text{plant})^*$ behave the same as just follow- (1590
ing the continuous dynamics of braking for unbounded time. It replaces the original game with (1591
 $a := -B; t := 0; \{p' = v, v' = a \ \& \ \wedge \ v \geq 0\}$, which is loop-free and easily symbolically (1592
executed. Symbolically executing this game to reach safety condition *safe* yields controllable (1593
invariant $e - p > \frac{v^2}{2B}$ to fill the blank of Line 2. (1594

Intuitively, this refinement (formalized in Section 5.3.4) captures situations where the con- (1595
troller stays safe forever by picking a single control action (braking). It generates the optimal (1596
solution for this example because braking forever is the dominant strategy: given any state, if (1597
braking forever does not keep the train safe, then certainly no other strategy will. However, there (1598
are other problems where the dominant control strategy requires the controller to strategically (1599
switch between actions, and this refinement misses some controllable invariant states. So we in- (1600
troduce a new refinement: bounded game unrolling via a recurrence (Section 5.3.5). A solution (1601
generated by unrolling n times captures states where the controller can stay safe by switching (1602
control actions up to n times. (1603

1604 Having synthesized the controllable invariant, CESAR fills the action guards (Line 3 and
 1605 Line 4). An action should be permissible when running it for one iteration maintains the control-
 1606 lable invariant. For example, acceleration is safe to execute exactly when $[a := A; \text{plant}]e - p >$
 1607 $\frac{v^2}{2B}$. We symbolically execute this game to synthesize the formula that fills the guard of Line 3.

1608 5.3 Approach

1609 This section formally introduces the *Control Envelope Synthesis via Angelic Refinements (CESAR)* approach for hybrid systems control envelope synthesis.

1611 5.3.1 Problem Definition

1612 We frame the problem of *control envelope synthesis* in terms of filling in holes $\sqcup$ in a problem
 1613 of the following shape:

$$1614 \quad \text{prob} \equiv \text{assum} \wedge \sqcup \rightarrow [((\cup_i (? \sqcup_i ; \text{act}_i)) ; \text{plant})^*] \text{safe}. \quad (5.1)$$

1615 Here, the control envelope consists of a nondeterministic choice between a finite number of
 1616 guarded actions. Each action act_i is guarded by a condition $\sqcup_i$ to be determined in a way that
 1617 ensures safety within a controllable invariant [9, 59] $\sqcup$ to be synthesized also. The plant is
 1618 defined by the following template:

$$1619 \quad \text{plant} \equiv t := 0 ; \{x' = f(x), t' = 1 \ \& \ \text{domain} \wedge t \leq T\}. \quad (5.2)$$

1620 This ensures that the plant must yield to the controller after time T at most, where T is assumed
 1621 to be positive and constant. In addition, we make the following assumptions:

- 1622 1. Components **assum**, **safe** and **domain** are propositional arithmetic formulas.
- 1623 2. Timer variable t is fresh (does not occur except where shown in template).
- 1624 3. Programs act_i are discrete dL programs that can involve choices, assignments and tests
 1625 with propositional arithmetic. Variables assigned by act_i must not appear in **safe**. In
 1626 addition, act_i must terminate in the sense that $\models \langle \text{act}_i \rangle \text{true}$.
- 1627 4. The modeling assumptions **assum** are invariant in the sense that

$$1628 \quad \models \text{assum} \rightarrow [(\cup_i \text{act}_i) ; \text{plant}] \text{assum}.$$

1629 This holds trivially for assumptions about constant parameters such as $A > 0$ in Model 7
 1630 and this ensures that the controller can always rely on them being true.

1631 **Definition 5.1.** A *solution* to the synthesis problem above is defined as a pair (I, G) where I is
 1632 a formula and G maps each action index i to a formula G_i . In addition, the following conditions
 1633 must hold:

- 1634 1. Safety is guaranteed: $\text{prob}(I, G) \equiv \text{prob}[\sqcup \mapsto I, \sqcup_i \mapsto G_i]$ is valid and $(\text{assum} \wedge I)$ is
 1635 a loop invariant that proves it so.
- 1636 2. There is always some action: $(\text{assum} \wedge I) \rightarrow \bigvee_i G_i$ is valid.

1637 Condition 2 is crucial for using the resulting nondeterministic control envelope, since it guar-
 1638 antees that safe actions are always available as a fallback.

5.3.2 An Optimal Solution

1639

Solutions to a synthesis problem may differ in quality. Intuitively, a solution is better than another if it allows for a strictly larger controllable invariant. In case of equality, the solution with the more permissive control envelope wins. Formally, given two solutions $S = (I, G)$ and $S' = (I', G')$, we say that S' is better or equal to S (written $S \sqsubseteq S'$) if and only if $\models \text{assum} \rightarrow (I \rightarrow I')$ and additionally either $\models \text{assum} \rightarrow \neg(I' \rightarrow I)$ or $\models (\text{assum} \wedge I) \rightarrow \bigwedge_i (G_i \rightarrow G'_i)$. Given two solutions S and S' , one can define a solution $S \sqcap S' = (I \vee I', i \mapsto (I \wedge G_i \vee I' \wedge G'_i))$ that is better or equal to both S and S' ($S \sqsubseteq S \sqcap S'$ and $S' \sqsubseteq S \sqcap S'$). A solution S' is called the *optimal solution* when it is the maximum element in the ordering, so that for any other solution S , $S \sqsubseteq S'$. The optimal solution exists and is expressible in dGL:

$$I^{\text{opt}} \equiv [(\bigcap_i \text{act}_i); \text{plant}]^* \text{ safe} \quad (5.3) \quad 1649$$

$$G_i^{\text{opt}} \equiv [\text{act}_i; \text{plant}] I^{\text{opt}}. \quad (5.4) \quad 1650$$

Intuitively, I^{opt} characterizes the set of all states from which an optimal controller (played here by Demon) can keep the system safe forever. In turn, G^{opt} is defined to allow any control action that is guaranteed to keep the system within I^{opt} until the next control cycle as characterized by a modal formula. Section 5.3.3 formally establishes the correctness and optimality of $S^{\text{opt}} \equiv (I^{\text{opt}}, G^{\text{opt}})$.

While it is theoretically reassuring that an optimal solution exists that is at least as good as all others and that this optimum can be characterized in dGL, such a solution is of limited practical usefulness since Eq. (5.3) cannot be executed without solving a game at runtime. Rather, we are interested in *explicit* solutions where I and G are quantifier-free real arithmetic formulas. There is no guarantee in general that such solutions exist that are also optimal, but our goal is to devise an algorithm to find them in the many cases where they exist or find safe approximations otherwise.

5.3.3 Controllable Invariants

1663

The fact that S^{opt} is a solution can be characterized in logic with the notion of a controllable invariant that, at each of its points, admits some control action that keeps the plant in the invariant for one round. All lemmas and theorems throughout this chapter are proved in Appendix D.

Definition 5.2 (Controllable Invariant). A *controllable invariant* is a formula I such that $\models I \rightarrow \text{safe}$ and $\models I \rightarrow \bigvee_i [\text{act}_i; \text{plant}] I$.

From this perspective, I^{opt} can be seen as the largest controllable invariant.

Lemma 5.1. I^{opt} is a controllable invariant and it is optimal in the sense that $\models I \rightarrow I^{\text{opt}}$ for any controllable invariant I .

Moreover, not just I^{opt} , but *every* controllable invariant induces a solution. Indeed, given a controllable invariant I , we can define $\mathcal{G}(I) \equiv (i \mapsto [\text{act}_i; \text{plant}] I)$ for the *control guards induced by I* . $\mathcal{G}(I)$ chooses as the guard for each action act_i the modal condition ensuring that act_i , preserves I after the plant.

Lemma 5.2. If I is a controllable invariant, then $(I, \mathcal{G}(I))$ is a solution (Def. 5.1).

Conversely, a controllable invariant can be derived from any solution.

Lemma 5.3. *If (I, G) is a solution, then $I' \equiv (\text{assum} \wedge I)$ is a controllable invariant. Moreover, we have $(I, G) \sqsubseteq (I', \mathcal{G}(I'))$.*

Solution comparisons w.r.t. $\sqsubseteq$ reduce to implications for controllable invariants.

Lemma 5.4. *If I and I' are controllable invariants, then $(I, \mathcal{G}(I)) \sqsubseteq (I', \mathcal{G}(I'))$ if and only if $\models \text{assum} \rightarrow (I \rightarrow I')$.*

Taken together, these lemmas allow us to establish the optimality of S^{opt} .

Theorem 5.1. *S^{opt} is an optimal solution (i.e. a maximum w.r.t. $\sqsubseteq$) of Def. 5.1.*

This shows the roadmap for the rest of the chapter: finding solutions to the control envelope synthesis problem reduces to finding controllable invariants that imply I^{opt} , which can be found by restricting the actions available to Demon in I^{opt} to guarantee safety, thereby *refining* the associated game.

5.3.4 One-Shot Fallback Refinement

The simplest refinement of I^{opt} is obtained when fixing a single fallback action to use in all states (if that is safe). A more general refinement considers different fallback actions in different states, but still only plays one such action forever.

Using the dGL axioms, any loop-free dGL formula whose ODEs admit solutions expressible in real arithmetic can be automatically reduced to an equivalent first-order arithmetic formula (in $\text{FOL}_{\mathbb{R}}$). An equivalent propositional arithmetic formula in $\mathcal{P}_{\mathbb{R}}$ can be computed via quantifier elimination (QE). For example:

$$\begin{aligned}
& [(v := 1 \cap v := -1); \{x' = v\}] x \neq 0 \\
& \equiv [v := 1 \cap v := -1] [\{x' = v\}] x \neq 0 && \text{by } [;] \\
& \equiv [v := 1] [\{x' = v\}] x \neq 0 \vee [v := -1] [\{x' = v\}] x \neq 0 && \text{by } [\cap] \\
& \equiv [\{x' = 1\}] x \neq 0 \vee [\{x' = -1\}] x \neq 0 && \text{by } [:=] \\
& \equiv (\forall t \geq 0 \ x + t \neq 0) \vee (\forall t \geq 0 \ x - t \neq 0) && \text{by } ['], [:=] \\
& \equiv x > 0 \vee x < 0 && \text{by QE} .
\end{aligned}$$

Even when a formula features nonsolvable ODEs, techniques exist to compute weakest preconditions for differential equations, with conservative approximations [143] or even exactly in some cases [22, 129]. In the rest of this section and for most of this chapter, we are therefore going to assume the existence of a **reduce** oracle that takes as an input a loop-free dGL formula and returns a quantifier-free arithmetic formula that is equivalent modulo some assumptions. Section 5.3.7 shows how to implement and optimize **reduce**.

Definition 5.3 (Reduction Oracle). *A reduction oracle is a function **reduce** that takes as an input a loop-free dGL formula F and an assumption $A \in \mathcal{P}_{\mathbb{R}}$. It returns a formula $R \in \mathcal{P}_{\mathbb{R}}$ along with a boolean flag **exact** such that the formula $A \rightarrow (R \rightarrow F)$ is valid, and if **exact** is true, then $A \rightarrow (R \leftrightarrow F)$ is valid as well.*

Back to our original problem, I^{opt} is not directly reducible since it involves a loop. However, conservative approximations can be computed by restricting the set of strategies that the Demon

player is allowed to use. One extreme case allows Demon to only use a single action act_i repeatedly as a fallback (e.g. braking in the train example). In this case, we get a controllable invariant $[(\text{act}_i; \text{plant})^*] \text{ safe}$, which further simplifies into $[\text{act}_i; \text{plant}_\infty] \text{ safe}$ with

$$\text{plant}_\infty \equiv \{x' = f(x), t' = 1 \ \& \ \text{domain}\}$$

a variant of plant that never yields control. For this last step to be valid though, a technical assumption is needed on act_i , which we call *action permanence*.

Definition 5.4 (Action Permanence). An action act_i is said to be *permanent* if and only if $(\text{act}_i; \text{plant}; \text{act}_i) \equiv (\text{act}_i; \text{plant})$, i.e., they are equivalent games.

Intuitively, an action is *permanent* if executing it more than once in a row has no consequence for the system dynamics. This is true in the common case of actions that only assign constant values to control variables that are read but not modified by the plant, such as $a := A$ and $a := -B$ in Model 7.

Lemma 5.5. If act_i is permanent, $\models [(\text{act}_i; \text{plant})^*] \text{ safe} \leftrightarrow [\text{act}_i; \text{plant}_\infty] \text{ safe}$.

Our discussion so far identifies the following approximation to our original synthesis problem, where $\mathbf{P}$ denotes the set of all indexes of permanent actions:

$$\begin{aligned} I^0 &\equiv [(\bigcap_{i \in \mathbf{P}} \text{act}_i); \text{plant}_\infty] \text{ safe}, \\ G_i^0 &\equiv [\text{act}_i; \text{plant}] I^0. \end{aligned}$$

Here, I^0 encompasses all states from which the agent can guarantee safety indefinitely with a single permanent action. G^0 is constructed according to $\mathcal{G}(I^0)$ and only allows actions that are guaranteed to keep the agent within I^0 until the next control cycle. Note that I^0 degenerates to false in cases where there are no permanent actions, which does not make it less of a controllable invariant.

Theorem 5.2. I^0 is a controllable invariant.

Moreover, in many examples of interest, I^0 and I^{opt} are equivalent since an optimal fallback strategy exists that only involves executing a single action. This is the case in particular for Model 7, where

$$\begin{aligned} I^0 &\equiv [a := -B; \{p' = v, v' = a \ \& \ v \geq 0\}] e - p > 0 \\ &\equiv e - p > v^2/2B \end{aligned}$$

characterizes all states at safe braking distance to the obstacle and G^0 associates the following guard to the acceleration action:

$$\begin{aligned} G_{a=A}^0 &\equiv [a := A; \{p' = v, v' = a, t' = 1 \ \& \ v \geq 0 \wedge t \leq T\}] e - p > v^2/2B \\ &\equiv e - p > vT + AT^2/2 + (v + AT)^2/2B \end{aligned}$$

That is, accelerating is allowed if doing so is guaranteed to maintain sufficient braking distance until the next control opportunity. Section 5.3.6 discusses automatic generation of a proof that (I^0, G^0) is an optimal solution for Model 7.

5.3.5 Bounded Fallback Unrolling Refinement

In Section 5.3.4, we derived a solution by computing an underapproximation of I^{opt} where the fallback controller (played by Demon) is only allowed to use a one-shot strategy that picks a single action and plays it forever. Although this approximation is always safe and, in many cases of interest, happens to be exact, it does lead to a suboptimal solution in others. In this section, we allow the fallback controller to switch actions a bounded number of times before it plays one forever. There are still cases where doing so is suboptimal (imagine a car on a circular race track that is forced to maintain constant velocity). But this restriction is in line with the typical understanding of a fallback controller, whose mission is not to take over a system indefinitely but rather to maneuver it into a state where it can safely get to a full stop [114].

For all bounds $n \in \mathbb{N}$, we define a game where the fallback controller (played by Demon) takes at most n turns to reach the region I^0 in which safety is guaranteed indefinitely. During each turn, it picks a permanent action and chooses a time θ in advance for when it wishes to play its next move. Because the environment (played by Angel) has control over the duration of each control cycle, the fallback controller cannot expect to be woken up after time θ exactly. However, it can expect to be provided with an opportunity for its next move within the $[\theta, \theta + T]$ time window since the plant can never execute for time greater than T . Formally, we define I^n as follows:

$$I^n \equiv [\text{step}^{\times \leq n}; \text{forever}] \text{ safe} \quad \text{forever} \equiv (\bigcap_{i \in \mathcal{P}} \text{act}_i); \text{plant}_{\infty}$$

$$\text{step} \equiv (\theta := *; ?\theta \geq 0)^d; (\bigcap_{i \in \mathcal{P}} \text{act}_i); \text{plant}_{\theta+T}; ?\text{safe}^d; ?t \geq \theta$$

where $\text{plant}_{\theta+T}$ is the same as plant , except that the domain constraint $t \leq T$ is replaced by $t \leq \theta + T$. Equivalently, we can define I^n by induction as follows:

$$I^{n+1} \equiv I^n \vee [\text{step}] I^n \quad I^0 \equiv [\text{forever}] \text{ safe}, \quad (5.5)$$

where the base case coincides with the definition of I^0 in Section 5.3.4. Importantly, I^n is a loop-free controllable invariant and so **reduce** can compute an explicit solution to the synthesis problem from I^n .

Theorem 5.3. *I^n is a controllable invariant for all $n \geq 0$.*

Theorem 5.3 establishes a nontrivial result since it overcomes the significant gap between the *fantasized* game that defines I^n and the *real* game being played by a time-triggered controller. Our proof critically relies on the action permanence assumption along with the following property of differential equations, which establishes that ODE programs preserve a specific form of reach-avoid property as a result of being deterministic.

Lemma 5.6. *Consider a property of the form $R(a, b) \equiv [\alpha_b] (S \wedge (t \geq a \rightarrow I))$ with $\alpha_b \equiv (t := 0; \{x' = f(x), t' = 1 \ \& \ Q \wedge t \leq b\})$. Then this formula is valid:*

$$c \leq b \wedge R(a, b) \rightarrow [\alpha_c] R(a - t, b - t).$$

Example. As an illustration, consider the example in Fig. 5.1 and Model 8 of a 2D robot moving in a corridor that forms an angle. The robot is only allowed to move left or down at a

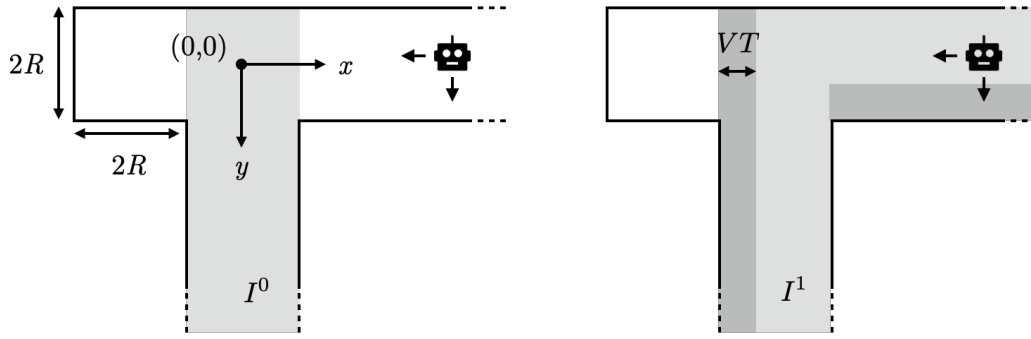


Figure 5.1: Robot navigating a corridor (Model 8). A 2D robot must navigate safely within a corridor with a dead-end without crashing against a wall. The corridor extends infinitely on the bottom and on the right. The robot can choose between going left and going down with a constant speed V . The left diagram shows I^0 in gray. The right diagram shows I^1 under the additional assumption $VT < 2R$ (I^1 and I^0 are otherwise equivalent). A darker shade of gray is used for regions of I^1 where only one of the two available actions is safe according to G^1 .

constant velocity and must not crash against a wall. Computing I^0 gives us the vertical section of the corridor, in which going down is a safe one-step fallback. Computing I^1 forces us to distinguish two cases. If the corridor is wider than the maximal distance travelled by the robot in a control cycle ($VT > 2R$), then the upper section of the corridor is controllable (with the exception of a dead-end that we prove to be uncontrollable in Section 5.3.6). On the other hand, if the corridor is too narrow, then I^1 is equivalent to I^0 . Formally, we have $I^1 \equiv (y > -R \wedge |x| < R) \vee (VT < 2R \wedge (x > -R \wedge |y| < R))$. Moreover, computing I^2 gives a result that is equivalent to I^1 . From this, we can conclude that I^1 is equivalent to I^n for all $n \geq 1$. Intuitively, it is optimal with respect to *any* finite fallback strategy (restricted to permanent actions).

Model 8 Robot navigating a corridor with framed solutions of holes.

assum	1	$V > 0 \wedge T > 0$
ctrlable	2	$\wedge \left[(y > -R \wedge x < R) \vee (VT < 2R \wedge (x > -R \wedge y < R)) \right] \rightarrow \{$
ctrl	3	$(\text{?}[x > -R + VT]; v_x := -V; v_y := 0)$
	4	$\cup (\text{?}[y < R - VT \vee x < R]; v_x := 0; v_y := V) \quad \};$
plant	5	$(t := 0; \{x' = v_x, y' = v_y, t' = 1 \ \& \ t \leq T\})$
safe	6	$\}^*]((x > -3R \wedge y < R) \vee (y > -R \wedge x < R))$

The controllable invariant unrolling I^n has a natural stopping criterion.

Lemma 5.7. *If $I^n \leftrightarrow I^{n+1}$ is valid for some $n \geq 0$, then $I^n \leftrightarrow I^m$ is valid for all $m \geq n$ and $I^n \leftrightarrow I^\omega$ is valid where $I^\omega \equiv [\text{step}^\times; \text{forever}] \text{ safe}$.*

5.3.6 Proving Optimality via the Dual Game

Suppose one found a controllable invariant I using techniques from the previous section. To prove it optimal, one must show that $\models \text{assum} \rightarrow (I^{\text{opt}} \rightarrow I)$. By contraposition and $[\alpha] P \leftrightarrow \neg \langle \alpha \rangle \neg P$ ($[\cdot]$), this is equivalent to proving that:

$$\models \text{assum} \wedge \neg I \rightarrow \underbrace{\langle (\cap_i \text{act}_i); \text{plant} \rangle^*}_{\neg I^{\text{opt}}} \neg \text{safe}. \quad (5.6)$$

We define the largest *uncontrollable* region $U^{\text{opt}} \equiv \neg I^{\text{opt}}$ as the right-hand side of implication 5.6 above. Intuitively, U^{opt} characterizes the set of all states from which the environment (played by Angel) has a winning strategy against the controller (played by Demon) for reaching an unsafe state. In order to prove the optimality of I , we compute a sequence of increasingly strong approximations U of U^{opt} such that $U \rightarrow U^{\text{opt}}$ is valid. We do so via an iterative process, in the spirit of how we approximate I^{opt} via bounded fallback unrolling (Section 5.3.5), although the process can be guided by the knowledge of I this time. If at any point we manage to prove that $\text{assum} \rightarrow (I \vee U)$ is valid, then I is optimal.

One natural way to compute increasingly good approximations of U^{opt} is via loop unrolling. The idea is to improve approximation U by adding states from where the environment can reach U by running the control loop once, formally, $\langle (\cap_i \text{act}_i); \text{plant} \rangle U$. This unrolling principle can be useful. However, it only augments U with new states that can reach U in time T at most. So it cannot alone prove optimality in cases where violating safety from an unsafe state takes an unbounded amount of time.

For concreteness, let us prove the optimality of I^0 in the case of Model 7. In [126] essentially the following statement is proved when arguing for optimality: $\models \text{assum} \wedge \neg I^0 \rightarrow \langle (a := -B; \text{plant})^* \rangle \neg \text{safe}$. This is identical to our optimality criterion from Eq. (5.6), except that Demon's actions are restricted to braking. Intuitively, this restriction is sound since accelerating always makes things worse as far as safety is concerned. If the train cannot be saved with braking alone, adding the option to accelerate will not help a bit. In this work, we propose a method for formalizing such arguments within dGL to arbitrary systems.

Our idea for doing so is to consider a system made of two separate copies of our model. One copy has all actions available whereas the other is only allowed a single action (e.g. braking). Given a safety metric m (i.e. a term m such that $\models m \leq 0 \rightarrow \neg \text{safe}$), we can then formalize the idea that “action i is always better w.r.t safety metric m ” within this joint system.

Definition 5.5 (Uniform Action Optimality). Consider a finite number of discrete dL programs α_i and $p \equiv \{x' = f(x) \ \& \ Q\}$. Let $V = \text{BV}(p) \cup \bigcup_i \text{BV}(\alpha_i)$ be the set of all variables written by p or some α_i . For any term θ and integer n , write $\theta^{(n)}$ for the term that results from θ by renaming all variables $v \in V$ to a fresh tagged version $x^{(n)}$. Using a similar notation for programs and formulas, define $p^{(1,2)} \equiv \{(x^{(1)})' = f(x^{(1)}), (x^{(2)})' = f(x^{(2)}) \ \& \ Q^{(1)} \wedge Q^{(2)}\}$. We say that action j is *uniformly optimal* with respect to safety metric m if and only if:

$$\models m^{(1)} \geq m^{(2)} \rightarrow [\alpha_j^{(1)}; (\cup_i \alpha_i^{(2)})]; p^{(1,2)} m^{(1)} \geq m^{(2)}.$$

$\text{best}_j((\alpha_i)_i, p, m)$ denotes that action j is uniformly optimal with respect to m for actions α_i and dynamics p .

With such a concept in hand, we can formally establish the fact that criterion Eq. (5.6) can be relaxed in the existence of uniformly optimal actions.

Theorem 5.4. *Consider a finite number of discrete dL programs α_i such that $\models \langle \alpha_i \rangle$ true for all i and $p \equiv \{x' = f(x) \ \& \ q \geq 0\}$. Then, provided that $\text{best}_j((\alpha_i)_i, p, m)$ and $\text{best}_j((\alpha_i)_i, p, -q)$ (no other action stops earlier because of the domain constraint), we have:*

$$\models \langle ((\cap \alpha_i); p)^* \rangle m \leq 0 \leftrightarrow \langle (\alpha_j; p)^* \rangle m \leq 0 .$$

A general heuristic for leveraging Theorem 5.4 to grow U automatically works as follows. First, it considers $R \equiv \text{assum} \wedge \neg I \wedge \neg U$ that characterizes states that are not known to be controllable or uncontrollable. Then, it picks a disjunct $\bigwedge_j R_j$ of the disjunctive normal form of R and computes a forward invariant region V that intersects with it: $V \equiv \bigwedge_j \{R_j : \text{assum}, R_j \vdash [(\cup_i \text{act}_i); \text{plant}] R_j\}$. Using V as an assumption to simplify $\neg U$ may suggest metrics to be used with Theorem 5.4. For example, observing $\models V \rightarrow (\neg U \rightarrow (\theta_1 > 0 \wedge \theta_2 > 0))$ suggests picking metric $m \equiv \min(\theta_1, \theta_2)$ and testing whether $\text{best}_j(\text{act}, p, m)$ is true for some action j . If such a uniformly optimal action exists, then U can be updated as $U \leftarrow U \vee (V \wedge \langle (\text{act}_j; \text{plant})^* \rangle m \leq 0)$. The solution I^1 for the corridor (Model 8) can be proved optimal automatically using this heuristic in combination with loop unrolling.

5.3.7 Implementing the Reduction Oracle

The CESAR algorithm assumes the existence of a *reduction oracle* that takes as an input a loop-free dGL formula and attempts to compute an equivalent formula within the fragment of propositional arithmetic. When an exact solution cannot be found, an implicant is returned instead and flagged appropriately (Def. 5.3). This section discusses our implementation of such an oracle.

As discussed in Section 5.3.4, exact solutions can be computed systematically when all ODEs are solvable by first using the dGL axioms to eliminate modalities (see Appendix D) and then passing the result to a *quantifier elimination algorithm* for first-order arithmetic [29, 151]. Although straightforward in theory, a naïve implementation of this idea hits two practical barriers. First, quantifier elimination is expensive and its cost increases rapidly with formula complexity [38, 161]. Second, the output of existing QE implementations can be unnecessarily large and redundant. In iterated calls to the reduction oracle, these problems can compound each other.

To alleviate this issue, our implementation performs *eager simplification* at intermediate stages of computation, between some axiom application and quantifier-elimination steps. This optimization significantly reduces output solution size and allows CESAR to solve a benchmark that would otherwise timeout after 20 minutes in 26s. Appendix A.4 further discusses the impact of eager simplification. Still, the doubly exponential complexity of quantifier elimination puts a limit on the complexity of problems that CESAR can currently tackle.

In the general case, when ODEs are not solvable, our reduction oracle is still often able to produce *approximate* solutions using differential invariants generated automatically by existing tools [143]. Differential invariants are formulas that stay true throughout the evolution of an ODE system.¹ To see how they apply, consider the case of computing $\text{reduce}(\{x' = f(x)\} P, A)$

¹dGL provides ways to reason about differential invariants without solving the corresponding differential equa-

where P is the postcondition formula that must be true after executing the differential equation, and A is the assumptions holding true initially. Suppose that formula $D(x)$ is a differential invariant such that $D(x) \rightarrow P$ is valid. Then, a precondition sufficient to ensure that P holds after evolution is $A \rightarrow D(x)$. For a concrete example, Appendix A.2 shows how our reduction oracle computes the precondition for the dynamics of the `parachute` benchmark. It first uses the Pegasus tool [143] to identify a Darboux polynomial, suggesting an initial differential invariant D_0 . Once we have D_0 , the additional information required to conclude post condition P is $D_0 \rightarrow P$. To get an invariant formula that implies $D_0 \rightarrow P$, eliminate all the changing variables $\{x, v\}$ in the formula $\forall x \forall v (D_0 \rightarrow P)$, resulting in a formula D_1 . D_1 is a differential invariant since it features no variable that is updated by the ODEs. Our reduction oracle returns $D_0 \wedge D_1$, an invariant that entails postcondition P . More details on our implementation of `reduce` and how it deals with ODEs in particular can be found in Appendix A.1.

5.3.8 The CESAR Algorithm

The CESAR algorithm for synthesizing control envelopes is summarized in Algorithm 1. It is expressed as a generator that yields a sequence of solutions with associated optimality guarantees. Possible guarantees include “*sound*” (no optimality guarantee, only soundness), “*k-optimal*” (sound and optimal w.r.t all k -switching fallbacks with permanent actions), “ *ω -optimal*” (sound and optimal w.r.t all finite fallbacks with permanent actions) and “*optimal*” (sound and equivalent to S^{opt}). Line 11 performs the optimality test described in Section 5.3.6. Finally, Line 10 performs an important soundness check for the cases where an approximation has been made along the way of computing (I^n, G^n) . In such cases, I is not guaranteed to be a controllable invariant and thus Case (2) of Def. 5.1 must be checked explicitly.

When given a problem with solvable ODEs and provided with a complete QE implementation within `reduce`, CESAR is guaranteed to generate a solution in finite time with an “*n-optimal*” guarantee at least (n being the unrolling limit).

5.4 Benchmarks and Evaluation

To evaluate our approach to the Control Envelope Synthesis problem, we curate a benchmark suite with diverse optimal control strategies. As Table 5.1 summarizes, some benchmarks have non-solvable dynamics, while others require a sequence of clever control actions to reach an optimal solution. Some have *state-dependent fallbacks* where the current state of the system determines which action is “safer”, and some are drawn from the literature. We highlight a couple of benchmarks here. See Appendix A.3 for a discussion of the full suite and the synthesized results, and [70] for the benchmark files and evaluation scripts.

`Power Station` is an example where the optimal control strategy involves two switches, corresponding to two steps of unrolling. A power station can either produce power or dispense it to meet a quota, but never give out more than it has produced. Charging is the fallback action that is safe for all time *after* the station has dispensed enough power. However, to cover all

tion. For example, for an invariant of the form $e = 0$, the differential invariant axiom is $[\{x' = f(x)\}] e = 0 \leftrightarrow (e = 0 \wedge [\{x' = f(x)\}] e' = 0)$.

Algorithm 1 CESAR: Control Envelope Synthesis via Angelic Refinements

```
1: Input: a synthesis problem (as defined in Section 5.3.1), an unrolling limit  $n$ .  
2: Remark:  $\text{valid}$  is defined as  $\text{valid}(F, A) \equiv (\text{first}(\text{reduce}(\neg F, A)) = \text{false})$ .  
3:  $k \leftarrow 0$   
4:  $I, e_I \leftarrow \text{reduce}([\text{forever}] \text{ safe}, \text{assum})$   
5: while  $k \leq n$  do  
6:    $e_G \leftarrow \text{true}$   
7:   for each  $i$  do  
8:      $G_i, e \leftarrow \text{reduce}([\text{act}_i; \text{plant}] I, \text{assum})$   
9:      $e_G \leftarrow e_G \text{ and } e$   
10:  if  $(e_G \text{ and } e_I)$  or  $\text{valid}(I \rightarrow \bigvee_i G_i, \text{assum})$  then  
11:    if  $e_G$  and  $\text{optimal}(I)$  then  
12:      yield  $((I, G), \text{"optimal"})$   
13:    return  
14:    else if  $e_G$  and  $e_I$  then yield  $((I, G), \text{"k-optimal"})$   
15:    else yield  $((I, G), \text{"sound"})$   
16:   $I', e \leftarrow \text{reduce}(I \vee [\text{step}] I, \text{assum})$   
17:   $e_I \leftarrow e_I \text{ and } e$   
18:  if  $e_G$  and  $e_I$  and  $\text{valid}(I' \rightarrow I, \text{assum})$  then  
19:    yield  $((I, G), \text{"}\omega\text{-optimal"})$   
20:  return  
21:   $I \leftarrow I'$   
22:   $k \leftarrow k + 1$ 
```

controllable states, we need to switch at least two times, so that the power station has a chance to produce energy and then dispense it, before settling back on the safe fallback. `Parachute` is an example of a benchmark with non-solvable, hyperbolic dynamics. A person jumps off a plane and can make an irreversible choice to open their parachute. The objective is to stay within a maximum speed that is greater than the terminal velocity when the parachute is open.

We implement CESAR in Scala, using Mathematica for simplification and quantifier elimination, and evaluate it on the benchmarks. Simplification is an art [78, 83]. We implement additional simplifiers with the `Egg` library [162] and SMT solver `z3` [39]. Experiments were run on a 32GB RAM M2 MacBook Pro machine. CESAR execution times average over 5 runs.

CESAR synthesis is automatic. The optimality tests were computed manually. Table 5.1 summarizes the result of running CESAR. Despite a variety of different control challenges, CESAR is able to synthesize safe and in some cases also optimal safe control envelopes within a few minutes. As an extra step of validation, synthesized solutions are checked by the hybrid system theorem prover `KeYmaera X` [51]. All solutions are proved correct, with verification time as reported in the last column of Table 5.1.

Table 5.1: Summary of CESAR experimental results

Benchmark	Synthesis Time (s)	Checking Time (s)	Optimal	Needs Unrolling	Non Solvable Dynamics
ETCS Train [126]	14	9	✓		
Sled	20	8	✓		
Intersection	49	44	✓		
Parachute [52]	46	8			✓
Curvebot	26	9			✓
Coolant	49	20	✓	✓	
Corridor	20	8	✓	✓	
Power Station	26	17	✓	✓	

5.5 Related Work

Hybrid controller synthesis has received significant attention [12, 90, 149], with popular approaches using temporal logic [5, 12, 165], games [107, 154], and CEGIS-like guidance from counterexamples [1, 36, 134, 144]. CESAR, however, solves the different problem of synthesizing control *envelopes* that strive to represent not one but *all* safe controllers of a system. Generating *valid* solutions is not an issue (a trivial solution always exists that has an empty controllable set). The real challenge is *optimality* which imposes a higher order constraint because it reasons about the relationship between possible valid solutions, and cannot, e.g., fit in the CEGIS quantifier alternation pattern $\exists\forall$. So simply adapting existing controller synthesis techniques does not solve symbolic control envelope synthesis.

Safety shields computed by numerical methods [2, 47, 80] serve a similar function to our *control envelopes* and can handle dynamical systems that are hard to analyze symbolically. However, they scale poorly with dimensionality and do not provide rigorous formal guarantees due to the need to discretize continuous systems. Compared to our symbolic approach, they cannot handle unbounded state spaces (e.g. our infinite corridor) nor produce shields that are parametric in the model’s parameters without hopelessly increasing dimensionality.

On the optimality side, a systematic but manual process was used to design a safe European Train Control System (ETCS) and justify it as optimal with respect to specific train criteria [126]. Our work provides the formal argument filling the gap between such case-specific criteria and end-to-end optimality. CESAR is more general and automatic.

5.6 Conclusion

This chapter presents the CESAR algorithm for Control Envelope Synthesis via Angelic Refinements. It is the first approach to automatically synthesize symbolic control envelopes for hybrid systems. The synthesis problem and optimal solution are characterized in differential game logic. Through successive refinements, the optimal solution in game logic is translated into a controllable invariant and control conditions. The translation preserves safety. For the many cases where refinement additionally preserves optimality, an algorithm to test optimality of the result post translation is presented. The synthesis experiments on a benchmark suite of diverse control problems demonstrate CESAR’s versatility.

Can CESAR be extended to handle more control shapes? How far does the idea of characterizing optimal control envelope solutions using game theory extend? Is it perhaps possible to solve control envelope synthesis for *all* the problems whose solutions are characterized by a hybrid game? Chapter 6 shows that the answer is yes. Since we become able to solve for *all* hybrid system sketches that mirror the structure of a game, it becomes convenient to treat *the hybrid game itself* as the specification of the shape of the control envelope. In the next chapter, we see control envelope synthesis for time-triggered control, event-triggered control and reach-avoid problems all handled by a single, extremely general framework.

Chapter 6

1966

Generalized Hybrid Game Control Envelope Synthesis

1967

1968

Chapter 5 introduced control envelope synthesis for a class of hybrid systems by identifying a connection between the solution of a hybrid system sketch and hybrid games. In fact, the shape of control problems can be modeled directly by two-player hybrid games, which only indicate what moves players can make (what control actions are possible) without specifying the actual strategy that the players follow. This chapter synthesizes control envelopes for *all* dGL games, greatly extending the generality of control envelope synthesis.

6.1 Introduction

1975

A hybrid *game* between the *controller* and the *environment* serves as a specification [103, 107, 154] for synthesis. The game models the *control possibilities* and the possible *environment behaviors*, with the controller winning if it meets the desired control contract. The synthesis task is to identify within the possible control space the control behaviors that win into the control contract. A control envelope corresponds to a *nondeterministic policy* in the specification game that indicates what actions a player can choose at any given decision point, such that *a player following it will never lose*, regardless of what the opponent does. Such a *nondeterministic policy* induces a *set* of control solutions because multiple actions are allowed at each decision point, and following each option leads to a *different* control solution. This chapter introduces *subvalue maps* as a symbolic, compositional representation of control envelopes that we can reason about recursively along the syntactic structure of the game. Via this representation we develop the theory to compare, verify, and synthesize symbolic control envelopes for hybrid games in their full generality.

Our formalization is in differential game logic (Chapter 2 [120]), which models two-player games with imperative program constructs like assignments, loops and branching. Subvalue map verification/synthesis composes at the level of such program constructs. This lets us leverage program analysis techniques like loop variants, invariants and their continuous analogs to solve control envelope synthesis. After defining subvalue maps, this chapter introduces the theory to verify that subvalue maps induce only policies that don't lose. To ensure that induced policies

1995 additionally never let the player get stuck, always allowing some action, the refined concept of
1996 *inductive* subvalue map is introduced.

1997 We address control envelope synthesis. Since our ultimate goal is to use control envelopes to
1998 derive/check control actions, we consider a restricted category of inductive subvalue maps where
1999 checking if a given control action lies within the envelope is efficient (polynomial time). We
2000 present an algorithmic framework to synthesize such subvalue maps. Next, we formalize when
2001 subvalue maps are more permissive, permitting more control options, through a partial order. It
2002 is always possible to construct an optimal inductive subvalue map for a given game, establishing
2003 completeness.

2004 The theoretical development needed is subtle. For example, at loops, a control envelope
2005 should allow *unboundedly* many repetitions so long as there is still a *finite* exit strategy *within*
2006 the same control envelope. The first few sections of the chapter build up the theoretical tools that
2007 make the theory for subvalue maps possible, and largely internalized within dGL. These include:
2008 1. a dGL analog to Brzozowski derivatives [25] that let us step through a game and characterize
2009 gameplay after a given subgame (Section 6.4), and 2. dGL characterizations checking whether
2010 a set of states is reachable by following *some* strategy or *any* strategy in a given subvalue map
2011 (Section 6.5).

2012 *Subvalue* maps are analogous to *value* functions in reinforcement learning [146]. A value
2013 function maps states to utility. Once it is known, an agent policy can be derived from it. Similarly,
2014 a player's subvalue map maps every subgame to a formula denoting a set of states from which
2015 that player can win the rest of the game (*subvalue* because some maps map to a set of *known*
2016 winning states smaller than the theoretical maximum winning set). In dGL, *subgames* specify
2017 game decision points where at most one player makes a play. These subgames compose together
2018 per a recursive syntax to form large, arbitrarily complex games. A player's policy set can then
2019 be reconstructed by syntactically stepping through the game: at every subgame where the player
2020 must take a decision, it accepts only those decisions that reach the next subgame in a mapped
2021 winning state for that subgame per the subvalue map.

2022 This chapter addresses control envelope synthesis for *all* of dGL: Synthesis for safety prop-
2023 erties, liveness properties, and any combination of these are all solved by a single, natural frame-
2024 work without special handling of either. A special case of our technical development, which
2025 simply excludes subgames relating to continuous dynamics, solves the problem for purely dis-
2026 crete games. The flexibility of our final synthesis framework is demonstrated through examples
2027 with diverse control challenges, and performance is evaluated on benchmarks from the literature
2028 and a procedurally generated benchmark suite.

2029 Our main contributions are to:

- 2030 1. Introduce *inductive subvalue maps* as symbolic, composable representations of control
2031 envelopes for hybrid games (Section 6.6).
- 2032 2. Present an algorithmic framework (Section 6.7) for *synthesis* of inductive subvalue maps
2033 where checking that control is within the induced nondeterministic policy is *computation-*
2034 *ally efficient*.
- 2035 3. Order subvalue maps by permissiveness and prove completeness for optimal hybrid game
2036 inductive subvalue map synthesis via the construction of a *maximal subvalue map* (Sec-
2037 tion 6.8).

4. Evaluate the approach, demonstrating its application to representatives of various control theory problem classes (Section 6.10). 2038
2039

6.2 Differential Game Logic Background 2040

This chapter synthesizes policies for games written in dGL. As discussed before (Chapter 2), a dGL game is played by two adversarial players, canonically called Angel and Demon. Various operators give either player opportunities to make control decisions to try to attain a winning condition, which is a dGL formula. *Unlike* Chapter 5, this chapter uses the convention that Angel is responsible for the controller’s decisions, and wins when she maintains the desired system properties (e.g. safety), and Demon plays for the environment. dGL games are determined, meaning exactly one player has a winning strategy from any given state [120], and players use Markov strategies, relying only on the current state with no information about the strategy of the other player. dGL games are generated by a recursive grammar. In many of the game constructs, either Angel or Demon can make a choice, deciding how the game should proceed, and try to choose in a way that is most favorable to them. These decisions are represented using *actions*, which will later be used to keep track of state change, construct game trees and represent policies¹. We recall the game constructs below and provide notation for *actions* that they correspond to. 2041
2042
2043
2044
2045
2046
2047
2048
2049
2050
2051
2052
2053

1. Angelic free assignment $x := *$ models Angel assigning any real value of her choice to variable x , and can be used, e.g., to model a controller’s ranged choices. Angel’s available actions are of the form $(x := e)$, where e is a (real polynomial) term. Dually, Demonic free assignment $x := \otimes$ models Demon assigning a real value of his choice to variable x . It models, e.g., environment disturbances that make control harder. Demon’s available actions are of the form $(x := e)^d$, where e is a term. 2054
2055
2056
2057
2058
2059
2. Loop α^* runs α as many times as Angel wants. At the end of every iteration, Angel gets to choose whether to run the loop again or to exit it. This can model, e.g., a controller that repeatedly performs a task until a criterion is met. Angel’s available actions are g to *go* repeat the loop for one more iteration, or s to *stop* and exit the loop. Dually, $\alpha^\times$ runs α as many times as Demon chooses. It can model a control loop that runs arbitrarily many times, and must stay safe forever. Demon’s actions are g^d to repeat the loop, or s^d to exit it. 2060
2061
2062
2063
2064
2065
2066
3. Branch choice $\alpha \cup \beta$ lets Angel choose to play either game α or game β . This can model a controller choosing between two modes of operation, for example accelerating or braking. Angel’s available actions are l to play *left* game α , or r to play *right* game β . Dually, $\alpha \cap \beta$ lets Demon choose to play either game α or game β . This can model different modes of environment behavior, such as a car in the environment, that the controller must avoid colliding with, accelerating or braking. Demon’s available actions are l^d to play game α , or r^d to play game β . 2067
2068
2069
2070
2071
2072
2073
4. Continuous evolution $\{x' = f(x) \ \& \ Q\}$ modifies x per the solution of differential equa- 2074

¹The action notation presented here is a simplified form of the operational semantics of dGL [120][Appendix C]. Sequential compositions are given an action for uniformity but do not have an action in the original operational semantics.

tion $x' = f(x)$. How long the ODE runs is determined by Angel, who must also maintain that domain constraint formula Q is true throughout. This can model, e.g., a control mode running till the controller interrupts. Angel's available actions are of the form $(x' = f(x) \& Q @ t)$, where t is a term representing the time Angel chooses to run the ODE for. Dually, in $\{x' = f(x) \& Q\}^d$, Demon chooses how long to run the ODE while ensuring that Q holds throughout. Demon's available actions are of the form $(x' = f(x) \& Q @ t)^d$, where t is the time Demon runs the ODE. This can model, e.g., control loop latency in controllers that periodically poll to take decisions. On omitting this construct and any cases it generates in definitions and theorems, we obtain results for purely discrete games.

Additionally, some game constructs do not involve Angel or Demon making any decisions.

1. Assignment $x := e$ assigns the expression e to variable x . The corresponding action that Angel performs to change state is written as $(x := e)$.
2. Sequential composition $\alpha; \beta$ runs game α followed by game β . The corresponding action that Angel performs is $\mathbf{c}$ to start α .
3. Test $?Q$ makes Angel immediately lose the current game if formula Q is false, but has no effect if Q is true. The action representing Angel going through the test (and possibly losing) is $(?Q)$. Dually, $!Q$ makes Demon lose when Q is false. It can model physical environment constraints. Demon's only action, to take the test, is $(!Q)$.

As usual, state σ is a mapping from variables to real numbers, and $\sigma(x)$ is the real value associated with variable x . Eq. (6.1) shows the grammar of dGL hybrid games that we will use in this chapter. An alternative notation uses duals for Demon controlled constructs, but for clarity in this chapter we use the explicit operators for some Demon controlled games shown below.

$$\begin{aligned} \alpha \equiv & x := e \mid \alpha; \beta \mid ?Q \mid \alpha^* \mid \alpha \cup \beta \mid x := * \mid \{x' = f(x) \& Q\} \\ & \mid !Q \mid \alpha^\times \mid \alpha \cap \beta \mid x := \otimes \mid \{x' = f(x) \& Q\}^d \end{aligned} \quad (6.1)$$

The *subgames* of dGL game α are the games that recursively compose to construct α , corresponding to the nodes of the abstract syntax tree induced by the grammar of Eq. (6.1).

We briefly describe the formal representation of actions and policies. Sections 6.4 and 6.5 will revisit these ideas with further detail. In any given initial state, a dGL game generates a game tree (Section 6.4.2) where each node corresponds to an action. A *play* of a game is a sequence of actions corresponding to a path from a root of the game tree to a leaf. Actions map old state to new state. For example, the action $(x := e)$ maps state σ to $\sigma(x \mapsto e)$, i.e., σ with value e replaced for x . Since agents and their gameplay are symmetric, in this chapter we often focus on the Angelic perspective without loss of generality.

An Angelic (nondeterministic) *policy* is a function that maps every subgame and state to the set of actions that Angel can take at that subgame and state. *Henceforth in this chapter, policy will be used to mean nondeterministic policy.* Angel *follows* an Angelic policy if at every subgame she must take a decision, she plays one of the actions permitted by the policy. As usual, formulas can be interpreted as the set of states in which they are true. So, formula $\langle \alpha \rangle \phi$ represents the *winning region* of Angel for game α and winning condition ϕ . Notation $\models \phi$ means formula ϕ is valid (true in all states).

6.3 Overview

2114

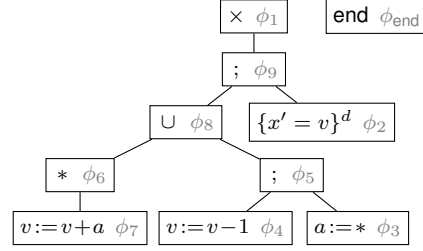
1. Input

Game: $((v := v + a)^* \cup (v := v - 1; a := *)) ; \{x' = v\}^d \times$ Angel's
Goal: $x > 0$

3. Policy for Angel derived from subvalue map

$\cup$	Taking the left (resp. right) branch is allowed if ϕ_6 (resp. ϕ_5) is true.
$*$	(Re-)entering the loop is allowed if ϕ_7 is true. Exiting is allowed if ϕ_2 is.
$a := *$	The value assigned to a must satisfy ϕ_2 .

2. Compute subvalue map



Computing the subvalue map:

Deduce $\phi_{\text{end}} \equiv x > 0$	(Goal)
Guess $\phi_1 \equiv \langle ((v := v + a)^* \cup (v := v - 1; a := *)) ; \{x' = v\}^d \rangle x > 0$	(Seek Invariant)
$\Leftrightarrow \langle ((n := *; ?n \geq 0; v := v + an) \cup (v := v - 1; a := *)) ; \{x' = v\}^d \rangle x > 0$	(Refinement)
$\Leftrightarrow x > 0 \wedge (v \geq 0 \vee a > 0)$	(Axioms, QE)
Deduce $\phi_2 \equiv \langle \{x' = v\}^d \rangle \phi_1 \Leftrightarrow \forall t (t \geq 0 \rightarrow \phi_1(x \mapsto x + vt))$	(Solve ODE)
$\Leftrightarrow x > 0 \wedge v \geq 0$	(QE)
Deduce $\phi_3 \equiv \exists a \phi_2 \Leftrightarrow x > 0 \wedge v \geq 0, \quad \phi_5 \equiv \phi_4 \equiv \phi_3(v - 1 \mapsto v) \Leftrightarrow x > 0 \wedge v \geq 1$	(Axioms, QE)
Guess $\phi_6 \equiv \langle (v := v + a)^* \rangle \phi_2 \Leftrightarrow \langle n := *; ?n \geq 0; v := v + an \rangle \phi_2$	(Invariant)
$\Leftrightarrow x > 0 \wedge (v \geq 0 \vee a > 0)$	(QE)
Deduce $\phi_7 \equiv \phi_6(v \mapsto v + a) \Leftrightarrow x > 0 \wedge (v + a \geq 0 \vee a > 0)$	
Check $\phi_6 \rightarrow \langle (? \phi_7; v := v + a)^* ; ? \phi_2 \rangle \phi_2$ valid	(Check Invariant)
Deduce $\phi_9 \equiv \phi_8 \equiv \phi_6 \vee \phi_5 \Leftrightarrow x > 0 \wedge (v \geq 0 \vee a > 0)$	
Check $\phi_1 \rightarrow \phi_9 \wedge x > 0$ valid	(Check Invariant)

Figure 6.1: An *inductive subvalue map* for a simple hybrid game and the associated policy. The subvalue formulas are numbered in the order in which they are derived by the algorithm presented in Section 6.7. We will revisit this example throughout the chapter, providing more explanations.

To illustrate our new concept of an *inductive subvalue map*, let us consider the simple hybrid game from Fig. 6.1. In this game, Angel aims to ensure $x > 0$ after each iteration of a loop ($\times$) controlled by Demon. At each iteration, Angel must choose ($\cup$) between two alternative paths, after which Demon continuously evolves x at rate v for a duration of his choice ($\{x' = v\}^d$). In the first path, Angel can increment v by a in a loop ($*$) as many times as she desires. In the second path, v is decremented by 1 and Angel gets an opportunity to arbitrarily set the value of a ($a := *$).

Our goal is to characterize not *one*, but *all* of the ways Angel can win (or at least *as many* as possible). At every choice point, we must determine which actions are compatible with Angel

still winning the game and which actions are not. This way, Angel can pursue additional, unmodeled and possibly changing secondary goals without threatening her mission-critical objective of ensuring $x > 0$. In the traditional setting of control theory [10, 11] and reinforcement learning [146, 147, 148], this information is typically presented in the form of a *value function* that splits the state space into winning and losing states. This work proposes a representation of such value functions for hybrid games which is *sound*, *symbolic*, and which respects their *composable* structure.

Subvalue maps (Section 6.5, Section 6.6). A *subvalue map* associates *each* subgame of a hybrid game to a formula that *conservatively* estimates the set of states from which the overall game can be won, *starting from* this specific subgame. In the example from Fig. 6.1, ϕ_1 provides a sufficient initial condition for winning the game. More interestingly, ϕ_5 provides a sufficient condition for winning the subgame starting from the second Angel branch within the loop.

$$\models \phi_5 \rightarrow \langle v := v - 1; a := *; \{x' = v\}^d; \alpha^\times \rangle x > 0 \quad (6.2)$$

where α stands for the full Demon loop body. At every control point for Angel, the subvalue map indicates which actions can be taken without forfeiting winning. For example, when reaching the $\cup$ decision point, Angel can safely take the left branch if ϕ_6 is true. Similarly, she can safely take the right branch if ϕ_5 is true. What if none of these hold though? Useful policies permit at least one action to be chosen. For this reason, Section 6.6 introduces the refined concept of an *inductive* subvalue map, whose local compatibility conditions ensure this property. In our example, these conditions mandate in particular that $\phi_8 \rightarrow \phi_6 \vee \phi_5$ is valid, ensuring that a safe action is always available to Angel at subgame $\cup$.

Internalizing policy reasoning (Section 6.5.2, 6.5.3). It is possible to reason about the policies represented by subvalue maps *within dGL itself*. The *universal projection* of a game onto a subvalue map constructs a dGL game that wins precisely when *all plays that respect the subvalue-map policy* are guaranteed to win. The *existential projection* of a game onto a subvalue map wins when *there exists some policy-compliant play* that wins. These projections let us use dGL to state and prove properties, keeping the theory tidy. The safety of the monitor derived from an inductive subvalue map that keeps the game on track for winning can similarly be expressed within dGL. In our example, the following formula is *guaranteed* to be valid:

$$\phi_1 \rightarrow \langle ((?(\phi_6 \vee \phi_5); (!\phi_6; ?\phi_6; (!\phi_7; v := v + a; ?\phi_6)^\times; !\phi_2) \cap (!\phi_5; v := v - 1; ?(\exists a \phi_2); a := \otimes; !\phi_2)); \{x' = v\}^d)^\times \rangle x > 0. \quad (6.3)$$

In the game above, every Angel choice from the original game has been transferred to Demon, on the condition that he complies with Angel's subvalue strategy monitor. Compliance is always possible, as ensured by proper Angel tests. A subtlety is the treatment of infinite gameplay. Consider a car operating within a control envelope ensuring that it avoids obstacles and can eventually stop at a parking spot. The car should not be limited to only strategies with some fixed maximum number of moves so that it can travel arbitrarily long routes. Still, the car should be able to eventually park. Inductive subvalue maps enforce not *finite control* but the *existence of a finite path to victory*. They do not force the controller to take this finite path if it still has a way to win in the future, *allowing gameplay of unbounded length*.

Solving hybrid games via symbolic execution (Section 6.7). For practical applications, it is useful for subvalue maps to check if an action is acceptable at a given state *efficiently*. To compute such subvalue maps, with modality-and-quantifier-free subvalues, Section 6.7 presents an algorithmic framework. Like precondition calculus [40], the framework is parametrized on an invariant generation procedure. We propose *game rewriting* as a general principle for implementing this procedure (Section 6.7.1). Fig. 6.1 illustrates the resulting algorithm. The algorithm leverages the fact that any dGL formula that is free of loops and whose ODEs have polynomial solutions can be automatically rewritten into an equivalent modality-and-quantifier-free formula, by using the axioms of dGL along with quantifier elimination (QE) [151]. However, whenever a loop is hit, a guess is needed for a subvalue of the associated subgame. Such a guess is typically made by using heuristics to conservatively rewrite it into a loop-free dGL formula that can be symbolically evaluated. The algorithm then proceeds with the loop body, after which the validity of the guess is checked retrospectively. Section 6.7 provides more details.

Existence of an optimal subvalue map (Section 6.8). Subvalue maps can be ordered by how permissive their policy is. Section 6.8 formalizes such an ordering, where the main subtlety is comparing monitor guards *in the context in which they are reachable*. An optimal subvalue map can be constructed using dGL modalities. In our example, this solution maps ϕ_5 to the right-hand side of the implication in Eq. (6.2). The optimal subvalue map construction achieves completeness for the subvalue map synthesis problem. In practical subvalue maps like in Fig. 6.1, where modal subvalues are not permitted, the maps are synthesized using sound *lower* bounds of a game’s value function. An *upper* bound can also be computed by considering the game from the opposing player’s perspective. We can compare the dual subvalue maps and thereby symmetrically prove the optimality of the solution derived in Fig. 6.1. The proofs in this chapter largely follow from structural induction. *Appendix B.5 provides all proofs throughout this chapter.*

Remark 1. Assertion/invariant maps [48] for static analysis of imperative programs annotate every program point with a formula that must hold whenever execution reaches that program point. Analogously, an *Angelic subvalue map* annotates every subgame with a formula, and constrains Angel to play such that the formula must hold whenever execution reaches that subgame. Analogous to assertion maps [64], subvalue maps will be computed by backward symbolic execution.

Remarks, such as Remark 1 above, draw analogies to familiar concepts but are not critical to understanding the chapter.

6.4 Prefixes and Suffixes

An *Angelic subvalue map* maps every dGL subgame to a formula denoting Angel’s known winning subregion for the rest of the game starting at that subgame. As it is not always possible to compute exact winning regions, the *known* winning subregions in a subvalue map are conservative, safe subsets of the actual theoretical winning regions per dGL semantics. Fig. 6.1 shows an example where each subgame, corresponding to nodes of the syntax tree, is mapped to its corresponding winning subregion $(\phi_1, \dots, \phi_9)$. In this example, the winning subregions are maximal, *equal* to the winning regions.

This section develops the formalism necessary for defining subvalue maps. A subvalue map maps each subgame to a formula that must logically imply the winning region for playing the rest of the game *starting at that subgame*. To logically characterize this, we first introduce the *game suffix*, or game remainder of a subgame. The game suffix of a subgame features *all* possible future behaviors of the overall game from *any* point where an instance of the subgame is encountered. Surprisingly, we can characterize all these behaviors with a finite dGL game. Then the exact winning region starting from a subgame is defined as the winning region for the game suffix.

6.4.1 Game Suffix Construction

For disambiguation, each subgame is given a unique label. Notation $a:\alpha$ refers to the *named* game α where each subgame has a unique label, and the entire game α (corresponding to the root node of the abstract syntax tree) is labeled a . Even subgames that are identical in structure but appear at different places in the game are given different labels to distinguish multiple occurrences. $\text{subgames}(a:\alpha)$ refers to the set of all named subgames of $a:\alpha$. Game suffix $a:\alpha_{b:}$ is a game modeling what remains to be played after seeing some occurrence of subgame $b:\beta$ while playing $a:\alpha$ (Def. 6.1)². This is a complex idea: unboundedly many different plays of a game can lead to playing a given subgame b , and even within a single play, b can occur an unbounded number of times. The ability to succinctly characterize all suffix plays (leveraging dGL) is crucial to the definition of subvalue maps.

Example 6.1. Let the overall game of Fig. 6.1 be $a:\alpha$, and let the label of each subgame be the index i of the winning subregion formula ϕ_i shown on its corresponding node. Consider subgame 5 inside loop α . All execution traces after any instance of subgame 5 have the remainder of the loop body ($\{x' = v\}^d$) followed by potential future iterations of the loop (α). So the game suffix $a:\alpha_{5:}$ of subgame 5 is $\{x' = v\}^d; \alpha$.

Remark 2. The suffix characterizing behavior of a dGL game after a subgame *itself as a dGL game* is analogous to how Brzowski derivatives reason about the behavior seen in a regular expression after a given string as a new regular expression [25].

Definition 6.1 (Game suffix). The game suffix $a:\alpha_{b:}$ of subgame $b:\beta$ in $\text{subgames}(a:\alpha)$ is constructed as follows. If $a:\alpha$ is $b:\beta$ then $a:\alpha_{b:} = a:\alpha$. Otherwise, if $a:\alpha$ has structure:

$$\begin{aligned}
 & a:(g:\gamma)^* \text{ or } a:(g:\gamma)^\times \text{ then } a:\alpha_{b:} = (g:\gamma_{b:}); a:\alpha \\
 & a:(g:\gamma \cup d:\delta) \text{ or } a:(g:\gamma \cap d:\delta) \text{ then } \begin{cases} a:\alpha_{b:} = g:\gamma_{b:} & b \in \text{subgames}(g:\gamma) \\ a:\alpha_{b:} = d:\delta_{b:} & b \in \text{subgames}(d:\delta) \end{cases} \\
 & a:(g:\gamma; d:\delta) \text{ then } \begin{cases} a:\alpha_{b:} = (g:\gamma_{b:}); d:\delta & b \in \text{subgames}(g:\gamma) \\ a:\alpha_{b:} = d:\delta_{b:} & b \in \text{subgames}(d:\delta) \end{cases}
 \end{aligned}$$

In the $a:(g:\gamma; d:\delta)$ and $a:(g:\gamma \cup d:\delta)$ cases, b is either in $\text{subgames}(g:\gamma)$ or $\text{subgames}(d:\delta)$ because b is a subgame of $a:\alpha$ (and not a itself which is already handled initially). Cases where $a:\alpha$ is

²The notation is analogous to Python list slicing. $a:\alpha_{b:}$ represents the part of $a:\alpha$ appearing *after* b . The complementary concept of *game prefix* is written as $a:\alpha_{b:}$, and consists of the part of the game appearing before subgame b .

atomic, i.e., $\alpha \in \{x := e, x := *, ?Q, !Q, \{x' = f(x) \ \& \ Q\}, \{x' = f(x) \ \& \ Q\}^d\}$, b must be a , so these cases are already handled initially.

6.4.2 Game Trees and Plays

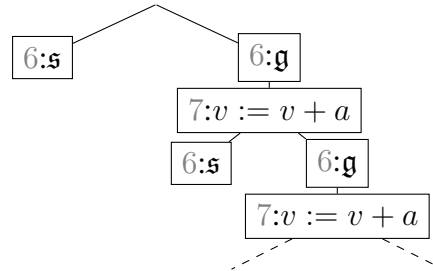
This section discusses dGL gameplay to precisely connect subgame suffixes to their defining property in dGL operational semantics. However, later in this chapter we will rely only on the syntactic suffix definition that has already been introduced. The background on dGL game trees is also useful for understanding *strategies*, which in Section 6.5 will provide a global view of the gameplay that a policy induces. This section first recounts which plays lie within a game and makes precise which suffix plays occur after a subgame. Then it defines a way to syntactically compare these play suffixes by the impact they have on state, *ignoring minor game structure differences*, to ensure that the *game suffix* has the same effect as these *suffix plays*, establishing that game suffixes reconstruct all gameplay that can occur after a subgame is seen. The definitions of game trees and plays that follow are from the existing operational semantics of dGL [120][Appendix C], extended to account for labels.

A labeled game $a:\alpha$, given initial state σ , corresponds to a *labeled game tree* $\mathcal{T}(a:\alpha, \sigma)$ where each node has a subgame label and an action. Per the conventions of descriptive set theory, the tree is a prefix-closed set of the paths originating at the root (Fig. 6.2).

Example 6.2. In Fig. 6.1 let the label of each subgame be the index i of the corresponding winning subregion formula ϕ_i . Consider subgame 6, the inner loop. The game tree for this subgame has the paths $6 \cdot s$ for 0 iterations, $6 \cdot g \cdot 7 \cdot (v := v + a) \cdot 6 \cdot s$ for one iteration, and so on. Operator $\cdot$ concatenates actions/sequences of actions. Fig. 6.2 visually represents this tree.

Paths $\equiv \{$

$(6:s), (6:g),$
 $(6:g \cdot 7:v := v + a),$
 $(6:g \cdot 7:v := v + a \cdot 6:s),$
 $(6:g \cdot 7:v := v + a \cdot 6:g \cdot 7:v := v + a),$
 $\dots \}$



(b) Visualization of the labeled game tree

(a) Representation of game tree as (prefix-closed) set of paths through the tree.

Figure 6.2: The game tree for subgame 6 of the game in Fig. 6.1, i.e., $6:(7:v := v + a)^*$

Def. B.2 in Appendix B.6.2 shows the full construction of trees.³ A path in a tree can be represented as the sequence of labels and actions encountered while traversing the path, where the subgame label appears immediately before the action that subgame corresponds to. Because

³This is similar to the standard dGL operational semantics [120, Appendix C], but with the addition of labels to keep track of subgames and some notation changes.

of loops, subgame labels can repeat even within a single path. For example, in Fig. 6.2 because of the Angelic loop, labels 6 and 7 repeat.

Tree nodes have actions, which correspond to transformations from old game state before the action to new state [120][Appendix C]. Appendix B.6.1 recalls the transformations. Notation $\llbracket \text{act} \rrbracket_\sigma$ is the state reached by running action act starting at state σ , e.g., $\llbracket (x := e) \rrbracket_\sigma = \sigma(x \mapsto e)$. For tests and ODEs, it is possible for the transition to be undefined, e.g., for tests:

$$\llbracket (?Q) \rrbracket_\sigma = \begin{cases} \sigma & \text{if } \sigma \in \llbracket Q \rrbracket \\ \text{not defined} & \text{otherwise} \end{cases} \quad \llbracket (!Q) \rrbracket_\sigma = \begin{cases} \sigma & \text{if } \sigma \in \llbracket Q \rrbracket \\ \text{not defined} & \text{otherwise} \end{cases}$$

In this case, the player responsible for the action loses. Labels, along with some actions whose only role is to restrict tree structure, have no impact on state, corresponding to the identity transformation. The set $\mathfrak{S}$ of such actions is $\{\mathfrak{g}, \mathfrak{s}, \mathfrak{l}, \mathfrak{r}, \mathfrak{c}, \mathfrak{g}^d, \mathfrak{s}^d, \mathfrak{l}^d, \mathfrak{r}^d\}$.

A path is executed by inductively applying the actions: $\llbracket a \cdot t \rrbracket_\sigma = \llbracket t \rrbracket_{\llbracket a \rrbracket_\sigma}$. We syntactically compare paths and trees per the reachable states resulting from gameplay, ignoring structural differences, *by erasing the nodes that have no impact on state*. This lets us formalize how subgame suffixes preserve reachable states, but not structure. The action projection operator $\pi_{-\mathfrak{S}}$ returns the path with all labels and actions belonging to $\mathfrak{S}$ removed (Appendix B.6.6, Def. B.8). For paths p_1 and p_2 , if $\pi_{-\mathfrak{S}}(p_1) = \pi_{-\mathfrak{S}}(p_2)$, then $\llbracket p_1 \rrbracket_\sigma = \llbracket p_2 \rrbracket_\sigma = \llbracket \pi_{-\mathfrak{S}}(p_1) \rrbracket_\sigma$. $\pi_{-\mathfrak{S}}$ is overloaded to apply to game trees. For game tree t , $\pi_{-\mathfrak{S}}(t)$ is a game tree where all paths have only actions that can impact state, produced by applying $\pi_{-\mathfrak{S}}$ to every path in t . Two trees result in the same reachable states (even if their structure differs) if their projections are the same. We can now show the defining property of game suffixes.

Lemma 6.1 (Game Suffixes are Operational Gameplay Suffixes). *For subgame $b:\beta$ in overall game $a:\alpha$, let the game suffix be $a:\alpha_b$. Let σ be some state. The states reachable from playing $a:\alpha$ such that σ is reached at subgame $b:\beta$ at some point during the gameplay are the same as the states reachable from playing $a:\alpha_b$ starting in σ . That is,*

$$\{\pi_{-\mathfrak{S}}(s) : p \cdot b \cdot s \in \mathcal{T}(a:\alpha, \sigma'), \llbracket p \rrbracket_{\sigma'} = \sigma \text{ and } \pi_{-\mathfrak{S}}(s) \neq ()\} = \pi_{-\mathfrak{S}}(\mathcal{T}(a:\alpha_b, \sigma)),$$

where $()$ is the empty path.

In Lemma 6.1, the left-hand side considers all paths in the original game that contain b . These must be of the form $p \cdot b \cdot s$ where p is a prefix of b and s is a suffix. A suffix set is built by collecting the projection of all such suffixes, ignoring sequences that are empty after applying $\pi_{-\mathfrak{S}}$. The quantity on the right-hand side is the projection of the game tree of the subgame suffix at state σ . Suffix $a:\alpha_b$ captures all remaining plays occurring after any occurrence of $b:\beta$ while playing $a:\alpha$, ignoring the impactless actions of $\mathfrak{S}$.

6.4.3 Game Prefix

The complementary concept of a *game prefix* computes *what contexts* a subgame can be seen in. For example, it will help to show that Angel, while following a subvalue map, only reaches a subgame within the region from which she can win the rest of the game (Section 6.5). The prefix of subgame b in the game $a:\alpha$, written $a:\alpha_b$, features *all* behaviors of game α that can

happen *before* an occurrence of subgame b within $a:\alpha$. Constructed analogously to the game suffix (Appendix B.6.8, Def. B.10), it satisfies an analogous property (Lemma 6.2). 2294

Lemma 6.2 (Execution Prefixes are Original Gameplay Prefixes). *For subgame $b:\beta$ in $a:\alpha$, for any initial state σ , the states reachable at $b:\beta$ while playing $a:\alpha$ are equal to those reachable by playing game prefix $a:\alpha.b$. That is, prefix set $\{\pi_{-\ominus}(p) : p \cdot b \cdot s \in \mathcal{T}(a:\alpha, \sigma)\}$ is equal, under prefix closure, to prefix game tree $\pi_{-\ominus}(\mathcal{T}(a:\alpha.b, \sigma))^4$.* 2295 2296 2297 2298 2299

6.5 Subvalue Maps 2300

We define subvalue maps, discuss their interpretation as policies, and these policies' properties. 2301

6.5.1 Subvalue Map Definition 2302

Def. 6.2 characterizes a subvalue map. Each subgame maps to a formula that must logically imply the winning region for playing the rest of the game starting at that subgame. The exact winning region starting from a subgame is defined as the winning region for the game suffix. A special **end** subgame additionally stores the terminal condition that the policy induced by the subvalue map tries to reach. Mathematically, when Angel is playing the overall game $a:\alpha$ with winning condition $S(\text{end})$, her winning region at the point when she is about to play subgame $b:\beta$ to achieve $S(\text{end})$ overall is $\langle a:\alpha.b. \rangle S(\text{end})$. 2303 2304 2305 2306 2307 2308 2309

Definition 6.2 (Subvalue maps). A map S from the subgames of game $a:\alpha$ and special subgame **end** to formulas is an *Angelic subvalue map* of $a:\alpha$ when $\models S(b) \rightarrow \langle a:\alpha.b. \rangle S(\text{end})$ for every subgame $b:\beta$ in $\text{subgames}(a:\alpha)$. Dually, S is a *Demonic subvalue map* for game $a:\alpha$ when $\models S(\beta) \rightarrow [a:\alpha.b.] S(\text{end})$ for every subgame $b:\beta$ in $\text{subgames}(a:\alpha)$. 2310 2311 2312 2313

Interpretation as policy. Angelic subvalue maps provide Angel with a policy that describes how to play the game. For example, the map in Fig. 6.1 says that Angel at her choice $\cup$ should not transition to $(v := v + a)^*$ when ϕ_6 is false, and not transition to $v := v - 1 ; a := *$ when ϕ_5 is false. Intuitively, at any subgame where Angel has a choice (e.g. Fig. 6.1 at $\cup$), the *next subgame* that she chooses to run (e.g., left branch $(v := v + a)^*$) should be such that the current state σ satisfies the winning subregion for that subsequent subgame (e.g., $\sigma \models \phi_6$). This next subgame to run after the current one ends is called the *successor* (Def. 6.3). 2314 2315 2316 2317 2318 2319 2320

Definition 6.3 (Successor). The successor of subgame $b:\beta$ within the overall game $a:\alpha$, written $\text{succ}(b, a:\alpha)$, is defined as follows. If $a:\alpha$ is $b:\beta$ then $\text{succ}(b, a:\alpha)$ is the special subgame **end**. Otherwise, if $a:\alpha$ has structure: 2321 2322 2323

⁴The prefix closure is required to recover tree structure. Unlike with the suffix construction, the prefix set builder collects only *parent* subgames of $b:\beta$ and *not all predecessors*. Expanding the prefix closure results in the condition: $\{\pi_{-\ominus}(t) : p \cdot b \cdot s \in \mathcal{T}(a:\alpha, \sigma) \text{ and } p = t \cdot u\} = \pi_{-\ominus}(\mathcal{T}(a:\alpha.b, \sigma))$

$a:(g:\gamma; d:\delta)$ then

$$\begin{cases} \text{succ}(b, a:\alpha) = d:\delta & b \in \text{subgames}(g:\gamma) \text{ and } \text{succ}(b, g:\gamma) = \text{end} \\ \text{succ}(b, a:\alpha) = \text{succ}(b, g:\gamma) & \text{otherwise if } b \in \text{subgames}(g:\gamma) \\ \text{succ}(b, a:\alpha) = \text{succ}(b, d:\delta) & \text{otherwise} \end{cases}$$

2324 $a:(g:\gamma)^*$ or $a:(g:\gamma)^\times$ then

$$\begin{cases} \text{succ}(b, a:\alpha) = a & b \in \text{subgames}(g:\gamma) \text{ and } \text{succ}(b, g:\gamma) = \text{end} \\ \text{succ}(b, a:\alpha) = \text{succ}(b, g:\gamma) & \text{otherwise if } b \in \text{subgames}(g:\gamma) \end{cases}$$

$$\begin{matrix} a:(g:\gamma \cup d:\delta) \text{ or} \\ a:(g:\gamma \cap d:\delta) \end{matrix} \text{ then } \begin{cases} \text{succ}(b, a:\alpha) = \text{succ}(b, g:\gamma) & b \in \text{subgames}(g:\gamma) \\ \text{succ}(b, a:\alpha) = \text{succ}(b, d:\delta) & \text{otherwise} \end{cases}$$

2325 Using the successor definition, Def. 6.4 shows how in general, Angel can interpret the sub-
2326 value map as a (nondeterministic) policy indicating what control decisions are safe to take to
2327 eventually win based on the current state⁵.

2328 **Definition 6.4** (Policy Interpretation). *Policy interpretation function* $\mathcal{P}_{a:\alpha}(S)$ transforms an An-
2329 gelic subvalue map S for game $a:\alpha$ to a nondeterministic policy function that maps current state σ
2330 and subgame $b:\beta$ to the set of acceptable control actions Angel can take. $\mathcal{P}_{a:\alpha}(S)(b, \sigma)$ is defined
2331 as below. When the current subgame $b:\beta$ has structure:

$$b:(d:\delta \cup g:\gamma) \text{ then } \begin{cases} \{\mathbf{l}, \mathbf{r}\} & \sigma \models S(d) \wedge S(g) \\ \{\mathbf{l}\} & \text{else if } \sigma \models S(d) \\ \{\mathbf{r}\} & \text{else if } \sigma \models S(g) \\ \emptyset & \text{otherwise} \end{cases}$$

$$2332 \quad b:(g:\gamma)^* \text{ then } \begin{cases} \{\mathbf{s}, \mathbf{g}\} & \sigma \models S(g) \wedge S(\text{succ}(b, a:\alpha)) \\ \{\mathbf{g}\} & \text{else if } \sigma \models S(g) \\ \{\mathbf{s}\} & \text{else if } \sigma \models S(\text{succ}(b, a:\alpha)) \\ \emptyset & \text{otherwise} \end{cases}$$

$$\begin{aligned} & b:x := * \text{ then } \{(x := e) \mid \sigma(x \mapsto e) \models S(\text{succ}(b, a:\alpha))\} \\ & b:\{x' = f(x) \ \& \ Q\} \text{ then } \{(x' = f(x) \ \& \ Q @ t) : t \geq 0, \forall s \in [0, t], \varphi(s) \models S(b) \text{ and} \\ & \quad \varphi(t) \models S(\text{succ}(b, a:\alpha)) \text{ where } \varphi : [0, t] \rightarrow \mathcal{S} \text{ is differentiable,} \\ & \quad \varphi(0) = \sigma, \text{ and } \forall s \in [0, t], \varphi(s) \models x' = f(x) \wedge Q\} \end{aligned}$$

2333 **Relationship with strategies.** We have focused on *policies* which provide local answers for
2334 what action to play next. *Strategies*, on the other hand, are global gameplay solutions. The
2335 Angelic strategy at a given initial state corresponds to a game tree at that state that has been
2336 pruned so that at every subgame where Angel has to take a decision, she takes exactly one

⁵The successor is not just the next label to occur in the game suffix. This is because the successor should not be a child of the current subgame. It should only start when the current subgame ends.

decision. Thus, a strategy plans out all the future moves of its player for all potential actions of the opponent for all parts of the game at once. Viewing policies from the lens of strategies is useful for understanding the global behaviors that a policy induces, and will be relevant in Section 6.5.2 and Section 6.5.3. The set of strategies that Angel can play for the game $a:\alpha$ while following the policy of a subvalue map S starting in a state σ is written as $\mathcal{S}_{a:\alpha}(S)(\sigma)$. The construction is shown in Appendix B.6.6, Def. B.7. The intuition is to generate the game tree while restricting Angel to play per the policy. When the policy has multiple options at a given Angelic decision point, $\mathcal{S}_{a:\alpha}(S)(\sigma)$ has different strategies for each way to proceed. When the policy is deterministic, the set contains a single strategy.

Winning games. Subvalue maps induce policies that are suitable to play games with *compatible* winning conditions, i.e., winning conditions that contain the terminal subregion of the policy that end maps to (Def. 6.5).

Definition 6.5 (Compatible winning conditions). An Angelic subvalue map S for game $a:\alpha$ is said to be *compatible with Angel winning condition* ϕ when $\models S(\text{end}) \rightarrow \phi$. Dually, a Demonic subvalue map S for game $a:\alpha$ is compatible with Demon winning condition ϕ when $\models S(\text{end}) \rightarrow \phi$.

If Angel plays a game for a compatible winning condition per a policy induced by an Angelic subvalue map, she always remains in a state where there *exists* a way for her to win the remaining game (Theorem 6.1 at the end of this section). However, the policy she needs to play to win *might not lie within the subvalue map*.

For example, in Fig. 6.1 at Angel choice $\cup$, consider the subvalue map in the figure except that ϕ_6 is $\perp$ instead of $x > 0 \wedge (v \geq 0 \vee a > 0)$. This is still an Angelic subvalue map. If we arrive at ϕ_8 in state $\sigma \equiv \{x = 0, v = 0, a = 0\}$, then Angel can win by choosing the left branch guarded by ϕ_6 , but will lose if she chooses the right branch. The policy induced by the map at this point does not let Angel into the losing play of choosing the right branch. However it does not let her choose the left branch either since $\sigma \not\models \perp$. Its returned action set is $\emptyset$. Angel is stuck. This motivates the definition of an *inductive* subvalue map (Section 6.6), where not only does the policy keep Angel in a state where she has a way to win, but this victory can be achieved by continuing to follow the subvalue map policy.

We next formulate within dGL the formalism to say *for all ways to play the policy of the subvalue map* (Section 6.5.3) up to any subgame, *there exists a way to continue to play within the subvalue map* (Section 6.5.2).

6.5.2 Existential Subvalue Map Projection

The *existential projection* of game $a:\alpha$ onto Angelic subvalue map S , written $a:\alpha \Downarrow S$, is the game that can be won exactly when there *exists* a way for Angel to win $a:\alpha$ while restricted to stay within the (nondeterministic) policy of S . Projection changes the game so that Angel is put on rails, limited to the policy induced by S while taking every decision, and is forced to manifest what she must do to win while playing within the policy.

Lemma 6.3 (Existential projection correspondence). *For any Angelic subvalue map S for game $a:\alpha$ and compatible winning condition ϕ , in initial state $\sigma \models \langle \alpha \rangle \phi$, Angel has a winning strategy*

2377 for the game $a:\alpha \Downarrow S$ if and only if Angel can win $\langle \alpha \rangle \phi$ while following the policy of S . That is,
 2378 $\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$ iff $\mathcal{S}_{a:\alpha}(S)(\sigma) \neq \emptyset$.

2379 Def. 6.6 defines existential projection by inserting Angelic tests to guard every Angelic deci-
 2380 sion that would make Angel lose if her choice would result in playing into the subsequent sub-
 2381 game in a state outside that subgame's winning subregion. For example, in Fig. 6.1, the subvalue
 2382 map says that ϕ_2 is the winning subregion of the subgame that is played after subgame $3:a := *$.
 2383 The projection of the subvalue map onto subgame 3 is $a := *; ?\phi_2$, thus *locally* ensuring that
 2384 Angel makes a correct assignment to a that will let Angel win the rest of the game. When Angel
 2385 has no decisions to take, e.g., $g:\gamma \cap d:\delta$ where *Demon* chooses between $g:\gamma$ and $d:\delta$, projection does
 2386 not interfere, only recursively adding guards for any Angelic decisions within subgames γ and δ .
 2387 During recursive calls, we sometimes have to update the end subvalue to reflect the appropriate
 2388 terminal condition for the subgame under consideration. Notation $S(\text{end} \mapsto Q)$ refers to S with
 2389 the subvalue of end updated to Q .

2390 **Definition 6.6** (Existential projection). The *existential projection* of dGL game $a:\alpha$ onto Angelic
 2391 subvalue map S , written $a:\alpha \Downarrow S$, is generated recursively from the structure of $a:\alpha$ as follows. If
 2392 $a:\alpha$ has structure:

2393 $a:x := * \text{ then } x := *; ?S(\text{end}). \quad a:(g:\gamma \cup d:\delta) \text{ then } (?S(g); g:\gamma \Downarrow S) \cup (?S(d); d:\delta \Downarrow S)$
 2394 $a:(g:\gamma)^* \text{ then } (?S(g); (g:\gamma \Downarrow (S(\text{end} \mapsto S(a))))^*; ?S(\text{end}))$
 2395 $a:(g:\gamma; d:\delta) \text{ then } (g:\gamma \Downarrow (S(\text{end} \mapsto S(d)))) ; (d:\delta \Downarrow S)$
 2396 $a:(g:\gamma \cap d:\delta) \text{ then } g:\gamma \Downarrow S \cap d:\delta \Downarrow S \quad a:(g:\gamma)^\times \text{ then } (g:\gamma \Downarrow (S(\text{end} \mapsto S(a))))^\times$
 2397 $a:\{x' = f(x) \ \& \ Q\} \text{ then } \{x' = f(x) \ \& \ Q\}; ?S(\text{end})$
 2398 atomic and not controlled by Angel, i.e.,
 2399 $\alpha \in \{x := e, x := \otimes, ?Q, !Q, \{x' = f(x) \ \& \ Q\}^d\}, \text{ then } \alpha$

2400 where game labels are preserved and fresh labels are used for newly introduced subgames. The
 2401 existential projection of a Demonic subvalue map is symmetric (Appendix B.6.4), with the addi-
 2402 tion of Demonic assertions after subgames controlled by Demon.

2403 *Remark 3.* $a:\alpha \Downarrow S$ is an *Angelic refinement* of $a:\alpha$, that is, for any winning condition ψ , $\models$
 2404 $\langle a:\alpha \Downarrow S \rangle \psi \rightarrow \langle \alpha \rangle \psi$ (Lemma B.2). Intuitively, if Angel has a winning strategy within S , she
 2405 can win with the same strategy in the unconstrained game.

2406 6.5.3 Universal Subvalue Map Projection

2407 The *universal projection*, written $a:\alpha \Downarrow S$, explores a subvalue map maximally flexibly by making
 2408 Demon play Angel's policy in Angelic subvalue map S to look for loopholes. It changes the game
 2409 to put *Demon* on rails forcing him to play within Angel's S , as he looks for *any* strategy in the
 2410 policy that falsifies Angel's winning condition.

2411 **Lemma 6.4** (Universal projection correspondence). *For any Angelic subvalue map S for game*
 2412 *$a:\alpha$, in initial state $\sigma \models \langle \alpha \rangle a$, for compatible winning condition ϕ , Angel has a winning strategy*
 2413 *for the game $a:\alpha \Downarrow S$ if and only if all ways to pursue the policy of S for game $a:\alpha$ that complete the*
 2414 *game end in Angel's winning region. That is, $\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$ iff all elements of set $\mathcal{S}_{a:\alpha}(S)(\sigma)$*
 2415 *are Angel winning strategies.*

Def. 6.7 defines universal projection by replacing all Angelic games ($\cup, x := *, \alpha^*, \{x' = f(x) \& Q\}$) by their dual Demonic games and inserting Demonic tests to ensure Demon plays these decisions within Angel's control envelope. Like in projection, control choices that were already Demon's remain unguarded. Eq. (6.3) shows universal projection for the example subvalue map and game in Fig. 6.1.

Definition 6.7 (Universal projection). The *universal projection* of Angelic subvalue map S onto dGL game $a:\alpha$, written $a:\alpha \Downarrow S$, is generated recursively per the structure of $a:\alpha$ as follows. If $a:\alpha$ has structure:

$$\begin{aligned}
& a:x := *, \text{ then } x := \otimes; !S(\text{end}). \quad a:(g:\gamma; d:\delta), \text{ then } (g:\gamma \Downarrow (S(\text{end} \mapsto S(d)))) ; (d:\delta \Downarrow S). \\
& a:(g:\gamma \cup d:\delta), \text{ then } (!S(g); g:\gamma \Downarrow S) \cap (!S(d); d:\delta \Downarrow S). \\
& a:(g:\gamma)^*, \text{ then } (!S(g); (g:\gamma \Downarrow (S(\text{end} \mapsto S(a))))^\times; !S(\text{end})). \\
& a:(g:\gamma \cap d:\delta), \text{ then } (g:\gamma \Downarrow S) \cap (d:\delta \Downarrow S). \quad a:(g:\gamma)^\times, \text{ then } (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times. \\
& a:\{x' = f(x) \& Q\}, \text{ then } \{x' = f(x) \& Q\}^d; !S(\text{end}). \\
& \text{atomic and not controlled by Angel, i.e.,} \\
& \alpha \in \{x := e, x := \otimes, ?Q, !Q, \{x' = f(x) \& Q\}^d\}, \text{ then } \alpha.
\end{aligned}$$

where game labels are preserved (including for Angel-controlled subgames transformed to their Demon-controlled equivalent) and fresh labels are used for newly introduced subgames. The universal projection of a Demonic subvalue map is symmetric (see Appendix B.6.5).

We can now show the defining property of subvalue maps.

Theorem 6.1 (Subvalue Map stays in Winning Region). Suppose S is an Angelic subvalue map for game $a:\alpha$ compatible with winning condition ϕ , i.e., for every subgame $b \in \text{subgames}(a:\alpha)$, $\models S(b) \rightarrow \langle a:\alpha_b \rangle S(\text{end})$ and $\models S(\text{end}) \rightarrow \phi$. Upon starting in any state $\sigma \in \llbracket S(a) \rrbracket$, and reaching subgame b by following any strategy induced by the subvalue map S , there exists a winning strategy for Angel to win the remainder of the game. That is, $S(a) \models \langle (a:\alpha \Downarrow S)_b \rangle \langle a:\alpha_b \rangle \phi$.

As discussed in Section 6.5.1, the property of Theorem 6.1 is critical but not alone strong enough, since it does not guarantee that following the policy will ensure Angel wins. The property it maintains is only that Angel always continues to have some winning strategy ($\langle a:\alpha_b \rangle \phi$), not that such a strategy is within the policy ($\langle a:\alpha \Downarrow S_b \rangle \phi$). Angel might get stuck without a valid choice permitted by the map S . The *inductive* subvalue map introduced next prevents this situation, disqualifying subvalue maps where Angel gets stuck.

6.6 Inductive Subvalue Maps

To correct for the shortcoming of subvalue maps by preventing players from getting stuck, we introduce *inductive* subvalue maps. Notation $a:\alpha \models S$ indicates S is an *inductive* subvalue map for game $a:\alpha$. Theorem 6.2 shows the characteristic property of an inductive subvalue map: its policy always keeps the player in a region where the player can win the rest of the game by following the subvalue map policy.

Theorem 6.2 (Inductive subvalue map ensures winning actions). Suppose S is an inductive Angelic subvalue map for game $a:\alpha$ and compatible with winning condition ϕ , i.e., $a:\alpha \models S$ and

2454 $S(\text{end}) \models \phi$. Upon starting in any state $\sigma \models S(a)$, and reaching subgame b by following any
 2455 strategy induced by the subvalue map S , there is a way for Angel to win by continuing to follow
 2456 S . That is, $S(a) \models \langle (a:\alpha \nabla S)_{:b} \rangle \langle (a:\alpha \nabla S)_{:b} \rangle \phi$.

2457 It is possible to characterize an inductive subvalue map using recursive conditions expressed
 2458 via dGL formulas (Def. 6.8).

2459 **Definition 6.8** (Inductive subvalue maps). Let S be a map from the subgames of $a:\alpha$ to winning
 2460 subregions. S is an *inductive Angelic subvalue map* for game $a:\alpha$, written $a:\alpha \models S$, when the
 2461 following holds. If $a:\alpha$ has structure:

- 2462 $a:(g:\gamma \cup d:\delta)$ then $\models S(a) \rightarrow S(g) \vee S(d)$ and $g:\gamma \models S$ and $d:\delta \models S$.
 2463 $a:(g:\gamma \cap d:\delta)$ then $\models S(a) \rightarrow S(g) \wedge S(d)$ and $g:\gamma \models S$ and $d:\delta \models S$.
 2464 $a:(g:\gamma ; d:\delta)$ then $\models S(a) \rightarrow S(g)$ and $g:\gamma \models S(\text{end} \mapsto S(d))$ and $d:\delta \models S$.
 2465 $a:(g:\gamma)^*$ then $\models S(a) \rightarrow \langle a:\alpha \nabla S \rangle S(\text{end})$ and $g:\gamma \models S(\text{end} \mapsto S(a))$.
 2466 $a:(g:\gamma)^\times$ then $\models S(a) \rightarrow S(g) \wedge S(\text{end})$ and $g:\gamma \models S(\text{end} \mapsto S(a))$.
 2467 atomic, i.e., $\alpha \in \{x := e, x := *, x := \otimes, ?Q, !Q, \{x' = f(x) \ \& \ Q\},$
 2468 $\{x' = f(x) \ \& \ Q\}^d\}$ then $\models S(a) \rightarrow \langle \alpha \rangle S(\text{end})$.

2469 Symmetric conditions characterize when S is an *inductive Demonic subvalue map* for game $a:\alpha$
 2470 (Appendix B.6.7).

2471 **Example 6.3.** Consider again the Angelic choice in Fig. 6.1. To establish $S \models 8:(6:(v := a)^* \cup$
 2472 $5:(v := v - 1 ; a := *))$, Def. 6.8 first checks that if $S(8)$ (i.e., ϕ_8) holds then either Angel can
 2473 win by choosing to run subgame 6 (when ϕ_6 holds) or subgame 5 (when ϕ_5 holds). It then also
 2474 recursively ensures that after going to subgame 5 or 6, S continues to provide a valid Angelic
 2475 strategy ($6:(v := v + a)^* \models S$ and $5:(v := v - 1 ; a := *) \models S$).

2476 We discuss the subtle loop cases of Def. 6.8. For Demonic loop validity $a:(g:\gamma)^\times \models S$, the
 2477 following conditions should hold.

- 2478 1. $\models S(a) \rightarrow S(\text{end})$, and $\models S(a) \rightarrow \langle \gamma \rangle S(a)$, i.e. $S(a)$ is an invariant that holds inductively
 2479 and implies the postcondition. Inductiveness condition $\models S(a) \rightarrow \langle \gamma \rangle S(a)$ does not
 2480 appear in Def. 6.8 because it is already implied by later conditions $\models S(a) \rightarrow S(g)$ and
 2481 $g:\gamma \models S(\text{end} \mapsto S(a))$.
- 2482 2. $g:\gamma \models S(\text{end} \mapsto S(a))$, i.e. the subvalue map for the loop body ensures finishing within
 2483 the safe invariant. Using instead the more liberal condition $g:\gamma \models S$ to finish the loop body
 2484 within the postcondition is unsound: Demon can choose to repeat the loop indefinitely,
 2485 so Angel must play to remain within the inductive region. Alternative condition $g:\gamma \models$
 2486 $S(\text{end} \mapsto S(g))$ is also unsound since then $S(g)$ is merely inductive and does not have to
 2487 imply reaching desired postcondition $S(\text{end})$.
- 2488 3. $\models S(a) \rightarrow S(g)$. In case Demon chooses to run the loop, Angel is guaranteed to not
 2489 get stuck while playing the loop body in winning subregion $S(g)$, so overall loop winning
 2490 subregion $S(a)$ should not exceed $S(g)$. On the other hand, $S(g)$ can be weaker than $S(a)$
 2491 as it can be one loop iteration away from implying $S(\text{end})$, while $S(a)$ must already imply
 2492 $S(\text{end})$.

2493 For Angelic loop validity $a:(g:\gamma)^* \models S$, the following conditions must hold.

1. $\models S(a) \rightarrow \langle a:\alpha \exists S \rangle S(\text{end})$, i.e., Angelic loop invariant $S(a)$ ensures that there is a *well-founded* strategy to exit while staying within the control envelope. Angel should not get stuck in a state where her only option is playing the loop forever. Using instead the more liberal condition $\models S(a) \rightarrow \langle \alpha \rangle S(\text{end})$ is not enough, since Angel can get stuck in a state where there is a way to exit the loop but only when going outside the subvalue map (see Appendix B.2.1).
2. $g:\gamma \models S(\text{end} \mapsto S(a))$, i.e., the subvalue map for the loop body ensures that after playing the loop body, there is still a well-founded exit strategy within the subvalue map $(S(a))$. Using condition $g:\gamma \models S(\text{end} \mapsto S(g) \vee S(\text{end}))$ instead is unsound as then $S(g)$ is merely inductive and does not have to imply an exit strategy to reach postcondition $S(\text{end})$ (see Appendix B.2.2).

Theorem 6.3 below proves that *all* inductive subvalue maps are subvalue maps, indicating that the validity conditions compose to produce correct Angel and Demon winning subregions. Thus, while following an inductive subvalue map, the player will never reach a state from which it cannot win (Theorem 6.1). Additionally, Theorem 6.2 at the start of this section shows that following the policy induced by an inductive subvalue map ensures that the agent will never reach a state where it is stuck. In combination, these mean that starting in a suitable initial state, while following an inductive subvalue map, a player cannot lose.

Theorem 6.3 (Inductive subvalue maps are subvalue maps). *For dGL game $a:\alpha$, if S is an inductive Angelic subvalue map ($a:\alpha \models S$), then it is also an Angelic subvalue map for $a:\alpha$, i.e., for every subgame $b:\beta$ in $\text{subgames}(a:\alpha)$, $\models S(b) \rightarrow \langle a:\alpha_b \rangle S(\text{end})$. Dually, if S is an inductive Demonic subvalue map for $a:\alpha$, then S is a Demonic subvalue map for $a:\alpha$.*

Def. 6.9 shows how an inductive subvalue map can capture strategies to win a dGL game from every state in the theoretical maximal winning region, indicating that Def. 6.2 is sufficiently liberal. It sets subvalues via modal dGL formulas. The winning subregion for the overall game is maximal.

Definition 6.9 (Model predictive subvalue map). For the game $a:\alpha$ and win condition ϕ , the *model predictive Angelic subvalue map* $\{\text{end} \mapsto \phi\} \cup \bigcup_{b \in \text{subgames}(a:\alpha)} \{b \mapsto \langle a:\alpha_b \rangle \phi\}$ maps every subgame b of $a:\alpha$ to the optimal winning regions of its game suffix. Dually, the *model predictive Demonic subvalue map* is $\{\text{end} \mapsto \phi\} \cup \bigcup_{b \in \text{subgames}(a:\alpha)} \{b \mapsto [a:\alpha_b] \phi\}$.

Example 6.4. For the game of Fig. 6.1, the model predictive Angelic subvalue map maps subgame 5 to $\langle \{x' = v\}^d; \alpha \rangle x > 0$, which per the semantics of dGL, precisely means the condition under which Angel can win by reaching her assigned winning conditions after playing the remainder of the game $\{x' = v\}^d; \alpha$.

Remark 4. Model Predictive Control (MPC) [34, 53, 136] is a well-known controller design principle that uses a mathematical model to predict controller behavior and back-computes the control conditions that respect the desired system constraints. Def. 6.9 extends this construction to hybrid games logic. Game suffix $\langle a:\alpha_b \rangle \phi$ serves as an exact model of future game behavior after subgame b ⁶.

Next, Section 6.7 defines a natural, symbolic execution based framework to synthesize subvalue maps that generate policy decisions in polynomial time. Then, Def. 6.10 provides a partial

⁶Though unlike standard MPC, as we are computing maximal control sets, there is no optimization step to identify the best control solution within the set.

ordering on inductive subvalue maps where more permissive maps are better, and Theorem 6.5 shows that Def. 6.9 is a maximally permissive inductive subvalue map.

6.7 Synthesis Framework for Inductive Subvalue Map with Efficient Checks

To check whether a given control decision lies within the policy of a subvalue map, Def. 6.4 checks whether the resulting state is within a subvalue (e.g., for $b:(d:\delta \cup g:\gamma)$, the policy checks $\sigma \models S(d)$ and $\sigma \models S(g)$). This section synthesizes inductive subvalue maps where state checks are efficient, for quick responses in real systems. In the model predictive solution from the previous section (Def. 6.9), such checking is not efficient because of subvalues that are dGL formulas, for which checking is undecidable [120]. In this section we restrict subvalues to formulas consisting of equations/inequalities of real polynomials along with propositional logic connectives ($\wedge, \vee, \neg$, etc.) but no quantifiers or modalities, henceforth, *propositional real arithmetic* formulas ($\mathcal{P}_{\mathbb{R}}$). For these, checking the truth value at a state takes polynomial time. *Given a dGL game and winning condition in $\mathcal{P}_{\mathbb{R}}$, this section synthesizes inductive subvalue maps where subvalues are in $\mathcal{P}_{\mathbb{R}}$.*

Synthesizing such a map is challenging because the subvalues must remain *mutually compatible*, indicating only the winning regions achievable while following the other subvalues. That is, they must follow the inductive subvalue map conditions Def. 6.8, where there are circular dependencies between subvalues. Consider, for example, the case of Demonic loop $a:(g:\gamma)^\times$. Subvalue $S(a)$ must imply $S(g)$, but $S(g)$ depends on $S(a)$ via condition $g:\gamma \models S(\text{end} \mapsto S(a))$. In particular, starting from the optimal subvalue map construction (Def. 6.9) and changing individual dGL formulas to under-approximations in $\mathcal{P}_{\mathbb{R}}$ would not alone work because cyclically interdependent soundness conditions would need to be reestablished. 1. We address the challenge of interdependent conditions by isolating their effect with a hybrid games analog of loop invariants. 2. To soundly lower formulas to $\mathcal{P}_{\mathbb{R}}$ while maintaining permissiveness, we use a *predicate transformer* based approach that given a subgame and winning condition in $\mathcal{P}_{\mathbb{R}}$ computes subvalues in $\mathcal{P}_{\mathbb{R}}$. This lets us present a backwards symbolic execution [8, 40] based *algorithmic framework* to compute inductive subvalue maps with subvalues in $\mathcal{P}_{\mathbb{R}}$.

Remark 5. Weakest precondition calculi [40] generate weak *preconditions*. To reason about loops, they generally *check* loop invariants, and rely on external loop invariant generation. Analogously, our framework generates permissive *subvalues*. To reason about loops, it *provides checks for invariants of Angel and Demon loop subvalue maps*. It is *parametric* in the generation of such invariants.

We define predicate transformers over subgames such that *given a postcondition in $\mathcal{P}_{\mathbb{R}}$, they output a subvalue in $\mathcal{P}_{\mathbb{R}}$* to recursively produce an entire subvalue map with subvalues in $\mathcal{P}_{\mathbb{R}}$. The transformers for the loop-free fragment of dGL are derived from a closely related definition in Chapter 5 [69], which we discuss later in this section. For loops, our framework uses the approach of heuristically generating *invariant candidates* and checking their correctness, which has worked well for hybrid systems [125, 143]. We characterize the necessary and sufficient conditions for a sound invariant (for Angel loops, $\models \text{Inv} \rightarrow \langle a:\alpha \exists S \rangle \phi$, for Demon loops, $\models$

$\text{Inv} \rightarrow S(g) \wedge \phi$. The framework generates invariant candidate Inv (details in (Section 6.7.1)), computes the subvalue map of the loop body, and then checks whether the invariant and loop body map are compatible⁷. Proposed invariants must be in $\mathcal{P}_{\mathbb{R}}$.

Algorithm 2 puts together the predicate transformers to define a recursive solving function $\langle \text{map} \rangle_{(a:\alpha, \phi)}$ that computes an inductive Angelic subvalue map with winning regions in $\mathcal{P}_{\mathbb{R}}$ for game $a:\alpha$ that is compatible with winning condition $\phi \in \mathcal{P}_{\mathbb{R}}$. Operator $\uplus$ computes the disjoint union of subvalue maps. $S \setminus \text{end}$ indicates the subvalue map S with the mapping for end removed. Such deletion is sometimes necessary to ensure that two maps to which we apply the disjoint union operator are genuinely disjoint. The algorithm is parametrized on *invariant candidate* generation function genInv . $\llbracket \text{map} \rrbracket$ similarly computes an inductive Demonic subvalue map (Appendix B.6.9).

Algorithm 2 Inductive subvalue map synthesis framework

```

1: function  $\langle \text{map} \rangle_{(a:\alpha, \phi)}$  ▷ Parametric Input:  $\text{genInv}$ 
2:    $S_e \leftarrow \{\text{end} \mapsto \phi\};$ 
3:   if  $\alpha \in \{x := e, x := *, x := \otimes, ?Q, !Q, \{x' = f(x) \& Q\}, \{x' = f(x) \& Q\}^d\}$  then
4:     return  $\{a \mapsto \text{simpl}(\langle \alpha \rangle \phi)\} \uplus S_e$ 
5:   else if  $\alpha = a:(g:\gamma \cup d:\delta)$  then
6:      $S_1 \leftarrow \langle \text{map} \rangle_{(g:\gamma, \phi)}; S_2 \leftarrow \langle \text{map} \rangle_{(d:\delta, \phi)}; \text{return } S_1 \uplus (S_2 \setminus \text{end}) \uplus \{a \mapsto S_1(g) \vee S_2(d)\}$ 
7:   else if  $\alpha = a:(g:\gamma \cap d:\delta)$  then
8:      $S_1 \leftarrow \langle \text{map} \rangle_{(g:\gamma, \phi)}; S_2 \leftarrow \langle \text{map} \rangle_{(d:\delta, \phi)}; \text{return } S_1 \uplus (S_2 \setminus \text{end}) \uplus \{a \mapsto S_1(g) \wedge S_2(d)\}$ 
9:   else if  $\alpha = a:(g:\gamma; d:\delta)$  then
10:     $S_1 \leftarrow \langle \text{map} \rangle_{(d:\delta, \phi)}; S_2 \leftarrow \langle \text{map} \rangle_{(g:\gamma, S_1(d))}; \text{return } S_1 \uplus (S_2 \setminus \text{end}) \uplus \{a \mapsto S_2(g)\}$ 
11:   else if  $\alpha = a:(g:\gamma)^*$  then
12:     while true do
13:        $\text{Inv} \leftarrow \text{genInv}(\langle (g:\gamma)^* \rangle \phi); S \leftarrow \langle \text{map} \rangle_{(g:\gamma, \text{Inv} \vee \phi)}$ 
14:       if  $\models \text{Inv} \rightarrow \langle a:\alpha \exists S \rangle \phi$  then return  $(S \setminus \text{end}) \uplus \{a \mapsto \text{Inv}\} \uplus S_e$ 
15:   else  $\alpha = a:(g:\gamma)^\times$ 
16:     while true do
17:        $\text{Inv} \leftarrow \text{genInv}(\langle (g:\gamma)^\times \rangle \phi); S \leftarrow \langle \text{map} \rangle_{(g:\gamma, \text{Inv})}$ 
18:       if  $\models \text{Inv} \rightarrow S(g) \wedge \phi$  then return  $(S \setminus \text{end}) \uplus \{a \mapsto \text{Inv}\} \uplus S_e$ 

```

$\langle \text{map} \rangle$ recursively computes the subvalue for each subgame. The terminal winning subregion $S(\text{end})$ is always set to the target winning condition ϕ . In the atomic cases, there is only one subgame, and thus only one additional subvalue to compute. Here, the predicate transformer sets the subvalue per dGL winning region semantics (e.g., for $a:x := e$, the winning subregion is $\langle x := e \rangle \phi$). But $\langle \alpha \rangle \phi$ is a modal dGL formula which needs to be further simplified to a $\mathcal{P}_{\mathbb{R}}$ formula (in this case, to $\phi(x \mapsto e)$, i.e., the formula $\phi \in \mathcal{P}_{\mathbb{R}}$ with every occurrence of variable

⁷Checking Angel loop invariants ($\models \text{Inv} \rightarrow \langle a:\alpha \exists S \rangle \phi$) can be difficult to automate using traditional invariant generation techniques. We use refinement based proofs in practice (Appendix B.4.5).

x substituted by e). This simplification is written as $\text{simpl}(\langle\alpha\rangle\phi)$. Chapter 5 and Appendix A.1 [69] show how to implement simpl ⁸.

While $\text{simpl}(\langle\alpha\rangle\phi)$ usually computes an exact expression in $\mathcal{P}_{\mathbb{R}}$, this is sometimes not possible for subgames with complicated ODEs. In such situations, a conservative, sound approximation is made such that $\models \text{simpl}(\langle\alpha\rangle\phi) \rightarrow \langle\alpha\rangle\phi$, i.e. the simplified expression is true only in the states where the original expression was true. Theorem 6.4 shows that the solving framework is robust to such conservative approximations, producing inductive subvalue maps despite them. In the compositional cases, e.g., $a:(g:\gamma \cup d:\delta)$, $\Diamond\text{map}$ first computes the subvalues of the composing subgames (in this case S_1 for $g:\gamma$ and S_2 for $d:\delta$), and then the subvalue for the outer subgame (in the case, $S(g) \vee S(d)$). For the loop cases, $\Diamond\text{map}$ generates invariants until it finds one that passes the check.

Fig. 6.1 shows an example of computing a subvalue map using $\Diamond\text{map}$. As before, label i denotes the subgame with winning subregion ϕ_i . At the top level, subgame 1 is a Demon loop. For this structure, after setting ϕ_{end} , in second step **Guess** ϕ_1 , $\Diamond\text{map}$ guesses invariant Inv using a *refinement* heuristic (discussed in Section 6.7.1), computing answer $x > 0 \wedge (v \geq 0 \vee a > 0)$. This invariant is used to recursively compute an inductive Angelic subvalue map for the loop body consisting of subgame 9. A final **Check** step ensures that $\models \psi \rightarrow \phi_9 \wedge x > 0$, i.e., the invariant is compatible with the inner subvalue map so Angel won't get stuck regardless of whether Demon runs the loop or exits.

Theorem 6.4 (Sound solving). $S := \Diamond\text{map}(a:\alpha, \phi)$ is an inductive Angelic subvalue map for game $a:\alpha$ ($a:\alpha \models S$) with all subvalues in $\mathcal{P}_{\mathbb{R}}$, compatible with Angel winning condition $\phi \in \mathcal{P}_{\mathbb{R}}$. Dually, $S := \Box\text{map}(a:\alpha, \phi)$ is an inductive Demonic subvalue map with all subvalues in $\mathcal{P}_{\mathbb{R}}$, compatible with Demon winning condition ϕ .

6.7.1 Invariant Generation

The subvalue map synthesis framework of Algorithm 2 provides the conditions to check if a candidate formula is a sound invariant. Any combination of invariant generation heuristics [69, 125, 143] can be used to generate candidate invariants. To solve for games where no other techniques apply, we propose a *game rewriting* framework. This framework uses a library of case-based syntactic transformations to rewrite loop game α^* or $\alpha^\times$ as a simpler game β that emulates aspects of its behavior. Then, if β is loop-free, $\text{simpl}(\langle\beta\rangle\phi)$ becomes an invariant guess, otherwise other invariant generation heuristics are tried on β . Checks in Algorithm 2, Line 14 and Line 18 recover soundness.

For example, the step **Guess** ϕ_1 in Fig. 6.1 shows a rewrite when computing an invariant of the outer loop (ϕ_1). The rewritten game (on the second line) emulates inner loop $(v := v + a)^*$ by letting Angel decide in advance how many loop iterations to run and assign her decision freely to new variable n . Then, setting $v := v + an$ conservatively estimates the effect of running the loop $\lceil n \rceil$ times (when n is fractional, v ends up smaller, and thus conservatively less favorable to Angel than after $\lceil n \rceil$ loop iterations). The rewrite also eliminates the outer loop using a generalized form of one-shot refinement (Chapter 5 [69]) where running the ODE

⁸The function $\text{reduce}(\langle\alpha\rangle\phi, A)$ (Appendix A.1) produces a $\mathcal{P}_{\mathbb{R}}$ formula that is equivalent to $\langle\alpha\rangle\phi$ under assumption formula A . $\text{simpl}(\langle\alpha\rangle\phi)$ is $\text{reduce}(\langle\alpha\rangle\phi, \top)$.

emulates the worst case behavior of multiple loop iterations. In general, different rewrite rules are tried heuristically based on the shape of a problem until an answer is found. New rewrite rules can be added to the library as needed. Appendix B.4.4 shows the rewrite rules that solve the problems of Section 6.10.

6.8 Maximal Solution

For a given game and winning condition, there are many possible correct inductive subvalue maps. Most useful are the *more permissive* subvalue maps that allow as many control actions as possible. More permissive subvalue maps leave more control options open, permitting more space to optimize for secondary goals when identifying a concrete, specialized controller within the envelope. We formalize subvalue map permissiveness with a partial ordering where more permissive subvalue maps are greater than less permissive subvalue maps.

A natural candidate for ordering the subvalue maps of game $a:\alpha$ would be to say that S is at least as permissive as S' ($S \sqsupseteq S'$) when for each subgame b in $\text{subgames}(a:\alpha)$, *every state in subregion $S'(b)$ is also within subregion $S(b)$* , i.e., $\models S'(b) \rightarrow S(b)$, so that $S(b)$ locally *permits* the agent to play into subgame b in all the states that $S'(b)$ does. However, this ordering fails to order cases like the following. Let S' set the winning region for the overall game to be empty ($S'(a) = \perp$), meaning that Angel is never allowed to play at all. For *any* other inductive subvalue map S , a good ordering should indicate that $S \sqsupseteq S'$ since S cannot possibly be less permissive, but the proposed naïve ordering does not. The winning subregion for some later subgame $b \neq a$ can be such that the check $\models S'(b) \rightarrow S(b)$ fails (see Appendix B.1 for details). An improved ordering makes the following modified check for every subgame b . $S \sqsupseteq S'$ when, *amongst the states that are reachable at subgame b while playing per subvalue map S'* , every state that is in subregion $S'(b)$ should also be within subregion $S(b)$. To express this logically, we first characterize the states that are reachable at subgame b while playing per subvalue map S' .

The game prefix (Def. B.10) of subgame b in the game $a:\alpha$, written $a:\alpha.b$, features *all* possible behaviors that can happen *before* an occurrence of subgame b within $a:\alpha$. To characterize all the states reachable at subgame b while following the policy induced by S' , we use the universal projection of S' onto the game prefix $(a:\alpha.b) \Downarrow S'$. Def. 6.10 compares the winning subregions of S and S' after running the projected game to filter out unreachable states.

Definition 6.10 (Permissiveness ordering). For two inductive Angelic subvalue maps S and S' for the game $a:\alpha$, S is at least as good as S' , written $S \sqsupseteq S'$, iff for each subgame b of $a:\alpha$, $\models \langle a:\alpha.b \Downarrow S' \rangle (S'(b) \rightarrow S(b))$ and $\models \langle a:\alpha \Downarrow S' \rangle (S'(\text{end}) \rightarrow S(\text{end}))$. Dually, for inductive Demonic subvalue maps, $S \sqsupseteq S'$ iff for subgame b of α , $\models [(a:\alpha.b) \Downarrow S'] (S'(b) \rightarrow S(b))$ and $\models [a:\alpha \Downarrow S'] (S'(\text{end}) \rightarrow S(\text{end}))$.

Remark 6. For every game and winning condition combination, the model predictive subvalue map (Def. 6.9) is a maximally permissive solution (Theorem 6.5). As Def. 6.9 always constructs an optimal subvalue map, it optimally solves the dGL control envelope synthesis problem, achieving completeness for symbolic hybrid games synthesis (when modal, dGL subvalues are permitted).

Theorem 6.5 (Maximal inductive subvalue map). *Amongst the inductive Angelic subvalue maps for game $a:\alpha$ compatible with winning condition ϕ , the model predictive Angelic map given by*

2672 Def. 6.9 is maximal under the ordering of Def. 6.10. That is, for all S such that $a:\alpha \models S$, model
 2673 predictive Angelic subvalue map S' satisfies $S' \sqsupseteq S$. Dually, the model predictive Demonic
 2674 subvalue map (Def. 6.9) is maximal amongst inductive Demonic subvalue maps per the ordering
 2675 of Def. 6.10.

2676 6.8.1 Recovering Solution Optimality

2677 When constructing subvalue maps where subvalues must be in $\mathcal{P}_{\mathbb{R}}$, Algorithm 2 is not guaranteed
 2678 to construct an optimal subvalue map. This section describes a retrospective check that can be
 2679 used to test whether the synthesis procedure failed to produce a maximally permissive envelope,
 2680 indicating that more aggressive synthesis/invariant generation is required for more permissive-
 2681 ness. First, use Algorithm 2 to independently construct strategies for Angel and Demon for the
 2682 same game $a:\alpha$. Let $S_{\diamond}$ be the computed Angelic subvalue map for winning condition ϕ and $S_{\square}$
 2683 be a Demonic subvalue map for the opposite objective $\neg\phi$. Cross-checking these subvalue maps
 2684 can recover whether they are optimal, capturing all possible ways for their players to win. If for
 2685 each subgame b , $S_{\square}(b) \vee S_{\diamond}(b)$ is valid, then intuitively, $S_{\diamond}(b)$ and $S_{\square}(b)$ together cover every
 2686 possible state, identifying it either as one where Angel has a winning strategy or where Demon
 2687 does. Further weakening $S_{\diamond}(b)$ would violate consistency (i.e., it is never possible for both Angel
 2688 and Demon to win)⁹. So $S_{\diamond}(b)$ must be a maximally permissive, optimal Angel winning region,
 2689 and $S_{\diamond}$, an optimal Angelic subvalue map.

2690 6.9 Relationship to Runtime Monitoring

2691 A runtime monitor (constructed similarly to existing work on monitoring dL [98] and construc-
 2692 tive dGL [20]) raises an alarm or intervenes to enforce a correct fallback when gameplay deviates
 2693 from the control envelope. It is possible to model the use of a subvalue map as a *runtime monitor*
 2694 in a single dGL formula. For an Angelic subvalue map, this *monitored game* is constructed by
 2695 1. giving Demon control over Angelic decisions while restricting him to play per the subvalue
 2696 map (like in universal projection) to mimic an unverified controller that plays unreliably while
 2697 being forced to stay within the control envelope by a runtime monitor and 2. ensuring before
 2698 each such monitored subgame that the subvalue map allows *some* control action using Angelic
 2699 tests. Eq. (6.3) shows the monitored game for the running example. The *monitored game* for dGL
 2700 game $a:\alpha$ and Angelic subvalue map S , written $a:\alpha \setminus S$, is generated recursively per the structure
 2701 of $a:\alpha$ as follows. If $a:\alpha$ has structure:

2702 $a:x := *, \text{ then } ?(\exists x S(\text{end})) ; x := \otimes ; !S(\text{end}). \quad a:(g:\gamma ; d:\delta), \text{ then } g:\gamma \setminus S(\text{end} \mapsto S(d)) ; d:\delta \setminus S.$
 2703 $a:(g:\gamma \cup d:\delta), \text{ then } ?(S(g) \vee S(d)) ; ((!S(g) ; g:\gamma \setminus S) \cap (!S(d) ; d:\delta \setminus S)).$
 2704 $a:(g:\gamma)^*, \text{ then } ?S(a) ; (!S(g) ; g:\gamma \setminus S(\text{end} \mapsto S(a))) ; ?S(a))^{\times} ; !S(\text{end}).$
 2705 $a:(g:\gamma \cap d:\delta), \text{ then } (g:\gamma \setminus S) \cap (d:\delta \setminus S). \quad a:(g:\gamma)^{\times}, \text{ then } (g:\gamma \setminus S(\text{end} \mapsto S(a)))^{\times}.$
 2706 $a:\{x' = f(x) \ \& \ Q\}, \text{ then } ?\langle \alpha \rangle S(\text{end}) ; \{x' = f(x) \ \& \ Q\}^d ; !S(\text{end}).$

⁹This is analogous to how in α - β pruning [79], if MIN's action results in a position where MAX wins, then that action cannot be part of MIN's strategy.

atomic and not controlled by Demon, i.e.,

$$\alpha \in \{x := e, x := \otimes, ?Q, !Q, \{x' = f(x) \ \& \ Q\}^d\}, \text{ then } \alpha.$$

Monitoring a Demonic subvalue map is symmetric (see Appendix B.6.5). Every inductive subvalue map S is a sound runtime monitor, i.e., $\models S(a) \rightarrow \langle \alpha \alpha^* S \rangle S(\text{end})$ (Theorem D.1).

6.10 Evaluation

To test the generality and flexibility of our approach, we demonstrate it on examples from classic control theory problem classes (event-triggered control, reach-avoid problems, nonlinear control, infinite horizon switching). We compare to CESAR (Chapter 5 [69]), which synthesizes symbolic control envelopes as well, but for hybrid systems with a *fixed, time-triggered template*. NYCS [13] is a tool that can synthesize control envelopes for hybrid games (expressed via hybrid automata), but does not solve parametrically for symbolic constraints¹⁰. We compare performance.

CESAR and NYCS cannot solve the new examples, each of which displays different problem features, demonstrating the greater generality of our approach, which solves them within a few minutes (Fig. 6.3a). We also evaluate our approach on the CESAR benchmark suite (Appendix A.3 [69]). Our implementation’s performance is similar to CESAR despite the significantly greater generality (Fig. 6.3b). To compare to NYCS, we procedurally generate 25 instantiations of a reach-avoid problem. The implementation finds control envelopes for 17 of them, timing out for the rest, while NYCS solves all of them, suggesting potential for future optimizations for this problem class.

The experiments were run on a 32GB RAM M2 MacBook Pro. The implementation uses Pegasus [143], CESAR (Chapter 5 [69]), and additional rewriting heuristics for invariant generation. It uses Mathematica for simplification and quantifier elimination. Our examples favor interesting control challenges over dynamic complexity to avoid the computer algebra bottleneck, which is an avenue for future research. Appendix B.4.3 provides dGL formulas for all the new examples.

Event-triggered ETCS (Model 9) is an example of an *event-triggered* control problem, in contrast with the time-triggered control handled by Chapter 5 [69]. In time-triggered control, the controller repeatedly polls with some maximum time latency. In event-triggered control, the controller is instead triggered at *events*, when specific conditions are met. Identifying what these events are that should trigger controller intervention is a part of the synthesis challenge. Our example problem (Model 9) translates the ETCS Train benchmark of the CESAR benchmark suite from the time-triggered to the event-triggered paradigm. It describes a train controller that must stop the train by point e , and has the choice to either accelerate or to brake. Line 3 shows the plant which Angel (emulating the controller) chooses how long to run. The synthesized subvalue map indicates when Angel must stop continuous evolution to enforce braking, characterizing the

¹⁰Our problems have symbolic parameters that can represent any real number, e.g., in event-triggered ETCS (Model 9), A , B and T . We generate an envelope that is parametric in these symbols, equivalent to making uncountably infinitely many calls to NYCS corresponding to every possible parameter value assignment. Such parametricity is crucial in some applications (e.g., when parameters are unknown statically and only identified at runtime [50]).

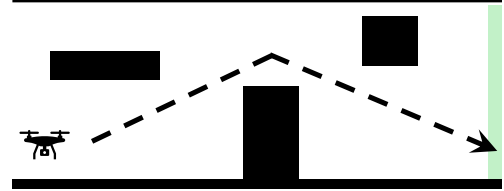
(a) Summary of new examples. Appendix B.4 discusses them.

Benchmark	Ours	CESAR	NYCS	Problem Type
Event ETCS	23s	∞	∞	Event-triggered control
Infinite Track	233s	∞	∞	Infinite horizon switching
Surgical Robot [81]	168s	∞	∞	Nonlinear control
Highway Driving [93]	91s	∞	∞	Time-triggered control with adversarial agent
Reach-avoid Robot	21s	∞	∞	Reach-avoid problem

Benchmark	Ours	CESAR	NYCS
ETCS Train	12s	15s	∞
Sled	64s	64s	∞
Intersection	104s	104s	∞
Curvebot	111s	112s	∞
Parachute	113s	123s	∞
Corridor	40s	40s	∞
Power Station	56s	56s	∞
Coolant	311s	312s	∞

(b) Running time comparison on the CESAR benchmark suite (Appendix A.3 [69]).

Tool	Success	Failure
Ours	17	8
NYCS	25	0
CESAR	0	25



(c) Setup and outcome of the quadcopter reach-avoid problem with procedurally generated obstacle placement.

Figure 6.3: Benchmark summaries and results.

event at which to trigger the controller. Modeling an event-triggered system as a game is subtle: the controller has the ability to choose how long to run the plant, and should not be permitted to win by interrupting infinitely often and inducing Zeno behavior. *dGL*'s flexibility allows a correct encoding using nested loops and time progress assertions.

Surgical Robot shows an example of nonlinear control (Model 19). It expresses the core control challenge in a case study [81] modeling force feedback in surgical robots. The objective is to dampen the force applied by the surgeon to ensure that a surgical tool stays within bounding planes called *virtual fixtures*. The damping factor g (Line 3) that we synthesize for can be *any* nonnegative real number, implying infinitely many control possibilities.

Highway Driving (Model 22) expresses the core control challenge of a case study [93] where on a highway, a controlled car must follow the car ahead at a safe distance. The car ahead can behave adversarially by braking suddenly. *Reach-avoid Robot* (Model 21) demonstrates an example of envelope synthesis for reach-avoid problems, where an agent must *reach* an objective while *avoiding* unsafe situations. *Infinite Track* (Model 20) is an example of infinite horizon switching. To remain safe for arbitrarily long time, the controlled vehicle must keep switching directions at the right moments, and make *infinitely* many switches. Appendix B.4.1 discusses these examples further, and Appendix B.4.3 lists the models. The synthesized control envelopes

for *Infinite Track*, *Surgical Robot*, and *Reach-avoid Robot* are maximally permissive. The envelopes for *Highway Driving* and *Event-triggered ETCS* are *not* maximally permissive because of the rewrite heuristics used.

To compare the algorithm to NYCS, we procedurally generate instantiations of *quadcopter*, replacing symbolic parameters with concrete numbers, to create a reach-avoid challenge. Each instantiation describes a scenario like in Fig. 6.3c, where a quadcopter must reach a target safe area while avoiding obstacles placed randomly in the environment. The quadcopter constantly moves forward and can choose to either move upwards or downwards, but cannot revise its decision, once made, for one second. The objective is to find a control envelope showing from which points the quadcopter has a way to reach the target safe zone while avoiding the obstacles. The implementation finds control envelopes for 17 problems. The other 8 problems time out after 20 minutes. CESAR cannot solve any of these problems because they do not fit its template. NYCS, with its polytope representation and optimizations for this problem class, solves all of the problems. Appendix B.4.2 provides further details.

6.11 Related Work

Subvalue maps represent nondeterministic policies for hybrid games in a way that is compositional at the level of imperative program constructs such as loops and branches. They can be verified/synthesized leveraging program analysis techniques like loop invariants, making it possible to answer control envelope synthesis questions that no previous work solves.

- *Program logics* like game logic and dGL express games as imperative programs with Angelic and Demonic nondeterminism [110, 113, 120]. Their semantics compositionally define when *there exists* a way to win a game, but not when a given policy wins.
- Refinement [16, 104], Hoare logic [94], *Differential Floyd-Hoare logic* [60], and constructive proof [20] based approaches soundly construct deterministic policies (i.e., programs with Angelic nondeterminism fully resolved) but do not study *maximal sets of solutions*. They do not have a direct representation of nondeterministic policies from which to derive *sets* of permissible control actions, or a formal comparison of permissiveness of such policies, both of which are crucial parts of the permissive policy synthesis problem that we solve.
- A line of research solves games specified by *hybrid automata* with objectives expressed in logics like LTL [13, 41, 62, 155, 167]. Our approach targets specifications at a higher

Model 9 Event-triggered ETCS

$$\begin{array}{ll}
 \text{assume} \mid &_1 \quad A > 0 \wedge B > 0 \wedge T > 0 \rightarrow \langle \\
 \text{choice} \mid &_2 \quad \left(t := 0 ; \left((a := A \cup a := -B) ; \right. \right. \\
 \text{plant} \mid &_3 \quad \left. \left. \{p' = v, v' = a, t' = 1 \ \& \ e - p > 0\} \right)^* ; ?t \geq 1 \right)^{\times} \rangle \text{True}
 \end{array}$$

level of abstraction, where the game has imperative programming constructs like loops and branches. Compositionality is also at the level of these constructs, and the theoretical development required is different (e.g., the subtle Angel loop case in Def. 6.8).

- Traditional *game theory* [140, 157], and *reinforcement learning* [146] have recursive solving techniques and representations of nondeterministic policies (value functions in RL and value tables in tabular games [10, 147, 148]). But these works operate at a different level of abstraction than our imperative program-like hybrid games. Also, our work is symbolic rather than numeric, and creates envelopes parametric in the symbolic parameters of the input game. Finally, our subvalue maps are logically formalized, suitable for computerized verification and formally justified synthesis.
- *Compositional game theory* [7, 56] also has a compositional representation of nondeterministic policies but also operates at a different level of abstraction, solving not hybrid games but abstract games with sequential and parallel composition.
- Compared to safe set and *barrier function* approaches [4, 6, 33, 35, 96, 131, 145], we again solve for the higher level of abstraction of dGL, with program structures such as loops, as well as the adversarial dynamics of games.
- Our control envelopes are also conceptually different from *permissive strategies* [15]. A permissive strategy typically needs to ensure the player wins, but a control envelope just needs to ensure that the player remains in states from where a winning strategy exists.

6.12 Conclusion

We introduce a generalized symbolic control envelope synthesis approach for hybrid games. Such envelopes identify what control action is safe to take when. We represent the envelopes using inductive subvalue maps and provide an algorithmic framework to generate them. Our approach allows the expression and synthesis of a variety of control problems. The approach should generalize for non-dGL Markovian games, where the policy depends only on state and not the history of previous moves, which is an avenue for future research.

An artifact is available [75]. This artifact contains:

1. A version of KeYmaera X extended with implementations of our synthesis framework, and of the CESAR algorithm from Chapter 5.
2. KeYmaera X Specification files for the examples introduced by this chapter.
3. NYCS specifications for the quadcopter instances.
4. Instructions to set up and run the code.

A limitation that arises in the chapter is the bottleneck of computer algebra, and the requirement to identify good rewrite heuristics to cover new problem classes. The next chapter addresses these challenges by leveraging large language models as a general heuristic for generating invariants.

Chapter 7

2827

Automatic Synthesis

2828

Chapter 6 introduced a framework for synthesizing symbolic control envelopes for differential game logic games. It implemented the framework by symbolic execution which relied on expensive computer algebra operations such as quantifier elimination. For performance, it used a library of rewrite rules for different problem types. This chapter increases scalability: it reduces dependence on expensive computer algebra operations and eliminates the need for case-by-case heuristics by using large language models (LLMs). It also introduces the ability to focus on a *specific class* of solutions, which often, in real-world applications, results in a sufficiently useful control envelope while making the synthesis process more efficient. In this chapter, the user provides the game and describes their desired solution informally, and our tool computes the corresponding verified control envelope.

7.1 Overview

2839

As before, we represent nondeterministic policies as *subvalue maps*. Once the LLM guesses a policy (subvalue map), we can verify if it is correct (non-losing, never stuck) by checking some dGL formulas identified by Chapter 6.

For synthesis, the same five case studies as Chapter 4 are used (Model 6, Model 23, Model 24, Model 25 and Model 26 listed in Appendix C.3) but with some initial assumptions missing (it will be the task of the synthesis pipeline to come up with all conditions necessary to make a strategy succeed). Additionally, each is associated with an informal natural language description of which of the many possible control solutions to synthesize (listed in Appendix C.3). Given the example of Model 6 (illustrated by Fig. 7.1) and a request to synthesize the solution that sets the system to an equilibrium point as soon as possible, our tool automatically identifies the corresponding subvalue map. It computes that Angel must pick x_add and y_add such that $x + x_add = g/d$ and $y + y_add = a/b$, and that this policy only works when, initially, $x \leq g/d$, $y \leq a/b$, $g/d \geq x_{mn}$, and $a/b \geq y_{mn}$.

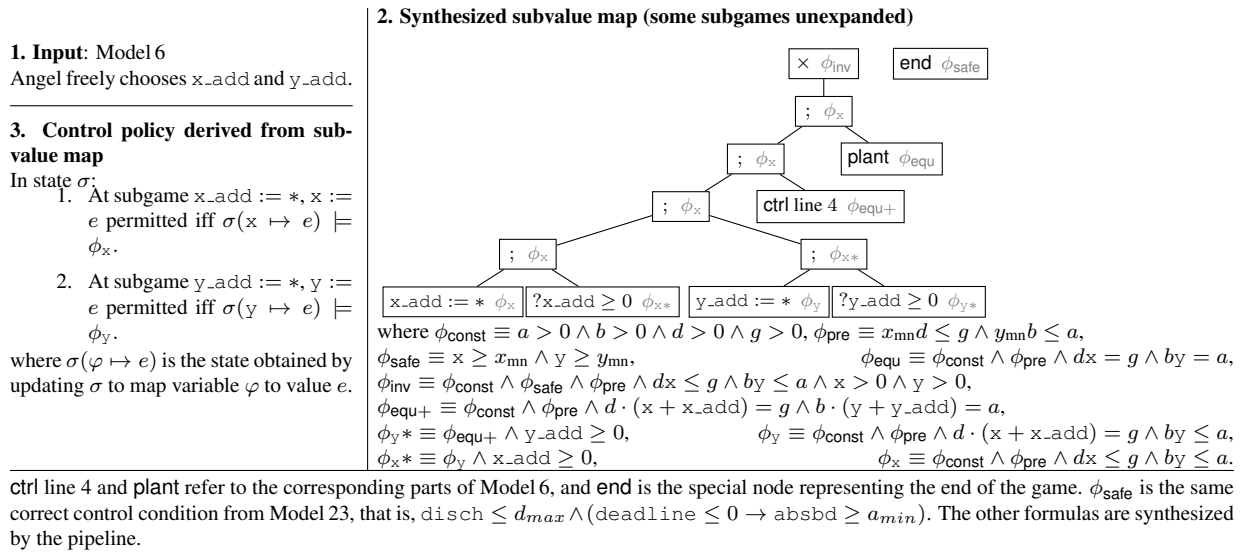


Figure 7.1: Example of envelope synthesis shown on Model6

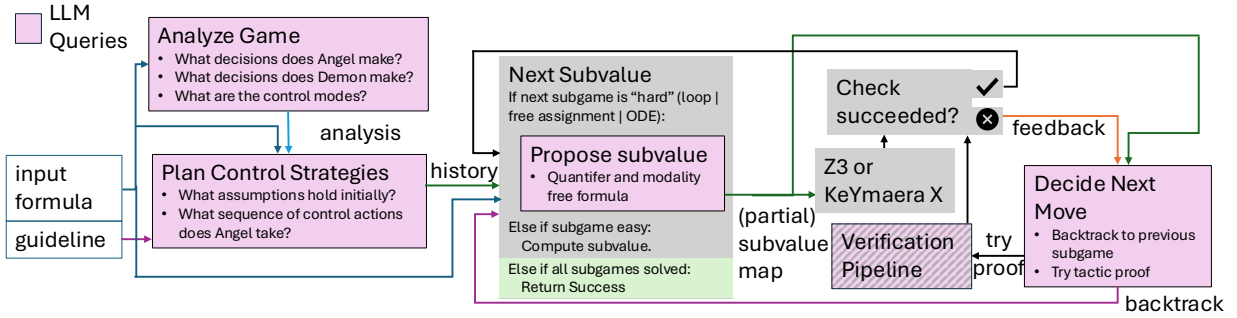


Figure 7.2: Synthesis Pipeline. Appendix C.5 shows prompts.

7.1.1 Pipeline Description

Weakest precondition calculus for normal programs [40] starts at the end of the game and works backwards, requiring loop invariant guesses and retrospective checks at loops. Similarly, our pipeline executes the precondition calculus for subvalue maps (Chapter 6) on the game backwards from end to start, computing subvalues as preconditions. Some subgames, such as simple assignments, are easy to symbolically execute. Computing the precondition of others: loops, continuous dynamics, and nondeterministic choices, is expensive or undecidable in general. For these, the LLM is prompted to guess subvalue map entries which encode Angel’s winning strategy. A retrospective check using Z3 [39] when possible and KeYmaera X when necessary recovers soundness. Fig. 7.2 shows the synthesis pipeline.

Initially, the LLM analyzes the game, identifying player actions and control modes, and comes up with a detailed strategy for Angel based on the informal guideline provided by the user. The strategy is expressed as a sequence of steps that Angel will take, stored to provide context in later steps. In the main subvalue map computation phase, the *Next Subvalue* step goes backwards to the next unprocessed subgame, and either symbolically computes the subvalue if

it is easy, or prompts the LLM to guess it otherwise. In the latter case, the subvalue is checked using Z3 or KeYmaera X with existing, LLM-free automation. Guesses and their outcomes are stored to provide context for future guesses¹. If the check succeeds, the pipeline continues to the next unprocessed subgame. Otherwise, there are two possible recovery strategies that the LLM chooses between. If the lemma was too complicated to verify using existing automation, it should try using the automatic theorem proving system from Section 4.3. Otherwise, the subvalue may be genuinely incorrect. The incorrectness may lie at the current subgame, or result from a bad earlier subvalue guess that makes the current subgame unsolvable (e.g., a non-inductive loop invariant causing problems in the loop body subvalues). The LLM should choose to backtrack to the subgame where the bad guess was made, and subvalue computation resumes from there.

7.1.2 Outcomes

Table 7.1 summarizes the results of our experiments. A case study is solved when one of four runs of the synthesis tool produces a correct subvalue map within a total budget of ten dollars for LLM calls. As a baseline, we compare to the tool associated with Chapter 6, the only previous work general enough to synthesize control envelopes for our problems. It fails, although it could succeed if extended with the right custom heuristics on a case-by-case basis. In contrast, the pipeline of this chapter is very general, free of the need for case-by-case encoding of heuristics, and solves four out of five case studies. We also perform an ablation experiment where we do not allow the LLM to use the proving system from Section 4.3 or perform custom backtracking. Instead, upon failure, we simply repeat LLM call points until the proof succeeds.

All experiments consist of four parallel pipeline runs that terminate when any one of them succeeds, or when they collectively run out of a ten dollar budget for LLM calls. The ten dollar budget is chosen because of diminishing returns beyond this point.

We also try an ablation with the initial analysis and control strategy planning step removed. The result is that three out of the five case studies receive a verified subvalue map, and even amongst these three, one has a synthesized subvalue map that is different than the desired solution per the provided guideline. This results in a lower success rate than the four problems solved by the full pipeline, indicating that the initial analysis segment is helpful.

We next report the number of LLM calls, dollar cost, and time taken for each case study by the full pipeline in Table 7.2.

In the ablation without proving or backtracking (recovery choice), as the LLM does not guide backtracking, we need a separate *search policy* to decide how to do retries. We use depth-first

¹Control flow is more complicated for the special case of loops, where the loop invariant is guessed first, then the subvalue map of the loop body is computed assuming the invariant holds at the end of the body, and finally the loop invariant is retrospectively checked for inductiveness given the computed subvalue map of the body.

Previous Work	Without Recovery and LLM-Assisted Proving	Full Pipeline
0/5	2/5	4/5

Table 7.1: Synthesis: ablation shows the benefit of the recovery segment of the pipeline (detailed proving and custom backtracking).

Model	Number of LLM Calls	Dollar Cost	Time (minutes)
Coolant	115	10.0	83.8
Chemical Reaction	16	0.7	6.8
Van der Pol	26	0.7	6.3
Lotka Volterra	58	2.8	23.7
Train	89	5.5	41.4

Table 7.2: Synthesis experiments: Number of LLM calls, time to run, and dollar cost for each case study.

Model	Number of LLM Calls	Dollar Cost
Chemical Reaction	17	0.9
Van der Pol	22	0.7

Table 7.3: Synthesis experiments without recovery choice: Number of LLM calls and dollar cost for each case study. The costs for all other case studies exceeds the budget.

search with a maximum branching factor of three. That is, when a subvalue check fails, we first retry it twice, and if both fail, we backtrack to the second most recent subvalue guess and retry it, and so on. The two benchmarks that succeed in this ablation are Chemical Reaction and Van der Pol, with the number of LLM calls and dollar costs listed in Table 7.3.

7.2 Conclusion

Notice that in solving the benchmark Model 24, we have come full-circle by effectively synthesizing the conservative train control envelope (Model 1 in Chapter 3) automatically. Synthesizing the more complex solutions later in Chapter 3 are beyond the current abilities of the pipeline, but remain interesting avenues for future work.

This concludes the second part of the dissertation, on automatic synthesis of control envelopes given the shape of the control problem. The remaining bottleneck for control envelope design becomes how to come up with this shape, the hybrid game specifying the control possibilities, in the first place. This is the subject of Chapter 8.

Chapter 8

2913

Autoformalization

2914

Previous chapters have shown how to synthesize control envelopes once the shape of the control problem has been formalized as a hybrid game. A final challenge that remains is coming up with the right hybrid game in the first place.

In fact, in general, writing a formal model of reliable cyber-physical systems (CPSs) is a bottleneck preventing broader industry adoption [138]. Unlike program verification, where a mathematically meaningful object to verify already exists in the form of a program, for CPSs, we must first create a formal model of the physical system and environment. This process is notoriously difficult, time-consuming, and error-prone.

Can large language models (LLMs) help autoformalize physical systems? They have shown autoformalization abilities for mathematics [88, 163] and CPS contracts [30, 109]. This chapter presents an initial, experiment-focused exploration of the ability of LLMs to autoformalize the underlying physics problems, faithfully preserving their exact continuous dynamics and discrete transitions in differential game logic (Chapter 2 [120]). We propose 20 benchmarks derived from hard undergraduate kinematics problems [66, 89, 95]. We find that, supported by few-shot prompting and parser feedback, OpenAI o3 has a top-5 accuracy of 70% (success rate on selecting the best out of 5 attempts per benchmark). The main cause of failure is checker limitations for the most complex problems.

Our benchmarks and pipeline are available as an online artifact at <https://doi.org/10.1184/R1/28934195>. Evaluating autoformalization of a physical system is challenging because every problem has many valid formal representations, and conversely, models can be incorrect for many subtle reasons. To automatically assess the semantic proximity of generated formal models to the original physics problem, we provide a checker that:

1. performs *symbolic execution* over the generated formal model, examining whether the expected symbolic solution to the original physics problem is recovered.
2. ensures the model is not in stasis by requiring that a minimum number of variables are mutated, indicating they are subject to physical effects.

Recovering the expected solution to the original natural language problem is a strong indication that the formalization models the problem correctly, though the checker may sometimes reject answers that are in principle correct but missed some implicit assumptions that the original solution made.

Autoformalization of physical models can have tremendous impact on reliable autonomy. Unlike in mathematics, validating a hypothesized CPS formalization is possible with comparisons between real measurements and predicted measurements during test runs [98]. Easy formalization would allow every modular component to be modeled formally, with all assumptions explicit. Verification could lead to safer systems and synthesis could make control system design easier while ensuring no edge cases are missed relative to the formal model.

8.1 Overview

This section explains by example what the process of autoformalizing a physics problem involves. Consider the following physics question [66, problem 1.2], which will serve as a running example:

“A point traversed half the distance with a velocity v_0 . The remaining part of the distance was covered with velocity v_1 for half the time, and with velocity v_2 for the other half of the time. Find the mean velocity of the point averaged over the whole time of motion.”

Solution: “ $v_{\text{avg}} = \frac{2v_0(v_1+v_2)}{2v_0+v_1+v_2}$,”

The physical motion described here is hybrid, with continuous dynamics (motion at velocity v_0 , v_1 , and finally v_2) as well as discrete transitions (changing velocity under specified conditions). Additionally, there are non-trivial conditions describing *when* the discrete transitions happen. We seek to symbolically model this motion formally.

We use hybrid games [63, 154] to express the control problems. Like in the rest of this thesis, we use the logic formalization of hybrid games [120] (as opposed to the automaton-based formalization [63]). This may be beneficial because it is closer to code, which LLMs have been more exposed to and fine-tuned for. Model 10 shows an example of a dGL hybrid game which formalizes the running example.

Model 10 Running Example: Finding Mean Velocity

setup		1	$\langle x := 0; d_{\text{avg}} := 0;$
phase 0		2	$\{x' = v_0, d'_{\text{avg}} = v_{\text{avg}}\}; d_h := x;$
phase 1		3	$t := 0; \{x' = v_1, d'_{\text{avg}} = v_{\text{avg}}, t' = 1\};$
phase 2		4	$t_h := t; \{x' = v_2, d'_{\text{avg}} = v_{\text{avg}}, t' = 1\};$
transitions		5	$?t = 2t_h; ?x = 2d_h$
win when		6	$\rangle d_{\text{avg}} = x$

In this game, a player (canonically called Angel) runs physical dynamics per the problem specification, moving the point’s position x with velocity v_0 in phase 0 on Line 2, with velocity v_1 in phase 1 on Line 3, and with velocity v_2 in phase 2 on Line 4. Angel is forced to transition between phases under the exact conditions specified by the problem because of *tests* (assertions) on Line 5. These assertions use *auxiliary variables* t_h to keep track of the time elapsed in phase 1,

t to keep track of time elapsed in phase 1 and phase 2 combined, and d_h to keep track of half the distance covered. On Line 5, test $?x = 2d_h$ ensures that the first phase transition occurred when half the distance was covered. The test $?t = 2t_h$ ensures that the second transition happened at half the total run time of phases 1 and 2. If Angel fails either test condition, she immediately loses the game.

Notice that there is a variable v_{avg} representing average velocity, and Line 6 says that Angel will win the game only if v_{avg} was correct, as checked using auxiliary variable d_{avg} , which tracks the displacement covered by a particle traveling at v_{avg} through all phases. The value that v_{avg} must hold for Angel to win matches the solution to the original problem, $v_{\text{avg}} = \frac{2v_0(v_1+v_2)}{2v_0+v_1+v_2}$. In this example, there happens to be no adversarial dynamics (Angel is the only player making decisions). However, in problems with adversarial nondeterminism such as unpredictable environmental factors or controller latency, a second, symmetric, adversarial player (canonically called Demon) models the situation by resolving nondeterminism with the goal to make Angel lose. Adversarial dynamics are also useful for optimization problems, e.g., “choose the launch angle that minimizes drift”, which can be modeled as Angel and Demon competing to find this angle.

Having formalized the problem as a dGL hybrid game, we next describe how symbolic execution enables a check for whether the model aligns with the natural language question. For the fragment of dGL where most of the chosen benchmark problems lie (loop-free, polynomial solutions), symbolic execution is decidable and Chapter 5 describes how to perform it. Backwards symbolic execution over a game lets us compute the conditions under which Angel (or dually, Demon) can win the game. For example, symbolic execution of the subgame $\langle \text{Line 5} \rangle d_{\text{avg}} = x$ at the end of Model 10 evaluates to $x = d_{\text{avg}} \wedge t = 2t_h \wedge x = 2d_h$ which is precisely the weakest precondition starting from which Angel passes the tests of Line 5 and ends the game in a state where she wins. Consider the precondition ϕ of Line 1 computed in this way. Angel should be able to win the game exactly when v_{avg} is set to the correct solution, i.e., $v_{\text{avg}} = \frac{2v_0(v_1+v_2)}{2v_0+v_1+v_2} \leftrightarrow \phi$. Such a check is implemented using Z3 and Mathematica.

Observe that even a minor modeling mistake can result in this check failing. In fact, symbolically executing Model 10 reveals that it is actually equivalent to $\top$. It suffers from a subtle exploit where Angel can always run each phase for 0 time, in which case $x_{\text{avg}} = x = 0$ regardless of what v_{avg} is set to. The problem is that the model lets Angel choose the distance that the particle will travel, as her choice at Line 2 determines the value of half-distance d_h , which in turn determines the distance that phases 1 and 2 must cover. The fix is simple: the assignment $d_h := x$ on Line 2 should be turned to a test $?d_h = x$. The automated check was able to catch this difference and is a strong signal of correctness, especially in combination with manual review. The automated checker has a second test to ensure the system is not in stasis, ruling out empty games that exploit the symbolic execution test by setting the win condition directly to the desired solution. For this example, the second test checks via static analysis that at least two variables are written to, since *any* reasonable model will at least modify the variables representing the particle’s position and time. The checker is provided with the minimum number of variables that must be written to and the expected solution for every benchmark problem.

Finally, we see the benefit of formal modeling when the model with Line 2 corrected is still not equivalent to the expected solution. The problem is that the textbook solution is correct *only*

3017 *under some unstated assumptions*: distance d is non-zero, velocity v_0 is positive, and $v_1 + v_2$ is
 3018 also positive. Finally, Model 11 makes these assumptions. Under the assumptions of Model 11,
 3019 Line 1, the modal formula from Model 11, Line 2 to Line 7 is equivalent to $\frac{2v_0(v_1+v_2)}{2v_0+v_1+v_2}$. In unver-
 3020 ified or informal physics, subtle modeling glitches are widespread. As a human, it is easy to
 3021 neglect a measure zero case like $d_h = 0$ during an intermediate calculation, but formalization
 3022 forces this case to be explicit.

Model 11 Repaired Model: Finding Mean Velocity

assum2		1	$(d_h > 0 \wedge v_0 > 0 \wedge v_1 + v_2 > 0) \rightarrow$
setup		2	$\langle x := 0; d_{\text{avg}} := 0;$
phase 0		3	$\{x' = v_0, d'_{\text{avg}} = v_{\text{avg}}\}; \text{ ?}d_h = x;$
phase 1		4	$t := 0; \{x' = v_1, d'_{\text{avg}} = v_{\text{avg}}, t' = 1\};$
phase 2		5	$t_h := t; \{x' = v_2, d'_{\text{avg}} = v_{\text{avg}}, t' = 1\};$
transitions		6	$\text{?}t = 2t_h; \text{?}x = 2d_h$
win when		7	$\rangle d_{\text{avg}} = x$

3023 This example begins to demonstrate why formalizing a hybrid physical problem is *not* a
 3024 simple translation task. Introducing the right auxiliary variables, absent in the specification, is
 3025 often necessary for a clean, tractable formalization. All the pieces of the model must fit together
 3026 precisely, e.g., even making a weak inequality strict can transform a correct model into one
 3027 admitting no runs [150]. Correctness requires careful attention to the subtleties of the problem
 3028 specification, e.g., benchmark problem 10 (available in the online artifact [72]) has one variable
 3029 representing a *displacement* and another, representing a *distance*. The former is a signed number
 3030 while the latter must be modeled as non-negative. Models must prune mathematically degenerate
 3031 cases using background knowledge of the physical world, e.g., by eliminating imaginary roots
 3032 of polynomials computing real quantities. Additionally, there are many ways to model the same
 3033 problem, and the right choice of coordinate system or frame of motion can be crucial to writing
 3034 a model for which symbolic reasoning is tractable.

3035 Given the complexity of the task, it is unclear whether LLMs can succeed at all, making a
 3036 careful evaluation necessary. We create a benchmark suite drawing from textbooks and problem
 3037 sets [66, 89, 95], choosing tricky problems with varying structures to systematically identify the
 3038 strengths and weaknesses of LLMs on autoformalization.

3039 8.2 Methodology

3040 The autoformalization system, shown by Fig. 8.1, takes as input the natural language specifi-
 3041 cation of the physical problem to model, and produces either a correct dGL formalization of a
 3042 failure (no formalization found that passed the checker). It uses two types of LLM queries: one
 3043 is *Propose Formalization*, and the second one is *Revise Formalization*. The artifact [72] shows
 3044 all the prompt templates used.

The autoformalization system first queries the LLM to propose the formalization using the *Propose Formalization* query, providing up to four solved examples as part of the prompt. It provides an LLM with the natural language description of the problem and asks it to produce a dGL formula, including in the prompt common dGL syntax mistakes and their remedies. An example of a problem description the prompt would use based on the running example is

“An object A starts from rest and travels a distance d . For the first half of the distance, it travels with velocity v_0 . For the second half of the distance, it spends half the time traveling with velocity v_1 and half the time traveling with velocity v_2 . Another object B tracks A. Its motion has the same duration, starting, and ending points as A, but it travels with uniform velocity v_{avg} . Let v_{avg} remain a free variable.”

The prompt also includes upto four solved examples. The returned formal model is passed to the KeYmaera X dGL parser for a syntax check.

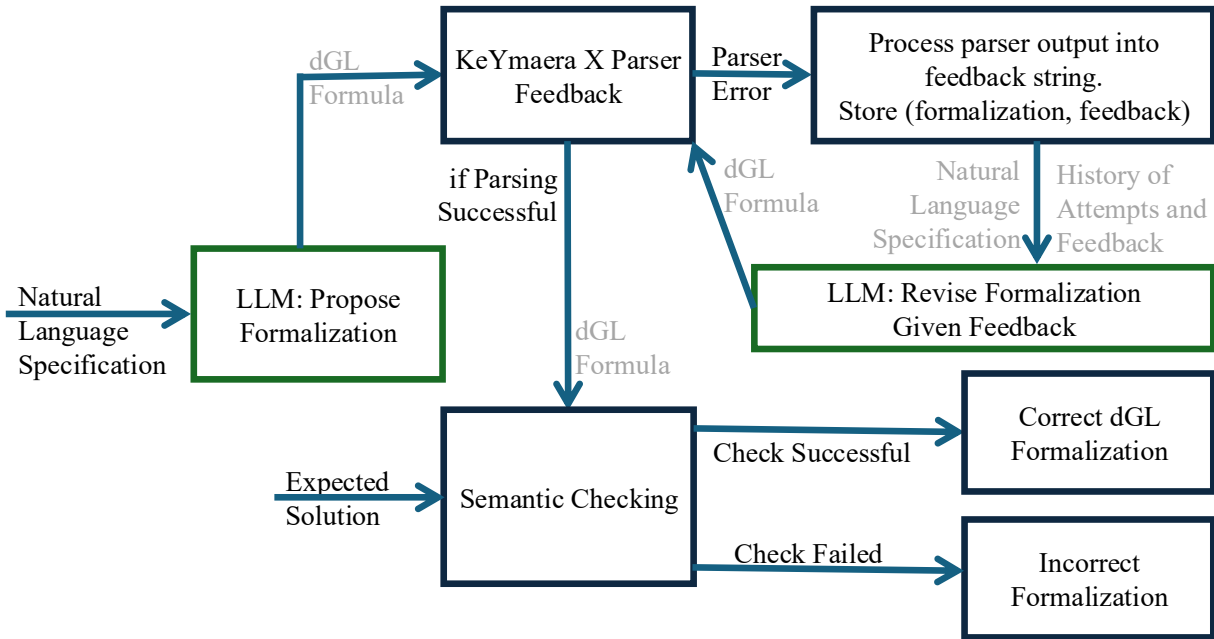


Figure 8.1: Autoformalization pipeline

If the syntax check fails, then the parser output is processed into short feedback strings. An example of a parser feedback string is

“The input formula contains an unsupported Unicode character (possibly {bad_char}). Use only ASCII characters.”

Where `bad_char` is a parameter extracted from parser output. The incorrect formalization along with the feedback is then appended to a list of previous failed formalization attempts.

The LLM is prompted to repair the formalization given past attempts and feedback using the *Revise Formalization* query. The prompt first provides up to two examples of repair. Then it provides the natural language question, and finally past incorrect proposals along with the feedback indicating why these proposals were wrong. The LLM replies with a new, repaired dGL game, which is again passed to the parser for checking. This syntactic repair loop is allowed

to run at most three times, after which autoformalization fails. A full implementation, along with examples, is available in the online artifact [72].

If the syntax check succeeds, then the dGL formula is passed onto the semantic evaluation system. This system takes as input the expected symbolic solution to the problem along with the proposed dGL formula. For our running example, the expected solution consists of the formula $v_{\text{avg}} = \frac{2v_0(v_1+v_2)}{2v_0+v_1+v_2}$, to test against for equivalence, and minimum expected number of variables that are written to, 2. If the checker finds that the dGL formula is equivalent to the expected solution under the initial assumptions of the formula and has enough variable writes, then it declares that the formalization is acceptable. The pipeline is implemented using Delphyne [86] and KeYmaera X [51], which in turn uses Z3 [39] and Mathematica.

8.3 Related Work

Given a natural language description, formal physics models are generally written manually. Prior work has studied the use of structured natural language to perform formalization by rule-based transformations [28, 108], but this limits flexibility in input specification. Recent progress in NLP (LLMs) raises the prospect of automating formalization directly from unstructured natural language.

The use of LLMs for autoformalizing mathematics has received significant research attention [88, 163], including for purely continuous systems with partial differential equation [42, 65]. Recent work also uses LLMs to synthesize contracts for cyber-physical systems [30, 106, 109]. However, we focus on LLM based autoformalization for *physics models* of hybrid systems, with discrete transitions and continuous dynamics faithfully represented, which to the best of our knowledge has not yet been studied. Compared to general mathematics and LTL specifications, we expect this domain to be more data-scarce, with few examples of differential dynamic logic system formalizations and even fewer examples of differential game logic games available on the Internet. Additionally, we expect CPS model formalization to have a more operational flavor, modeling physical realities required to make a system work rather than abstract concepts that can be simplified for theoretical convenience (mathematics) or focused on only output behavior (LTL contracts). This difference makes it worth investigating the hybrid systems modeling domain.

8.4 Evaluation

We propose 20 benchmarks derived from challenging problems in undergraduate physics textbooks and problem sets [66, 89, 95]. The online artifact [72] lists them. Phrasing is changed to focus on modeling the physical situation described in the problem (rather than on solving for a variable) and the numbers are changed to induce rational answers to avoid failures of the automated check over numerical imprecision. Each benchmark is accompanied by an expected symbolic solution that the model should be equivalent to and the minimum expected number of variables that must be written to. Four additional problems, separate from the benchmarks, serve as examples for few-shot prompting.

We run the LLM autoformalization system using OpenAI GPT 4o, GPT 4.1 (best available

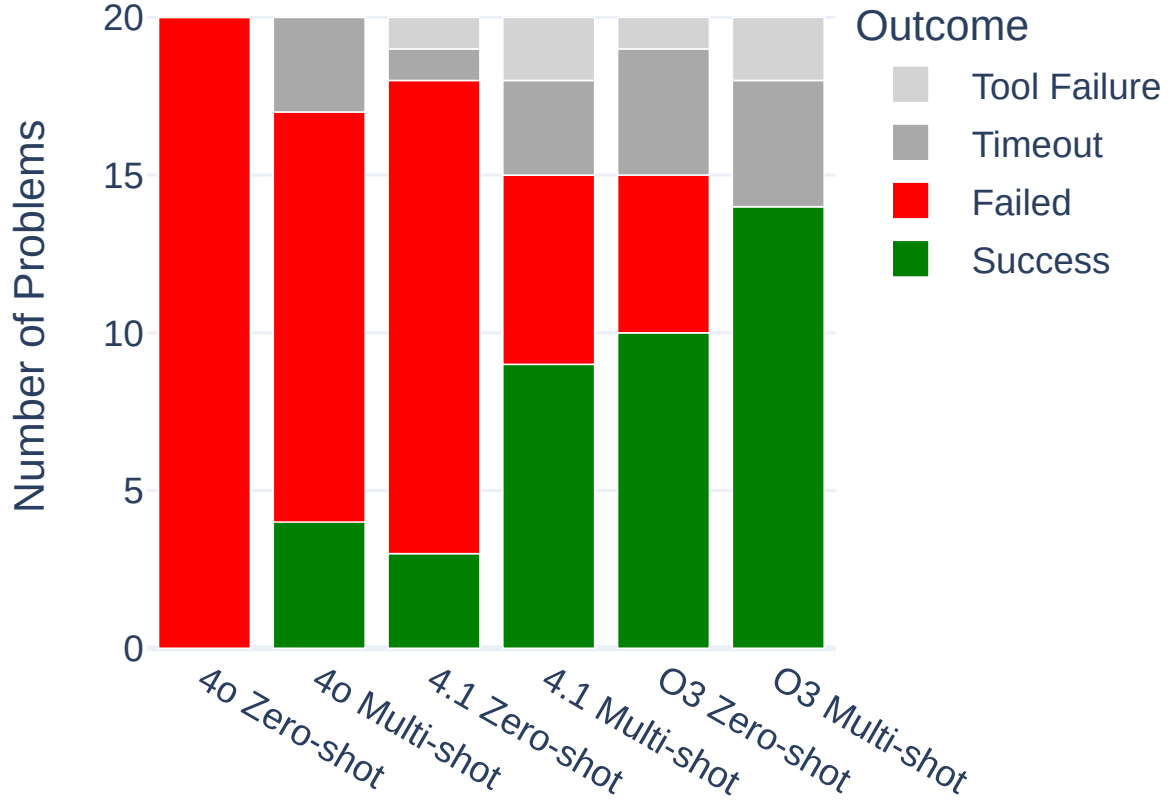


Figure 8.2: Outcomes of autoformalization by GPT 4o, GPT-4.1 and o3, with zero-shot and multi-shot prompting.

OpenAI non-reasoning model) and OpenAI o3 (best available OpenAI reasoning model) on the benchmark set, getting successively improving results. Outcomes are classified into 4 buckets:

1. *Success*: Correct, checker-certified formalization.
2. *Failed*: autoformalization failed either the syntax or semantic check.
3. *Timeout*: The solver timed out after 3 minutes of symbolic execution. Symbolic execution uses real quantifier elimination, which is doubly exponential [38], and can result in timeouts.
4. *Tool Failure*: The solver encountered an expression that it could not automatically symbolically execute (e.g., ODE with a square roots on the right hand side).

Fig. 8.2 shows the outcome. The autoformalization system is sampled five times independently with temperature 1, and the best outcome is reported (where tool failure and solver timeout are considered better than failure). For each sample, the LLM is given three chances to repair syntax if the KeYmaera X parser reported an error. We evaluate two different prompts, one using few-shot prompting considering four examples (*multi-shot*) and another one without examples (*zero-shot*).

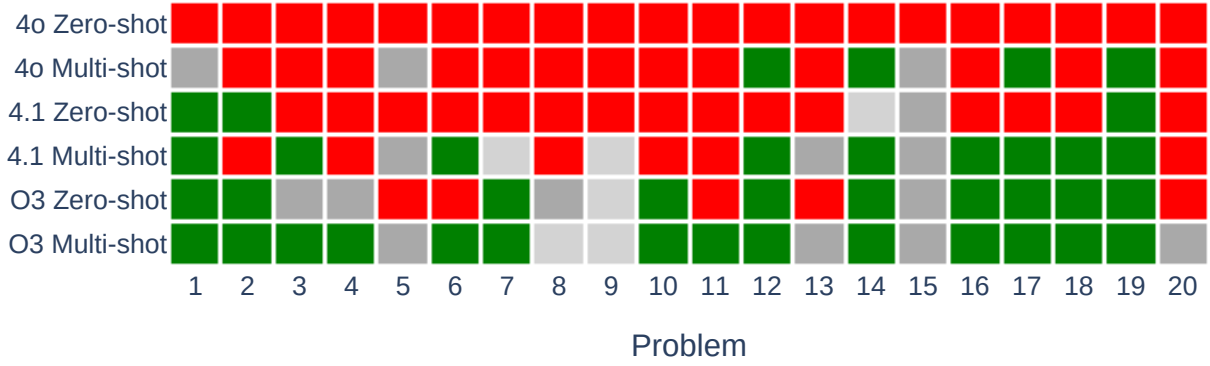


Figure 8.3: Problem outcome by model.

GPT 4o without few-shot prompting fails on every benchmark. Few-shot prompting significantly improves performance for all models. With few-shot prompting o3 has a success rate of 70%, which is higher than GPT 4.1’s 45% and GPT 4o’s 20%. Our results support that the improvements and the introduction of reasoning LLM models significantly increase their success rate on mathematical tasks, enabling strong performance in the autoformalization of hybrid games.

8.5 Discussion

Fig. 8.3 shows the outcome of each problem by model. We can observe that models found the same problems hard. There are six problems that no model solves. For these, the best performing model produces a formalization that could potentially be correct but is too complex for the checker to verify. The problems can be classified into three groups that explain where their complexity arises from.

1. Optimization Problems: Problem 5 and 20.
2. Non-polynomial continuous dynamics: Problem 8 and 9.
3. Complicated dynamics: Problem 13 and 15.

The first category, for example, includes problem 5, which is about identifying the launch angle of a boat that minimizes drift, modeled by adversarial Angel trying to pick a better angle than Demon, who must set the optimal angle to win. The interplay of the two players leads to combinatorial branching over possible sequence of events, while continuous dynamics occur under constraints preventing control blocking. These constraints make symbolic reasoning about the differential equations particularly computationally expensive.

An example of the second category is problem 8, which is about drones arranged in an equilateral triangle that cyclically pursue each other with constant radial velocity, collapsing inwards in increasingly rapid revolutions, till they crash. Their dynamics, often modeled with a non-polynomial square root expression on the right side of an ODE, cannot be handled automatically by the tool. Problem 13 is an example from the third category. It has three phases of dynamics, including one where acceleration has a continuous dependence on time. Such a differential

equation is more expensive to symbolically evaluate, making a timeout the likely outcome. 3148

Sometimes a problem fundamentally requires capabilities that a solver lacks, and the solver 3149
must be extended. Other times a difficult but crucial part of formalizing problems is choosing 3150
the right abstractions and representations that let proofs succeed. For the six failing problems, 3151
the solution likely lies in a combination of these methods. The solver must be extended to bet- 3152
ter automatically handle, for instance, non-polynomial dynamics by using continuous invariant 3153
generation [143]. Complementing this, the LLM should be guided to use tricks like polar co- 3154
ordinates and auxiliary variables to rewrite dynamics in friendlier forms, and to also provide 3155
checking hints such as continuous invariants. 3156

Unsolved benchmarks provide a roadmap for improvement in both these directions. Addi- 3157
tionally, the rapid improvement in success rate displayed by better LLM models suggests that 3158
autoformalization of hybrid games will continue to improve with advancements in LLMs. 3159

8.6 Conclusion 3160

This work experimentally studies the ability of LLMs to formally model physical behavior given 3161
a natural language specification. The results (70% success rate) are encouraging, and suggest that 3162
CPS model autoformalization using LLMs could be an impactful direction for future research. 3163

Beyond the broader adoption of formal methods in offline controller and control envelope de- 3164
sign, physics autoformalization enables a new class of applications, where autonomous systems 3165
autoformalize unfamiliar situations they face in the open world and derive formally justified con- 3166
trol decisions to respond to them. For such applications, where the *physical environment* must 3167
be formalized at runtime, the input to autoformalization would be vision and sensor data rather 3168
than a natural language description. LLM autoformalization from such multimodal input data 3169
provides an interesting future research direction. 3170

With this chapter, we explore the possibility of extending the synthesis pipeline all the way 3171
back to an informal natural language description of the control problem, asking whether the for- 3172
mal game specification itself can be generated automatically before the envelope is synthesized 3173
and verified. An open challenge remains recovering soundness guarantees for this step. In the 3174
course of this exploration, interesting future research directions are revealed, which we discuss 3175
in Chapter 9. 3176

Chapter 9

3177

Conclusion

3178

Control envelopes are sets of control solutions. Verified control envelopes provide a path to verifiably safe control. When developing a formally verified controller (e.g., a train control system) one generally has to consider primary, mission-critical requirements (e.g., safety) along with secondary or evolving requirements (e.g., smoothness of the ride). A possible approach to synthesizing a controller that is formally correct with respect to the primary requirements is: (1) first identify a maximal *set* of controllers formally verified to satisfy the primary requirement, i.e., a *verified control envelope* for the requirement, and then (2) optimize within this set for the controller choice that best meets the secondary requirements (e.g., using machine learning [49, 132, 153]).

This thesis made symbolic verified control envelope synthesis possible for richly expressive control system models, and developed methods to increase scalability. Given the physically possible control possibilities and environment behavior, its techniques computed a set of control solutions that satisfied the control contract. It introduced the first approach applicable at the level of imperative program-like specifications¹, enabling synthesis for richly expressive control system models.

Train Control As a concrete example of control envelopes for real-world applications, we considered a train controller that must stay within its *motion authority* (stop before some stopping points) to avoid collisions with other trains (Chapter 3 [68]). The controller can freely choose some engine acceleration or dynamic braking, but is not able to revise its decision for some bounded time latency (typically on the order of seconds) after its decision. We created a control envelope that answered: given the current state, what values of acceleration/braking are acceptable so that the train has a way to always remain safe? We had to account for all the competing influences that govern train motion. For example, uphill slope decreases velocity, which decreases resistance, which permits a more rapid increase in velocity, slope and curve resistance effects. These complex interactions illustrate why safe and efficient control envelope synthesis is hard, and why it requires precise formal reasoning.

¹There is envelope synthesis work in the automaton community [13], though the input is at a different level of abstraction (automata, not program-like logic). It is interesting future work to investigate what ideas can transfer between the very different formalisms in the future.

Our envelope was *symbolic*, supporting unbounded state space using variables that can represent any real number, and *parametric* in these variables (e.g., maximum train braking magnitude $b_{\max}$). The control envelope captured an entire *set* of control solutions, so in the same state, it allowed multiple different choices for acceleration or braking, and each choice led to a different correct controller.

The thesis then turned to making control envelope synthesis possible, which involved answering questions such as the following. Given a control system as input², how can we *represent* its control envelope, a potentially infinite control solution set, in a way that is *compositional* (synthesized/verified by the synthesis/verification of its parts)? How do we *define mathematically* what it means for this control envelope to be *correct*? How can we perform *automatic synthesis*? Are there ways to make the synthesis *scalable*? The thesis answered these questions as summarized below.

Subvalue Maps The input to the control envelope synthesis problem is a *two-player game* between the controller and the environment [69, 107] (Chapter 6 [73]). This is because, intuitively, a game defines the shape of a control problem without specifying control strategies (which must be synthesized). The controller has some control moves, the environment adversarially resolves nondeterminism, and the controller’s winning condition consists of meeting some critical control requirements. Synthesizing a control envelope corresponds to identifying the nondeterministic policy indicating what plays the controller can make at each step so that it can still win the game. Input games are represented in an imperative program-like language, *differential game logic* [120], that decomposes into subgames recursively along familiar syntactic structures: loops, sequential composition, and branching. As a symbolic, compositional representation of a control envelope that we can reason about recursively along the syntactic structure of the game, the thesis proposes the *inductive subvalue map* (Chapter 6 [73]), a map from subgames to formulas that hold exactly when it is possible to play the corresponding subgame and win. This map induces a nondeterministic policy for the controller: to check if playing into a subgame g is permitted in a given state, we only need to check that the subvalue associated with g holds in that state. We define a *correct* control envelope using recursive logical constraints on the subvalue formulas that ensure playing per *any* controller in the envelope guarantees that the controller player never gets stuck with no acceptable next action and can *always continue towards winning the game within the envelope*. We identify a backward precondition calculus that derives the subvalue map for a given differential game logic game and winning condition.

Scaling with LLMs The precondition calculus gives us a general, formally justified symbolic control envelope synthesis framework. But scalability remains an issue because of dependence on difficult subroutines like quantifier elimination (doubly exponential) [38], and invariant generation (undecidable, infinite search space). We attack the scalability bottlenecks in three different ways (Chapter 7). (1) Invariant generation is performed by LLMs, leveraging their physics problem-solving abilities, coupled with backtracking search. (2) Quantifier elimination and simplification are also replaced with LLM queries, with soundness maintained by retrospective for-

²We considered a very general class of input control problems: those that can be expressed as hybrid games, in differential game logic [120].

mal verification. (3) Verification can itself become a bottleneck, for which our solution is to have the LLM use an interactive theorem prover to break down the proof into tractable subgoals (Chapter 4).

Control Envelope Synthesis via Angelic Refinement (CESAR). We identify another useful technique to synthesize for complicated games: *refinement*, or rewriting the game so that it is harder for the controller (maintaining soundness) but easier to symbolically reason about. For controllers that poll repeatedly in a single loop of some maximum time latency, we in fact identify systematic refinements via the CESAR algorithm (Chapter 5 [69]). We find that for controllers with *action permanence* (a hybrid analog of idempotence) and dynamics with polynomial solutions, CESAR is decidable.

9.1 Future Work

Control envelopes provide a practical path towards verifiably safe autonomy. They make it possible to safeguard controllers, even those based on legacy code or machine learning, by runtime monitoring against the verified envelope. Synthesis removes a large barrier to adoption: designing envelopes in the first place. However, some barriers remain.

Formalizing for the Complexities of Reality. Formalization work with a similar flavor to this thesis will let control envelopes model increasingly complex real-world situations (partial observability, multiplayer coordination) with increasing scalability (efficiency improvements to the synthesis procedure for various problem classes, harnessing advances in AI via intersymbolic algorithms). How should we define probabilistic control envelopes that assure safety with some confidence interval given a probabilistic model of the environmental uncertainties? Extending the precondition calculus to follow the rules of stochastic reasoning [118] and statistics provides a path. How do we model control envelope synthesis for multi-agent, distributed systems where each agent may be open to collaborating? Statically agreed-upon contracts can be synthesized by treating all the collaborative control agents as a *single* controller. After carefully formalizing the answer to these questions, how do we perform scalable synthesis for the new kinds of envelopes? Building on Chapter 7, an answer lies in careful algorithm design that combines the use of AI and verification (intersymbolic AI [124]).

Control Envelope Synthesis for the Masses. Reliable autoformalization tools can put envelopes within reach of audiences without formal methods expertise. Before synthesizing/verifying an envelope, one still needs a *formal model* of their system (the environment, control possibilities, and the desired guarantees). Today, writing such models is hard, a bottleneck for broader adoption of control envelopes, and more generally, formal methods. Meanwhile, potential users often already have the relevant information in a different form, e.g., design documents, log data, or simulation code. LLMs now show promise at writing formal models from informal natural language descriptions (Chapter 8 [71]). If we can use LLMs to *automatically* create formal models, with iterative feedback, users outside formal methods could access verified control envelopes

and use them for downstream applications. Two flavors of challenges arise: for future work making autoformalization *reliable*, e.g., with formal consistency checks and comparison to logs, and making it *usable*, e.g., with visual representations that help users spot errors.

Applications. Initial case studies demonstrate the benefits and costs of using control envelopes, and pave a path for future adopters. The verified *train* control envelope [68] provides a concrete example of a control envelope with the complexities of real-world dynamics. Similar projects would make control envelopes useful for other applications. Another future application is, when *fine tuning foundation models* for robot control using reinforcement learning (RL), feedback on the model’s control decisions should reason over unbounded time to catch a mistake now even if its consequences won’t be seen till far in the future. Control envelopes batch-compute the solutions for the entire control problem over infinite time once and for all, so that later checking a single action for RL feedback is efficient.

Finally, the phase of retrieving a controller from the synthesized symbolic envelope, by optimizing for secondary requirements, is an interesting future research direction. A technical path lies in extending work that quantitatively estimates how well states adhere to an end objective in a dL system [98] to also handle games. The answer to this question, together with the work of this thesis, will deliver controllers that are formally guaranteed to be correct while being optimized and high-performance, even for complex systems.

Appendix A

3299

CESAR (Appendix)

3300

A.1 Reduce Operation

3301

We define `reduce` after first introducing two helper functions that it requires. Function $\triangleright(a, b)$ attempts to simplify FOL_ℝ formula a to $\mathcal{P}_\mathbb{R}$ assuming that b holds. The second helper function, `odereduce` (Def. A.1), isolates the action of `reduce` on differential equations. Since it is solely continuous programs that could lead to `reduce` failing to produce an exact solution, the `exact` bit of `reduce` depends on `odereduce`. Fig. A.1 shows the definition of `reduce`, eliding the `exact` bit, which is simply true if all of the `odereduce` calls that `reduce` makes are exact, and false otherwise.

Definition A.1 (ODE reduction). Let $\langle\!\langle\alpha\rangle\!\rangle \in \{\langle\alpha\rangle, [\alpha]\}$, $\bowtie \in \{\leftrightarrow, \rightarrow\}$, and A , Q , and P be formulas in quantifier-free real arithmetic. An *ODE reduction oracle* `odereduce` is a function such that

$$A \rightarrow (\text{odereduce}(\langle\!\langle\{x' = f(x) \& Q\}\!\rangle P, A) \bowtie \langle\!\langle\{x' = f(x) \& Q\}\!\rangle P)$$

is valid. When $\bowtie$ is $\leftrightarrow$ then `odereduce` is exact, otherwise it is approximating.

For solvable ODEs, `odereduce` is implemented as an exact oracle in Eq. (A.1).

$$\begin{aligned} \text{odereduce}(\langle\!\langle\{x'_1 = \theta_1, \dots, x'_n = \theta_n \& Q\}\!\rangle P, A) = \\ \forall t (\text{subst}(A, t) \wedge t \geq 0 \wedge \forall 0 \leq s \leq t \text{subst}(Q, s)) \rightarrow \text{subst}(P, t) \\ \text{odereduce}(\langle\!\langle\{x'_1 = \theta_1, \dots, x'_n = \theta_n \& Q\}\!\rangle P, A) = \\ \exists t (\text{subst}(A, t) \wedge t \geq 0 \wedge \forall 0 \leq s \leq t \text{subst}(Q, s)) \wedge \text{subst}(P, t) \end{aligned} \quad (\text{A.1})$$

where $\text{subst}(f, t) = f\{\int_{x_i}^t \theta_i \cdot dt / x_i\}$, t fresh, $\langle\!\langle\alpha\rangle\!\rangle \in \{\langle\alpha\rangle, [\alpha]\}$

In the general case, Pegasus [143], a tool that automatically generates *ODE invariants*, can often produce a formula satisfying the specification of `odereduce`. This may come at the cost of lost precision, possibly requiring `reduce` to set `exact` to false.

Theorem A.1 (Correctness of `reduce`). For any loop-free dGL formula F and assumptions $A \in \mathcal{P}_\mathbb{R}$ the function `odereduce` either sets `exact=true` and the formula $A \rightarrow (\text{reduce}(F, A) \leftrightarrow F)$ is valid, or else it sets `exact=false` and the formula $A \rightarrow (\text{reduce}(F, A) \rightarrow F)$ is valid.

$$\begin{aligned}
& \text{Let } \langle\!\langle\alpha\rangle\!\rangle \in \{\langle\alpha\rangle, [\alpha]\} \\
& \text{reduce}(\langle\!\langle\alpha; \beta\rangle\!\rangle P, A) = \text{reduce}(\langle\!\langle\alpha\rangle\!\rangle (\text{reduce}(\langle\!\langle\beta\rangle\!\rangle P, \top)), A) \\
& \text{reduce}([\alpha \cup \beta] P, A) = \text{reduce}([\alpha] P, A) \wedge \text{reduce}([\beta] P, A) \\
& \text{reduce}(\langle\alpha \cup \beta\rangle P, A) = \text{reduce}(\langle\alpha\rangle P, A) \vee \text{reduce}(\langle\beta\rangle P, A) \\
& \text{reduce}(\langle\!\langle x := e \rangle\!\rangle P, A) = \triangleright(P\{e/x\}, A) \\
& \text{reduce}([\![?f]\!] P, A) = \triangleright(f \rightarrow P, A) \quad \text{reduce}(\langle\!\langle ?f \rangle\!\rangle P, A) = \triangleright(f \wedge P, A) \\
& \text{reduce}([\![x' = f(x) \& Q]\!] P, A) = \triangleright(\text{odereduce}([x' = f(x) \& Q] P, A), \top) \\
& \text{reduce}(\langle\!\langle x' = f(x) \& Q \rangle\!\rangle P, A) = \triangleright(\text{odereduce}(\langle\!\langle x' = f(x) \& Q \rangle\!\rangle P, A), \top) \\
& \text{reduce}([\alpha^d] P) = \text{reduce}(\langle\alpha\rangle P) \quad \text{reduce}(\langle\alpha^d\rangle P) = \text{reduce}([\alpha] P) \\
& \text{reduce}(P \wedge Q, A) = \triangleright(P \wedge Q, A) \quad \text{reduce}(P \vee Q, A) = \triangleright(P \vee Q, A) \\
& \text{reduce}(P \rightarrow Q, A) = \triangleright(P \rightarrow Q, A) \quad \text{reduce}(\neg P, A) = \triangleright(\neg P, A)
\end{aligned}$$

Figure A.1: Definition of **reduce** (exact elided). Notation $P\{e/x\}$ indicates P with unbound occurrences of x replaced by expression e . **odereduce** isolates the effect of **reduce** on ODEs. $\triangleright$ simplifies and quantifier eliminates P assuming A .

A.2 Parachute: reduce of Non-solvable Dynamics

The parachute benchmark (Appendix A.3.3) has non-solvable dynamics whose exact solution involves hyperbolic functions. The differential equations are $\{x' = -v, v' = -rv^2 + g\}$ and admit the solution:

$$\left\{ v(t) \rightarrow \frac{\sqrt{g} \tanh(\sqrt{g}\sqrt{r}t + c_1\sqrt{g}\sqrt{r})}{\sqrt{r}}, x(t) \rightarrow c_2 - \frac{\log(\cosh(\sqrt{g}\sqrt{r}(t + c_1)))}{r} \right\}.$$

This solution does not belong to the decidable fragment of arithmetic, and QE is not guaranteed to terminate. Reduce is able to still return preconditions over this ODE by using differential invariants generated by Pegasus.

The first call to reduce is for program $\{[x' = -v, v' = -rv^2 + g, t' = 1 \& v > 0 \wedge t \leq T](x < 0 \vee v < m)\}$. A call to Pegasus identifies the first degree Darboux polynomials of this ODE to be $\{\sqrt{g}, \sqrt{g} - \sqrt{r}v, \sqrt{g} + \sqrt{r}v\}$. As a heuristic to retain only the most useful polynomials, we filter out terms that are purely constant. The Darboux rule of **dl** tells us that the following formulas are now invariants of the differential equation system: $\{\sqrt{g} - \sqrt{r}v \geq 0, \sqrt{g} + \sqrt{r}v \geq 0\}$.

When do these invariants imply the desired post condition, $x < 0 \vee v < m$? Mathematically, the answer is the formula

$$\sqrt{g} - \sqrt{r}v \geq 0 \wedge \sqrt{g} + \sqrt{r}v \geq 0 \rightarrow x < 0 \vee v < m.$$

To find an invariant that implies this formula, we eliminate variables x and v using quantifier elimination. The resulting expression is $m > 0 \wedge g \geq 0 \wedge g < m^2r$. Thus, **reduce** returns $\sqrt{g} - \sqrt{r}v \geq 0 \wedge \sqrt{g} + \sqrt{r}v \geq 0 \wedge m > 0 \wedge g \geq 0 \wedge g < m^2r$ as a precondition. This expression is invariant under the problem's dynamics and implies the postcondition $x < 0 \vee v < m$.

A.3 Benchmarks

3342

This section lists all the benchmarks proposed. It also shows the solution of each benchmark, annotated with the meaning of the solution expressions. The algebraic formulas presented are synthesized by CESAR automatically. The annotations are added manually for the convenience of the reader. Table A.1 provides a one-sentence summary for each benchmark problem.

3343
3344
3345
3346

Table A.1: Benchmark listing.

Benchmark	Description
ETCS Train	Kernel of the European Train Control System case study [126].
Sled	Swerve to avoid a wall.
Intersection	Car must either cross an intersection before the light turns red, or stop before the intersection. It may never stop at the intersection.
Curvebot	A Dubin’s car must avoid an obstacle.
Parachute	Use the irreversible choice to open a parachute with some air resistance to stay at a safe velocity. Drawn form [52].
Corridor	Navigate a corridor with a side passage and dead end.
Power Station	Choose between producing and distributing power while dealing with resistance loss and trying to meet a quota.
Coolant	A coolant system in an energy plant must maintain sufficient heat absorption while meeting coolant discharge limits.

A.3.1 ETCS Train

3347

The European Train Control System has been systematically but manually modeled and proved safe in the literature [126]. We consider the central model of this study and apply CESAR to automate design. It is listed as the running example Model 7.

3348
3349
3350

A.3.2 Sled

3351

Model 12 Sled must swerve to avoid an obstacle.

$$\begin{array}{ll}
 \text{assum} & | \quad 1 \quad \boxed{I} \wedge T_x > 0 \wedge T_y > 0 \wedge V > 0 \wedge T > 0 \rightarrow \{ \\
 \text{ctrl} & | \quad 2 \quad \quad \quad (? \boxed{G_1}; v_x := -V \cup ? \boxed{G_2}; v_x := V); \\
 \text{plant} & | \quad 3 \quad \quad \quad (t := 0; \{x' = v_x, y' = V, t' = 1 \ \& \ t \leq T\}) \\
 \text{safe} & | \quad 4 \quad \quad \quad \}^* \} (y < T_y \vee x < -T_x \vee T_x < x)
 \end{array}$$

Where I , G_1 and G_2 are to be synthesized.

This benchmark displays CESAR’s ability to reason about state-dependent fallbacks. The slope of a hill is pushing a sled forward along the y axis with constant speed v_y . However, there

3352
3353

is a wall blocking the way. It starts at y axis position T_y and extends along the x axis from $-T_x$ to T_x . The sled must swerve to avoid the obstacle. It can either go left (with velocity $-V$) or right (with velocity V). Which action is best depends on where the sled already is. Swerving left is a safe strategy when the sled can pass from the $-x$ side, mathematically, $T_y > T_x + x + y$. Likewise, swerving right is a safe strategy when $T_y + x > T_x + y$. Neither action alone gives the optimal invariant, but CESAR's I^0 characterization correctly captures the disjunction to find it. In synthesizing the guards for the actions, CESAR also identifies when it is still safe to switch strategies.

CESAR finds the solution below. The algebraic formulas presented are synthesized by CESAR automatically. The annotations describing their meaning are added manually for the convenience of the reader.

$$I \equiv y < T_y \wedge \overbrace{(T_y > T_x + x + y \vee T_y + x > T_x + y)}^{\text{can swerve left} \vee \text{can swerve right}} \vee \overbrace{T_x < x \vee T_x + x < 0}^{\text{already safe}}$$

The guard for going left is

$$\begin{aligned} G_1 \equiv & \overbrace{T_x + 2TV + y < T_y + x \wedge T_x + T_y > x + y \wedge x \geq 0}^{\text{despite going left one time period, can still pass from right}} \\ & \vee \overbrace{T_x + x + y < T_y \wedge \neg T_x + x = 0}^{\text{can pass from left}} \vee \overbrace{0 = T_x + x}^{\text{at the left boundary}} \\ & \vee \overbrace{T_x + TV < x \wedge T_x + T_y \leq x + y}^{\text{far enough right to stay right after this cycle}} \vee \overbrace{T_x + x < 0}^{\text{already left}} \end{aligned}$$

The guard for going right is

$$\begin{aligned} G_2 \equiv & \overbrace{x < 0 \wedge T_x + 2TV + x + y < T_y \wedge T_x + T_y + x > y}^{\text{despite going right one time period, can still pass from left}} \vee \overbrace{T_x < x}^{\text{already right}} \\ & \vee \overbrace{T_x + y < T_y + x \wedge T_x > x}^{\text{can pass from right}} \vee \overbrace{T_x + TV + x < 0 \wedge T_x + T_y + x \leq y}^{\text{far enough left to stay right after this cycle}} \\ & \vee \overbrace{T_y > y \wedge T_x \leq x}^{\text{right at the right boundary but there is time still to swerve}} \end{aligned}$$

There are some redundancies in these expressions, but they are correct, comprehensible, and can be simplified further.

A.3.3 Parachute

The parachute benchmark presents the challenge of dynamics with a solution that departs from the decidable fragment of real arithmetic. A person is free falling. At most once, they are allowed to take the action of opening a parachute. Once they do, their air resistance changes to p . The objective is to land at a speed no greater than m . The benchmark is inspired by one that appears in the literature (running example in [52]). CESAR uses Pegasus's Darboux Polynomial generation to solve the problem.

Model 13 Parachute benchmark from [52].

```

assum | 1  ( $T > 0 \wedge p > 0 \wedge g > 0 \wedge r > 0 \wedge m > 0 \wedge v > 0 \wedge \boxed{I}$ )  $\rightarrow \{$ 
ctrl   | 2  ( $(\boxed{G_1}; \text{skip}) \cup (\boxed{G_2}; r := p)$ );
plant  | 3  ( $t := 0; \{x' = -v, v' = -r \cdot v^2 + g, t' = 1 \ \& \ v > 0 \wedge t \leq T\}$ )
safe   | 4   $\}^*(x \geq 0 \rightarrow v < m)$ 

```

Where I , G_1 and G_2 are to be synthesized.

CESAR finds the solution below. The algebraic formulas presented are synthesized by CE- 3379
SAR automatically. The annotations describing their meaning are added manually for the con- 3380
venience of the reader. 3381

$$\begin{aligned}
I \equiv & \quad \text{either start below terminal velocity and terminal velocity without parachute is already safe} \\
& \quad \overbrace{g \geq rv^2 \wedge m > (gr^{-1})^{1/2}} \\
& \quad \text{or air terminal velocity with parachute is safe and start below terminal velocity} \\
\vee & \quad \overbrace{m > (gp^{-1})^{1/2} \wedge pv^2 \leq g}
\end{aligned}$$
3382

The guard for not opening the parachute is that already terminal velocity without the parachute 3383
is safe, and the person has not exceeded terminal velocity yet. 3384

$$G_1 \equiv m > (gr^{-1})^{1/2} \wedge rv^2 \leq g$$
3385

Likewise, the guard for opening the parachute is that terminal velocity with the parachute is safe. 3386

$$G_2 \equiv m > (gp^{-1})^{1/2} \wedge pv^2 \leq g$$
3387

A.3.4 Intersection 3388

Model 14 Intersection benchmark.

```

assum | 1  ( $B > 0 \wedge T > 0 \wedge v \geq 0 \wedge \boxed{I}$ )  $\rightarrow \{$ 
ctrl   | 2  ( $(\boxed{G_1}; a := 0) \cup (\boxed{G_2}; a := -B)$ );
plant  | 3  ( $t := 0; \{x' = v, v' = a, \text{timeToRed}' = -1, t' = 1 \ \& \ t \leq T \wedge v \geq 0\}$ )
safe   | 4   $\}^*((\text{timeToRed} > 0 \wedge v \neq 0) \vee \neg(x = 0))$ 

```

Where I , G_1 and G_2 are to be synthesized.

The intersection benchmark is a simple example of a system with free choice and state depen- 3389
dent fallback. A car sees a yellow light, and must decide whether to coast past the intersection, 3390
or to stop before it. It may never stop at the intersection, which is located at $x = 0$. Whether it 3391
would be safe to stop or to coast depends on the car's current position and velocity. 3392

3393 CESAR finds the solution below. The algebraic formulas presented are synthesized by CE-
 3394 SAR automatically. The annotations describing their meaning are added manually for the con-
 3395 venience of the reader.

$$\begin{aligned}
 I \equiv & \overbrace{x > 0}^{\text{safe if already past the intersection}} \\
 & \vee \overbrace{x < 0 \wedge (v \leq 0 \vee v(\text{timeToRed} \cdot v + x) > 0)}^{\text{otherwise, before the intersection, safe if velocity is already 0 or could coast past intersection}} \\
 & \vee v > 0 \wedge (\text{timeToRed} > 0 \wedge (\\
 & \quad \overbrace{B < \frac{v}{\text{timeToRed}} \wedge B \cdot \text{timeToRed}^2 < 2(\text{timeToRed} \cdot v + x)}^{\text{signal flips before car stops. Even braking, car crosses intersection before signal flips}} \\
 & \quad \vee \overbrace{v^2 + 2 \cdot B \cdot x \neq 0 \wedge B \geq \frac{v}{\text{timeToRed}})}^{\text{stop somewhere that's not the intersection, the signal flips after the car stops}}
 \end{aligned}$$

3397 The guard for coasting has many repeated clauses, so we first explain them before presenting
 3398 the expression. Assuming v positive and $x \leq 0$, $C \equiv v^3 + 2Bv(Tv + x) < 0$ means that after one
 3399 time period of coasting, the car still stops before the intersection. $D_1 \equiv 0 = 3\text{timeToRed} + 2v^{-1}x$
 3400 means that the signal flips when the car is 2/3rds along the way of coasting to the intersection.
 3401 $D_2 \equiv v(3\text{timeToRed} \cdot v + 2x) > 0$ means that the signal flips after the car is 2/3rds along the way
 3402 of coasting to the intersection. $D_3 \equiv v(3\text{timeToRed} \cdot v + 2x) < 0$ means that the signal flips before
 3403 the car is 2/3rds along the way of coasting to the intersection. $E \equiv B = \text{timeToRed}^{-1}v$ means
 3404 that the signal flips exactly when the car halts if it starts braking now. $F \equiv 0 = \text{timeToRed} + v^{-1}x$
 3405 means that the signal flips exactly when the car reaches the intersection by coasting. $G \equiv$
 3406 $2B + v^2(\text{timeToRed} \cdot v + x)^{-1} = 0$ means that the car will be at the intersection when the signal
 3407 flips if it starts braking now. $H_1 \equiv v(\text{timeToRed} \cdot v + x) < 0$ means that if the car coasts, it will
 3408 be before the intersection when the signal flips. $H_2 \equiv v(\text{timeToRed} \cdot v + x) > 0$ means that if
 3409 the car coasts, it will be after the intersection when the signal flips.

$$\begin{aligned}
 G_1 \equiv & \overbrace{v = 0}^{\text{already still and safe}} \vee (v > 0 \wedge (\\
 & \quad \overbrace{x > 0}^{\text{already past intersection}} \vee x \leq 0 \wedge (\\
 & \quad \quad \overbrace{D_1 \wedge ((T < \text{timeToRed} \wedge E) \vee (C \wedge \neg E))}^{\text{can stop before intersection after cycle of coasting}} \\
 & \quad \vee \overbrace{C \wedge (F \vee \neg G \wedge (D_3 \vee D_2 \wedge H_1))}^{\text{can stop before intersection after cycle of coasting (but } D_1 \text{ doesn't have to hold)}} \\
 & \quad \vee \overbrace{(D_2 \wedge H_1 \vee D_3) \wedge T < \text{timeToRed} \wedge G}^{\text{after one cycle of coasting and then braking, car passes intersection}} \\
 & \quad \vee \overbrace{H_2}^{\text{can coast past intersection before signal switches}}
 \end{aligned}$$

3411 Likewise, the guard for braking has many repeated clauses. $P \equiv 2B + v^2/x \leq 0$ means that
 3412 the car won't stop before the intersection. $Q \equiv \text{timeToRed} > \frac{BT^2 + 2x}{2(BT - v)}$ means that even after one

braking cycle, the car can cross the intersection by coasting. $R \equiv x(v^2 + 2Bx) > 0$ means that the car will stop before the intersection. $S \equiv B \cdot \text{timeToRed} + (v^2 + 2Bx)^{1/2} > v$ means that the car will stop before the intersection before the signal turns red. $U \equiv 2(Tv + x) < BT^2$ means that the car will stop before time T. $V \equiv BT^2 = 2(Tv + x)$ means that if it were to brake, the car will come to a stop at time T. $W \equiv BT^2 < 2(Tv + x)$ means that if it were to brake, the car will come to a stop after time T. Because of structural similarities with G_1 , we do not provide a full annotation.

$$\begin{aligned}
G_2 \equiv & x < 0 \wedge (v(Tv + x) < 0 \wedge (P \wedge Q \vee R) \\
& \vee v(Tv + x) > 0 \wedge T < -2v^{-1}x \wedge (W \wedge S \vee P \wedge Q \wedge U \\
& \vee R \vee V \wedge \text{timeToRed} > T) \\
& \vee 0 = T + v^{-1}x \wedge (P \wedge Q \wedge U \vee R) \\
& \vee T \geq -2v^{-1}x \wedge (R \vee S \wedge x(v^2 + 2Bx) < 0)) \vee x \geq 0
\end{aligned}$$

A.3.5 Curvebot

Model 15 Curvebot benchmark.

```

assum | 1  (T > 0 ∧ [I]) → {
  ctrl | 2  (([G1]; om := 1) ∪ ([G2]; om := 0) ∪ ([G3]; om := -1));
  plant | 3  (t := 0; {x' = v, y' = w, v' = om · w, w' = -om · v, t' = 1 & t ≤ T})
  safe | 4  }*]¬(x = 0 ∧ y = 0)

```

Where I , G_1 , G_2 and G_3 are to be synthesized.

Curvebot models a Dubin's car that must avoid an obstacle at (0,0). The dynamics result in a solution that is not in the decidable fragment of arithmetic, so CESAR again uses Pegasus to find a controllable invariant.

Our implementation generates the optimal invariant, consisting of everywhere except the origin. The algebraic formulas presented are synthesized by CESAR automatically. The annotations describing their meaning are added manually for the convenience of the reader.

$$I \equiv \neg x = 0 \vee \neg y = 0$$

The guard for setting om to 1 is simply that the origin does not lie on the resulting circular path.

$$G_1 \equiv y = 0 \wedge -2w + x = 0 \vee \neg y(2wx + x^2 + y((-2)v + y)) = 0$$

The guard for setting om to 0 is that the origin does not lie in the straight line segment of length

3432 $T(v^2 + w^2)^{(1/2)}$ ahead.

$$\begin{aligned}
& \overbrace{G_2 \equiv v < 0 \wedge (\neg y = 0 \wedge (0 = w \vee 0 = x))}^{\text{going in the -x direction } \dots} \\
& \quad \overbrace{\vee x > 0 \wedge (\neg w = 0 \wedge \neg y = wx/v \vee v(Tv + x) < 0) \vee x < 0}^{\dots \text{ is safe}} \\
& \quad \overbrace{\vee v = 0 \wedge (\neg x = 0 \vee y > 0 \wedge (w \geq 0 \vee w(Tw + y) < 0))}^{\text{purely y motion } \dots} \\
& \quad \quad \overbrace{\vee y < 0 \wedge (w(Tw + y) < 0 \vee w \leq 0))}^{\dots \text{ is safe}} \\
& \quad \quad \overbrace{\vee v > 0 \wedge (\neg y = 0 \wedge (0 = w \vee 0 = x))}^{\text{going in the x direction } \dots} \\
& \quad \quad \quad \overbrace{\vee x > 0 \vee (\neg w = 0 \wedge \neg y = wx/v \vee v(Tv + x) < 0) \wedge x < 0}^{\dots \text{ is safe}}
\end{aligned}$$

3434 Like G_1 , the guard for setting om to -1 is that the origin does not lie on the resulting circular
3435 path.

$$3436 \quad G_3 \equiv y = 0 \wedge \neg 2w = x \vee \neg y(x^2 + y(2v + y)) = 2wxy$$

3437 This solution is almost optimal. It only misses the cases for G_1 and G_2 where despite the obstacle
3438 lying on the circular path, T is small enough that there is time to switch paths before collision.

3439 A.3.6 Corridor

3440 Corridor, shown in Model 8, is an example of a system requiring unrolling.

3441 A.3.7 Power Station

Model 16 Power Station benchmark.

$$\begin{array}{ll}
\text{assum} & \left| \begin{array}{l} 1 \quad (T = 1 \wedge i \geq 0 \wedge \text{chargeRate} = 7000000 \wedge J = 100 \\ 2 \quad \wedge V = 2000 \wedge R = 5 \wedge \text{quota} = 3000 \wedge [I]) \rightarrow [\{ \\ \\ \end{array} \right. \\
\text{ctrl} & \left| \begin{array}{l} 3 \quad (([G_1]; i := 0; \text{slope} := \text{chargeRate}) \\ 4 \quad \cup ([G_2]; i := J; \text{slope} := -i \cdot V)); \\ \\ \end{array} \right. \\
\text{plant} & \left| \begin{array}{l} 5 \quad (t := 0; \{\text{produced}' = i \cdot V - i^2 \cdot R, \\ 6 \quad \text{stored}' = \text{slope}, t' = 1, gt' = -1 \ \& \ t \leq T\}) \\ \\ \end{array} \right. \\
\text{safe} & \left| \begin{array}{l} 7 \quad \}^*](\text{stored} > 0 \wedge (gt > 0 \vee \text{produced} > \text{quota})) \end{array} \right.
\end{array}$$

Where I , G_1 and G_2 are to be synthesized.

3442 Power station is an example of a system that needs two steps of unrolling in order to reach the
3443 optimal invariant. A power station capable of producing 7000 kW can choose between charging
3444 (Line 3) and distributing out stored power at current 100 A and voltage of 2000 V (Line 4). Its

objective is to meet an energy quota of 3000 J, excluding loss due to resistance of 5Ω by the time that timer gt counts down to 0. The station must never reach a state where it has no stored power left. The system is modeled in Model 16. The zero-shot invariant corresponds to the case where the station has already met its quota and is now charging. Discharge is not included in the invariant because regardless of how high stored energy is, in the unbounded time of the zero-shot invariant, a station that has chosen to distribute power will eventually run out of it, thus violating the condition $stored > 0$. The one shot invariant catches the case where the station first chooses to discharge till it meets its quota, and then flips to charging for infinite time. The two shot invariant, which is finally optimal, catches the case where the station first chooses to charge till it has enough energy stored to meet its quota, then discharges, and finally switches back to charging mode for infinite time.

CESAR finds the solution below. The algebraic formulas presented are synthesized by CESAR automatically. The annotations describing their meaning are added manually for the convenience of the reader.

$$\begin{aligned}
 I \equiv & \text{produced} \leq 3000 \wedge \overbrace{(3000(-51 + 50gt) + \text{produced} > 0)}^{\text{enough time to charge} \dots} \\
 & \wedge \overbrace{70000000gt + 48\text{produced} + \text{stored} > 7344000 \wedge 437500gt + 3\text{produced} < 459000}^{\dots \text{ then distribute}} \\
 & \wedge \overbrace{150000gt + \text{produced} > 3000 \wedge 4\text{produced} + 3\text{stored} > 612000)}^{\text{enough stored power to distribute and meet quota}} \\
 & \wedge \overbrace{\text{stored} > 0 \wedge (437500gt + 3\text{produced} \geq 459000 \vee \text{produced} > 3000)}^{\text{without drawing stored energy, enough time to charge then distribute}}
 \end{aligned}$$

The guard for charging checks that the choice to charge for a cycle still leaves enough time for distributing the power.

$$\begin{aligned}
 G_1 \equiv & \overbrace{gt > 1 \wedge 3000(gt \cdot 50 - 51) + \text{produced} > 0}^{\text{enough time to charge then distribute}} \\
 & \vee \overbrace{gt \cdot 35 < 36 \wedge \text{produced} > 3000}^{\text{quota already met}}
 \end{aligned}$$

The guard for distributing basically checks that the choice to discharge for a cycle still leaves enough stored energy.

$$\begin{aligned}
 G_2 \equiv & \overbrace{\text{produced} \leq 3000 \wedge 3000(50gt - 51) + \text{produced} \leq 0}^{\text{must distribute to stay safe}} \\
 & \vee \overbrace{3000(50gt - 51) + \text{produced} > 0 \wedge \text{stored} > 200000 \wedge gt > 1}^{\text{enough stored energy to distribute for a cycle, can produce enough afterwards}} \\
 & \vee \overbrace{gt \leq 1 \wedge \text{stored} > 200000 \wedge \text{produced} > 3000}^{\text{enough stored energy for a cycle, quota satisfied}}
 \end{aligned}$$

A.3.8 Coolant

The coolant benchmark is an example of a system that requires unrolling. A coolant system draws water at either a rate of F or $0 \text{ m}^3/\text{s}$ from a reservoir. It runs the water through a heat exchanger,

Model 17 Coolant benchmark.

assum	1 $(T > 0 \wedge F > 0 \wedge \text{minAbsorbed} > 0 \wedge \text{maxDischarge} > 0$ 2 $\wedge \text{tempDiff} > 0 \wedge c > 0 \wedge \boxed{I}) \rightarrow \{$
ctrl	3 $((\boxed{G_1}; f := 0)$ 4 $\cup (\boxed{G_2}; f := F));$
plant	5 $(t := 0; \{ \text{absorbed}' = f \cdot c \cdot \text{tempDiff},$ 6 $\text{discharged}' = f, t' = 1, gt' = -1 \ \& \ t \leq T \}$
safe	7 $\}^*](\text{discharged} < \text{maxDischarge} \wedge (gt > 0 \vee \text{absorbed} \geq \text{minAbsorbed}))$

Where I , G_1 and G_2 are to be synthesized.

3469 drawing heat proportional to the specific heat of water c and the temperature difference between
 3470 the water and the system. It then discharges the water into a river at the rate that it draws water.
 3471 The cooling system must absorb at least minAbsorbed heat by the time that timer gt counts
 3472 down to 0. But it must also not discharge more that maxDischarged water into the river to
 3473 avoid environmental damage. The benchmark is modeled in Model 17.

3474 CESAR finds the solution below. The algebraic formulas presented are synthesized by CE-
 3475 SAR automatically. The annotations describing their meaning are added manually for the con-
 3476 venience of the reader.

$$\begin{aligned}
 & \text{already absorbed enough heat} \\
 I \equiv & \boxed{\text{absorbed} \geq \text{minAbsorbed} \wedge \text{discharged} < \text{maxDischarge}} \\
 & \text{drawing water one cycle will not exceed discharge limit} \\
 \vee & \boxed{gt > 0 \wedge (\text{discharged} + F \cdot T < \text{maxDischarge}} \\
 & \text{but drawing water till timer } gt \text{ runs out will} \\
 & \boxed{\wedge \text{maxDischarge} \leq \text{discharged} + F \cdot (gt + T)} \\
 & \text{drawing as much water as discharge limit allows} \dots \\
 & \boxed{\wedge c > (\text{absorbed} - \text{minAbsorbed}) \cdot \text{tempDiff}} \\
 & \dots \text{ absorbs enough heat} \\
 & \boxed{\cdot (\text{discharged} - \text{maxDischarge} + F \cdot T)^{-1}} \\
 & \text{drawing water now} \dots \\
 \vee & \boxed{\text{discharged} + F \cdot (gt + T) < \text{maxDischarge}} \\
 & \dots \text{ till the timer runs out is safe} \\
 & \boxed{\wedge gt^{-1} \cdot (\text{absorbed} - \text{minAbsorbed} + c \cdot F \cdot gt \cdot \text{tempDiff}) \geq 0)}
 \end{aligned}$$

3478 The guard for drawing no water just needs to check that it still has enough time to absorb enough

heat.

3479

$$\begin{aligned}
G_1 \equiv & \overbrace{absorbed \geq minAbsorbed}^{\text{already absorbed enough heat}} \\
& \vee \overbrace{gt > T \wedge ((F = (maxDischarge - discharged) \cdot gt^{-1} \\
& \quad \quad \quad \dots \text{exceed the } \dots \\
& \quad \quad \quad \vee discharged + T \cdot F \geq maxDischarge \\
& \quad \quad \quad \dots \text{discharge limit} \\
& \quad \quad \quad \vee gt \cdot (discharged + gt \cdot F - maxDischarge) > 0) \\
& \quad \quad \quad \text{despite deferring one cycle } \dots \\
& \quad \quad \quad \vee gt \cdot (discharged + gt \cdot F - maxDischarge) < 0 \\
& \quad \quad \quad \dots \text{can still absorb enough heat} \\
& \quad \quad \quad \wedge (absorbed - minAbsorbed) \cdot (gt - T)^{-1} + F \cdot (c \cdot tempDiff) \geq 0)
\end{aligned}$$

3480

The guard for drawing water checks that drawing water for one cycle will not exceed the discharge limit.

3481

3482

$$G_2 \equiv discharged + F \cdot T < maxDischarge$$

3483

A.4 Simplification Impact Example

Reduction of even a loop free dGL formula, if done naïvely, can produce very large expressions. Not only are large expression hard to reason about, but they also slow down quantifier elimination. The effect on QE is so stark that the benchmark *Curvebot*, which completes in 26s with our simplification stack, times out after 20 minutes without simplification. Our simplification uses *Mathematica*'s *Simplify* function. Additionally, to catch common patterns that *CESAR* produces but *Mathematica* does not handle, we implemented a custom simplifier with *Egg* and *z3*.

The impact of simplification can be dramatic. For example, the I_0 invariant produced by *CESAR* for *Curvebot* *without* simplification is the following expression:

$$\begin{aligned} & ((v \neq wx/y \wedge y \neq 0 \vee y = 0 \wedge (w = 0 \wedge (v \geq 0 \wedge x > 0 \vee v \leq 0 \wedge x < 0) \\ & \vee w \neq 0 \wedge x \neq 0) \vee w \geq 0 \wedge y > 0 \vee w \leq 0 \wedge y < 0) \\ & \vee x = 0 \wedge 2v \neq y \wedge y \neq 0 \vee x \neq 0 \wedge (2w + x \neq 0 \vee y \neq 0) \\ & \wedge (y = 0 \vee 2vx \cdot y \neq x \cdot (2wx + x^2 + y^2))) \vee (2w \neq x \vee y \neq 0) \\ & \wedge x \neq 0 \wedge (y = 0 \vee x^3 + x \cdot y \cdot (2v + y) \neq 2wx^2) \\ & \vee x = 0 \wedge 2v + y \neq 0 \wedge y \neq 0 \end{aligned}$$

Running this formula through our simplifier produces the expression $y \neq 0 \vee (y = 0 \wedge 0 \neq x)$. Thus, the expression is actually just $\neg(x = 0 \wedge y = 0)$.

Besides being much easier to read, simpler expressions are also much faster to eliminate quantifiers from. Since doubly exponential quantifier elimination is the bottleneck in scaling to increasingly complex problems, simplification is crucial to the scalability of *CESAR*. In the case of our example above, without simplification, *Curvebot* gets stuck on the following QE problem.

$$\begin{aligned} & \forall v_1, w_1, x_1, y_1 (((((v^2 + w^2 = v_1^2 + w_1^2 \wedge vw + vx - wy - xy \\ & = v_1w_1 + v_1x_1 - w_1y_1 - x_1y_1) \wedge w^2/2 + vy - y^2/2 = w_1^2/2 + v_1y_1 - y_1^2/2) \\ & \wedge v - y = v_1 - y_1) \wedge w^2/2 + wx + x^2/2 = w_1^2/2 + w_1x_1 + x_1^2/2) \\ & \wedge w + x = w_1 + x_1 \rightarrow (((v_1 \neq w_1x_1/y_1 \wedge y_1 \neq 0 \\ & \vee y_1 = 0 \wedge (w_1 = 0 \wedge (v_1 \geq 0 \wedge x_1 > 0 \vee v_1 \leq 0 \wedge x_1 < 0) \\ & \vee w_1 \neq 0 \wedge x_1 \neq 0) \vee w_1 \geq 0 \wedge y_1 > 0 \vee w_1 \leq 0 \wedge y_1 < 0) \\ & \vee x_1 = 0 \wedge 2v_1 \neq y_1 \wedge y_1 \neq 0 \vee x_1 \neq 0 \wedge (2w_1 + x_1 \neq 0 \\ & \vee y_1 \neq 0) \wedge (y_1 = 0 \vee 2v_1x_1 \cdot y_1 \neq x_1 \cdot (2w_1x_1 + x_1^2 + y_1^2))) \\ & \vee (2w_1 \neq x_1 \vee y_1 \neq 0) \wedge x_1 \neq 0 \wedge (y_1 = 0 \\ & \vee x_1^3 + x_1 \cdot y_1 \cdot (2v_1 + y_1) \neq 2w_1x_1^2) \\ & \vee x_1 = 0 \wedge 2v_1 + y_1 \neq 0 \wedge y_1 \neq 0) \wedge \neg(x_1 = 0 \wedge y_1 = 0)) \end{aligned}$$

Table A.2: Simplification impact on solution size.

Benchmark	% Solution Size Reduction
ETCS Train	62%
Sled	41%
Intersection	38%
Curvebot	∞
Parachute	5%
Coolant	92%
Corridor	85%
Power Station	55%

But with simplification applied regularly, instead the corresponding call to QE is

$$\begin{aligned}
& \forall v_1, w_1, x_1, y_1 (((((v^2 + w^2 = v_1^2 + w_1^2 \wedge vw + vx - wy - xy \\
& = v_1w_1 + v_1x_1 - w_1y_1 - x_1y_1) \wedge w^2/2 + vy - y^2/2 = w_1^2/2 + v_1y_1 - y_1^2/2) \\
& \wedge v - y = v_1 - y_1) \wedge w^2/2 + wx + x^2/2 = w_1^2/2 + w_1x_1 + x_1^2/2) \\
& \wedge w + x = w_1 + x_1 \rightarrow \neg(x_1 = 0 \wedge y_1 = 0))
\end{aligned}$$

which terminates in 70 milliseconds instead of timing out.

Finally, even on benchmark examples that would terminate without the use of *eager simplification*, enabling this optimization often results in shorter solutions. Table A.2 shows the percentage reduction in size of solutions due to simplification, where size is measured in terms of number of characters. A 0% reduction means no change and a 50% reduction means cutting formula-size in half.

Appendix B

3511

Generalized Hybrid Game Synthesis (Appendix)

3512

3513

B.1 Naïve Ordering Counterexample

3514

For game $a:\alpha$, let S' set the winning region for the overall game to be empty ($S'(a) = \perp$), meaning that Angel is never allowed to play at all. Now the winning subregions for later subgames are irrelevant because Angel cannot reach them, but S' nevertheless sets them to the theoretical optimal (for every subgame $b \neq a$, $S(b) \mapsto \langle a:\alpha_b \rangle \phi$). For *any* other inductive subvalue map S , a good ordering should indicate that $S \sqsupseteq S'$ since S cannot possibly be less permissive, but the proposed naïve ordering does not. When S sets the winning subregion for any of the later subgames $b \neq a$ to anything less than the optimal solution, the check $\models S'(b) \rightarrow S(b)$ fails.

For a concrete visualization, consider the two inductive subvalue maps shown in Fig. B.1: S on the left and S' on the right. Because S' immediately blocks off *all* strategies at the root, a good ordering should indicate $S \sqsupseteq S'$, but the proposed naïve ordering does not because at the subgame α , S has a strictly stronger formula ($y = 1 \rightarrow \top$ is false). The problem in the naïve, point wise definition (Def. B.1) is it does not recognize that the $\top$ at subgame α is vacuous: S' permits no traces to reach subgame α in the first place, so the permissiveness of $\top$ is unimpressive.

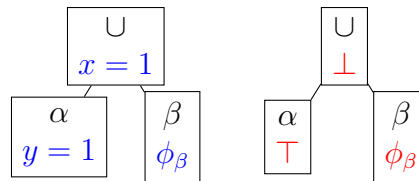


Figure B.1: Inductive subvalue maps S and S' . Locally at subgame α , S' seems more permissive, but global analysis reveals S is more permissive.

B.2 Inductive Subvalue Map Loop Condition Discussion

We discuss the details of the definition of an inductive subvalue maps (Def. 6.8) for the case $a:(g:\gamma)^* \models S$, and correspondingly the synthesis framework $\langle \text{map} \rangle$ (Algorithm 2) for the case of $\langle \text{map} \rangle(a:(g:\gamma)^*, \phi)$.

B.2.1 Counterexample for Projection Condition Weakening

Def. 6.8 and Algorithm 2 for the case $(g:\gamma)^*$ use the check $\models S(a) \rightarrow \langle a:\alpha \Downarrow S \rangle \phi$, i.e., winning subregion $S(a)$ should ensure that there is a *finite* strategy to exit while staying within the control envelope. The subvalue map should ensure that Angel does not get stuck in a state where her only option is playing the loop forever. It might seem that the more liberal, simpler condition $\models S(a) \rightarrow \langle \alpha \rangle \phi$ also ensures Angel has a finite exit strategy, but this condition is subtly incorrect. Angel can get stuck in a state where there is theoretically a way to exit the loop but this requires going outside the control envelope. Eq. (B.1) shows a counterexample.

$$\begin{aligned} & \langle a:(b:(c:x := x + 1 \cup d:x := x - 1))^* \rangle x \geq 0 \\ & S := \{a \mapsto \top, b \mapsto \top, c \mapsto \perp, d \mapsto \top\} \end{aligned} \quad (\text{B.1})$$

For this game, if Def. 6.8 were modified such that $\langle a:\alpha \Downarrow S \rangle \phi$ is replaced by $\langle \alpha \rangle \phi$, the subvalue map S would be inductive. However, it would also let Angel get stuck iterating the loop forever. Consider starting state $\{x \mapsto -1\}$. This state belongs to the winning subregion $S(a)$ of the overall game, so Angel should be able to win. And indeed she can win the game by running one iteration of the loop where she increments x . The new state would be $\{x \mapsto 0\}$, so Angel would exit the loop and win since winning condition $x \geq 0$ holds. However, if Angel is forced to actually respect the subvalue map (modeled by projection of S onto the game as shown in Eq. (B.2)), she cannot win.

$$\langle ?\top ; (? \top ; ((? \perp ; x := x + 1) \cup (? \top ; x := x - 1)))^* ; ?x \geq 0 \rangle x \geq 0 \quad (\text{B.2})$$

Starting at $\{x \mapsto -1\}$, Angel has to repeat the loop since she has not reached the winning region $x \geq 0$. But inside the loop, the subvalue map says she can never choose to increment x (subgame c). So she is forced to decrement x , reaching new state $\{x \mapsto -2\}$. In principle, if she were allowed to ignore the subvalue map, she could still exit with finite iterations by incrementing x twice. But with subgame c blocked off, she is only ever allowed to decrement x and is stuck in the loop forever. To prevent such subvalue maps from counting as inductive, the check $\models S(a) \rightarrow \langle a:\alpha \Downarrow S \rangle \phi$ is necessary.

B.2.2 Counterexample for Recursive Condition Weakening

Def. 6.8 and Algorithm 2 for the case $(g:\gamma)^*$ also use the check $g:\gamma \models S$, i.e., the subvalue map for the loop body ensures that after playing the loop body, either there is still a finite exit strategy or Angel can already exit the loop. Eq. (B.3) shows a counterexample where using $g:\gamma \models S$ instead is not sound, since the inner subvalue map permit reaching a state without a finite exit strategy.

$$\begin{aligned} & \langle a:(b:(c:x := x + 1 \cup d:x := x - 1))^* \rangle x \geq 0 \\ & S := \{a \mapsto x \geq -1, b \mapsto x \geq -1, c \mapsto x \geq -1, d \mapsto \top\} \end{aligned} \quad (\text{B.3})$$

Starting at $\{x \mapsto -1\}$, it is possible for Angel to choose subgame d and decrement x . The new state is $\{x \mapsto -2\}$, from which there is now no finite exit strategy because to increment x , per the guard of subgame c , x must be at least -1 . Angel is now stuck decrementing x forever.

B.3 Example of Subvalue Map Shortfall for Runtime Monitoring

We show how for a (non-inductive) subvalue map, the property $\models S(a) \rightarrow \langle a:\alpha \setminus S\phi \rangle \phi$ does not always hold. Consider the subvalue map of Fig. 6.1, but with ϕ_6 set to the empty subregion $\perp$ instead of $x > 0 \wedge (v \geq 0 \vee a > 0)$. As before, let the overall game be $a:\alpha$, and subgame labels be the numbers indexing the winning subregions. The new map is still a subvalue map since $\models \perp \rightarrow \langle a:\alpha_5 \rangle x > 0$. The untrusted controller monitoring correctness property is Eq. (6.3) with ϕ_6 replaced by $\perp$. The result has shape

$$x > 0 \wedge (v \geq 0 \vee a > 0) \rightarrow \langle ((\perp \vee (x > 0 \wedge v \geq 1)); \dots \cap \dots); \dots \rangle^\times x > 0.$$

This formula is not valid. Consider the starting state $\{x \mapsto 1, v \mapsto 0\}$. Demon first chooses to run the loop, then Angel must pass the test $x > 0 \wedge v \geq 1$, which fails. The *inductive* subvalue map prevents this situation by requiring that ϕ_8 implies $\phi_6 \vee \phi_5$, thus disqualifying this counterexample where Angel gets stuck.

B.4 Benchmarks

This appendix discusses the benchmarks, provides their full listings, and discusses the rewrite heuristics used to solve them.

B.4.1 New Examples Discussion

We discuss the examples representing diverse control challenges introduced in this chapter to demonstrate the flexibility of our dGL control envelope synthesis approach.

Event-triggered ETCS (Model 9) models a train from a case study modeling the European Train Control System (ETCS) [126], but with a key modification: the train is modeled event triggered instead of time triggered. The control envelope computed by our implementation finds the winning subregion for the overall loop to be $p < e \wedge (2Ae + v^2 < 2Ap \vee v \leq 0) \wedge (A \leq 0 \vee v > 0) \vee p < e \wedge (2(-B)e + v^2 < 2(-B)p \vee v \leq 0) \wedge (-B \leq 0 \vee v > 0)$. After simplification, the key term ends up being $e - p + v^2/2B > 0$, i.e., there is still enough time for the train to come to a stop before the end of motion authority e if it starts braking now. Hence, the control envelope has a strategy to keep the train safe whenever it is possible for the train to be safe by braking.

Surgical Robot (Model 19) is based on a case study from the literature [81] modeling force feedback in surgical robots. The control envelope computed by our implementation finds the winning subregion for the overall loop to be $(q_x - p_x)n_x + (q_y - p_y)n_y \geq 0$, i.e., so long as the robot is not already outside the fixture, the subvalue map has a strategy for safe control.

Infinite Track (Model 20) is an example of infinite horizon switching. A vehicle in a looping track can choose to move in any of the four coordinate directions, but must remain moving at all times. To remain safe for arbitrarily long time, this vehicle must keep switching directions at the right moments, and make *infinitely* many switches. Our approach synthesizes a control envelope that indicates when it is safe to switch to any given direction. CESAR does not solve this problem as it does not handle infinite control strategies. The control envelope computed by our implementation finds the winning subregion for the overall loop to be $2R > x \wedge x > -2R \wedge 2R > y \wedge y > -2R \wedge (x > R \vee x < -R \vee y > R \vee y < -R)$, i.e., the subvalue map has a safe strategy for the vehicle at every point in the track.

Reach-avoid Robot (Model 21) demonstrates an example of envelope synthesis for reach-avoid problems, where an agent must *reach* an objective while *avoiding* unsafe situations. Such problems can model the simultaneous requirement of safety and liveness. In this example, the robot is safe in the square $[-2R, 2R] \times [-2R, 2R]$ (enforced in domain constraint of Line 3) and must reach the target region $[R, 2R] \times [R, 2R]$ (on Line 4). The robot can either travel upward ($v_x = 0, v_y = V$) or leftward ($v_x = -V, v_y = 0$). The synthesized subvalue map identifies that it is safely possible to reach the target region starting in $[R, 2R] \times [-2R, 2R]$.

Highway Driving (Model 22) is a time-triggered problem based on the core control challenge from a case study on highway driving [93]. Two cars, are driving on a highway. The lead car, with velocity v_l and acceleration a_l is driven by some external agent with the physical limitation that the car's accelerates is bounded above by A and its braking is bounded below by $-B$ (Line 2). The following car, with velocity v_f and acceleration a_f , is driven by a controller that must follow the lead car while avoiding collisions (Line 3). The controllers revise decisions in a time triggered fashion, with maximum latency T . Regardless of the lead car's behavior, the following car must always be able to stop in time to avoid collision (Line 5). This problem does not fit CESAR's template, which does not support an adversarial agent like the lead car. In the computed control envelope, the winning region for the overall game is $p_f < p_l \wedge v_f \leq v_l$, i.e. the subvalue map has a strategy to avoid the collisions whenever that velocity of the controlled car is less than that of the lead car and hasn't already collided with it.

The descriptions of the CESAR benchmarks can be found in Appendix A.3 [69]. The envelopes that we synthesize for these benchmarks are the same as those computed by CESAR.

B.4.2 Quadcopter

The procedurally generated Quadcopter benchmark suite is based on the following template with the addition of randomly generated values for velocity and obstacles.

The Quadcopter chooses whether to go up or down and runs continuous dynamics. The catch is that the quadcopter has a slow processor and cannot make a new choice in less than 1 second (Line 4). However, it can choose to run the dynamics for any time of its choice that is greater than 1 second (Line 3 has an Angel ODE). It can repeat this process of choosing and then sticking with a choice any number of times (Angel loop in Line 4). While it flies, it must avoid obstacles at all times. It must also not crash into the floor at $y > 0$. It must eventually reach the target region $x > 20$. CESAR cannot solve this problem since it lies outside the template. NYCS solves all instances.

An example of the formula that can be generated for `<avoid obstacles>` obstacles is

```

assume | 1   $x > 0 \wedge V = \langle \text{velocity} \rangle \rightarrow \langle$ 
direction | 2   $\left( (v_y := 1) \cup (v_y := -1) \right);$ 
plant | 3   $t := 0; \{x' = V, y' = v_y, t' = 1 \ \& \ \langle \text{avoid obstacles} \rangle \wedge y \geq 0\};$ 
safe | 4   $?t \geq 1 \rangle^* \rangle x > 20$ 

```

$(x < 2 \vee 2 < x \vee y < 0 \vee 3 < y) \wedge (x < 5 \vee 6 < x \vee y < 0 \vee 3 < y) \wedge (x < 5 \vee 8 < x \vee y < 9 \vee 9 < y) \wedge (x < 8 \vee 10 < x \vee y < 7 \vee 7 < y)$. This formula has four obstacles. The first one is a line segment from (2, 0) to (2, 3). The second is a rectangle from (5, 0) to (6, 3). The third is a rectangle from (5, 9) to (8, 9). The fourth is a rectangle from (8, 7) to (10, 7). An example of a generated number that fills `<velocity>` is 1. The procedural generation parameters allow up to 5 obstacles with widths and heights of at most 3 distributed in the region $[0, 10] \times [0, 10]$. Velocity is an integer between 1 and 5.

As expected, the implementation solves problems with fewer obstacles quickly, with the fastest solutions for one obstacle completing in a couple of seconds. However, the time taken increases with the number of obstacles, and more importantly, when obstacles are placed in such a way that they interact to make the solution space more complex. Table B.1 summarizes the generated problems and their solving outcomes.

As a demonstrative example, we discuss the control envelope computed for problem the 23 in the benchmark suite. Fig. B.2 shows a Mathematica plot of the winning subregion for the overall loop of this example. The problem has three obstacles, two of which overlap. It is naturally unsafe to start at an obstacle. But it is further unsafe to start right before an obstacle because collision with the obstacle becomes unavoidable. Notice a triangle of unsafe space right after the lowest obstacle. This unsafe triangle is an example how obstacles can interact to create unsafe regions. In this triangle, it is neither safe to go upwards because of collision with the upper obstacle, nor is it safe to go downwards because of collision with the floor. This envelope is computed in 68 seconds.

B.4.3 Full Benchmark Listings

We list the dGL formulas for all the new examples in Fig. 6.3a.

Fig. 6.3b compares the performance of our approach to CESAR on the CESAR benchmark suite. The performance is similar at most benchmarks with only a few seconds of overhead. An exception is Intersection, where CESAR is significantly more expensive. The observed differences are likely a consequence of different simplification heuristics rather than any fundamental algorithmic reasons. Simplification is itself an expensive operation, but if done at the right juncture, can make the even more expensive quantifier eliminations that occur later in symbolic execution cheaper. Thus simplification has a large influence on performance. Further code tuning and heuristics optimizations can likely reduce the overheads. LLM assistance was used to

Table B.1: Quadcopter benchmark suite generated parameters and solving outcomes. In the result column, $\checkmark$ indicates that a nonempty control envelope was computed, while $\times$ indicates that algorithm timed out after 20 minutes, failing to compute an envelope.

No.	Velocity	Obstacles	Result
1	5	$x < 9 \vee 11 < x \vee y < 1 \vee 3 < y$	$\checkmark$
2	4	$x < 9 \vee 9 < x \vee y < 3 \vee 3 < y$	$\checkmark$
3	1	$(x < 2 \vee 2 < x \vee y < 0 \vee 3 < y) \wedge (x < 5 \vee 6 < x \vee y < 0 \vee 3 < y) \wedge (x < 5 \vee 8 < x \vee y < 9 \vee 9 < y) \wedge (x < 8 \vee 10 < x \vee y < 7 \vee 7 < y)$	$\times$
4	1	$x < 3 \vee 6 < x \vee y < 9 \vee 10 < y$	$\checkmark$
5	2	$x < 0 \vee 1 < x \vee y < 2 \vee 4 < y$	$\checkmark$
6	2	$(x < 7 \vee 10 < x \vee y < 8 \vee 8 < y) \wedge (x < 3 \vee 5 < x \vee y < 8 \vee 9 < y)$	$\checkmark$
7	4	$(x < 7 \vee 8 < x \vee y < 1 \vee 3 < y) \wedge (x < 5 \vee 5 < x \vee y < 6 \vee 8 < y)$	$\times$
8	1	$(x < 3 \vee 3 < x \vee y < 8 \vee 11 < y) \wedge (x < 0 \vee 0 < x \vee y < 1 \vee 1 < y)$	$\checkmark$
9	5	$(x < 1 \vee 3 < x \vee y < 7 \vee 9 < y) \wedge (x < 4 \vee 4 < x \vee y < 0 \vee 3 < y)$	$\times$
10	1	$\top$	$\checkmark$
11	2	$x < 3 \vee 4 < x \vee y < 6 \vee 8 < y$	$\checkmark$
12	2	$(x < 10 \vee 13 < x \vee y < 10 \vee 11 < y) \wedge (x < 5 \vee 5 < x \vee y < 7 \vee 8 < y) \wedge (x < 7 \vee 10 < x \vee y < 10 \vee 12 < y) \wedge (x < 3 \vee 4 < x \vee y < 7 \vee 8 < y) \wedge (x < 2 \vee 2 < x \vee y < 2 \vee 4 < y)$	$\times$
13	3	$(x < 2 \vee 2 < x \vee y < 7 \vee 7 < y) \wedge (x < 8 \vee 11 < x \vee y < 7 \vee 8 < y) \wedge (x < 1 \vee 2 < x \vee y < 9 \vee 12 < y)$	$\times$
14	4	$x < 9 \vee 11 < x \vee y < 4 \vee 7 < y \wedge x < 3 \vee 3 < x \vee y < 8 \vee 8 < y \wedge x < 9 \vee 11 < x \vee y < 8 \vee 11 < y \wedge x < 8 \vee 11 < x \vee y < 4 \vee 7 < y \wedge x < 7 \vee 8 < x \vee y < 9 \vee 10 < y$	$\checkmark$
15	3	$(x < 1 \vee 4 < x \vee y < 9 \vee 10 < y) \wedge (x < 7 \vee 10 < x \vee y < 0 \vee 3 < y) \wedge (x < 8 \vee 10 < x \vee y < 8 \vee 11 < y) \wedge (x < 3 \vee 6 < x \vee y < 3 \vee 4 < y)$	$\times$
16	4	$(x < 1 \vee 1 < x \vee y < 8 \vee 11 < y) \wedge (x < 8 \vee 10 < x \vee y < 6 \vee 9 < y) \wedge (x < 5 \vee 8 < x \vee y < 8 \vee 11 < y) \wedge (x < 6 \vee 6 < x \vee y < 3 \vee 4 < y) \wedge (x < 10 \vee 13 < x \vee y < 10 \vee 13 < y)$	$\times$
17	4	$\top$	$\checkmark$
18	3	$(x < 5 \vee 5 < x \vee y < 4 \vee 4 < y) \wedge (x < 1 \vee 2 < x \vee y < 5 \vee 8 < y) \wedge (x < 8 \vee 11 < x \vee y < 9 \vee 12 < y) \wedge (x < 2 \vee 2 < x \vee y < 1 \vee 3 < y) \wedge (x < 6 \vee 7 < x \vee y < 1 \vee 2 < y)$	$\times$
19	3	$(x < 4 \vee 6 < x \vee y < 0 \vee 0 < y) \wedge (x < 8 \vee 8 < x \vee y < 9 \vee 10 < y) \wedge (x < 2 \vee 2 < x \vee y < 2 \vee 2 < y) \wedge (x < 3 \vee 3 < x \vee y < 0 \vee 2 < y) \wedge (x < 6 \vee 9 < x \vee y < 8 \vee 11 < y)$	$\checkmark$
20	2	$(x < 2 \vee 3 < x \vee y < 9 \vee 11 < y) \wedge (x < 10 \vee 13 < x \vee y < 4 \vee 5 < y) \wedge (x < 9 \vee 11 < x \vee y < 8 \vee 11 < y)$	$\checkmark$
21	3	$(x < 4 \vee 4 < x \vee y < 5 \vee 5 < y) \wedge (x < 5 \vee 6 < x \vee y < 7 \vee 9 < y) \wedge (x < 10 \vee 13 < x \vee y < 3 \vee 4 < y)$	$\checkmark$
22	4	$(x < 6 \vee 8 < x \vee y < 3 \vee 4 < y)$	$\checkmark$
23	3	$(x < 5 \vee 8 < x \vee y < 8 \vee 9 < y) \wedge (x < 0 \vee 3 < x \vee y < 3 \vee 6 < y) \wedge (x < 6 \vee 8 < x \vee y < 6 \vee 9 < y)$	$\checkmark$
24	4	$(x < 3 \vee 3 < x \vee y < 7 \vee 10 < y) \wedge (x < 9 \vee 9 < x \vee y < 1 \vee 4 < y) \wedge (x < 3 \vee 5 < x \vee y < 10 \vee 12 < y) \wedge (x < 4 \vee 4 < x \vee y < 6 \vee 9 < y)$	$\checkmark$
25	1	$(x < 6 \vee 6 < x \vee y < 9 \vee 10 < y)$	$\checkmark$

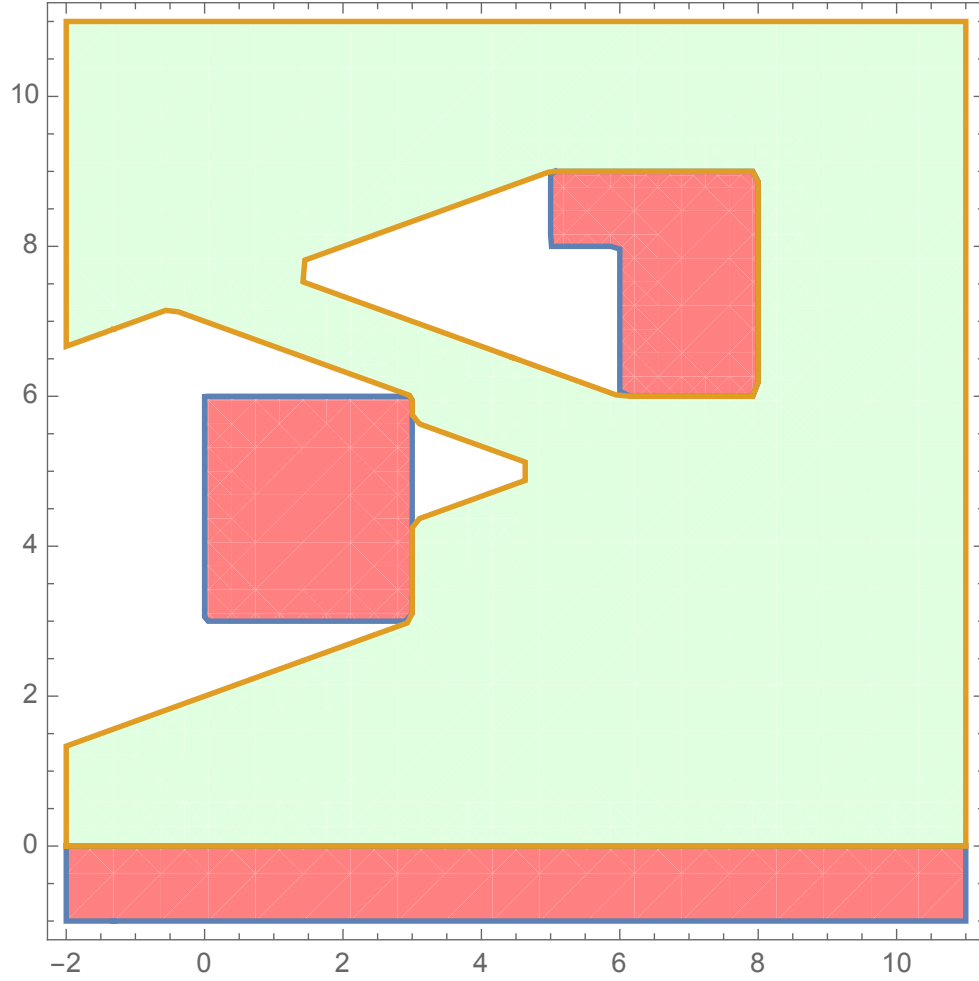


Figure B.2: Computed control envelope for quadcopter problem 23. Red regions are unsafe (obstacles or floor). Green regions are safe starting points per the computed control envelope. The x and y coordinates represent the quadcopter's position.

Model 19 Surgical Robot example based on [81]

```

assume | 1   $T > 0 \wedge K > 0 \wedge n_x^2 + n_y^2 = 1 \rightarrow \langle$ 
input  | 2   $(f_{xp} := \otimes; f_{yp} := \otimes;$ 
damping | 3   $g := *; ?g \geq 0;$ 
plant  | 4   $t := 0; \{q'_x = K f_x g, q'_y = K f_y g, f'_x = f_{xp}, f'_y = f_{yp}, t' = 1 \ \& \ t \leq T\}^d$ 
safe   | 5   $\rangle^\times \rangle ((q_x - p_x)n_x + (q_y - p_y)n_y \geq 0)$ 

```

Model 20 Infinite Track

```

assume | 1   $T > 0 \wedge V > 0 \wedge R > 0 \wedge 2VT < R \rightarrow \langle$ 
east/west | 2   $((v_x := V; v_y := 0) \cup (v_x := -V; v_y := 0) \cup$ 
north/south | 3   $(v_x := 0; v_y := -V) \cup (v_x := 0; v_y := V)) ;$ 
plant  | 4   $t := 0; \{x' = v_x, y' = v_y, t' = 1 \ \& \ t \leq T\}^d \rangle^\times$ 
safe   | 5   $\rangle (2R > x \wedge x > -2R \wedge 2R > y \wedge y > -2R \wedge$ 
        | 6   $(x > R \vee x < -R \vee y > R \vee y < -R))$ 

```

3672 generate some of the boilerplate code of the implementation.

3673 B.4.4 Rewrite Heuristics

3674 Our implementation uses rewriting heuristics to simplify problems into shapes where they can
3675 either be symbolically executed or solved using other invariant generation heuristics. Table B.2
3676 lists the rewrite heuristics and which evaluation problem each is used in.

3677 We discuss how the rewrite heuristics work.

- 3678 • *Extremal assignment rewrite.* This rewrite replaces a guarded assignment (e.g., $x :=$
3679 $\otimes; ?x > 0$) with an assignment to an extremal value identified from the guard (e.g., $x := 0$).
3680 The rewritten program is now much easier to reason about, since a free assignment other-

Table B.2: Rewrite heuristics used in the evaluation problems.

Heuristic	Problems
Extremal assignment rewrite	Surgical robot, Highway driving
Adversarial one-shot rewrite	Event-triggered ETCS
Loop unrolling	Reach-avoid robot, Quadcopter
CESAR rewrite	Highway driving, CESAR benchmarks

Model 21 Reach-avoid Robot

assume |₁ $V > 0 \wedge R > 0 \rightarrow \langle$
direction |₂ $\left((v_x := -V; v_y := 0) \cup (v_x := 0; v_y := V) \right);$
plant |₃ $\{x' = v_x, y' = v_y \& (2R \geq x \wedge x \geq -2R \wedge 2R \geq y \wedge y \geq -2R)\}^*$
safe |₄ $R \leq x \wedge x \leq 2R \wedge R \leq y \wedge y \leq 2R$

Model 22 Highway Driving

assume |₁ $A > 0 \wedge B > 0 \wedge T > 0 \wedge v_f > 0 \wedge v_l > 0 \rightarrow \langle$
lead car |₂ $(a_l := \otimes; !-B < a_l \wedge a_l < A;$
controlled car |₃ $a_f := *; ?-B < a_f \wedge a_f < A;$
plant |₄ $t := 0; \{p'_f = v_f, v'_f = a_f, p'_l = v_l, v'_l = a_l, t' = 1 \& t \leq T\}^d \rangle^x$
safe |₅ $p_f < p_l$

wise requiring a quantifier to reason about has been replaced by a constant expression. 3681

- *Adversarial one-shot rewrite.* This rewrite is similar in spirit to the *one-shot unrolling* of 3682
CESAR [69]. It takes an Angel loop ending with an Angel ODE (e.g., the inner loop of 3683
Model 9) and replaces it with a single iteration of the loop but with the ODE changed to a 3684
Demon ODE guarded at the end by an Angel test for the domain constraint of the original 3685
ODE. The intuition is that the rewritten program emulates running any number of iterations 3686
of the original loop with the ODE run for any amount of time, so a precondition for the 3687
harder rewritten program is likely to also be a precondition for the original program. 3688
- *Loop unrolling.* This heuristic unrolls loops a fixed number of times. For example, $\langle \alpha^* \rangle \phi$ 3689
is rewritten as $\langle \alpha \rangle \phi$ or $\langle \alpha \rangle \langle \alpha \rangle \phi$ or $\langle \alpha \rangle \langle \alpha \rangle \langle \alpha \rangle \phi$ and so on. 3690
- *CESAR rewrite.* This is the one-shot refinement and multi-shot unrolling form CESAR 3691
(Chapter 5 [69]) where a time-triggered control loop is replaced by running the ODE for 3692
an unbounded amount of time. Because of the of our framework’s retrospective checks, 3693
the requirement that the original program should have an idempotence property (“action 3694
permanence”) which was essential to the soundness of CESAR is relaxed. 3695

B.4.5 Refinements 3696

While synthesizing subvalue maps for loops, Algorithm 2 uses checks to test invariant candidates 3697
before accepting them. It is possible to use *refinements* to perform these checks. A game β is an 3698

3699 *Angelic refinement* of game α when for any formula ψ , $\models \langle \beta \rangle \psi \rightarrow \langle \alpha \rangle \psi$. Some rewrites result in
 3700 refinements, making the invariant candidate generated correct by construction. For instance, for
 3701 the Angelic loop in Reach-avoid robot (Model 21), let α be the loop body (Line 2 and Line 3), and
 3702 ϕ the postcondition (Line 4). The inductiveness check is $\text{Inv} \rightarrow \langle (a : \alpha^*) \Downarrow S \rangle \phi$. If Inv is computed
 3703 by rewriting loop $\langle \alpha^* \rangle \phi$ as $\langle \alpha \rangle \phi$ (loop unrolling rewrite) and then setting $\text{Inv} = \text{simpl}(\langle \alpha \rangle \phi)$
 3704 while writing all the intermediate symbolic execution steps of simpl into the subvalue map S ,
 3705 then Inv will pass the check by construction. The reason is that $\langle a : \alpha \Downarrow S \rangle \phi$ is a refinement of
 3706 $\langle (a : \alpha)^* \Downarrow S \rangle \phi$. Since $\models \text{Inv} \rightarrow \langle a : \alpha \Downarrow S \rangle \phi$ by construction, it follows that $\models \text{Inv} \rightarrow \langle (a : \alpha)^* \Downarrow S \rangle \phi$.
 3707 We exploit such refinement properties in our implementation to soundly reduce the number of
 3708 expensive checks performed.

3709 B.5 Proofs

3710 We prove the correctness of the auxiliary theorems and lemmas in the chapter. The proofs for
 3711 the theorems and lemmas that appear in the main text are in Appendix D. Notation $S|_U$ is the
 3712 restriction of the subvalue map S to the set of subgames $U \cup \{\text{end}\}$.

3713 **Definition B.1** (Weak ordering). We define a weaker ordering relation $\succsim$ on inductive subvalue
 3714 maps that compares formulas at each subgame separately. For two inductive subvalue maps S_1
 3715 and S_2 for the game $a : \alpha$, S_1 is at least as good as S_2 , written $S_1 \succsim S_2$, iff for each subgame b in
 3716 $\text{subgames}(a : \alpha)$, $\models S_2(b) \rightarrow S_1(b)$ and $\models S_2(\text{end}) \rightarrow S_1(\text{end})$.

3717 **Lemma B.1** (Ordering subsumption). *If $S \succsim S'$, then $S \supseteq S'$.*

3718 *Proof.* By definition of $\succsim$, for each subgame $b \in \text{subgames}(\alpha) \cup \{\text{end}\}$, $\models S'(b) \rightarrow S(b)$.
 3719 Using the dGL Gödel generalization rule G, we infer $\models \langle (a : \alpha : b) \Downarrow S' \rangle (S(b) \rightarrow S'(b))$ showing
 3720 that $S \supseteq S'$ when interpreted as an Angelic strategy. When interpreted as a Demonic strat-
 3721 egy, the same argument holds: using the dGL Gödel generalization rule G we can infer that
 3722 $\models [(a : \alpha : b) \Downarrow S'] (S'(b) \rightarrow S(b))$. $\square$

3723 **Lemma B.2** (Subvalue Map Projection is a Refinement). *The projection of an Angelic subvalue*
 3724 *map onto a game α is an Angelic refinement of the game, i.e.,*

$$3725 \quad \models \langle a : \alpha \Downarrow S \rangle \psi \rightarrow \langle \alpha \rangle \psi.$$

3726 *Dually, the projection of a Demonic subvalue map is a Demonic refinement.*

3727 *Proof.* We show this for Angelic subvalue maps, the proof for Demonic subvalue maps is sym-
 3728 metric. Intuitively this lemma is true because the projection of an Angelic subvalue map only
 3729 adds Angelic tests, which make the game monotonically harder. The proof uses structural induc-
 3730 tion based on the shape of $a : \alpha$. When $a : \alpha$ is:

- 3731 • Atomic and not controlled by Angel, i.e., $\alpha \in \{x := e, x := *, ?Q, !Q, \{x' =$
 3732 $f(x) \ \& \ Q\}^d\}$, the result is immediate since $a : \alpha \Downarrow S = \alpha$.
- 3733 • Atomic and controlled by Angel, i.e., $\alpha \in \{x := \otimes, \{x' = f(x) \ \& \ Q\}\}$, then $a : \alpha \Downarrow S =$
 3734 $\alpha ; ?S(\text{end})$. By the dGL axioms $\langle ? \rangle$ and $\langle ; \rangle$, $\models \langle \alpha ; ?S(\text{end}) \rangle \psi \leftrightarrow \langle \alpha \rangle (\psi \wedge S(\text{end}))$.
 3735 By the dGL axiom M, $\models \langle \alpha \rangle (\psi \wedge S(\text{end})) \rightarrow \langle \alpha \rangle \psi$. By transitivity of implication,
 3736 $\models \langle a : \alpha \Downarrow S \rangle \psi \rightarrow \langle \alpha \rangle \psi$.

- $g:\gamma \cup d:\delta$, then $a:\alpha \Downarrow S = (?S(g); g:\gamma \Downarrow S) \cup (?S(d); d:\delta \Downarrow S)$. 3737

By the dGL axiom $\langle \cup \rangle$, 3738

$\models \langle (?S(g); g:\gamma \Downarrow S) \cup (?S(d); d:\delta \Downarrow S) \rangle \psi \leftrightarrow \langle ?S(g); g:\gamma \Downarrow S \rangle \psi \vee \langle ?S(d); d:\delta \Downarrow S \rangle \psi$. By the 3739

dGL axioms $\langle ; \rangle$ and $\langle \cup \rangle$, 3740

$\models \langle ?S(g); g:\gamma \Downarrow S \rangle \psi \vee \langle ?S(d); d:\delta \Downarrow S \rangle \psi \leftrightarrow (S(g) \wedge \langle g:\gamma \Downarrow S \rangle \psi) \vee (S(d) \wedge \langle d:\delta \Downarrow S \rangle \psi)$. 3741

By conjunction elimination, 3742

$\models (S(g) \wedge \langle g:\gamma \Downarrow S \rangle \psi) \vee (S(d) \wedge \langle d:\delta \Downarrow S \rangle \psi) \rightarrow \langle a:\gamma \Downarrow S \rangle \psi \vee \langle d:\delta \Downarrow S \rangle \psi$. 3743

By the inductive hypothesis, 3744

$\models \langle g:\gamma \Downarrow S \rangle \psi \rightarrow \langle \gamma \rangle \psi$ and $\models \langle d:\delta \Downarrow S \rangle \psi \rightarrow \langle \delta \rangle \psi$. 3745

By the transitivity of implication, $\models \langle g:\gamma \Downarrow S \rangle \psi \vee \langle d:\delta \Downarrow S \rangle \psi \rightarrow \langle \gamma \rangle \psi \vee \langle \delta \rangle \psi$. 3746

By the dGL axiom $\langle \cup \rangle$, 3747

$\models \langle \gamma \rangle \psi \vee \langle \delta \rangle \psi \leftrightarrow \langle \gamma \cup \delta \rangle \psi$. 3748
- $g:\gamma \cap d:\delta$, then $a:\alpha \Downarrow S = g:\gamma \Downarrow S \cap d:\delta \Downarrow S$. By the dGL axiom $[\cap]$, $\models \langle g:\gamma \Downarrow S \cap d:\delta \Downarrow S \rangle \psi \leftrightarrow$ 3749

$\langle g:\gamma \Downarrow S \rangle \psi \wedge \langle d:\delta \Downarrow S \rangle \psi$. By the inductive hypothesis, $\models \langle g:\gamma \Downarrow S \rangle \psi \rightarrow \langle \gamma \rangle \psi$ and $\models$ 3750

$\langle d:\delta \Downarrow S \rangle \psi \rightarrow \langle \delta \rangle \psi$. Thus, $\models \langle g:\gamma \Downarrow S \rangle \psi \wedge \langle d:\delta \Downarrow S \rangle \psi \rightarrow \langle \gamma \rangle \psi \wedge \langle \delta \rangle \psi$. By the dGL ax- 3751

iom $[\cap]$, $\models \langle \gamma \rangle \psi \wedge \langle \delta \rangle \psi \leftrightarrow \langle \gamma \cap \delta \rangle \psi$. 3752
- $g:\gamma; d:\delta$, then $a:\alpha \Downarrow S = g:\gamma \Downarrow S(\text{end} \mapsto S(d)); d:\delta \Downarrow S$. 3753

Using the dGL axiom $\langle ; \rangle$, 3754

$\models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)); d:\delta \Downarrow S \rangle \psi \leftrightarrow \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)) \rangle \langle d:\delta \Downarrow S \rangle \psi$. By the inductive 3755

hypothesis, $\models \langle d:\delta \Downarrow S \rangle \psi \rightarrow \langle \delta \rangle \psi$. 3756

Thus, $\models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)) \rangle \langle d:\delta \Downarrow S \rangle \psi \rightarrow \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)) \rangle \langle \delta \rangle \psi$. Further, by the 3757

inductive hypothesis, $\models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)) \rangle \langle \delta \rangle \psi \rightarrow \langle \gamma \rangle \langle \delta \rangle \psi$. By the dGL axiom $\langle ; \rangle$, 3758

$\models \langle \gamma \rangle \langle \delta \rangle \psi \leftrightarrow \langle \gamma; \delta \rangle \psi$. 3759
- $a:(g:\gamma)^*$, then $a:\alpha \Downarrow S = (?S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(s)))^*; ?S(\text{end})$. By dGL axioms $\langle ; \rangle$, $\langle ? \rangle$, 3760

and M, $\models \langle a:\alpha \Downarrow S \rangle \psi \rightarrow \langle (?S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^* \rangle \psi$. By the dGL proof rule FP, 3761

$$\frac{\psi \vee \langle (?S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a))) \rangle \langle \gamma^* \rangle \psi \rightarrow \langle \gamma^* \rangle \psi}{\langle (?S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^* \rangle \psi \rightarrow \langle \gamma^* \rangle \psi} \quad . \quad 3762$$

Now, $\langle (?S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a))) \rangle \langle \gamma^* \rangle \psi$, per the inductive hypothesis, implies 3763

$\langle \gamma \rangle \langle \gamma^* \rangle \psi$. Per dGL axiom $\langle * \rangle$, $\langle \gamma \rangle \langle \gamma^* \rangle \psi \leftrightarrow \langle \gamma^* \rangle \psi$. This allows us to show the premise, 3764

since $\models \psi \vee \langle \gamma^* \rangle \psi \rightarrow \langle \gamma^* \rangle \psi$. 3765
- $a:(g:\gamma)^\times$, then $a:\alpha \Downarrow S = (g:\gamma \Downarrow S)^\times$. By the dGL proof rule loop, 3766

$$\frac{\langle (g:\gamma \Downarrow S)^\times \rangle \psi \vdash J \quad J \vdash \langle \gamma \rangle J \quad J \vdash \psi}{\langle (g:\gamma \Downarrow S)^\times \rangle \psi \vdash \langle \gamma^\times \rangle \psi} \quad . \quad 3767$$

Setting invariant J to $\langle (g:\gamma \Downarrow S)^\times \rangle \psi$, the first premise is immediate. 3768

The second premise is now 3769

$\langle (g:\gamma \Downarrow S)^\times \rangle \psi \vdash \langle \gamma \rangle \langle (g:\gamma \Downarrow S)^\times \rangle \psi$. 3770

By $\langle \times \rangle$, $\langle (g:\gamma \Downarrow S)^\times \rangle \psi \rightarrow \langle g:\gamma \Downarrow S \rangle \langle (g:\gamma \Downarrow S)^\times \rangle \psi$. 3771

By the inductive hypothesis, $\models \langle g:\gamma \Downarrow S \rangle \langle (g:\gamma \Downarrow S)^\times \rangle \psi \rightarrow \langle \gamma \rangle \langle (g:\gamma \Downarrow S)^\times \rangle \psi$, thus proving 3772

the second premise. 3773

3774

The third premise is proved as follows.

$$\frac{\frac{\text{id} \frac{*}{\psi \vdash \psi}}{\text{WL} \frac{\psi \wedge \langle g:\gamma \Downarrow S \rangle \langle (g:\gamma \Downarrow S)^\times \rangle \psi \vdash \psi}{\langle \times \rangle \frac{\psi \wedge \langle g:\gamma \Downarrow S \rangle \langle (g:\gamma \Downarrow S)^\times \rangle \psi \vdash \psi}{\langle (g:\gamma \Downarrow S)^\times \rangle \psi \vdash \psi}}}{\langle \times \rangle \frac{\psi \wedge \langle g:\gamma \Downarrow S \rangle \langle (g:\gamma \Downarrow S)^\times \rangle \psi \vdash \psi}{\langle (g:\gamma \Downarrow S)^\times \rangle \psi \vdash \psi}}$$

3776

□

Lemma B.3 (MPC Subvalue Map Projection Equivalence). *The projection game $\langle a:\alpha \Downarrow S \rangle \phi$ of the MPC Angelic subvalue map (Def. 6.9) is equivalent to the original game $\langle \alpha \rangle \phi$. Dually, the projection game $[a:\alpha \Downarrow S] \phi$ of the MPC Demonic subvalue map (Def. 6.9) is equivalent to the original game $[\alpha] \phi$.*

Proof. We show this for Angelic subvalue maps, the proof for Demonic subvalue maps is symmetric. Intuitively this lemma is true because the projection of an Angelic subvalue map only adds Angelic tests, which make the game monotonically harder. The proof uses structural induction based on the shape of $a:\alpha$. Note that for the MPC solution, end is ϕ . When $a:\alpha$ is:

- Atomic and not controlled by Angel, i.e., $\alpha \in \{x := e, x := *, ?Q, !Q, \{x' = f(x) \ \& \ Q\}^d\}$, the result is immediate since $a:\alpha \Downarrow S = \alpha$.
- Atomic and controlled by Angel, i.e., $\alpha \in \{x := \otimes, \{x' = f(x) \ \& \ Q\}\}$, then $a:\alpha \Downarrow S = \alpha; ?\phi$. By the dGL axioms $\langle ? \rangle$ and $\langle ; \rangle$, $\models \langle \alpha; ?\phi \rangle \phi \leftrightarrow \langle \alpha \rangle (\phi \wedge \phi) \leftrightarrow \langle \alpha \rangle \phi$.
- $g:\gamma \cup d:\delta$, then $a:\alpha \Downarrow S = (?S(g); g:\gamma \Downarrow S) \cup (?S(d); d:\delta \Downarrow S)$.

$$\begin{aligned} & \langle (?S(g); g:\gamma \Downarrow S) \cup (?S(d); d:\delta \Downarrow S) \rangle \phi \\ & \leftrightarrow \langle ?S(g); g:\gamma \Downarrow S \rangle \phi \vee \langle ?S(d); d:\delta \Downarrow S \rangle \phi \quad (\text{dGL axiom } \langle \cup \rangle) \\ & \leftrightarrow (S(g) \wedge \langle g:\gamma \Downarrow S \rangle \phi) \vee (S(d) \wedge \langle d:\delta \Downarrow S \rangle \phi) \quad (\text{dGL axioms } \langle ; \rangle, \langle \cup \rangle) \\ & \leftrightarrow (S(g) \wedge \langle \gamma \rangle \phi) \vee (S(d) \wedge \langle \delta \rangle \phi) \quad (\text{inductive hypothesis}) \\ & \leftrightarrow (\langle \gamma \rangle \phi \wedge \langle \gamma \rangle \phi) \vee (\langle \delta \rangle \phi \wedge \langle \delta \rangle \phi) \quad (\text{MPC definition}) \\ & \leftrightarrow \langle \gamma \rangle \phi \vee \langle \delta \rangle \phi. \end{aligned}$$
- $g:\gamma \cap d:\delta$, then $a:\alpha \Downarrow S = g:\gamma \Downarrow S \cap d:\delta \Downarrow S$.

$$\begin{aligned} & \langle g:\gamma \Downarrow S \cap d:\delta \Downarrow S \rangle \phi \\ & \leftrightarrow \langle g:\gamma \Downarrow S \rangle \phi \wedge \langle d:\delta \Downarrow S \rangle \phi \quad (\text{dGL axiom } [\cap]) \\ & \leftrightarrow \langle \gamma \rangle \phi \wedge \langle \delta \rangle \phi \quad (\text{inductive hypothesis}) \\ & \leftrightarrow \langle \gamma \cap \delta \rangle \phi \quad (\text{dGL axiom } [\cap]). \end{aligned}$$
- $g:\gamma; d:\delta$, then $a:\alpha \Downarrow S = g:\gamma \Downarrow S(\text{end} \mapsto S(d)); d:\delta \Downarrow S$.

$$\begin{aligned} & \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)); d:\delta \Downarrow S \rangle \phi \\ & \leftrightarrow \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)) \rangle \langle d:\delta \Downarrow S \rangle \phi \quad (\text{dGL axiom } [;]) \\ & \leftrightarrow \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)) \rangle \langle \delta \rangle \phi \quad (\text{inductive hypothesis}) \\ & \leftrightarrow \langle \gamma \rangle \langle \delta \rangle \phi \quad (\text{inductive hypothesis}) \\ & \leftrightarrow \langle \gamma; \delta \rangle \phi \quad (\text{dGL axiom } [;]). \end{aligned}$$
- $a:(g:\gamma)^*$

By Lemma B.2, $\models \langle a:\alpha \Downarrow S \rangle \phi \rightarrow \langle \gamma^* \rangle \phi$.

We now show the opposite direction, i.e., $\models \langle \gamma^* \rangle \phi \rightarrow \langle a:\alpha \Downarrow S \rangle \phi$.

Since $S(a) = \langle \gamma^* \rangle \phi$, we have $\models \langle \gamma^* \rangle \phi \rightarrow S(a)$. From $a:\alpha \models S$, we get $\models S(a) \rightarrow$ 3810
 $\langle a:\alpha \Downarrow S \rangle \phi$. 3811

Thus, by transitivity of implication, $\models \langle \gamma^* \rangle \phi \rightarrow \langle a:\alpha \Downarrow S \rangle \phi$. 3812

• $a:(g:\gamma)^\times$ 3813

By Lemma B.2, $\models \langle a:\alpha \Downarrow S \rangle \phi \rightarrow \langle \gamma^\times \rangle \phi$. We now show the opposite direction, i.e., $\models$ 3814
 $\langle \gamma^\times \rangle \phi \rightarrow \langle a:\alpha \Downarrow S \rangle \phi$. 3815

Since $S(a) = \langle \gamma^\times \rangle \phi$, we have $\models \langle \gamma^\times \rangle \phi \rightarrow S(a)$. We now show that $\models S(a) \rightarrow$ 3816
 $\langle a:\alpha \Downarrow S \rangle \phi$, so that by transitivity of implication, $\models \langle \gamma^\times \rangle \phi \rightarrow \langle a:\alpha \Downarrow S \rangle \phi$. 3817

By the dGL proof rule loop, 3818

$$\frac{S(a) \vdash J \quad J \vdash \langle g:\gamma \Downarrow S(\text{end} \mapsto S(a)) \rangle J \quad J \vdash \phi}{S(a) \vdash \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times \rangle \phi} \quad . \quad 3819$$

Setting invariant J to $S(a)$, the first premise is immediate. The last premise is $S(a) \vdash \phi$ 3820
which follows from $S(a) \leftrightarrow \phi \wedge \langle \gamma \rangle \langle \gamma^\times \rangle \phi$ per dGL axiom $\langle \times \rangle$. Only the middle premise 3821
remains. It is $S(a) \vdash \langle g:\gamma \Downarrow S(\text{end} \mapsto S(a)) \rangle S(a)$. 3822

Per the inductive hypothesis, $\models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(a)) \rangle S(a) \leftrightarrow \langle \gamma \rangle S(a)$. Thus, the middle 3823
premise can be rewritten as $S(a) \vdash \langle \gamma \rangle S(a)$. Recall that $S(a) = \langle \gamma^\times \rangle \phi$, so that the middle 3824
premise is $\langle \gamma^\times \rangle \phi \vdash \langle \gamma \rangle \langle \gamma^\times \rangle \phi$. By the dGL axiom $\langle \times \rangle$, $\langle \gamma^\times \rangle \phi \rightarrow \langle \gamma \rangle \langle \gamma^\times \rangle \phi$, thus proving 3825
the middle premise. 3826

□ 3827

Lemma B.4 (MPC Valid). *The MPC solution (Def. 6.9) S for a game $a:\alpha$ for Angel winning* 3828
condition ϕ is an inductive subvalue map compatible with ϕ : $a:\alpha \models S$. Dually, the MPC solution 3829
for a game $a:\alpha$ for Demon winning condition ϕ is an inductive subvalue map: $a:\alpha \Downarrow \models S$. 3830

Proof. Proved by induction on the structure of the game $a:\alpha$. We show this for the Angel MPC 3831
solution; the proof for the Demonic MPC solution is analogous. Firstly, for the MPC solution, 3832
 $S(\text{end}) = \phi$, so ϕ is a compatible winning condition. The base case is when $a:\alpha$ is atomic, i.e., 3833
 $\alpha \in \{x := e, x := *, ?f, !f, \{x' = f(x) \ \& \ Q\}, \{x' = f(x) \ \& \ Q\}^d\}$. In this case the MPC 3834
subvalue map consists of the mapping from a to $\langle \alpha \rangle \phi$ (along with $\text{end} \mapsto \phi$), which matches the 3835
corresponding condition for $a:\alpha \models S$ in Def. 6.8. In the recursive case, if $a:\alpha$ has the structure: 3836

1. $g:\gamma \cup d:\delta$: by the inductive hypothesis, as $S|_{\text{subgames}(g:\gamma)}$ is the MPC solution for game 3837
 γ and $S|_{\text{subgames}(d:\delta)}$ is the MPC solution for game δ , $g:\gamma \models S|_{\text{subgames}(g:\gamma)}$ and $d:\delta \models$ 3838
 $S|_{\text{subgames}(d:\delta)}$. So, $g:\gamma \models S$ and $d:\delta \models S$. It remains to show that $S(a) \rightarrow S(g) \vee S(d)$. By 3839
the definition of the MPC solution, $S(a) = \langle g:\gamma \cup d:\delta \rangle \phi$, $S(g) = \langle g:\gamma \rangle \phi$ and $S(d) = \langle d:\delta \rangle \phi$. 3840
By the dGL rule $\langle \cup \rangle$, $\langle g:\gamma \cup d:\delta \rangle \phi = \langle g:\gamma \rangle \phi \vee \langle d:\delta \rangle \phi$. Thus, $S(a) \rightarrow S(g) \vee S(d)$. 3841
2. $g:\gamma \cap d:\delta$: by the inductive hypothesis, as $S|_{\text{subgames}(g:\gamma)}$ is the MPC solution for game 3842
 γ and $S|_{\text{subgames}(d:\delta)}$ is the MPC solution for game δ , $g:\gamma \models S|_{\text{subgames}(g:\gamma)}$ and $d:\delta \models$ 3843
 $S|_{\text{subgames}(d:\delta)}$. So, $g:\gamma \models S$ and $d:\delta \models S$. It remains to show that $S(a) \rightarrow S(g) \wedge S(d)$. By 3844
the definition of the MPC solution, $S(a) = \langle g:\gamma \cap d:\delta \rangle \phi$, $S(g) = \langle g:\gamma \rangle \phi$ and $S(d) = \langle d:\delta \rangle \phi$. 3845
By the dGL rule $\langle \cap \rangle$, $\langle g:\gamma \cap d:\delta \rangle \phi = \langle g:\gamma \rangle \phi \wedge \langle d:\delta \rangle \phi$. Thus, $S(a) \rightarrow S(g) \wedge S(d)$. 3846
3. $g:\gamma ; d:\delta$: by the inductive hypothesis, as $S|_{\text{subgames}(d:\delta)}$ is the MPC solution for game δ and 3847
Angel winning condition ϕ , $d:\delta \models S|_{\text{subgames}(d:\delta)}$. So $d:\delta \models S$. Further, as $S|_{\text{subgames}(g:\gamma)}$ is 3848
the MPC solution for game γ and Angel winning condition $\langle d:\delta \rangle \phi = S(\delta)$, by the inductive 3849

hypothesis, $g:\gamma \models S(\text{end} \mapsto S(\delta))|_{\text{subgames}(g:\gamma)}$. So $g:\gamma \models S(\text{end} \mapsto S(\delta))$. It remains to show that $S(a) \rightarrow S(g)$ which is immediate from the definition of the MPC solution since $a:\alpha_a = a:\alpha_g$ in this case.

4. $g:\gamma^*$: observe that $S|_{\text{subgames}(g:\gamma)}$ is the MPC solution for game γ and Angel winning condition $\langle \gamma^* \rangle \phi$. $S(a) = \langle \gamma^* \rangle \phi$ and $S(a) \rightarrow \phi$, so $\langle \gamma^* \rangle \phi = S(a) \vee \phi$, so $S|_{\text{subgames}(g:\gamma)}$ is the MPC solution for game γ and Angel winning condition $S(a) \vee \phi$. Then, by the inductive hypothesis, $g:\gamma \models S(\text{end} \mapsto S(a))|_{\text{subgames}(g:\gamma)}$. So $g:\gamma \models S(\text{end} \mapsto S(a))$. The remaining condition is $S(a) \rightarrow \langle a:\alpha \Downarrow S \rangle \phi$. For MPC solution S , $\langle a:\alpha \Downarrow S \rangle \phi = \langle \gamma^* \rangle \phi$ (Lemma B.3). Since $S(a) = \langle \gamma^* \rangle \phi$, the condition is satisfied.

5. $g:\gamma^\times$: observe that $S|_{\text{subgames}(g:\gamma)}$ is the MPC solution for game γ and Angel winning condition $\langle \gamma^\times \rangle \phi$.

By definition, $S(a) = \langle \gamma^\times \rangle \phi$.

So, by the inductive hypothesis, $g:\gamma \models S(\text{end} \mapsto S(a))|_{\text{subgames}(g:\gamma)}$.

Thus $g:\gamma \models S(\text{end} \mapsto S(a))$.

Next we show that $\models S(a) \rightarrow S(g) \wedge \phi$. Using the dGL axiom $\langle^\times \rangle$, $\langle \gamma^\times \rangle \phi = \langle \gamma \rangle \langle \gamma^\times \rangle \phi \wedge \phi$.

Substituting back the MPC definitions of $S(a) = \langle \gamma^\times \rangle \phi$ and $S(g) = \langle \gamma \rangle S(a)$, this means $S(a) = S(g) \wedge \phi$. So, $\models S(a) \rightarrow S(g) \wedge \phi$.

□

Lemma B.5 (Subvalue maps win). *For any game $a:\alpha$, Angelic subvalue map S , and compatible Angel winning condition ϕ , $\langle \alpha \rangle \phi \models \langle a:\alpha \Downarrow S \rangle \phi$.*

Proof. The proof follows from structural induction along with the application of the usual dGL axioms and proof rules.

- If $a:\alpha$ is atomic and not controlled by Angel, i.e., $\alpha \in \{x := e, x := \otimes, ?Q, !Q, \{x' = f(x) \ \& \ Q\}^d\}$, then $\langle \alpha \rangle \phi = \langle a:\alpha \Downarrow S \rangle \phi$ and the result is immediate.
- If $a:\alpha = x := *$, then $a:\alpha \Downarrow S$ is $x := \otimes; !S(\text{end})$. Thus we must show that

$$\langle \alpha \rangle \phi \models \langle x := \otimes; !S(\text{end}) \rangle \phi.$$

By the dGL axioms $\langle ; \rangle$ and $[?]$, this is

$$\langle \alpha \rangle \phi \models \langle x := \otimes \rangle (S(\text{end}) \rightarrow \phi).$$

Since ϕ is a compatible Angel winning condition, $S(\text{end}) \rightarrow \phi$ holds. Thus, we need to show

$$\langle \alpha \rangle \phi \models \langle x := \otimes \rangle \top.$$

By the semantics of nondeterministic assignments in dGL, the succedent is $\top$, completing the proof.

- If $a:\alpha$ is $\{x' = f(x) \ \& \ Q\}$, then the argument is similar to the previous case. $a:\alpha \Downarrow S$ is $\{x' = f(x) \ \& \ Q\}^d; !S(\text{end})$. Thus we must show that

$$\langle \alpha \rangle \phi \models \langle \{x' = f(x) \ \& \ Q\}^d; !S(\text{end}) \rangle \phi.$$

Applying the dGL axioms $\langle ; \rangle$, $\langle^d \rangle$, and $[?]$, this is

$$\langle \alpha \rangle \phi \models \langle \{x' = f(x) \ \& \ Q\}^d \rangle (S(\text{end}) \rightarrow \phi).$$

Since ϕ is a compatible Angel winning condition, $S(\text{end}) \rightarrow \phi$ holds. Thus, we need to show

$$\langle \alpha \rangle \phi \models \langle \{x' = f(x) \ \& \ Q\}^d \rangle \top.$$

By the semantics of differential games in dGL, the succedent is $\top$, completing the proof.

- If $a:\alpha = g:\gamma \cup d:\delta$, then $a:\alpha \Downarrow S$ is $(!S(g); g:\gamma \Downarrow S) \cap (!S(d); d:\delta \Downarrow S)$. Thus we must show that

$$\langle \alpha \rangle \phi \models \langle (!S(g); g:\gamma \Downarrow S) \cap (!S(d); d:\delta \Downarrow S) \rangle \phi.$$

By the dGL axioms $\langle ; \rangle$ and $[\cup]$, this is

$$\langle \alpha \rangle \phi \models (S(g) \rightarrow \langle g:\gamma \Downarrow S \rangle \phi) \wedge (S(d) \rightarrow \langle d:\delta \Downarrow S \rangle \phi).$$

But by the inductive hypothesis, we have $\langle \gamma \rangle \phi \models \langle g:\gamma \Downarrow S \rangle \phi$ and $\langle \delta \rangle \phi \models \langle d:\delta \Downarrow S \rangle \phi$. $S(g) \models \langle \gamma \rangle \phi$ and $S(d) \models \langle \delta \rangle \phi$ by the definition of a subvalue map. By the proof rules $\rightarrow R$ and WL, the following two formulas hold: $S(g) \rightarrow \langle g:\gamma \Downarrow S \rangle \phi$ and $S(d) \rightarrow \langle d:\delta \Downarrow S \rangle \phi$. Thus, we have $\langle \alpha \rangle \phi \models (S(g) \rightarrow \langle g:\gamma \Downarrow S \rangle \phi) \wedge (S(d) \rightarrow \langle d:\delta \Downarrow S \rangle \phi)$, completing the proof for this case.

- If $a:\alpha = g:\gamma^*$, then $\langle a:\alpha \Downarrow S \rangle \phi = \langle (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times; !S(\text{end}) \rangle \phi$. Applying axioms $\langle ; \rangle$ and $[?]$, this is

$$\langle (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times \rangle (S(\text{end}) \rightarrow \phi)$$

Since ϕ is a compatible Angel winning condition, $\models S(\text{end}) \rightarrow \phi$.

Thus, we need to show

$$\langle \alpha \rangle \phi \models \langle (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times \rangle (\top)$$

This we do using the loop rule with invariant $\langle \alpha \rangle \phi$. The inductive step is shown using the inductive hypothesis, i.e., $\langle \gamma \rangle \phi \models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(a)) \rangle \langle \alpha \rangle \phi$. Here, $\langle \alpha \rangle \phi$ is a compatible winning condition because $S(a) \rightarrow \langle \alpha \rangle \phi$ holds by the definition of a subvalue map.

- If $a:\alpha = g:\gamma \cap d:\delta$, then $a:\alpha \Downarrow S$ is $(g:\gamma \Downarrow S) \cap (d:\delta \Downarrow S)$. We must show that

$$\langle \gamma \cap \delta \rangle \phi \models \langle (g:\gamma \Downarrow S) \cap (d:\delta \Downarrow S) \rangle \phi.$$

By the dGL axiom $\langle \cup \rangle$, this is

$$\langle \gamma \rangle \phi \wedge \langle \delta \rangle \phi \models \langle g:\gamma \Downarrow S \rangle \phi \wedge \langle d:\delta \Downarrow S \rangle \phi.$$

Applying the proof rules $\wedge R$ and WL, the goals to show are

$$\langle \gamma \rangle \phi \models \langle g:\gamma \Downarrow S \rangle \phi \text{ and } \langle \delta \rangle \phi \models \langle d:\delta \Downarrow S \rangle \phi.$$

Both of these follow from the inductive hypotheses.

- If $a:\alpha = g:\gamma ; d:\delta$, then $a:\alpha \Downarrow S$ is $(g:\gamma \Downarrow S(\text{end} \mapsto S(d))) ; (d:\delta \Downarrow S)$. We must show that

$$\langle \gamma ; \delta \rangle \phi \models \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(d))) ; (d:\delta \Downarrow S) \rangle \phi.$$

By the dGL axiom $[\cdot]$, this is

$$\langle \gamma \rangle \langle \delta \rangle \phi \models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)) \rangle \langle d:\delta \Downarrow S \rangle \phi.$$

By the inductive hypothesis, we know that $\langle \gamma \rangle \langle \delta \rangle \phi \rightarrow \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)) \rangle S(d)$ holds because $\langle \delta \rangle \phi$ is compatible with $S(\text{end} \mapsto S(d))$, since by the definition of a subvalue map, we have $S(d) \rightarrow \langle \delta \rangle \phi$. Per inductive hypothesis, we also know that $\langle \delta \rangle \phi \models \langle d:\delta \Downarrow S \rangle \phi$. Thus by the transitivity of implication along with the monotonicity rule, we complete the proof.

- If $a:\alpha = g:\gamma^\times$, then $a:\alpha \Downarrow S$ is $(g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times$. We must show that

$$\langle \gamma^\times \rangle \phi \models \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times \rangle \phi.$$

We show this by using the dGL loop rule loop with invariant $\langle \gamma^\times \rangle \phi$. The inductive stop is shown using the inductive hypothesis, i.e.,

$$\langle \gamma \rangle \langle \gamma^\times \rangle \phi \models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(a)) \rangle \langle \gamma^\times \rangle \phi,$$

where $\langle \gamma^\times \rangle \phi$ is a compatible winning condition because $S(a) \rightarrow \langle \gamma^\times \rangle \phi$ holds by the definition of a subvalue map. The invariant holds initially by id and implies the postcondition by the dGL axiom $\langle^\times \rangle$.

□

Lemma B.6 (Inductive subvalue maps have some strategy). *For any game $a:\alpha$, Angelic inductive subvalue map S , and compatible Angel winning condition ϕ , $S(a) \models \langle a:\alpha \Downarrow S \rangle \phi$.*

Proof. This follows from structural induction, unwrapping the inductive subvalue map definition, and applying the usual dGL axioms and proof rules.

- If $a:\alpha$ is atomic and not controlled by Angel, i.e., $\alpha \in \{x := e, x := \otimes, ?Q, !Q, \{x' = f(x) \ \& \ Q\}^d\}$, then $a:\alpha \Downarrow S = a:\alpha$. In these cases, by the definition of a inductive subvalue map, $\models S(a) \rightarrow \langle a:\alpha \rangle \text{end}$. Further since ϕ is a compatible winning condition, $\models S(\text{end}) \rightarrow \phi$. By the monotonicity rule, $S(a) \models \langle a:\alpha \rangle \phi$. This allows us to conclude that $S(a) \models \langle a:\alpha \Downarrow S \rangle \phi$.
- If $a:\alpha = a:x := *,$ then $a:\alpha \Downarrow S = x := * ; ?S(\text{end})$. Thus we must show that

$$S(a) \models \langle x := * ; ?S(\text{end}) \rangle \phi.$$

By the dGL axioms $\langle ; \rangle$ and $\langle ? \rangle$, this is

$$S(a) \models \langle x := * \rangle S(\text{end}) \wedge \phi.$$

Since ϕ is a compatible Angel winning condition, $S(\text{end}) \rightarrow \phi$ holds. Applying the monotonicity rule, we must show that

$$S(a) \models \langle x := * \rangle S(\text{end}). \quad 3953$$

By the definition of an inductive subvalue map for Angelic free assignments along with the $\rightarrow R$ rule, we conclude that $S(a) \models \langle x := * \rangle S(\text{end})$.

- If $a:\alpha = a:\{x' = f(x) \ \& \ Q\}$, the proof follows exactly the same steps as the previous case.
- If $a:\alpha = g:\gamma \cup d:\delta$, then $a:\alpha \Downarrow S = (?S(g); g:\gamma \Downarrow S) \cup (?S(d); d:\delta \Downarrow S)$. Thus we must show that

$$S(a) \models \langle (?S(g); g:\gamma \Downarrow S) \cup (?S(d); d:\delta \Downarrow S) \rangle \phi. \quad 3958$$

By the dGL axioms $\langle ; \rangle$ and $\langle \cup \rangle$, this is

$$S(a) \models (S(g) \rightarrow \langle g:\gamma \Downarrow S \rangle \phi) \vee (S(d) \rightarrow \langle d:\delta \Downarrow S \rangle \phi). \quad 3961$$

By the definition of an inductive subvalue map, we have $\models S(a) \rightarrow S(g) \vee S(d)$. Thus by proof rules cut, $\vee L$, $\vee R$, and WR it suffices to show the following two goals:

$$S(g) \models S(g) \rightarrow \langle g:\gamma \Downarrow S \rangle \phi \text{ and } S(d) \models S(d) \rightarrow \langle d:\delta \Downarrow S \rangle \phi. \quad 3964$$

Applying the rule $\rightarrow R$, the remaining goals hold per the inductive hypotheses, $S(g) \models \langle g:\gamma \Downarrow S \rangle \phi$ and $S(d) \models \langle d:\delta \Downarrow S \rangle \phi$.

- We show the case of Angel loop, which follows directly from the definition of an inductive subvalue map. If $a:\alpha = g:\gamma^*$, then by the definition of an inductive subvalue map, $S(a) \models \langle a:\alpha \Downarrow S \rangle S(\text{end})$. Since ϕ a compatible winning condition, we have $S(\text{end}) \models \phi$. Using dGL rule M, we can conclude that $S(a) \models \langle a:\alpha \Downarrow S \rangle \phi$.
- If $a:\alpha = g:\gamma; d:\delta$, then $a:\alpha \Downarrow S = g:\gamma \Downarrow S(\text{end} \mapsto S(d)); d:\delta \Downarrow S$. Thus we must show that

$$S(a) \models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)); d:\delta \Downarrow S \rangle \phi. \quad 3972$$

By the dGL axioms $\langle ; \rangle$, this is

$$S(a) \models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)) \rangle \langle d:\delta \Downarrow S \rangle \phi. \quad 3974$$

By the inductive hypothesis along with the fact that $\models S(a) \rightarrow S(g)$ from the definition of inductive subvalue maps,

$$S(a) \models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)) \rangle S(d). \quad 3977$$

Additionally, by the inductive hypothesis,

$$S(d) \models \langle d:\delta \Downarrow S \rangle \phi. \quad 3979$$

Applying the M rule, we can conclude that

$$S(a) \models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)) \rangle \langle d:\delta \Downarrow S \rangle \phi, \quad 3981$$

completing the proof for this case.

3983 • If $a:\alpha = g:\gamma \cap d:\delta$, then $a:\alpha \Downarrow S = g:\gamma \Downarrow S \cap d:\delta \Downarrow S$. Thus we must show that

$$3984 \quad S(a) \models \langle g:\gamma \Downarrow S \cap d:\delta \Downarrow S \rangle \phi.$$

3985 By the dGL axiom $\langle \cap \rangle$, this is

$$3986 \quad S(a) \models \langle g:\gamma \Downarrow S \rangle \phi \wedge \langle d:\delta \Downarrow S \rangle \phi.$$

3987 By the definition of an inductive subvalue map, we have $S(a) \models S(g) \wedge S(d)$. Applying
3988 the proof rules $\wedge R$, cut, $\wedge L$ and WL, the goals to show are

$$3989 \quad S(g) \models \langle g:\gamma \Downarrow S \rangle \phi \text{ and } S(d) \models \langle d:\delta \Downarrow S \rangle \phi.$$

3990 Both of these follow from the inductive hypotheses, completing the proof for this case.

3991 • If $a:\alpha = g:\gamma^\times$, then $a:\alpha \Downarrow S = (g:\gamma \Downarrow (S(\text{end} \mapsto S(a))))^\times$. Thus we must show that

$$3992 \quad S(a) \models \langle (g:\gamma \Downarrow (S(\text{end} \mapsto S(a))))^\times \rangle S(\text{end}).$$

3993 We use the loop rule with invariant $S(a)$. $S(a)$ holds initially by assumption. We show
3994 that it holds inductively, i.e., that

$$3995 \quad S(a) \models \langle g:\gamma \Downarrow (S(\text{end} \mapsto S(a))) \rangle S(a).$$

3996 By the definition of an inductive subvalue map, $\models S(a) \rightarrow S(g)$. Thus, after applying cut,
3997 we must show

$$3998 \quad S(g) \models \langle g:\gamma \Downarrow (S(\text{end} \mapsto S(a))) \rangle S(a).$$

3999 This holds by the inductive hypothesis. Finally, the invariant implies the postcondition by
4000 the definition of an inductive subvalue map, which requires that $\models S(a) \rightarrow S(\text{end})$.

4001 □

4002 **Lemma B.7** (Inductive subvalue maps win). *For any game $a:\alpha$, Angelic inductive subvalue map*
4003 *S , and compatible Angel winning condition ϕ , $S(a) \models \langle a:\alpha \Downarrow S \rangle \phi$.*

4004 *Proof.* Follows from Lemma B.5 because an inductive subvalue map is a subvalue map (Theo-
4005 rem 6.3) and $S(a) \rightarrow \langle \alpha \rangle \phi$ holds per the definition of a subvalue map. □

4006 B.6 Additional Definitions

4007 This appendix provides additional definitions and constructions including the Demonic version
4008 for definitions whose Angelic subvalue map version is already defined in the main text, and dGL
4009 operational semantics concepts.

B.6.1 dGL Action Semantics

4010

dGL actions can be interpreted as transformations from old state to new state. The definition of the functions is a part of dGL operational semantics [120][Appendix C], but we recall the definitions in Eq. (B.4), extended to labels and the extra action for sequence that this chapter uses in its presentation.

4014

$$\begin{aligned}
 \llbracket (x := e) \rrbracket_\sigma &= \llbracket (x := e)^d \rrbracket_\sigma = \sigma(x \mapsto e) \quad \llbracket \text{act} \rrbracket_\sigma = \sigma \text{ for } \text{act} \in \{\mathbf{g}, \mathbf{s}, \mathbf{l}, \mathbf{r}, \mathbf{c}, \mathbf{g}^d, \mathbf{s}^d, \mathbf{l}^d, \mathbf{r}^d, \text{label}\} \\
 \llbracket (?Q) \rrbracket_\sigma &= \begin{cases} \sigma & \text{if } \sigma \in \llbracket Q \rrbracket \\ \text{not defined} & \text{otherwise} \end{cases} \quad \llbracket (!Q) \rrbracket_\sigma = \begin{cases} \sigma & \text{if } \sigma \in \llbracket Q \rrbracket \\ \text{not defined} & \text{otherwise} \end{cases} \\
 \llbracket (\{x' = f(x) \& Q @ t\}) \rrbracket_\sigma &= \llbracket (\{x' = f(x) \& Q @ t\}^d) \rrbracket_\sigma = \\
 &\quad \varphi(t) \text{ for the unique (differentiable) } \varphi : [0, t] \rightarrow \text{States}, \varphi(0) = \sigma, \\
 &\quad \text{for all } s \in [0, t] \left(\frac{d\varphi(r)(x)}{dr}(s) = f(\varphi(s)(x)) \wedge \varphi(s) \models Q \right). \\
 &\quad \text{Not defined if no such } \varphi \text{ exists.}
 \end{aligned}$$

(B.4) 4015

B.6.2 Labeled Game Trees

4016

We construct the labeled game tree, extending the operational game tree of dGL [120][Appendix C] with labels. The operator $\hat{}$ denotes concatenation under prefix closure. That is:

4017

1. For two actions/labels act_1 and act_2 , $\{\text{act}_1 \hat{} \text{act}_2\}$ is the set $\{\text{act}_1, \text{act}_1 \cdot \text{act}_2\}$.
2. For an action act and a sequence of actions b , $\{\text{act} \hat{} b\}$ is the set $\{\text{act}, \text{act} \cdot b\}$.
3. For two actions/labels act_1 and act_2 , and sequence b , $\{\text{act}_1 \hat{} \text{act}_2 \hat{} b\}$ is the set $\{\text{act}_1, \text{act}_1 \cdot \text{act}_2, \text{act}_1 \cdot \text{act}_2 \cdot b\}$.
4. For an action act and a set of sequences $\{a, b\}$, $\{\text{act} \hat{} \{a, b\}\}$ is the set $\{\text{act}, \text{act} \cdot a, \text{act} \cdot b\}$.

4019

4020

4021

4022

4023

Thus, $\hat{}$ is overloaded to denote the concatenation of actions, sequences, and sets of sequences under prefix closure.

4025

Definition B.2 (Labeled Semantics). For any game $a:\alpha$ and state σ , the labeled semantics $g(a:\alpha)(\sigma)$ is defined as follows, where $\hat{}$ denotes concatenation under prefix closure (Appendix B.6.2).

4026

4027

4028

$$\begin{aligned}
 \mathcal{T}(x := \theta, \sigma) &\equiv \{a \hat{} x := \theta\} \\
 \mathcal{T}(x := \otimes, \sigma) &\equiv \{a \hat{} x := \theta^d : \theta \in \mathbb{R}\} \\
 \mathcal{T}(x := *, \sigma) &\equiv \{a \hat{} x := \theta : \theta \in \mathbb{R}\} \\
 \mathcal{T}(\{x' = f(x) \& Q\}, \sigma) &\equiv \{a \hat{} (x' = f(x) \& Q @ r) : r \in \mathbb{R}, r \geq 0, \varphi(0) = \sigma \text{ for some} \\
 &\quad \text{(differentiable) } \varphi : [0, r] \rightarrow \mathcal{S} \text{ such that } \frac{d\varphi(t)(x)}{dt}(\zeta) = \llbracket \theta \rrbracket_{\varphi(\zeta)} \\
 &\quad \text{and } \varphi(\zeta) \in \llbracket Q \rrbracket^I \text{ for all } \zeta \leq r\} \\
 \mathcal{T}(?Q, \sigma) &\equiv \{a \hat{} ?Q\}
 \end{aligned}$$

4029

4030

4031

4032

4033

4034

4035

$$\begin{aligned}
\mathcal{T}(!Q, \sigma) &\equiv \{a^{\wedge}!Q\} \\
\mathcal{T}(g:\gamma \cup d:\delta, s) &\equiv \{a^{\wedge}l^{\wedge}b : b \in g(g:\gamma)(\sigma)\} \cup \\
&\quad \{a^{\wedge}r^{\wedge}b : b \in g(d:\delta)(\sigma)\} \\
\mathcal{T}(g:\gamma \cap d:\delta, \sigma) &\equiv \{a^{\wedge}l^d b : b \in g(g:\gamma)(\sigma)\} \cup \\
&\quad \{a^{\wedge}r^d b : b \in g(d:\delta)(\sigma)\} \\
\mathcal{T}(g:\gamma; d:\delta, \sigma) &\equiv a^{\wedge}c^{\wedge}\mathcal{T}(\gamma, g) \cup \bigcup_{t \in \text{leaf}(a^{\wedge}\mathcal{T}(\gamma, g))} \mathcal{T}(d:\delta, [t]_{\sigma}) \\
\mathcal{T}((g:\gamma)^*, \sigma) &\equiv \bigcup_{n < \omega} f^n(\{(a^{\wedge}s), (a^{\wedge}g)\}) \\
&\quad \text{where } f^n \text{ is the } n\text{-fold composition of the function} \\
f(Z) &\stackrel{\text{def}}{=} Z \cup \bigcup_{t^{\wedge}g \in \text{leaf}(Z)} t^{\wedge}g^{\wedge}\mathcal{T}(\alpha, [t^{\wedge}g]_{\sigma})^{\wedge}\{(a^{\wedge}s), (a^{\wedge}g)\} \\
\mathcal{T}((g:\gamma)^{\times}, \sigma) &\equiv \bigcup_{n < \omega} f^n(\{(a^{\wedge}s^d), (a^{\wedge}g^d)\}) \\
&\quad \text{where } f^n \text{ is the } n\text{-fold composition of the function} \\
f(Z) &\stackrel{\text{def}}{=} Z \cup \bigcup_{t^{\wedge}g^d \in \text{leaf}(Z)} t^{\wedge}g^d^{\wedge}\mathcal{T}(\alpha, [t^{\wedge}g^d]_{\sigma})
\end{aligned}$$

Leaves are the nodes of a tree with no children, i.e., any sequence l such that there is no sequence $l \cdot s$ where s is a single action/label in the tree.

B.6.3 Label-free Game Trees

To get back unlabeled dGL games operational semantics, we erase labels from the labeled semantics.

Definition B.3 (Erasing Labels). The function π_{act} takes in a node or game tree and returns it with all subgame labels removed. First we define erasing labels for a single node. Here, π_{act} accepts a node s (which is a sequence) and returns a label-free node (the sequence with labels dropped).

$$\pi_{\text{act}}(s) \equiv (a_i : a_i \in s, a_i \notin \text{subgame labels})$$

Erasing labels for a game tree t proceeds by erasing labels for every node, producing a label-free tree.

$$\pi_{\text{act}}(t) \equiv \{\pi_{\text{act}}(s) : s \in t \text{ and } \pi_{\text{act}}(s) \neq ()\}$$

If erasing labels results in an empty sequence $()$, it is discarded.

Definition B.4 (Label-free Semantics). The label-free game tree of a dGL game $a:\alpha$ results from erasing labels in the labeled semantics, i.e., $\pi_{\text{act}}(\mathcal{T}(a:\alpha, \sigma))$.

B.6.4 Demonic Subvalue Projection

Definition B.5 (Demonic subvalue projection). The *projection* of Demonic subvalue map S onto dGL game $a:\alpha$ with Demon winning condition ϕ , written $a:\alpha \Downarrow S$, is generated recursively per the structure of $a:\alpha$ as follows. If $a:\alpha$ has structure:

$$\begin{aligned}
 & a:x := \times \text{ then } x := \times ; !\phi. \\
 & a:\{x' = f(x) \ \& \ Q\}^d \text{ then } \{x' = f(x) \ \& \ Q\}^d ; !\phi \\
 & a:(g:\gamma \cap d:\delta) \text{ then } (!S(g) ; g:\gamma \Downarrow S) \cap (!S(d) ; d:\delta \Downarrow S) \\
 & a:(g:\gamma)^\times \text{ then } (!S(g) ; g:\gamma \Downarrow S)^\times ; !\phi \\
 & a:(g:\gamma ; d:\delta) \text{ then } g:\gamma \Downarrow S ; d:\delta \Downarrow S \\
 & a:(g:\gamma \cup d:\delta) \text{ then } g:\gamma \Downarrow S \cup d:\delta \Downarrow S \quad a:(g:\gamma)^* \text{ then } g:\gamma \Downarrow S^* \\
 & \text{atomic and not controlled by Demon, i.e.,} \\
 & \alpha \in \{x := e, x := \otimes, ?Q, !Q, \{x' = f(x) \ \& \ Q\}^d\}, \text{ then } \alpha
 \end{aligned}$$

B.6.5 Universal Projection of Demonic Subvalue Maps

Definition B.6 (Universal Projection). The *universal projection* of Demonic subvalue map S onto dGL game $a:\alpha$ with Demon winning condition ϕ , written $a:\alpha \Downarrow S$, is generated recursively per the structure of $a:\alpha$ as follows. If $a:\alpha$ has structure:

$$\begin{aligned}
 & a:x := \otimes \text{ then } !(\exists x \phi) ; x := * ; ?\phi. \\
 & a:\{x' = f(x) \ \& \ Q\}^d, \text{ then } !\langle \alpha \rangle \phi ; \{x' = f(x) \ \& \ Q\} ; ?\phi. \\
 & a:(g:\gamma \cap d:\delta), \text{ then } !(S(g) \vee S(d)) ; ((?S(g) ; g:\gamma \Downarrow S) \cup (?S(d) ; d:\delta \Downarrow S)). \\
 & a:(g:\gamma)^\times, \text{ then } !(S(g) \vee \phi) ; (?S(g) ; \Downarrow \gamma S ; ?S(a) \vee \phi)^* ; ?\phi. \\
 & a:(g:\gamma ; d:\delta), \text{ then } g:\gamma \Downarrow S ; d:\delta \Downarrow S. \\
 & a:(g:\gamma \cap d:\delta), \text{ then } g:\gamma \Downarrow S \cap d:\delta \Downarrow S. \\
 & a:(g:\gamma)^\times, \text{ then } g:\gamma \Downarrow S^\times. \\
 & \text{atomic and not controlled by Demon, i.e.,} \\
 & \alpha \in \{x := e, x := *, ?Q, !Q, \{x' = f(x) \ \& \ Q\}^d\}, \text{ then } \alpha.
 \end{aligned}$$

B.6.6 Strategy Set Generation

Def. 6.4 shows how an agent's Subvalue map lets it decide what actions to take at a given state. This section shows how to generate a set of strategies (as defined in dGL operational semantics) given a subvalue map by tracing through the game while keeping track of which strategy leads to what state, and using the policy to predict what actions to take at decision points.

Definition B.7 (Transformation to Strategy Set). The function $\mathcal{S}_{a,\alpha}(S)(b, \sigma)$ produces the set of Angel strategies to play game $a:\alpha$ following subvalue map S starting at subgame b , where σ represents the current state, where $\hat{\cdot}$ is prefix-closed concatenation as shown in Appendix B.6.2.

4098 If the structure of b is:

4099 $b.x := e$ then $\{\{(x := e)\}\}$
 4100 $b.x := *$ then $\{p : p \in \mathcal{P}_{a,\alpha}(S)(b, \sigma)\}$
 4101 $b.x := \otimes$ then $\{\{(x := e^d) : e \in \mathbb{R}\}\}$
 4102 $b.\{x' = f(x) \ \& \ Q\}$ then $\{\{p\} : \mathcal{P}_{a,\alpha}(S)(b, \sigma)\}$
 4103 $b.\{x' = f(x) \ \& \ Q\}^d$ then
 4104 $\{\{(x' = f(x) \ \& \ Q^d @ t) : t \in \mathbb{R} \text{ where}$
 4105 $\varphi(0) = \sigma(x) \text{ and } \forall s \in [0, t] \varphi'(s) = f(\varphi(s))\}\}$
 4106 $b.?Q$ then $\{\{(?Q)\}\}$
 4107 $b.!Q$ then $\{\{(!Q)\}\}$
 4108 $b.(g:\gamma \cup d:\delta)$ then
 4109
$$\begin{cases} \{\} & \text{if } \mathcal{P}_{a,\alpha}(S)(b) = \{\} \\ \{\{l^{\wedge} t\} : t \in \mathcal{S}_{g:\gamma}(S)(g, \sigma)\} & \text{if } \mathcal{P}_{a,\alpha}(S)(b) = \{l\} \\ \{\{r^{\wedge} t\} : t \in \mathcal{S}_{d:\delta}(S)(d, \sigma)\} & \text{if } \mathcal{P}_{a,\alpha}(S)(b) = \{r\} \\ \{\{l^{\wedge} t\} : t \in \mathcal{S}_{g:\gamma}(S)(g, \sigma)\} \cup & \text{otherwise if } \mathcal{P}_{a,\alpha}(S)(b) = \{l, r\} \\ \{\{r^{\wedge} t\} : t \in \mathcal{S}_{d,\sigma:\delta}(S)(d)\} \end{cases}$$

 4110 $b.(g:\gamma \cap d:\delta)$ then $\{l^d \wedge t_l \cup r^d \wedge t_r : t_l \in \mathcal{S}_{g:\gamma}(S)(g, \sigma) \text{ and } t_r \in \mathcal{S}_{d:\delta}(S)(d, \sigma)\}$
 4111 $b.(g:\gamma ; d:\delta)$ then $\{t \cup \bigcup_{v \in \text{leaf}(t)} (v \wedge u) : t \in \mathcal{S}_{g:\gamma}(S(\text{end} \mapsto S(d)))(g, \sigma) \text{ and}$
 4112 $u \in \mathcal{S}_{d:\delta}(S)(d)(\lfloor v \rfloor_{\sigma})\}$
 4113 $b.(g:\gamma)^*$ then $\bigcup_{n < \omega} \{t : t \in f^n(\mathcal{P}_{a,\alpha}(S)(b, \sigma)) \text{ and } \nexists u (u \wedge g \in \text{leaf}(t))\}$
 4114 where f^n is n -fold composition of the function :
 4115
$$f(Z) = \bigcup_{t \in Z} \{t \cup \bigcup_{u \wedge g \in \text{leaf}(t)} \begin{cases} \{u \wedge g \wedge v_u\} & \text{if } \lfloor u \wedge g \wedge v_u \rfloor_{\sigma} = \text{undefined} \\ \{u \wedge g \wedge v_u \wedge o_{uv}\} & \text{otherwise} \end{cases} \} :$$

 4116 $v_u \in \mathcal{S}_{g:\gamma}(S(\text{end} \mapsto S(a)))(g)(\lfloor u \rfloor_{\sigma})$ and
 4117 $o_{uv} \in \mathcal{P}_{a,\alpha}(S)(b, \lfloor u \wedge g \wedge v_u \rfloor_{\sigma})\}$
 4118 $b.(g:\gamma)^{\times}$ then $\bigcup_{n < \omega} f^n(\{\{\mathfrak{s}^d, \mathfrak{g}^d\}\})$
 4119 where f^n is n -fold composition of the function :
 4120
$$f(Z) = \bigcup_{t \in Z} \{t \cup \bigcup_{u \wedge \mathfrak{g}^d \in \text{leaf}(t)} \{u \wedge \mathfrak{g}^{d \wedge} v^{\wedge} \{\mathfrak{s}^d, \mathfrak{g}^d\} : v \in \mathcal{S}_{a,\alpha}(S)(g)(\lfloor u \rfloor_{\sigma})\}\}$$

4121 with implicit closure under prefixes.

4122 To reason only about the states reachable by playing the game trees, we should introduce a
 4123 version of the game trees free of actions that do nothing.

Definition B.8 (Erasing skip Actions). Let the set of skip actions $\mathfrak{S}$ be

$$\mathfrak{S} = \{g, s, l, r, c, g^d, s^d, l^d, r^d\}$$

. The function $\pi_{-\mathfrak{S}}$ takes in a node or game tree and returns it with all labels and skip actions removed. First we define $\pi_{-\mathfrak{S}}$ for a single node. Here, $\pi_{-\mathfrak{S}}$ accepts a node s (which is a sequence) and returns a node (the sequence with labels dropped).

$$\pi_{-\mathfrak{S}}(s) \equiv (a_i : a_i \in s, a_i \notin \text{subgame labels and } a_i \notin \mathfrak{S})$$

Erasing labels and skip actions for a game tree t proceeds by erasing these for every node, producing a new tree.

$$\pi_{-\mathfrak{S}}(t) \equiv \{\pi_{-\mathfrak{S}}(s) : s \in t \text{ and } \pi_{-\mathfrak{S}}(s) \neq ()\}.$$

B.6.7 Inductive Demonic Subvalue Maps

Definition B.9 (Inductive Demonic subvalue maps). Let S be a map from the subgames of $a:\alpha$ to winning subregions. S is an *inductive Demonic subvalue map* for game $a:\alpha$, written $a:\alpha \Vdash S$, when the following holds. If $a:\alpha$ has structure:

atomic, i.e., $\alpha \in \{x := e, x := *, x := \otimes, ?Q, !Q, \{x' = f(x) \ \& \ Q\}, \{x' = f(x) \ \& \ Q\}^d\}$ then $\models S(a) \rightarrow [\alpha]S(\text{end})$.

$a:(g:\gamma \cup d:\delta)$ then $\models S(a) \rightarrow S(g) \wedge S(d)$ and $g:\gamma \Vdash S$ and $d:\delta \Vdash S$.

$a:(g:\gamma \cap d:\delta)$ then $\models S(a) \rightarrow S(g) \vee S(d)$ and $g:\gamma \Vdash S$ and $d:\delta \Vdash S$.

$a:(g:\gamma ; d:\delta)$ then $\models S(a) \rightarrow S(g)$ and $g:\gamma \Vdash S(\text{end} \mapsto S(d))$ and $d:\delta \Vdash S$.

$a:(g:\gamma)^*$ then $\models S(a) \rightarrow S(\text{end}) \wedge S(g)$ and $g:\gamma \Vdash S(\text{end} \mapsto S(a))$.

$a:(g:\gamma)^\times$ then $\models S(a) \rightarrow [a:\alpha \Vdash S]S(\text{end})$ and $g:\gamma \Vdash S(\text{end} \mapsto S(a))$.

B.6.8 Game Prefix

This definition uses *the empty game skip* which can be treated like the game $?true$ in terms of effect on plays, except with no corresponding action in the operational semantics.

Definition B.10 (Game Prefix). The game suffix $a:\alpha_b$ of subgame $b:\beta$ in $\text{subgames}(a:\alpha)$ is constructed as below. If $a:\alpha$ is $b:\beta$ and α is not a loop, then $a:\alpha_b$ is the empty game, **skip**. If $b = a$ and $a:\alpha$ is a loop, then $a:\alpha_b$ is $a:\alpha$. If b is **end** then $a:\alpha_b$ is the entire game $a:\alpha$. Otherwise, if $a:\alpha$ has structure:

$$\begin{aligned} & a:(g:\gamma)^* \text{ or } a:(g:\gamma)^\times \text{ then } a:\alpha_b = a:\alpha ; (g:\gamma_b) \\ & a:(g:\gamma \cup d:\delta) \text{ or } a:(g:\gamma \cap d:\delta) \text{ then } \begin{cases} a:\alpha_b = g:\gamma_b & b \in \text{subgames}(g:\gamma) \\ a:\alpha_b = d:\delta_b & \text{otherwise} \end{cases} \\ & a:(g:\gamma ; d:\delta) \text{ then } \begin{cases} a:\alpha_b = g:\gamma_b & b \in \text{subgames}(g:\gamma) \\ a:\alpha_b = g:\gamma ; d:\delta_b & \text{otherwise} \end{cases} \end{aligned}$$

4152 **B.6.9 Solving Function for Inductive Demonic Subvalue Maps**

4153 $\llbracket \text{map}(a:\alpha, \phi) \rrbracket$, defined below, computes an inductive Demonic subvalue map for game $a:\alpha$ and
 4154 Demon winning condition ϕ . Operator $\uplus$ denotes the disjoint union of two subvalue maps.

$$\begin{aligned} \llbracket \text{map}(a:\alpha, \phi) \rrbracket &:= \{a \mapsto \text{simp1}([\alpha]\phi), \text{end} \mapsto \phi\} \text{ where } \alpha \in \{x := e, x := *, x := \otimes, \\ &\quad ?Q, !Q, \{x' = f(x) \& Q\}, \{x' = f(x) \& Q\}^d\} \\ \llbracket \text{map}(a:(g:\gamma \cap d:\delta), \phi) \rrbracket &:= S_1 \uplus S_2 \uplus \{a \mapsto S_1(g) \vee S_2(d)\} \\ &\quad \text{where } S_1 := \llbracket \text{map}(g:\gamma, \phi) \rrbracket, S_2 := \llbracket \text{map}(d:\delta, \phi) \rrbracket \setminus \text{end} \\ \llbracket \text{map}(a:(g:\gamma \cup d:\delta), \phi) \rrbracket &:= S_1 \uplus S_2 \uplus \{a \mapsto S_1(g) \wedge S_2(d)\} \\ &\quad \text{where } S_1 := \llbracket \text{map}(g:\gamma, \phi) \rrbracket, S_2 := \llbracket \text{map}(d:\delta, \phi) \rrbracket \setminus \text{end} \\ 4155 \llbracket \text{map}(a:(g:\gamma; d:\delta), \phi) \rrbracket &:= S_1 \uplus S_2 \uplus \{a \mapsto S_2(g)\} \\ &\quad \text{where } S_1 := \llbracket \text{map}(d:\delta, \phi) \rrbracket, S_2 := \llbracket \text{map}(g:\gamma, S_1(d)) \rrbracket \setminus \text{end} \\ \llbracket \text{map}(a:(g:\gamma)^\times, \phi) \rrbracket &:= S \uplus \{a \mapsto \text{Inv}, \text{end} \mapsto \phi\} \text{ if } \langle a:\alpha \vdash S \rangle \phi \\ &\quad \text{where } S := \llbracket \text{map}(g:\gamma, \text{Inv} \vee \phi) \rrbracket \setminus \text{end} \\ \llbracket \text{map}(a:(g:\gamma)^*, \phi) \rrbracket &:= S \uplus \{a \mapsto \text{Inv}, \text{end} \mapsto \phi\} \text{ if } \models \text{Inv} \rightarrow S(g) \wedge \phi \\ &\quad \text{where } S := \llbracket \text{map}(g:\gamma, \text{Inv}) \rrbracket \setminus \text{end} \end{aligned}$$

Appendix C

4156

Automatic Verification and Synthesis (Appendix)

4157

4158

C.1 Details of Verification Experiments

4159

We report further details on the verification experiments. All experiments consist of two sets of four parallel, independent pipeline runs (where each pipeline run consists of many iterations of the inner verification loop). These four parallel runs terminate only when any one of them succeeds, or when they collectively run out of a twelve dollar budget. The choice of twelve dollars is because beyond this point, we find reasoning to diverge, suggesting limited benefit to further iterations of the proving loop. Thus for each case study the verification experiment allows eight attempts with a total budget of twenty four dollars.

4160

4161

4162

4163

4164

4165

4166

Summing over the eight runs, we list the number of LLM calls made and the dollar budget consumed for LLM calls in Table C.1. Smaller numbers indicate that the proof was reached early. The 12.00 in the Lotka Volterra case study indicates that the proof was only reached in the second set of runs. We found that the time taken to run was dominated by LLM reasoning model response time.

4167

4168

4169

4170

4171

Model	Number of LLM Calls	Dollar Cost	Time (minutes)
Coolant	89 + 64	6.9 + 5.8	47.0+39.9
Train	87 + 63	7.0 + 5.2	43.9+31.4
Lotka Volterra	176 + 57	12.0 + 3.6	93.0+27.5
Chemical Reaction	31 + 49	2.0 + 3.7	16.7+31.7
Van der Pol	44 + 59	3.3 + 5.5	26.9+41.3

Table C.1: Verification experiments: Number of LLM calls, time to run, and dollar cost for two sets of four parallel runs per case study.

In the ablation where we disable the summary step, only the Chemical Reaction and Van der Pol case studies complete successfully. For Chemical Reaction, the number of LLM calls actually decreases, while for Van der Pol, it increases (Table C.2).

4172

4173

4174

Model	Number of LLM Calls	Dollar Cost
Chemical Reaction	28 + 24	2.7 + 2.4
Van der Pol	51 + 99	4.9 + 10.3

Table C.2: Verification experiments without summary step: Number of LLM calls and dollar cost for two sets of four parallel runs per case study. The costs for all other case studies exceeds the budget.

C.2 Tactic for Model 6

The tactic that the LLM discovers to prove Model 6 is as follows:

```

4177 unfold;
4178 loop("x>=xmin & y>=ymin & x<=g/d & y<=a/b & g/d>=xmin &
4179     a/b>=ymin & a>0 & b>0 & d>0 & g>0", 1); <(
4180     /* init */
4181     auto; print("Init subgoal after auto"),
4182     /* step */
4183     unfold; unfold;
4184     print("Step: after exposing existentials/diamond");
4185     /* witness for xadd and discharge ?xadd>=0 */
4186     existsR(1, "g/d - x");
4187     unfold; <(
4188         QE("Z3"); print("Step: discharged ?xadd>=0"),
4189         /* continue with yadd */
4190         unfold; existsR(1, "a/b - y");
4191         unfold; <(
4192             QE("Z3"); print("Step: discharged ?yadd>=0"),
4193             /* push deterministic assignments into box over ODE */
4194             unfold; unfold;
4195             print("Step: after pushing assignments");
4196             /* split post x>=xmin & y>=ymin */
4197             boxAnd(1); andR(1); <(
4198                 /* prove [ODE] x>=xmin */
4199                 dC("x=g/d & y=a/b", 1); <(
4200                     /* use: with equality in domain, finish */
4201                     cut("1=1"); <(
4202                         dW(1); QE("Z3"),
4203                         QE("Z3") using "1=1"
4204                     ),
4205                     /* show: [{LV}] (x=g/d & y=a/b) via dRI */
4206                     dC("d*x - g = 0 & a - b*y = 0", 1); <(
4207                         /* use: from q1=0 & q2=0 derive result */
4208                         cut("1=1"); <(

```

```

        dW(1); QE("Z3"),
        QE("Z3") using "1=1"
    ),
    /* show: [{LV}](q1=0 & q2=0) */
    boxAnd(1); andR(1); <(
        dRI(1); QE("Z3"),
        dRI(1); QE("Z3")
    )
)
)
/* prove [ODE] y>=ymin: symmetric */
dC("x=g/d & y=a/b", 1); <(
    cut("1=1"); <(
        dW(1); QE("Z3"),
        QE("Z3") using "1=1"
    ),
    /* same fix for the show branch here */
    dC("d*x - g = 0 & a - b*y = 0", 1); <(
        cut("1=1"); <(
            dW(1); QE("Z3"),
            QE("Z3") using "1=1"
        ),
        boxAnd(1); andR(1); <(
            dRI(1); QE("Z3"),
            dRI(1); QE("Z3")
        )
    )
)
)
)
/* post */
auto; print("Post subgoal after auto")
)

```

Here, the seemingly redundant `cut("1=1")` steps are a workaround to a bug in KeY-
maera X that incorrectly prevents application of `dW` on certain subgoal shapes. The print state-
ments are for the LLM to track progress and get information about the proof state in case of
failure. The use of `QE("Z3")` indicates that the Z3 SMT solver is used to discharge the corre-
sponding proof obligation. In the prompt, the LLM is encouraged to break down the proof rather
than using `auto` so that the proof remains stable despite changes to the automation and quicker to
check.

Model 23 Case study: Coolant System. ϕ_{init} is a long formula listed in Fig. C.1. It is not provided in the synthesis variant of the problem, and must be computed.

assume	1	$\phi_{\text{init}} \wedge T > 0 \wedge F > 0 \wedge a_{\text{min}} > 0 \wedge d_{\text{max}} > 0 \wedge c_1 > 0 \wedge c_2 > 0 \rightarrow$
ctrl	2	$\langle (?\text{absbd} < a_{\text{min}}^d ; \text{flow} := F) \cap (\text{absbd} \geq a_{\text{min}}^d ; \text{flow} := 0) ;$
plant	3	$\emptyset := 0 ; \left\{ \begin{array}{l} \text{absbd}' = \text{flow} \cdot c_1 \cdot \text{temp_diff}, \\ \text{disch}' = \text{flow}, \\ \text{temp_diff}' = -\text{flow} \cdot c_2 \cdot \text{temp_diff} + g, \\ \emptyset' = 1, \\ \text{deadline}' = -1 \ \& \ \emptyset \leq T \end{array} \right\}^d$
safe	4	$\rangle^\times (\text{disch} \leq d_{\text{max}} \wedge (\text{deadline} \leq 0 \rightarrow \text{absbd} \geq a_{\text{min}}))$

C.3 Other Case Studies

Case Study: Coolant. Consider a coolant system in a nuclear power plant that must achieve the control objective of absorbing *at least* a_{min} units of heat while expending *at most* d_{max} units of coolant in a given time frame. The coolant system controller chooses how much coolant to send through the system, and then goes to sleep for some amount of time bounded by maximum latency T , repeatedly polling in a control loop¹. Model 23 shows the formula modeling this system, that we must prove valid. We must reason directly in terms of symbolic parameters such as required absorption a_{min} and maximum flow rate F . The system has different control modes, can run for arbitrarily long (because of loop iterations), and dynamical behavior qualitatively changes depending on the initial temperature, all of which our reasoning will precisely account for.

Model 23 Explanation. We provide an explanation of the dGL model for this case study. Line 1 shows initial assumptions expressing physical constraints, e.g., that the maximum flow rate F is positive. Formula ϕ_{init} is a complex formula listed in Fig. C.1 representing initial conditions that make safe control possible: the discharge limit should not already have been exceeded, and if the heat absorption quota is not already met, then there is enough time and discharge capacity left to still meet it. At Line 2, the control loop begins.

The controller assigns a coolant flow rate. The choice $\text{flow} := F$ is chosen in states where the absorption quota has not yet been met ($\text{absbd} < a_{\text{min}}$), and the choice $\text{flow} := 0$ is chosen otherwise. At Line 3, the controller goes to sleep and the state evolves per the physical dynamics, where the heat absorbed is proportional to temperature difference between the coolant and the system, total coolant discharged is tracked by variable disch , the system generates heat at rate g , and the auxiliary deadline variable counts down to when the system must have met its

¹We use the typographic convention of monospaced font for variables that change over the course of a game and math font for each *symbolic parameter* P that can represent any real number which remains constant over the course of a game.

$$\begin{aligned}
\phi_{\text{inv}} \equiv & \overbrace{\text{disch} \leq d_{\max}}^{\text{discharge within limit}} \wedge \overbrace{F > 0 \wedge c_1 > 0 \wedge c_2 > 0 \wedge T > 0}^{\text{physics constraints}} \wedge (\\
& \overbrace{\text{absbd} \geq a_{\min}}^{\text{already absorbed enough}} \vee \overbrace{(\text{deadline} \geq 0)}^{\text{there's time left}} \wedge (\\
\text{Current} & \quad \overbrace{\text{temp_diff} \leq g/(F \cdot c_2)}^{\text{temp_diff only increases}} \wedge \overbrace{c_1 \cdot \text{temp_diff} \cdot (d_{\max} - \text{disch} - F \cdot T)}^{\text{can absorb enough} \dots} \\
\text{temp_diff} & \quad \dots \text{ without exceeding discharge budget} \\
\text{lower} & \quad \text{disch} - F \cdot T \geq a_{\min} - \text{absbd} \wedge \\
\text{bounds} & \quad \overbrace{\text{deadline} \cdot c_1 \cdot \text{temp_diff} \cdot F \geq a_{\min} - \text{absbd}}^{\text{can absorb enough before deadline}} \vee \\
\text{all future} & \quad \text{temp_diff} > g/(F \cdot c_2) \wedge c_1 \cdot (g/(F \cdot c_2)) \cdot \\
\text{temp_diff} & \quad \dots \text{ without exceeding discharge budget} \\
& \quad \overbrace{(d_{\max} - \text{disch} - F \cdot T) \geq a_{\min} - \text{absbd}}^{\text{temp_diff bound}} \wedge \\
& \quad \overbrace{\text{deadline} \cdot c_1 \cdot g/c_2 \geq a_{\min} - \text{absbd}}^{\text{can absorb enough before deadline}})))))
\end{aligned} \tag{C.1}$$

Figure C.1: Formula ϕ_{init} for the Coolant case study.

absorption obligations. The controller stays asleep for at most T time units, though it may wake up earlier, with variable $\emptyset$ tracking time elapsed. Finally, at Line 4, the control loop closes, with the syntax $\langle \alpha \rangle^\times$. The control loop repeats arbitrarily many times, and we must verify that all possible number of iterations are safe. The modal formula $\langle \alpha \rangle \phi$ is true when *all* possible runs result in a state satisfying postcondition ϕ . Therefore, the overall formula of Model 23 is valid when, given the assumptions of Line 1, all executions lead to a state where the control objective is satisfied, that is, $\text{disch} \leq d_{\max} \wedge (\text{deadline} \leq 0 \rightarrow \text{absbd} \geq a_{\min})$ holds. This case study is an example of a hybrid *system*, since there is no adversarial game play.

Case Study: Train. A train must stop before its *end of motion authority* e , beyond which may lie other trains. The train can brake using two mechanisms: a normal brake that decelerates at most at rate b_0 , and an air brake that ramps up deceleration gradually, as pressure leaks from brake pipes, at rate p until it reaches maximum deceleration A_{pb} . This case study is inspired by a previous case study on verification of trains with air brakes [100], but with the addition of air brakes [68] and adversarial game dynamics.

Model 24 Explanation. Model 24 shows the dGL formula modeling the train control problem via a game between Angel (the train controller) and Demon (environmental uncertainties). Angel chooses between the control modes of normal operation and air brake application (Line 5). In normal operation, she may choose any acceleration between $-b_0$ and a_0 . Then, Demon chooses how long the train will run before Angel gets to re-evaluate her decision, bounded by a maximum

Model 24 Case study: Train Control. Gray parts are not included in the synthesis variant of the problem, and must be computed. **Brown** text is not a part of the verification problem, but added to the synthesis problem.

assume	1	$e - \text{trainPos} - \frac{\text{vel}^2}{2 \cdot b_0} > 0 \wedge a_0 > 0 \wedge b_0 > 0 \wedge a_1 < 0 \wedge a_2 < 0 \wedge T > 0$
	2	$\wedge A_{pb} < 0 \wedge p < 0 \wedge \text{vel} \geq 0 \wedge m_b = 0 \wedge \text{airBrake} = 0 \rightarrow$
	3	$\langle (t := 0;$
ctrl	4	$((\text{trainAcc} := *; ? - b_0 \leq \text{trainAcc} < a_0;$
	5	$m_b := 0; \text{airBrake} := 0) \cup$
	6	$(\text{trainAcc} := -b_0; m_b := p))^d;$
plant	7	$\left(\left\{ \begin{array}{l} \text{trainPos}' = \text{vel}, \\ \text{vel}' = \text{trainAcc} + a_1 \cdot \text{vel} + a_2 \cdot \text{vel}^2 + \text{airBrake}, \\ \text{airBrake}' = m_b, \\ t' = 1 \ \& \ t \leq T \wedge \text{vel} \geq 0 \wedge \text{airBrake} \geq A_{pb} \end{array} \right\}^d \cap \right.$
	8	$\left. \left\{ \begin{array}{l} \text{trainPos}' = \text{vel}, \\ \text{vel}' = \text{trainAcc} + a_1 \cdot \text{vel} + a_2 \cdot \text{vel}^2 + \text{airBrake}, \\ \text{airBrake}' = 0, \\ t' = 1 \ \& \ t \leq T \wedge \text{vel} \geq 0 \wedge \text{airBrake} \leq A_{pb} \end{array} \right\}^d \right)^{\times}$
safe	9	$)^{\times} \rangle (e - \text{trainPos} > 0)$

Model 25 Case study: Coupled Van der Pol Oscillators. Gray parts are not included in the synthesis variant of the problem, and must be computed.

assume		1	$x_{max} > 0 \rightarrow$
ctrl		2	$\langle x1 := *; x2 := *; y1 := *; y2 := *;$
		3	$?m > 0 \wedge b > 0^d; ($
plant		4	$\left\{ \begin{array}{l} x1' = y1, \\ y1' = m \cdot (1 - x1^2) \cdot y1 + b \cdot (x2 - x1) - x1, \\ x2' = y2, \\ y2' = m \cdot (1 - x2^2) \cdot y2 - b \cdot (x2 - x1) - x2 \end{array} \right\}^d$
safe		5	$)^\times)(x1 + x2 \leq x_{max})$

time latency T (Line 7 and 8). The physical dynamics operate in two modes, depending on whether the air brakes are ramping up (Line 7) or have stabilized at their maximum (Line 8). This is modeled by ODEs for each mode within an inner loop that allows switching between them. Demon controls the number of iterations of the outer control loop (Line 3 to 8), forcing Angel to maintain safety for arbitrarily long. The overall modal formula of form $\langle \alpha \rangle \phi$ is true in states where Angel has a strategy to play so that regardless of how Demon plays, she will win, that is, the train controller has a way to maintain safety (Line 9). To reason correctly about this problem we must again reason over the nonlinear dynamics of the train, and ensure safety over unbounded time horizons. The mode switching inner-loop structure adds a further layer of complexity.

Case Study: Van der Pol. A pair of coupled Van der Pol oscillators must be set so that the sum of their positions $x1$ and $x2$ never exceeds a given bound x_{max} . The system dynamics are governed by parameters m and b . The oscillators can run arbitrarily long, and safety must be ensured throughout. This case study requires reasoning about nonlinear dynamics over unbounded time horizons. Model 25 shows the full control formula. Angel sets initial conditions at Line 2 and Line 3, then Demon chooses how long the oscillators run at Line 4, after which safety is checked at Line 5. Then Demon chooses how long to run the oscillators at Line 4, and whether to repeatedly further run them in the loop at Line 5. Angel wins in the end if the sum of the positions never exceeds x_{max} (Line 5). This case study is inspired by one appearing in ARCH-COMP [54], but is modified to add adversarial game dynamics and a loop structure that makes formal reasoning more challenging.

Case Study: Chemical Reaction. An exothermic chemical reaction $A + B \rightarrow C$ is taking place. The reaction rates depend on the temperature of the reactor $Temp$, which increases in proportion to the reaction rate. The reaction rate also depends on the concentrations of the reactants A and B . The safety requirement is that the temperature never exceeds a maximum safe temperature T_{max} . A time-triggered controller can turn the reaction on and off, getting to

Model 26 Case study: Chemical Reaction Control. Gray parts are not included in the synthesis variant of the problem, and must be computed.

assume	1	$k_A > 0 \wedge k_B > 0 \wedge k_C > 0 \wedge k_T \geq 0 \wedge T > 0 \wedge A \geq 0 \wedge B \geq 0 \wedge C \geq 0$
	2	$\wedge \text{Temp} \geq 0 \wedge (k_B \cdot A \leq k_A \cdot B \wedge k_A \cdot \text{Temp} + k_T \cdot A \leq k_A \cdot T_{max}$
	3	$\vee k_A \cdot B \leq k_B \cdot A \wedge k_B \cdot \text{Temp} + k_T \cdot B \leq k_B \cdot T_{max}) \rightarrow$
ctrl	4	$\langle ((\text{isOn} := 0 \cap \text{isOn} := 1); \emptyset := 0;$
plant	5	$\left. \begin{array}{l} A' = -A \cdot B \cdot \text{Temp} \cdot \text{isOn} \cdot k_A, \\ B' = -A \cdot B \cdot \text{Temp} \cdot \text{isOn} \cdot k_B, \\ C' = A \cdot B \cdot \text{Temp} \cdot \text{isOn} \cdot k_C, \\ \text{Temp}' = A \cdot B \cdot \text{Temp} \cdot \text{isOn} \cdot k_T, \\ \emptyset' = 1 \ \& \ \emptyset \leq T \wedge A \geq 0 \wedge B \geq 0 \wedge C \geq 0 \wedge \text{Temp} \geq 0 \end{array} \right\}^d$
safe	6	$\rangle^\times)(\text{Temp} \leq T_{max})$

revise its decision in at most every T time units. However, we do not have direct access to the controller, and instead conservatively safeguard the system by not putting in too much reactant to begin with. The verification challenge is to check whether regardless of what the controller does, safety is always ensured. The physical reactor setup in this case study is similar to one from the literature [19], but with an important change in setup: the controller is not directly accessible, and we want to verify safety for all possible controller actions.

Model 26 shows the full control formula. We make some initial assumptions about the quantities of reactant and signs of the rate constants at Line 1 and Line 2. Then Demon chooses whether to turn the reaction on or off at Line 4 in a time-triggered control loop, also controlled by Demon. In the end, Angel wins if the temperature never exceeds T_{max} (Line 6). This system again requires reasoning about complex dynamics over different, arbitrarily alternating control modes over unbounded iterations. The solution to the dynamics lies outside the decidable (polynomial) fragment of real arithmetic, requiring non-trivial deductive reasoning about differential equations via invariants.

Note: In the synthesis variants of problems, assumptions are written not in the form

$$\text{assumptions} \rightarrow \langle \alpha \rangle \text{postcondition},$$

but rather as

$$\langle ?\text{assumptions (excluding gray parts)}^d; \alpha \rangle \text{postcondition}.$$

This does not make a semantic difference, but allows subvalue map computation to account for these assumptions as it passes over the assumptions subgame, and produce a subvalue for the overall game that does not redundantly include the assumptions.

In the *synthesis* variants of the problems, the following guidelines are used:

- For *Lotka Volterra*: Set the system to its equilibrium point as soon as possible, and assume what you must so that the equilibrium point is good enough to maintain the post condition.

- For *train*: Find the control solution for when the train can stop by braking continuously, even conservatively ignoring rolling resistance and air brakes.
- For *chemical reaction*: Find the control solution for the situation where the reactant runs out before temperature can become too large.
- For *coolant*: Find the solution where initially enough heat has not been absorbed, but over time, enough is absorbed before the timer requirement or discharge capacity can be exceeded.
- For *Van der Pol*: Keep the two oscillators in antiphase, maintaining the postcondition this way.

C.4 Verification Prompts

This section lists all the prompts used for the verification case studies. The prompts use the OpenAI api format with system and user queries, and sometimes include few-shot prompts. Here we have shown the queries for the chemical reaction case study as an example. Some Unicode characters have been replaced with LaTeX commands and line breaks have been added for compilation. Prompts are listed in the order in which they are called in the verification progress, and have a one-to-one correspondence with the LLM queries depicted in Fig. 4.2.

1. AnalyzeGame is the first query in the pipeline that produces an analysis of the game used as context in later prompts. One example is shown as a few-shot example.

```
role: system
content: "Your task is to analyze a game and identify which
actions are taken by the Angel (controllable player) versus
the Demon (adversarial player)."
```

```
## Background Information
```

```
In differential game logic (dGL), there are two players:
- **Angel**: The controllable player who makes choices we can
  control. Angel constructs include:
  - Angel choices: 'a ++ b' (we choose between a and b)
  - Angel loops: '{a}*' (we choose how many iterations)
  - Angel tests: '?P' (we can proceed if P holds, otherwise we
    immediately lose)
  - Angel ODEs: '{x'=f(x) & Q}' (we control the duration)
  - Angel nondeterministic assignments: 'x := *' (we choose the
    value)
```

There is the dual operator '^@' that switches Angel and Demon roles in a game construct. Duals can be nested arbitrarily, so carefully check nesting levels to determine which player controls each action.

4382 Being within an odd number of `^@` operators indicates Demon
4383 control, while an even number (including zero) indicates Angel
4384 control.

4385 For example, in the game `{(a ++ b)^@; (c++d)}^@`, the choice
4386 between `a` and `b` is controlled by Angel, being within two
4387 `^@` operators, while the choice between `c` and `d` is
4388 controlled by Demon.

4389 Using this notation, we go over the Demon constructs below,
4390 indicating that they are Demon-controlled by wrapping them in a
4391 single `^@` operator.

4392

- 4393 - ****Demon****: The adversarial player who makes choices we cannot
4394 control. Demon has all the same constructs as Angel.
- 4395 - Demon choices: `{a ++ b}^@` (adversary chooses between a and
4396 b)
- 4397 - Demon loops: `{a\*}^@`
4398 (adversary chooses how many iterations)
- 4399 - Demon tests: `{?P}^@` (adversary can proceed if P holds, and
4400 immediately loses otherwise. So Angel can **assume** in
4401 subsequent game play that P held at this point; otherwise
4402 she would already have won.)
- 4403 - Demon ODEs: `{x'=f(x) & Q}^@` (adversary controls the
4404 duration)
- 4405 - Demon nondeterministic assignments: `{x := \*}^@` (adversary
4406 chooses the value)

4407

4408 Additionally, the non-controlled operations are:

- 4409 - Assignment: `x := e` (deterministic assignment like in regular
4410 programs)
- 4411 - Sequential composition: `a; b` (first play a, then b)

4412

4413 ## Notes

- 4414 - ODEs can sometimes introduce unmatched single quotes, so
4415 account for that when parsing.
- 4416 - In syntax such as `{ a++b; x:=\* }^@`,
4417 the `^@` applies to the entire loop including its body.
4418 In this example, Demon controls the choice within the loop
4419 between a and b, as well as the assignment to x,
4420 because both are inside one `^@`.

4421

4422 Some subgames are labeled with unique identifiers with the
4423 following syntax:

4424 <subgame_id>: <subgame>

4425

These subgame ids will help make your answers more specific. 4426

Input 4427

You will be provided a dGL game. 4428

Output 4429

Your analysis should identify: 4430

1. What actions or decisions are under Angel's control in the game. 4431
2. What actions or decisions are under Demon's control in the game. 4432
3. The control modes in the game, that is, regions of different qualitative behavior (e.g. growth vs. oscillation vs. decay vs. equilibrium). 4433

The modes are typically defined by two things: (1) initial conditions inducing different qualitative behavior in the dynamics, and (2) control actions leading to different possible values of variables. Comment about the asymptotes or polynomial invariants of each mode, when relevant. 4434

4. The overall control pattern in the game. 4435

Further Instructions 4436

Provide your analysis as clear, readable text describing the Angel and Demon actions, and control modes. Be specific about what choices each player makes and how they might interact. Structure it into the sections corresponding to the points above. 4437

Be as concise as possible while still being clear, with no superfluous or filler text or hedging. 4438

- Carefully consider the differential equation. There can be coupling between variables that affects your reasoning. 4439
- Based on the dynamics, you may some times need to come up with upper/lower bounds of quantities and use that to derive sound preconditions. It's best to use simplistic, polynomial bounds so that they verify more easily. 4440

You will now be provided with 0 or more examples of games and their analysis, followed by a new game that you must analyze." 4441

- role: user 4442

content: |- 4443

```

4470 Game to analyze:
4471 {? x >= 0 & g > 0 & air > 0 & parachute > air &
4472   T > 0 & m > 0;^@};
4473 {{subgame_c: {{{{{r := air;} ++ {r := parachute;}}^@};
4474 {t := 0;}}}; {subgame_j:
4475   {x' = v, v' = r * v ^ 2 - g, t' = 1 &
4476     x >= 0 & v < 0 & t <= T}
4477 }}*}^@}
4478
4479 Please analyze this game and identify the requested
4480 items (Angel vs. Demon control actions, control modes,
4481 overall control pattern).
4482 - role: assistant
4483 answer:
4484 mode: null
4485 content: |
4486   1) Angel-controlled actions
4487   - Choice {{{r := air;} ++ {r := parachute;}}^@
4488     | Angel chooses drag coefficient r each iteration,
4489     either r := air (low
4490     drag) or r := parachute (high drag).
4491
4492   2) Demon-controlled actions
4493   - Initial test
4494     {? x >= 0 & g > 0 & air > 0 & parachute > air &
4495       T > 0 & m > 0}^@ | Demon test. Angel can assume these
4496       preconditions hold initially; otherwise, play won't
4497       proceed.
4498   - Looping subgame_c
4499     | Demon chooses how many iterations of subgame_c
4500     to perform (including possibly zero; finite but
4501     unbounded).
4502     He chooses whether to repeat for one more cycle
4503     or stop at the end of each iteration.
4504   - ODE duration: subgame_j
4505     - Demon controls how long to flow
4506       along the ODE  $x' = v$ ,  $v' = r v^2 - g$ ,  $t' = 1$ 
4507       within the evolution domain  $x \geq 0$ ,  $v < 0$ ,  $t \leq T$ .
4508       He may stop at any time before the domain boundary
4509       is reached, including zero time; he cannot continue
4510       past any boundary (for example x cannot go below 0).
4511       Since t is reset to 0 at the start of each iteration,
4512       the duration is always capped by T
4513       (though the other boundary conditions may force

```

an earlier stop). 4514

3) Control modes (qualitative behaviors) 4515

- Mode AIR HIGH VEL 4517
 - (Angel chose $r = \text{air}$, low drag, 4518
 - $v < -(g/\text{air})^{(1/2)}$): 4519
 - $\text{air} * v^2 > g$, so $v' > 0$. 4520
 - Velocity increases (becomes less negative) 4521
 - over time, approaching terminal velocity 4522
 - $-((g / \text{air})^{(1/2)})$ from below. Will not cross it. 4523
- Mode PARACHUTE HIGH VEL 4524
 - (Angel chose $r = \text{parachute}$, high drag, 4525
 - $v < -(g/\text{parachute})^{(1/2)}$): 4526
 - $\text{parachute} * v^2 > g$, so $v' > 0$. 4527
 - Velocity increases (becomes less negative) 4528
 - over time, approaching terminal velocity 4529
 - $-((g / \text{parachute})^{(1/2)})$ from below, 4530
 - which is a higher (less negative) terminal 4531
 - velocity than in AIR mode. 4532
- Mode AIR EQU VEL 4533
 - (Angel chose $r = \text{air}$, low drag, 4534
 - and v is terminal velocity $-((g / \text{air})^{(1/2)})$): 4535
 - $v' = \text{air} * v^2 - g = 0$, 4536
 - so velocity remains constant at terminal velocity. 4537
- Mode PARACHUTE EQU VEL 4538
 - (Angel chose $r = \text{parachute}$, high drag, 4539
 - and v is terminal velocity $-((g / \text{parachute})^{(1/2)})$): 4540
 - $v' = \text{parachute} * v^2 - g = 0$, 4541
 - so velocity remains constant at terminal velocity, 4542
 - slower than in AIR mode. 4543
- Mode AIR LOW VEL 4544
 - (Angel chose $r = \text{air}$, low drag, 4545
 - and $0 > v > -((g / \text{air})^{(1/2)})$): 4546
 - $\text{air} * v^2 < g$, so $v' < 0$. 4547
 - Velocity decreases (becomes more negative) 4548
 - over time. It approaches terminal velocity 4549
 - $-((g / \text{air})^{(1/2)})$ from above but will not cross it. 4550
- Mode PARACHUTE LOW VEL 4551
 - (Angel chose $r = \text{parachute}$, high drag, 4552
 - and $0 > v > -((g / \text{parachute})^{(1/2)})$): 4553
 - $\text{parachute} * v^2 < g$, so $v' < 0$. 4554
 - Velocity decreases (becomes more negative) 4555
 - over time, approaching terminal velocity 4556
 - $-((g / \text{parachute})^{(1/2)})$ from above, 4557

```

4558         but will not cross it.
4559
4560     4) Overall control pattern
4561     - Some global preconditions hold initially ( $x \geq 0$ ,  $g > 0$ ,
4562       air  $> 0$ , parachute  $> \text{air}$ ,  $T > 0$ ,  $m > 0$ ).
4563     - Per iteration:
4564       1) Demon decides whether to continue for another
4565         iteration or stop (Demon loop).
4566       2) Angel picks drag coefficient  $r \in \{\text{air}, \text{parachute}\}$ 
4567         (Angel choice).
4568       3) Reset timer  $t := 0$  (deterministic assignment).
4569       4) Demon chooses ODE flow duration
4570         within domain (Demon ODE).
4571     - Across iterations: Demon decides how many iterations to
4572       perform and length of evolution per iteration.
4573     Angel's control lever is her binary choice of drag
4574     coefficient.
4575 - role: user
4576   content: |-
4577     Game to analyze:
4578     ( $k_A > 0 \ \& \ k_B > 0 \ \& \ k_C > 0 \ \& \ k_T \geq 0 \ \& \ T > 0 \ \&$ 
4579      $A \geq 0 \ \& \ B \geq 0 \ \& \ C \geq 0 \ \& \ \text{Temp} \geq 0 \ \&$ 
4580     ( $k_B \cdot A \leq k_A \cdot B \ \& \ k_A \cdot \text{Temp} + k_T \cdot A \leq k_A \cdot T_{\max} \mid$ 
4581      $k_A \cdot B \leq k_B \cdot A \ \& \ k_B \cdot \text{Temp} + k_T \cdot B \leq k_B \cdot T_{\max}$ ))  $\rightarrow$ 
4582     < { { { isOn:=0; ++ isOn:=1; }; t:=0;
4583         {  $A' = -A \cdot B \cdot \text{Temp} \cdot \text{isOn} \cdot k_A$ ,
4584            $B' = -A \cdot B \cdot \text{Temp} \cdot \text{isOn} \cdot k_B$ ,
4585            $C' = A \cdot B \cdot \text{Temp} \cdot \text{isOn} \cdot k_C$ ,
4586            $\text{Temp}' = A \cdot B \cdot \text{Temp} \cdot \text{isOn} \cdot k_T$ ,
4587            $t' = 1 \ \& \ t \leq T \ \& \ A \geq 0 \ \& \ B \geq 0 \ \& \ C \geq 0 \ \& \ \text{Temp} \geq 0$ 
4588         };
4589     }* }^@ > Temp<=Tmax
4590
4591     Please analyze this game and identify the requested
4592     items
4593     (Angel vs. Demon control actions, control modes, overall
4594     control pattern).
4595
4596 2. GetTactic is the second prompt in the pipeline (first in the proof loop). It gets a Bellerophon
4597 proof tactic from the LLM.
4598
4599 - role: system
4600   content: |
4601     You are trying to prove a differential game logic (dGL)
4602     theorem in KeYmaera X.

```

You must provide a Bellerophon tactic to complete the proof.

Background Information

Here is some information about dGL.

In differential game logic (dGL), there are two players:

- ****Angel****: The controllable player who makes choices we can control. Angel constructs include:
 - Angel choices: `'a ++ b'` (we choose between a and b)
 - Angel loops: `'{a}*` (we choose how many iterations)
 - Angel tests: `'?P'` (we can proceed if P holds, otherwise we immediately lose)
 - Angel ODEs: `'{x'=f(x) & Q}'` (we control the duration)
 - Angel nondeterministic assignments: `'x := *'` (we choose the value)

There is the dual operator `'^@'` that switches Angel and Demon roles in a game construct.

Duals can be nested arbitrarily, so carefully check nesting levels to determine which player controls each action.

Being within an odd number of `'^@'` operators indicates Demon control, while an even number (including zero) indicates Angel control.

For example, in the game `'{(a ++ b)^@; (c++d)}^@'`, the choice between 'a' and 'b' is controlled by Angel, being within two `'^@'` operators, while the choice between 'c' and 'd' is controlled by Demon.

Using this notation, we go over the Demon constructs below, indicating that they are Demon-controlled by wrapping them in a single `'^@'` operator.

- ****Demon****: The adversarial player who makes choices we cannot control. Demon has all the same constructs as Angel.
 - Demon choices: `'{a ++ b}^@'` (adversary chooses between a and b)
 - Demon loops: `'{a*}^@'` (adversary chooses how many iterations)
 - Demon tests: `'{?P}^@'` (adversary can proceed if P holds, and immediately loses otherwise. So Angel can **assume** in subsequent game play that P held at this

```

4645     point; otherwise she would already have won.)
4646 - Demon ODEs: `{x'=f(x) & Q}^@` (adversary controls the
4647     duration)
4648 - Demon nondeterministic assignments: `{x := *}^@`
4649     (adversary chooses the value)
4650
4651 Additionally, the non-controlled operations are:
4652 - Assignment: `x := e` (deterministic assignment like in
4653     regular programs)
4654 - Sequential composition: `a; b` (first play a, then b)
4655
4656 ## Notes
4657 - ODEs can sometimes introduce unmatched single quotes,
4658     so account for that when parsing.
4659 - In syntax such as { a++b; x:=* }*^@, the `^@` applies
4660     to the entire loop including its body. In this
4661     example, Demon controls the choice within the loop
4662     between a and b, as well as the assignment to x,
4663     because both are inside one `^@`.
4664
4665 Here is some information about Bellerophon, the tactic
4666 language in which you must output.
4667 # Bellerophon Tactic Guide (KeYmaera X 5.1)
4668
4669 ## 1. Overview
4670
4671 **KeYmaera X** proves properties of hybrid systems using
4672 differential dynamic logic (dL) or hybrid games using
4673 differential game logic (dGL).
4674 A **tactic** is a small program that automatically applies
4675 proof rules.
4676 **Bellerophon** is the language for writing these tactics.
4677 The following are instructions to write one Bellerophon
4678 script, ideally that closes the proof in one run, or at
4679 least provides new information to write such a tactic
4680 later.
4681
4682 ---
4683
4684 ## 2. Proof State Basics
4685
4686 To write good tactics, you should internally keep track
4687 of proof state. A tactic maps one proof state to another.
4688 A proof state consists of zero or more goals (subgoals).

```


Each goal is a sequent consisting of an antecedent (assumptions) and a succedent (formulas to prove).

2.1 Sequents and Antecedents in the KeYmaera X CLI

A **sequent** in the CLI is written as:

```

\ \
-1:  x>=0 GreaterEqual
-2:  y=0 Equal
==> 1:  x>=0 GreaterEqual
\ \

```

which corresponds to the following sequent in mathematical notation:

```

\ \
x>=0, y=0  |-  x>=0
\ \

```

Formulas are indexed.
Negative indices indicate antecedent formulas, and
positive indices indicate succedent formulas.

The antecedent lists assumptions.
The succedent lists formulas to prove.

We can assume all antecedent formulas are true and must use them to prove at least one succedent formula. You just need to prove one succedent formula to close the goal.

2.2 Goals and Subgoals

Tactics transform the current goal into zero or more subgoals.
You apply tactics until no subgoals remain, which would indicate the theorem is proved.

2.3 Formula Indexing

Many tactics require a formula index.
A common failure mode is using the wrong index.
Use printing (Section 3.2) to inspect the state and indices while debugging.

```

4733 ---
4734
4735
4736 ## 3. Proving Interactions
4737
4738 ### 3.1 Branches (Multiple Goals)
4739
4740 When a tactic creates multiple goals (e.g., from orL), the
4741 proof state has separate branches inside one printed
4742 block.
4743
4744 Example pattern:
4745
4746 ===== State of Hard Cases ===== ElidingProvable(Provable{
4747     -1: <formula for case 1> Or
4748     -2: <assumption>          Greater
4749     ...
4750     ==> 1: <modal or post-condition> Box
4751     from
4752     -1: <formula for first disjunct> And
4753     ...
4754     ==> 1: <some goal 1> Box
4755     with
4756     -1: <formula for remaining disjuncts> Or
4757     ...
4758     ==> 1: <some goal 2> Box}) =====
4759 print done (no progress, 1s)
4760 Done .
4761
4762 Here:
4763   • The from part shows the first branch (e.g., first
4764     disjunct, such as  $a \geq 0$  & ...).
4765   • The with part shows the other branch.
4766   • Each branch has its own sequent (==> line) with
4767     indexed formulas.
4768
4769 Each branch is a separate goal that must be closed.
4770
4771 You should check if a subgoal is closed by printing it. If
4772 the subgoal is proved, you should see something like:
4773
4774 ===== subgoal state ===== ElidingProvable(Provable{
4775     -1:  x>0 Greater
4776     ==> 1:  x>=0 GreaterEqual proved}) =====

```

```

    print done (proved, 1s)
4777
4778
### 3.2 Running and Rerunning Tactics; Printing State
4779
4780
If you know of a tactic that you want to slightly modify,
4781
simply reuse the same text, and edit it as needed.
4782
4783
To inspect the current proof state at any point, use the
4784
tactic:
4785
4786
print("State of Hard Cases")
4787
4788
This prints the current provable state (all goals and
4789
their indexed formulas) at the point where the tactic is
4790
invoked.
4791
Multiple print statements can be used in one tactic, to
4792
print the state at different points in the proof.
4793
The print message helps identify which part of the script
4794
produced that state.
4795
4796
### 3.3 Printing for Debugging
4797
4798
For debugging, when the proof state has been reduced to
4799
one or more purely arithmetic subgoals, you can:
4800
    1. Use print to display the current arithmetic
4801
       constraints.
4802
    2. Copy these constraints into an external tool such as
4803
       Z3 to search for counterexamples by checking if the
4804
       formula (AND of all antecedent constraints) -> (OR of
4805
       all succedent constraints) is valid (i.e., its
4806
       negation is unsatisfiable).
4807
4808
---
4809
4810
## 4. Bellerophon Essentials
4811
    • A script is a sequence of tactic expressions.
4812
    • t1 ; t2 runs t1 then t2.
4813
    • <(t1, t2) applies t1 to the first branch and t2 to
4814
      the second branch. Note the absence of a closing angle
4815
      bracket.
4816
    • A tactic either:
4817
        • succeeds and may produce subgoals,
4818
        • fails, or
4819
        • closes the current goal.
4820

```

```

4821
4822 ---
4823
4824 ## 5. Primitive Tactics
4825
4826 ### 5.1 Logical Tactics
4827
4828 Common examples (indices refer to formulas in the sequent):
4829 - Implication: implyR(i)
4830     This transforms a goal of form  $\vdash A \rightarrow B$  into
4831      $A \vdash B$ .
4832 - Conjunction: andR(i),
4833     This transforms a goal of form  $\vdash A \& B$  into two
4834     subgoals:  $\vdash A$  and  $\vdash B$ .
4835 - andL(i)
4836     This transforms a goal of form
4837      $A \& B, \Gamma \vdash \Delta$  into
4838      $A, B, \Gamma \vdash \Delta$ ,
4839     where the  $i$ -th antecedent formula in the original
4840     subgoal was  $A \& B$ .
4841 - Disjunction: orR
4842     This transforms a goal of form  $\vdash A \mid B$  into two
4843     subgoals:  $\vdash A$  and  $\vdash B$ .
4844 - orL(i)
4845     This transforms a goal of form
4846      $A \mid B, \Gamma \vdash \Delta$  into
4847     two subgoals:
4848      $A, \Gamma \vdash \Delta$  and  $B, \Gamma \vdash \Delta$ ,
4849     where the  $i$ -th antecedent formula in the original
4850     subgoal was  $A \mid B$ .
4851 - Universal: allR
4852     This transforms a goal of form
4853      $\vdash \forall x. P(x)$ 
4854     into  $\vdash P(y)$  for a fresh variable  $y$ .
4855 - allL(i):
4856     This transforms a goal of form
4857      $\forall x. P(x), \Gamma \vdash \Delta$ 
4858     into  $P(t),$ 
4859      $\Gamma \vdash \Delta$  for a term  $t$ , where the  $i$ -th
4860     antecedent formula in the original subgoal was
4861      $\forall x. P(x)$ .
4862 - allL2R(i)
4863     This transforms a goal of form
4864      $x=e, \Gamma \vdash \Delta, \forall x.$ 

```

$P(x)$ into $\Gamma \vdash \Delta, P(e)$,
 where the i -th antecedent formula in the
 original subgoal was $x=e$.
 - cut(formula)
 This introduces formula as an intermediate lemma
 to prove.
 It transforms a goal of form $\Gamma \vdash \Delta$
 into two subgoals:
 (1) Use subgoal: formula, $\Gamma \vdash \Delta$
 (2) Show subgoal: $\Gamma \vdash$ formula.

5.2 Program / Modal Tactics

For modal formulas like $[\alpha]\varphi$ and
 $\langle \alpha \rangle \varphi$,
 tactics handle:
 - assignments,
 - tests,
 - nondeterministic choice,
 - sequential composition,
 - duals.

These reduce program structure to simpler logical or
 arithmetic goals. You will almost never need to use these
 low-level tactics directly, as they are all handled by the
 high-level 'unfold' tactic.

A structure not handled by unfold, that you will need to
 handle directly is loops. Usually this needs the loop
 rule, 'loop("invariant", i)', which transforms a goal of
 form $\Gamma \vdash [\alpha^*]\varphi, \Delta$ where
 $[\alpha^*]\varphi$ is at index i into three
 subgoals:

1. init subgoal: $\Gamma \vdash$ invariant, Δ
2. step subgoal: invariant $\vdash [\alpha]$ invariant
3. post subgoal: invariant $\vdash \varphi$

Here, invariant is the (literal) formula you provide as a
 loop invariant.

There is a second class of modality tactics used to
 simplify not the program inside the modality, but the
 postcondition after the modality or otherwise modify the
 modality. The most common are:

- boxAnd(i)
 Transforms a goal of form $[a](P \& Q)$ to $[a]P \& [a]Q$.

```

4909         Usually followed by andR(i) to split into two
4910         subgoals.
4911     - GV(i)
4912         Transforms a goal of form
4913         \Gamma |- [a]P, \Delta to \Gammaconst |- P,
4914         \Deltaconst, where \Gammaconst and \Deltaconst are the
4915         subsets of \Gamma and \Delta that contain only formulas
4916         with constants (no variables).
4917     - discreteGhost("x", "x0", i)
4918         Adds a new variable x0 and the antecedent formula
4919         x = x0. The position, i, should be that of the
4920         modality formula (usually 1).
4921
4922 ### 5.3 Differential Equation Tactics
4923
4924 For goals involving differential equations of the form:
4925
4926 ```
4927 [{x' = f(x) & Q}] P
4928 ```
4929
4930 you can use the following high-level ODE tactics:
4931 - dIRule(i) (Differential Invariant)
4932     Applies a differential invariant rule to the ODE
4933     formula at index i.
4934     This proves formulas like [ODE] P where P may
4935     contain any comparison operator (<, <=, =, >=, >).
4936     The rule generates two proof obligations:
4937     1. The postcondition P holds initially, and
4938     2. The derivative condition P' satisfies the correct
4939        sign constraint (depending on the operator)
4940        throughout the ODE.
4941     For example, if in '[{x' = f(x) & Q}] P', P is
4942     p <= 0, the check is
4943     'Q |= [x':=f(x)] p' <= 0';
4944     if P is p > 0, the check is
4945     'Q |= [x':=f(x)] p' > 0';
4946     and similarly for the other operators.
4947     Along with Q, any assumptions from the
4948     antecedents that had only constants (no
4949     variables) are also kept.
4950     Note that you cannot use the invariant that
4951     you are trying to prove as an assumption in
4952     this subgoal, that is, you will not be able to

```

assume $p > 0$ to prove $p' > 0$.
 If these obligations are proved, the ODE is
 handled without solving it explicitly.
 Note how to be able to prove subgoal 2, we may
 want to strengthen the domain constraint Q using
 dC first before applying dIRule.

- dC("\\varphi", i) (Differential Cut)
 Adds an additional domain constraint φ to the ODE at
 index i .
 This transforms a goal with $\{x'=f(x) \ \& \ Q\} \ P$ at
 position i into two subgoals:

1. Use:
 The formula at position i becomes
 $\{x'=f(x) \ \& \ Q \ \& \ \varphi\} \ P$
2. Show:
 The formula at position i becomes
 $\{x'=f(x) \ \& \ Q\} \ \varphi$

- dW(i) (Differential Weakening)
 Applies weakening to an ODE goal at index i .
 If the formula at index i was $\{x'=f(x) \ \& \ Q\} \ P$, the
 new goal becomes $Q \vdash P$, which removes the ODE
 entirely. Along with Q , any assumptions from the
 antecedents that had only constants (no variables) are
 also kept.
 It makes sense to apply dW only after sufficiently
 strengthening the domain constraint Q using dC first.

These tactics reduce ODE proof obligations to simpler
 goals.
 A common pattern is to cut in *solutions* or *bounds*
 of the variables involved in the ODE and postcondition
 using dC.
 In the branch where we need to show that the bound
 holds, it is common to use dIRule.
 In the other, "use" branch, we typically apply further
 dC cuts to accumulate more bounds.
 Then after accumulating sufficiently many such cuts, we
 apply dW followed by QE to finish the proof.
 Sometimes the order of cuts really matters, as earlier
 cuts may become important to closing the dIRule
 obligations of later cuts.

Often, bounds are best expressed in terms of the initial values of variables before the ODE starts, e.g., $x \leq x_0 + v \cdot t$ for a variable x with initial value x_0 and constant velocity v over time t .

In this case, if x_0 does not already exist, a useful tactic is `'discreteGhost("x", "x0", i)'`, which adds a new variable x_0 and the antecedent formula $x = x_0$. The position, i , should be that of the ODE formula (usually 1).

Besides `dIRule`, another useful tactic that is especially useful for ODEs with exponential solutions, is the Darboux tactic:

`'dbx("g", 1)'`. Given a goal of form `'A |- [x'=f(x) & Q] p >= 0'`, this tactic generates two subgoals: `'A |- p >= 0'` and `'Q |- p' >= g * p'`, where p' is the Lie derivative of p along the ODE $x' = f(x)$. Here, $\geq$ can be replaced by any comparison operator. `dbx` is usually the way to go when we want to prove something about a variable whose derivative is dependent on itself.

In cases where you are trying to prove, e.g., $p' \geq 0$ and it is not immediate that $p' > 0$, but it is true that $p' \geq g \cdot p$, you should use Darboux and not `dIRule` (here $\geq$ and $>$ can be replaced by the other comparison operators).

Unlike `dIRule`, `dbx` does not create further subgoals, and is itself a terminal tactic.

5.4 Arithmetic Tactics

When there are no modalities left, the goal is purely arithmetic.

Then, it is possible to use the `'QE'` tactic to close the goal.

It is recommended to use `'QE("Z3")'` which runs QE with the Z3 SMT solver backend, as it is generally most effective.

If QE times out, try hiding irrelevant formulas using `'hide(i)'` to remove formula with index i from the subgoal permanently.

It can also sometimes help to use the <code>'fullSimplify'</code>	5041
tactic to make expressions simpler.	5042
	5043
If QE fails, there is likely an error in the proof or	5044
the theorem is not valid. It can help to look for a	5045
counterexample as described in Section 3.3.	5046
	5047
---	5048
	5049
## 6. Working With Indices	5050
	5051
### 6.1 Index Semantics	5052
	5053
Indices refer to the current goal only:	5054
• 1 is the first succedent formula.	5055
• 2 is the second succedent formula, etc.	5056
• -1 is the first antecedent formula.	5057
• -2 is the second antecedent formula, etc.	5058
	5059
After some tactics, the sequent may change, so indices	5060
may point to different formulas.	5061
	5062
### 6.2 Common Failure: Incorrect Positions	5063
	5064
A common reason for tactic failure is incorrect indices,	5065
e.g.:	5066
• the index no longer refers to the formula you expect,	5067
or	5068
• the selected formula does not match the rule (e.g.,	5069
andL on a non- $\&$ formula).	5070
	5071
Debugging pattern:	5072
1. Insert print before the failing step.	5073
2. Check the exact index and formula.	5074
3. Update the index or tactic accordingly.	5075
	5076
---	5077
	5078
## 7. Combinators	5079
	5080
Useful control constructs:	5081
• Sequential composition: <code>'t1 ; t2'</code>	5082
Run t1 then t2 on the resulting goals.	5083
• Branching <code>'<(t1, t2)'</code>	5084

5085 Apply t1 to the first branch and t2 to the second,
5086 after a rule that splits the goal.

5087
5088 Combining these is generally enough to express proofs.

5089
5090 ---

5091
5092 ## 8. High-Level Tactics

5093
5094 High-level tactics:

5095 • perform several logical steps at once,
5096 • handle common patterns in hybrid programs,
5097 • push the proof toward arithmetic goals.

5098
5099 The two most common ones are:

5100 - unfold

5101 Unfolds simpler constructs (assignments, tests,
5102 choices, sequences, antecedent conjunctions, dual,
5103 implication). Often changes sequent structure and
5104 might create multiple branches.

5105 - auto

5106 Applies a library of rules to try to automatically
5107 prove the goal. Generally succeeds on simpler goals.

5108
5109 You can use these to make the tactic shorter, but the
5110 tradeoff is less control and harder debugging.

5111
5112 ---

5113
5114 ## 9. Debugging

5115
5116 Common debugging workflows:

5117 1. Insert print between suspicious steps to see the
5118 goal after each one. This can help identify where
5119 the proof diverges from expectations.

5120 2. Remove steps until the error disappears. Once
5121 behavior is as expected, add more steps.

5122
5123 _____

5124
5125 ## 10. Additional Tips:

5126
5127 ### 10.1 Syntactic:

5128

- Comments in Bellerophon are written using the syntax `/* This is a comment */`. 5129
 - Do not use a concluding semicolon after the last tactic in your script. The semicolon is only a separator, not a terminator. Write `t1 ; t2`, not `t1 ; t2 ;`. 5130
 - The branching combinator `<( ... )` does not have a closing angle bracket. Write `<(t1, t2)`, not `<(t1, t2)>`. 5131
 - When using branching tactics like `dC` or `cut`, further development should be nested inside the branches using `<( ..., ... )` syntax. Write `dC("formula", pos); <(t1; dC("formula", pos); <(t2, t3), t4)`, not `dC("formula", pos); <(t1, t4) ; dC("formula", pos); <(t2 ; t3)`. Here, the first `dC` splits the goal into two branches. On the first branch we apply `t1` followed by `dC` that further splits into two branches where we apply `t2` and `t3` respectively. On the second branch of the first `dC`, we apply `t4`. 5132
 - Output along the lines of "Exception in thread "main" 6:2 Lexer 6:2 Lexer does not recognize input at ..." indicates that there is a syntax error in the Bellerophon tactic. There can still be syntax errors that do not get caught by the lexer, though. 5133
 - The Bellerophon parser supports only ASCII characters. 5134
 - You can apply tactics like `andL` and `orL` only to formulas that have the corresponding logical connective at the top level. For example, `andL` would not apply to `a | b & c` because the top-level connective is `|`, since `&` binds more tightly. A good pattern is to use `unfold` to split up all top-level conjunctions at once, and then apply `orL` as needed. 5135
- ### 10.2 Common Proof Patterns: 5136
- Sometimes, it is helpful to split out disjunctive cases in the antecedent using `orL`, because the disjuncts correspond to different control modes requiring different reasoning. 5137
 - When there are top-level conjunctions in the antecedent, you can either split them one-by-one using `andL`, or use `unfold` to split all at once. The latter is often more concise, while the former gives more control. It is often necessary to split conjunctions to expose individual underlying assumptions to which 5138

5173 further rules apply, e.g., disjunctions for orL case
5174 splitting, or equations on which to apply allL2R.

- 5175 - When you see errors about inapplicability of a tactic
5176 due to formula shape mismatch, try to work around by
5177 using a different tactic or by introducing cuts to
5178 reshape the goal into a more convenient form.
- 5179 - For time-triggered controllers, `'t>=0'` is often a
5180 useful invariant to cut in using dC, as it ensures we
5181 only have to reason about ODE properties after
5182 nonnegative time durations.

5183

5184 ### 10.3 Other Advice:

- 5185 - Do not assume a branch is closed until you print and
5186 verify that it shows "proved". Therefore, after
5187 applying a terminal tactic like auto or QE, it is good
5188 practice to print the state right after to check that
5189 the tactic indeed closed the goal.
- 5190 - `'unfold'` steps are important and should not be
5191 forgotten. They often change the shape of the subgoal,
5192 and their can lead subsequent tactics to fail because
5193 the positions of sequents they expect are no longer
5194 present. It is a good idea to start tactic scripts
5195 with `'unfold'` to normalize the formula first.
- 5196 - For the full list of tactics, see [https://github.com/LS-Lab/KeyMaeraX-release/tree/master/keymaerax-webui](https://github.com/LS-Lab/KeyMaeraX-release/tree/master/keymaerax-webui/src/main/resources/help/axiomsrules)
5197 [/src/main/resources/help/axiomsrules](https://github.com/LS-Lab/KeyMaeraX-release/tree/master/keymaerax-webui/src/main/resources/help/axiomsrules).
- 5198 - Duals are usually handled by `'unfold'`, which generally
5200 eliminates the outermost dual by flipping the modality
5201 (box to diamond or vice versa). Otherwise the
5202 individual tactics are `'duald(i)'` that handles diamond
5203 modality containing a dual game at position i and
5204 similarly and `'dualb(i)'` for box modality.
- 5205 - `unfold` generally handles implications, so `implyR(1)` is
5206 usually not necessary after `unfold`.
- 5207 - `dbx` is strict about the shape it allows: the
5208 postcondition formula after the box must be a single
5209 inequality or equation with 0 on the RHS. Rewrite the
5210 goal using a differential cut if needed to get it into
5211 this shape.
- 5212 - When you have two subsequent formulas, e.g., $\Gamma \vdash$
5213 $[\alpha_1] \varphi$, $[\alpha_2] \varphi$,
5214 (i.e., $[\alpha_1] \varphi$ at position 1 and
5215 $[\alpha_2] \varphi$ at position 2), remember that you
5216 only need to prove one of the subsequents.

You can think carefully about which one is more promising to prove, and get rid of the other using `hide(2)` or `hide(1)` as appropriate. But don't eliminate options too early: if you have a disjunction `'f1 | f2'` in your assumptions, it may be that `[\alpha1*]\varphi` is provable when `f1` holds, and `[\alpha2*]\varphi` is provable when `f2` holds, so you need to keep both options until you have split on the disjunction using `orL`.

- If an arithmetic proof obligation times out on `auto` or `QE`, it is a good idea to check if the goal is even valid by internally checking if a solver like `Z3` can find a counterexample within some small timeout.

10.4 On Loop Invariants:

identifying the right loop invariant for proving properties of loops (using tactic `'loop("invariant", i)'`) can be tricky, yet critical to allowing the subsequent proof branches succeed. A common pattern is to start with an invariant, and then strengthen or weaken it as required as one encounters counterexamples or difficulties in closing the subgoals, `init`, `step` and `post`, out of which `step` is usually the hardest to close.

The invariant can be no stronger than the initial assumptions, that is, `'\Gamma1 & \Gamma2'` in `'\Gamma1, \Gamma2 |- [\alpha*]\varphi'`. Anything stronger than `'\Gamma1 & \Gamma2'` would not allow the `init` subgoal to close.

The invariant cannot be weaker than `\varphi`, the postcondition. Anything weaker than `\varphi` would not allow the `post` subgoal to close.

The challenge is then to weaken the assumptions enough, or strengthen the postcondition enough, that the resulting invariant is **inductive**, allowing the `step` subgoal to close.

For example, consider the simple loop goal `'x=1 |- [{?x>=1;^@; {x'=1}}*]x>=0'`. `x=1` is too strong as an invariant because as time elapses, `x` becomes larger than 1, so it is not true that `x=1 |- [{?x>=1;^@; {x'=1}}]x=1`. `x>=0` is too weak as an invariant because `Demon` loses

within the loop at the test when, e.g., $x=0$, so it is not true that $x \geq 0 \mid - [\text{?}x \geq 1; \text{^@}; \{x' = 1\}]x \geq 0$.

A suitable invariant is $x \geq 1$, which is weaker than the assumptions, stronger than the postcondition, and inductive in that if it holds initially, after running one iteration of the loop, it will continue to hold.

Other advice for identifying an invariant:

- A good heuristic is to start with the initial assumptions as the invariant, and then weaken them as needed, or start with the postcondition φ and strengthen it. Ask yourself: do the initial assumptions already look inductive? Does the postcondition already look inductive?
- It is usually a good idea to retain all assumptions on signs of constants. These often end up being critical to inductiveness. You lose nothing by including assumptions in the invariant, unless they are non-inductive and stop holding after some time.
- It is easy to get the invariant subtly wrong because of small inaccuracies or missing edge cases. But in the course of doing the proof, if you carefully analyze proof failures, you will discover the missing requirements, so do not hesitate to modify the invariant accordingly.
- Finding the right invariant is the only way to do proofs about Angel loops in Demon modality games or Dually, Demon loops in Angel games (i.e. $[\alpha^*]\varphi$ or $\langle \alpha^* \rangle \varphi$). The only way to prove the latter is to first convert to $[\alpha^*]\varphi$ form, e.g. using unfold, and then apply the loop tactic.

10.5 On proving Angel choices in Angel games:

This section applies dually also to Demon choices in Demon games.

Consider an Angel choice in an Angel game, e.g.,

$x = 1 \mid x = -1 \mid - \langle \{v := 1; ++ v := -1;\}^@; \{x' = v\} \rangle^@ (x \geq 1 \mid x \leq -1)$.

After applying 'unfold', we get a two subsequents:

$x = 1 \mid x = -1 \mid - [\{x' = 1\}](x \geq 1 \mid x \leq -1),$
 $[\{x' = -1\}](x \geq 1 \mid x \leq -1)$

This reflects how we can choose which subsequent to prove

based on which choice Angel would have strategically made.

In this case, Angel would choose the first branch if $x=1$ initially, and the second branch if $x=-1$ initially. Thus we could apply the following tactic to complete the proof, with the `orL` splitting on initial conditions and `hideR` choosing the right subsequent to then prove:

```

'''
unfold; orL(-1); <(
  hideR(2); dC("x>=1", 1); <(
    dW(1); QE,
    dIRule(1); <(
      QE,
      unfold; QE
    )
  ),
  hideR(1); dC("x<=-1", 1); <(
    dW(1); QE,
    dIRule(1); <(
      QE,
      unfold; QE
    )
  )
)
'''

```

10.6 Common Error Messages:

Subgoal mismatch.

Error messages of the form "`<(e)(v)` is only defined when `len(e) = len(v)`, but `2!=0` subgoals `(v)`" indicates a branching combinator "`<( ... )`" has the wrong number of branches.

In this example, a branching combinator for two branches was used, but the goal was actually already closed (0 subgoals), and the fix is to remove the branching combinator since the subgoal is already proved.

Sometimes, this error message is also reached by `auto`. This is a bug in the implementation of `auto` that is triggered on some formula shapes. If this happens, since you cannot control `auto`, just try to work around by using a different tactic or breaking down the proof into smaller steps.

```

5349
5350 ##### dW Bug.
5351 Sometimes, dW fails with an error message like
5352 ```
5353     Not found: locator 'R=="a&b->c" of position tactic ...
5354     does not match anywhere in succedent of ... from ==>
5355     1: a -> c
5356 ```
5357 where typically a is the domain constraint of the ODE to
5358 which we are applying dW, c is the postcondition, and b
5359 consists of some formulas from the antecedent with only
5360 constants.
5361
5362 This is a bug in KeYmaera X.
5363 A trick that often works around this bug is cutting in
5364 "1=1". Replace `dW(i); *subsequent tactic*` with
5365 `cut("1=1"); <(dW(i); *subsequent tactic*, QE("Z3") using
5366 "1=1")`.
5367 The syntax "using "1=1"" tells QE to ignore all antecedent
5368 and consequent formulas in this subgoal except "1=1".
5369
5370 If the error persists, you can try first removing
5371 antecedent formulas with only constants that are
5372 appearing in the error message, and only then apply dW.
5373 Use hide(i) to hide the formula at position i.
5374
5375 ---
5376 End of Bellerophon Tactic Guide
5377 ---
5378
5379 ## Output Instructions
5380
5381 ### Output Format
5382
5383 In your response, first briefly summarize you reasoning.
5384 Then write a newline.
5385 Then, in a code block, write the new Bellerophon tactic
5386 you want KeYmaera X to try.
5387
5388 ### Advice
5389
5390 If you are not sure what to how to complete the proof, you
5391 can provide some initial steps.
5392 KeYmaera X will run these and you can review the open

```


branches to complete the proof in later steps. 5393
Remember to use the print tactic at the end of branches to 5394
find out what their state is for future reference. 5395

Tactics can be sequentially combined using semicolons 5396
(e.g., 'unfold; auto'. Note that there is no semicolon 5397
after the last tactic in a sequence). 5398

If the proof branches, e.g., after dC, then use the 5399
combinator '<(,)', e.g., 'dC; <(print("this is branch 5400
one."); print("this is a second print still in branch 5401
one."), print("this is branch two.")) \'. 5402
5403

Common mistakes to avoid: 5404
5405

- When expanding on a previous tactic script, do not 5406
forget the initial 'unfold' tactic because this can 5407
lead to subsequent tactic failing. You should retain 5408
the entire previous tactic and expand on it, unless 5409
there is an error requiring you to backtrack. 5410
- Do not modify the old tactic any more than strictly 5411
necessary. Tactics can be fragile and a single 5412
misplaced andL can break the proof. 5413
- Tactics should almost always begin with unfold to 5414
normalize the formula first. 5415
- Do not try more than one or two proof steps at once 5416
without checking if the proof state is like you expect. 5417
Do not try to solve more than one branch at a time 5418
because the complexity can quickly become unmanageable. 5419
- Do not forget to print the state of branches after QE or 5420
auto to confirm that they are indeed closed. 5421
- No unicode. Only ascii characters are allowed. 5422
5423

You will next be shown 0 or more examples of how to answer 5424
such a prompt, which you can learn from, followed by the 5425
current question which you should answer. 5426

- role: user 5427
content: |- 5428

The formula that you are trying to prove is: 5429
(t <= T & v = 0 & w = 0 & !(x = 0 & y = 0)) -> 5430
<{x' = v, y' = w, v' = om * w, w' = - om * v, t' = 1 & 5431
t <= T}^@>(!(x = 0 & y = 0) & (!x*w - y*v = 0 | (v = 0 & 5432
w = 0))) 5433
5434

Now, please provide a Bellerophon tactic to try next. 5435
You should retain the entire previous tactic and expand 5436

5437 on it, unless there is an error requiring you to
5438 backtrack.

5439 In your response, first briefly summarize your reasoning.
5440 Then write a newline.

5441 Then, in a code block, write the entire Bellerophon tactic
5442 you want KeYmaera X to try.

5443 - role: assistant
5444 answer:
5445 mode: null
5446 content: |

5447 Intuitively we expect this formula to be valid because
5448 if initially $v=w=0$, then v and w remain 0, so x and y
5449 remain constant, so if we start not at the origin we
5450 stay not at the origin. In this way, it should in the
5451 end hold that $!(x = 0 \ \& \ y = 0) \ \& \ (v = 0 \ \& \ w = 0)$ which
5452 implies the postcondition.

5453 First lets unfold and differential-cut in the invariant
5454 $v=0 \ \& \ w=0$.

5455 We can then print the state of the second branch to
5456 check that things look right so far.

5457
5458 ```
5459 unfold;
5460 dC("v = 0 & w = 0", 1) <(
5461 label("Use cut"),
5462 label("Show cut");
5463 print("Show branch state.")
5464)
5465 ```

5466 - role: user
5467 content: |-

5468 The formula that you are trying to prove is:
5469 $(t \leq T \ \& \ v = 0 \ \& \ w = 0 \ \& \ !(x = 0 \ \& \ y = 0)) \rightarrow$
5470 $\langle \{x' = v, y' = w, v' = \text{om} * w, w' = - \text{om} * v, t' = 1 \ \& \$
5471 $t \leq T\}^{\text{@}} \rangle (!(x = 0 \ \& \ y = 0) \ \& \ (!x*w - y*v = 0 \mid (v = 0 \ \& \$
5472 $w = 0)))$

5473

5474 This is what you have been trying so far and the results.
5475 Pay careful attention to not repeat past mistakes and make
5476 progress based on what you have learned.

5477

5478 We expect this formula to be valid because if initially
5479 $v=w=0$, then v and w remain 0, so x and y remain constant,
5480 so if we start not at the origin we stay not at the

origin. In this way, it should in the end hold that
 $!(x = 0 \ \& \ y = 0) \ \& \ (v = 0 \ \& \ w = 0)$ which implies the
 postcondition.

We have differential-cut in the invariant $v=0 \ \& \ w=0$ and
 shown that it holds using the tactic below. We have
 printed the proof state of the "Show cut" branch and it
 was successfully closed.

```

unfold;
dC("w=0&v=0", 1); <(
  label("Use cut"); print("Use branch state."),
  label("Show cut");
  dC("v^2+w^2=0", 1); <(
    label("Use second cut"); dW(1) ; QE,
    label("Show second cut"); dIRule(1); <(
      unfold; QE("Z3");
      print("Show branch of dC, dI init. State after QE."),
      unfold; QE("Z3");
      print("Show branch of dC, dI inductiveness subgoal.
        State after QE.")
    )
  )
)

```

Both dIRule branches are proved.

The proof state corresponding to the "Use cut" branch is:

```

-1:  t<=T      LessEqual
-2:  v=0       Equal
-3:  w=0       Equal
-4:  !(x=0&y=0) Not
    ==> 1:  [{x'=v,y'=w,v'=om*w,w'=-om*v,t'=1&t<=T&w=0&v=0}]
           ( !(x=0&y=0) & (!x*w-y*v=0 | v=0&w=0) ) Box}

```

So it remains to prove this goal.

Now, please provide a Bellerophon tactic to try next.

You should retain the entire previous tactic and expand
 on it, unless there is an error requiring you to
 backtrack.

In your response, first briefly summarize your reasoning.
 Then write a newline.

Then, in a code block, write the entire Bellerophon tactic
 you want KeYmaera X to try.

- role: assistant
 answer:

```

5525 mode: null
5526 content: |
5527     We want to show that after the ODE,  $!(x = 0 \ \& \ y = 0) \ \&$ 
5528      $(v = 0 \ \& \ w = 0)$  and already have  $v=0 \ \& \ w=0$  in the domain
5529     constraint. Intuitively,  $!(x = 0 \ \& \ y = 0)$  because this
5530     is true initially neither  $x$  nor  $y$  changes. To cut in
5531     the fact that  $x$  and  $y$  do not change, lets dC in
5532      $(x=x_0 \ \& \ y=y_0)$  where  $x_0$  and  $y_0$  are the initial values of
5533      $x$  and  $y$ , which we can introduce via discreteGhost
5534     (which should also automatically change the initial
5535     assumption  $!(x = 0 \ \& \ y = 0)$  into  $!(x_0 = 0 \ \& \ y_0 = 0)$ ).
5536     Later, we can then show the "show" branch by dIRule
5537     because  $x$  and  $y$  do not change, and the "use" branch by
5538     dW followed by QE because  $x=x_0$ ,  $y=y_0$  and  $v=0 \ \& \ w=0$ 
5539     should be in the antecedents from the domain constraint
5540     and  $!(x_0=0 \ \& \ y_0=0)$  should be in the antecedents because
5541     it has only free variables  $x_0$  and  $y_0$  which are not
5542     changed by the ODE.
5543     But first, lets print to make sure the subgoals look as
5544     expected.
5545
5546     ```
5547     unfold;
5548     dC("w=0&v=0", 1); <(
5549         "Use":
5550         discreteGhost("x", "x0", 1);
5551         discreteGhost("y", "y0", 1);
5552         dC("x=x0&y=y0", 1); <(
5553             "Use-use":
5554             print("State of the use-use branch dW."),
5555             "Use-Show":
5556             print("State of use-show branch.")
5557         ),
5558         "Show":
5559         dC("v^2+w^2=0", 1); <(
5560             "Use":
5561             dW(1) ; QE,
5562             "Show":
5563             dIRule(1); <(
5564                 unfold; QE("Z3"),
5565                 unfold; QE("Z3")
5566             )
5567         )
5568     ```

```

```

- role: user
content: |-
  The formula that you are trying to prove is:
  ((a >= 0 & a >= (-v)/T & end - (p + v*(T - t) + 1/2*a*(
  (T - t)^2 + (v + a*(T - t))^2/(2*B)) >= 0) | (a < 0 &
  a >= (-B) & end - (p - v^2/(2*a)) >= 0) | (a < (-B) &
  end - (p + v^2/(2*B)) >= 0)) & A > 0 & B > 0 & T > 0 &
  v >= 0) -> <{p' = v, v' = a, t' = 1 & t <= T & v >= 0}^@>
  (end - (p + v^2/(2*B)) >= 0 & A > 0 & B > 0 & T > 0 &
  v >= 0)

  This is what you have been trying so far and the results.
  Pay careful attention to not repeat past mistakes and make
  progress based on what you have learned.

  Proposed Tactic:
  unfold;
  boxAnd(1); andR(1);
  print("State")
  .
  Outcome:      ===== State ===== ElidingProvable(Provable{
  ==> 1:  (a>=0&a>=(-v)/T&end-(p+v*(T-t)+1/2*a*(T-t)^2+
          (v+a*(T-t))^2/(2*B))>=0|a < 0&a>=-B&end-(p-v^2/
          (2*a))>=0|a < -B&end-(p+v^2/(2*B))>=0)&A>0&B>0&
          T>0&v>=0-><{{p'=v,v'=a,t'=1&t<=T&v>=0}}^@>
          (end-(p+v^2/(2*B))>=0&A>0&B>0&T>0&v>=0)  Imply
  from
    -1:  a>=0&a>=(-v)/T&end-(p+v*(T-t)+1/2*a*(T-t)^2+
          (v+a*(T-t))^2/(2*B))>=0|a < 0&a>=-B&end-(p-v^2/
          (2*a))>=0|a < -B&end-(p+v^2/(2*B))>=0  Or
    -2:  A>0 Greater
    -3:  B>0 Greater
    -4:  T>0 Greater
    -5:  v>=0 GreaterEqual
  ==> 1:  [{p'=v,v'=a,t'=1&t<=T&v>=0}]end-(p+v^2/(2*B))>=0
          Box
  with
    -1:  a>=0&a>=(-v)/T&end-(p+v*(T-t)+1/2*a*(T-t)^2+
          (v+a*(T-t))^2/(2*B))>=0|a < 0&a>=-B&end-(p-v^2/
          (2*a))>=0|a < -B&end-(p+v^2/(2*B))>=0  Or
    -2:  A>0 Greater
    -3:  B>0 Greater
    -4:  T>0 Greater
    -5:  v>=0 GreaterEqual

```

```

5613 ==> 1:  [{p'=v,v'=a,t'=1&t<=T&v>=0}](A>0&B>0&T>0&v>=0)
5614         Box}) =====
5615 print done (no progress, 1s)
5616 Done .
5617
5618 Now, please provide a Bellerophon tactic to try next.
5619 You should retain the entire previous tactic and expand
5620 on it, unless there is an error requiring you to
5621 backtrack.
5622 In your response, first briefly summarize your reasoning.
5623 Then write a newline.
5624 Then, in a code block, write the entire Bellerophon tactic
5625 you want KeYmaera X to try.
5626 - role: assistant
5627 answer:
5628     mode: null
5629     content: |
5630         We see the proof has split into two goals, the first
5631         with consequent `[p'=v,v'=a,t'=1&t<=T&v>=0]end-(p+v^2/
5632         (2*B))>=0` and the second with consequent
5633         `[p'=v,v'=a,t'=1&t<=T&v>=0}](A>0&B>0&T>0&v>=0)`. The
5634         second one should be easy to prove, so we will try that
5635         first with the auto tactic.
5636         We will print the state after trying auto to see if it
5637         was successful, or if more work is required. We should
5638         also print the state of the harder branch so that we
5639         can plan our next steps there.
5640         Our tactic will carefully retain the initial
5641         `unfold; boxAnd(1); andR(1)` to ensure we don't lose the
5642         progress we have already made.
5643
5644         ```
5645         unfold; boxAnd(1); andR(1) <(
5646             label("harder");
5647             print("State in harder branch"),
5648             label("easier");
5649             auto; print("State after auto on easier branch")
5650         )
5651         ```
5652 - role: user
5653 content: |-
5654     The formula that you are trying to prove is:
5655     (kA > 0 & kB > 0 & kC > 0 & kT >= 0 & T > 0 &
5656     A >= 0 & B >= 0 & C >= 0 & Temp >= 0 &

```

```

(kB*A <= kA*B & kA*Temp + kT*A <= kA*Tmax |
  kA*B <= kB*A & kB*Temp + kT*B <= kB*Tmax)) ->
< { { { isOn:=0; ++ isOn:=1; }; t:=0;
    { A' = - A*B*Temp*isOn*kA,
      B' = - A*B*Temp*isOn*kB,
      C' = A*B*Temp*isOn*kC,
      Temp' = A*B*Temp*isOn*kT,
      t' = 1 & t<=T & A>=0 & B>=0 & C>=0 & Temp>=0
    };
  }* }^@ > Temp<=Tmax

```

Now, please provide a Bellerophon tactic to try next.
 You should retain the entire previous tactic and expand
 on it, unless there is an error requiring you to
 backtrack.
 In your response, first briefly summarize your reasoning.
 Then write a newline.
 Then, in a code block, write the entire Bellerophon tactic
 you want KeYmaera X to try.

3. Summarize summarizes the proof state in preparation for the next iteration of the verification loop.

```

- role: system
content: |
  You are given a step in an attempt to prove a
  differential game logic (dGL) formula in KeYmaera X,
  along with context about the previous state of the proof
  before the step, including the global plan and the
  things we have tried so far.
  Your task is to provide a new summary of the current
  state of the proof that now accounts for the latest step
  towards completing the proof and its outcome.

  Your proof summary will be used to then decide what the
  next step in the proof should be (if you decide to
  continue the proof) or how to change the formula (if you
  choose to revise the formula). Therefore it should
  remember all relevant information: previous mistakes and
  how you resolved them, location of sequents in subgoals
  that currently remain open, counterexamples discovered at
  the end of previous proof directions, etc. Expect that
  after many steps, the summary will be long. Be sure to
  include all relevant information.

```

5700

5701 ## Input:

5702 1. The dGL formula that you are trying to prove.

5703 2. The summary of the proof state before this step.

5704 3. The proof tactic that you then ran this time to get

5705 one step closer to completing the proof.

5706 4. The outcome of running that tactic.

5707

5708 Here, "step" means not a literal single step in the

5709 proof, but rather a new tactic that you applied that

5710 modifies the last one that was applied, either resolving

5711 errors or making progress towards completing the proof.

5712

5713 ## More information:

5714

5715 Here is more information about dGL:

5716 [This section is the same as in the GetTactic query.]

5717

5718 Here is more information about KeYmaera X tactics:

5719 [This section is that same as in the Bellerophon Tactic

5720 guide in the GetTactic query.]

5721

5722 ## Output Shape:

5723

5724 You should return a detailed summary of the proof state.

5725 In the summary, you should include the following

5726 sections, many of which should carry over from the

5727 previous summary provided to you.

5728

5729 1. The global proof plan so far.

5730 Mostly carries over from previous summary unless the

5731 latest step surfaced an issue or new insight.

5732 2. (New) Previous Tactic (that is, the currently applied

5733 tactic in the latest step).

5734 Just copies over from input item 3. Do not make any

5735 changes or omissions.

5736 3. The current open subgoals with exact sequent locations

5737 if we know them.

5738 Copies over from input item 4.

5739 4. Any mistakes made so far and how we fixed them. This

5740 is important to avoid repeating the same mistakes in

5741 future steps. Guide the reader so they know what they

5742 must do to avoid repeating these mistakes.

5743 Partly carries over from previous summary. Might need to

add new information if the latest step surfaced a new mistake or insight on how to fix a mistake.
Remember to include all relevant context, such as the tactic leading up to the issue, the reason for the problem, and how to avoid it in the future.

5. Previous proof directions that failed, if any, along with details such as the counterexamples discovered at the end of those directions.

Mostly carries over from previous summary unless the latest step surfaced a new issue or insight.

6. Any other relevant information about the proof state that will let a reader work on the proof from here assuming no other information besides the original formula to prove is provided to them.

Partly carries over from previous summary. Might need to add new information if the latest step surfaced new relevant information. Remember that your summary and the original formula will be all the reader has to work from to proceed with the proof, so do not omit useful details. Some of these sections may be empty if the proof is still in early stages, but you should still include the section headers for completeness."

- role: user

content: |

The formula that you are trying to prove is:

$$(kA > 0 \ \& \ kB > 0 \ \& \ kC > 0 \ \& \ kT \geq 0 \ \& \ T > 0 \ \& \ A \geq 0 \ \& \ B \geq 0 \ \& \ C \geq 0 \ \& \ Temp \geq 0 \ \& \ (kB \cdot A \leq kA \cdot B \ \& \ kA \cdot Temp + kT \cdot A \leq kA \cdot Tmax \mid kA \cdot B \leq kB \cdot A \ \& \ kB \cdot Temp + kT \cdot B \leq kB \cdot Tmax)) \rightarrow \langle \{ \{ \{ isOn:=0; ++ isOn:=1; \}; t:=0; \{ A' = - A \cdot B \cdot Temp \cdot isOn \cdot kA, B' = - A \cdot B \cdot Temp \cdot isOn \cdot kB, C' = A \cdot B \cdot Temp \cdot isOn \cdot kC, Temp' = A \cdot B \cdot Temp \cdot isOn \cdot kT, t' = 1 \ \& \ t \leq T \ \& \ A \geq 0 \ \& \ B \geq 0 \ \& \ C \geq 0 \ \& \ Temp \geq 0 \}; \}^* \}^{\omega} \rangle$$

$$Temp \leq Tmax$$

Remember that nested duals alternate control back. For example, in the game $\{(a ++ b)^{\omega}; (c ++ d); \}^{\omega}$, the choice between 'a' and 'b' is controlled by Angel, while the choice between 'c' and 'd' is controlled by Demon.

Here is an analysis of the formula, discussing what Angel vs. Demon controls and the game's control modes.

1) Angel-controlled actions

- None within the game body. The entire loop is dualized, so all otherwise-Angel constructs inside it are under

5788 Demon control.

5789

5790 2) Demon-controlled actions

5791 - Looping: { ... }*^@

5792 | Demon chooses how many iterations

5793 to execute (including zero; any finite number).

5794 - Discrete choice each iteration: { isOn := 0; ++

5795 isOn := 1; } | Within the single ^@, Demon chooses

5796 whether to set isOn to 0 (OFF) or 1 (ON) for that

5797 iteration.

5798 - ODE duration each iteration: { A' = -A B Temp isOn kA,

5799 B' = -A B Temp isOn kB, C' = A B Temp isOn kC,

5800 Temp' = A B Temp isOn kT, t' = 1 & t <= T & A >= 0 &

5801 B >= 0 & C >= 0 & Temp >= 0 }

5802 | Demon controls how long

5803 to flow, subject to the evolution domain. He may stop

5804 immediately or at any time before hitting t = T or a

5805 state-constraint boundary (he cannot cross

5806 A,B,C,Temp < 0 or t > T).

5807

5808 - Non-controlled deterministic assignments

5809 - t := 0 at the start of each iteration (resets the

5810 per-iteration clock; not a choice).

5811 - The chosen isOn := 0 or 1 is a deterministic assignment

5812 after Demon picks the branch.

5813

5814 3) Control modes (qualitative behaviors)

5815 The per-iteration discrete choice creates two qualitative

5816 modes; Demon may switch modes between iterations by

5817 choosing isOn.

5818

5819 - OFF mode (isOn = 0)

5820 - Dynamics: A' = B' = C' = Temp' = 0, t' = 1.

5821 - Behavior: State (A,B,C,Temp) is frozen; only the timer

5822 t increases up to T. This is an equilibrium for the

5823 chemistry/temperature.

5824 - Invariants: All polynomial combinations below remain

5825 constant (trivially, since derivatives are zero).

5826

5827 - ON mode (isOn = 1)

5828 - Dynamics: A' = -kA A B Temp <= 0, B' = -kB A B Temp <= 0,

5829 C' = kC A B Temp >= 0, Temp' = kT A B Temp >= 0, with

5830 t' = 1 and domain t <= T, A,B,C,Temp >= 0.

5831 - Monotonicity:

$A(t)$, $B(t)$ are nonincreasing; $C(t)$, $Temp(t)$ are nondecreasing. If any of A , B , or $Temp$ is zero, all chemical rates are zero (equilibrium subset).
 - Polynomial invariants (valid in both modes because every derivative carries the common factor $A B Temp$ isOn):
 - Stoichiometry: $k_B A - k_A B = \text{const}$ (so the sign of $k_B A - k_A B$ never changes; the "limiting reagent" relation stays fixed).
 - Thermal-reagent invariants:
 - $k_A Temp + k_T A = \text{const}$
 - $k_B Temp + k_T B = \text{const}$
 - Yield invariants (optional but useful):
 $k_C A + k_A C = \text{const}$ and $k_C B + k_B C = \text{const}$.
 - Bounds from invariants (safety envelopes chosen by initial branch):
 - If initially $k_B A \leq k_A B$ and $k_A Temp + k_T A \leq k_A T_{\max}$, then for all future time $Temp \leq T_{\max}$ because $A \geq 0$ and $k_A Temp + k_T A$ stays constant.
 - If initially $k_A B \leq k_B A$ and $k_B Temp + k_T B \leq k_B T_{\max}$, then likewise $Temp \leq T_{\max}$ from the second invariant.
 - These two cases are disjoint branches selected by the initial stoichiometric ordering; the ordering is preserved by $k_B A - k_A B = \text{const}$.
 - Asymptotes/equilibria: Trajectories move along invariant manifolds until one reagent depletes ($A \rightarrow 0$ or $B \rightarrow 0$), then all rates drop to zero. Temperature approaches a finite limit bounded by the corresponding invariant, never exceeding $T_{\max}$ under the given initial branch. If $k_T = 0$, $Temp$ is constant.

4) Overall control pattern

- Preconditions (outside the game) ensure positivity and pick one of two invariant safety envelopes: either $k_A Temp + k_T A \leq k_A T_{\max}$ when $k_B A \leq k_A B$, or $k_B Temp + k_T B \leq k_B T_{\max}$ when $k_A B \leq k_B A$. The stoichiometric ordering is invariant, so the chosen envelope remains applicable.

- Iteration (repeats a Demon-chosen finite number of times):

```

5876 1) Demon picks isOn \in {0, 1}
5877     (OFF or ON for this iteration).
5878 2) Deterministic reset  $t := 0$ .
5879 3) Demon chooses how long to flow within the ODE domain
5880     ( $0 \leq \text{duration} \leq T$ ).
5881     - OFF: no state change except  $t$ .
5882     - ON:  $A, B$  decay;  $C, \text{Temp}$  grow; all along the invariant
5883           manifolds keeping  $\text{Temp}$  within the selected bound.
5884 - Demon controls both switching (isOn) and dwell time
5885   (ODE duration) as well as the number of repetitions;
5886   Angel has
5887   no in-game choices. The qualitative modes are OFF
5888   (equilibrium) and ON (reaction with monotone decay/growth
5889   along invariant manifolds), with safety enforced by the
5890   polynomial invariants regardless of Demon's switching and
5891   timing.
5892
5893 Carefully account for who controls which choice while
5894 constructing any counterexamples.
5895 The counterexample cannot,
5896 e.g., require Demon to make a choice that Angel controls.
5897
5898 Then you performed the following proof attempt:
5899 We need an inductive invariant for the Demon loop. A
5900 natural invariant is the conserved linear bounds: either
5901  $k_A * \text{Temp} + k_T * A \leq k_A * T_{\max}$  when  $k_B * A \leq k_A * B$ , or
5902  $k_B * \text{Temp} + k_T * B \leq k_B * T_{\max}$  when  $k_A * B \leq k_B * A$ . Together with sign
5903 assumptions  $k_A > 0, k_B > 0, k_T \geq 0$  and  $A \geq 0, B \geq 0$ , this invariant
5904 implies  $\text{Temp} \leq T_{\max}$  for the post branch. We set up the loop
5905 with this invariant, automatically discharge the init and
5906 post branches (splitting the disjunction in the post), and
5907 print the step branch after unfolding so we can handle ODE
5908 preservation next.
5909
5910 ```
5911 unfold;
5912 loop("kA>0 & kB>0 & kT>=0 & A>=0 & B>=0 &
5913      ( (kB*A <= kA*B & kA*Temp + kT*A <= kA*Tmax) |
5914        (kA*B <= kB*A & kB*Temp + kT*B
5915        <= kB*Tmax) )", 1); < (
5916 /* init: assumptions imply invariant */
5917 unfold; auto; print("Init branch after auto."
5918                    ),
5919 /* step: unfold body, we will handle preservation next */

```

```

unfold; print("
    Step branch after unfolding body."),
/* post: invariant implies Temp<=Tmax */
andL(-1); andL(-1); andL(-1); andL(-1); andL(-1);
orL(-1); <(
    andL(-1); QE("
    Z3"); print("Post branch case 1 closed (from kA*Temp +
    kT*A <= kA*Tmax)."),
    andL(-1); QE("Z3"); print("Post branch case 2 closed
    (from kB*Temp + kT*B <= kB*Tmax).")
)
)
)
'''

The outcome was:
Tactic andL applied at -1 on a non-matching expression in
ElidingProvable(Provable{
-1:  kA>0&kB>0&kT>=0&A>=0&B>=0&(kB*A<=kA*B&kA*Temp+kT*A<=
kA*Tmax|kA*B<=kB*A&kB*Temp+kT*B<=kB*Tmax)
And
-2:  kA>0 Greater
-3:  kB>0 Greater
-4:  kC>0 Greater
-5:  kT>=0
GreaterEqual
-6:  T>0 Greater
==> 1:  [{isOn:=0;++isOn:=1;}t:=0;{A'=-A*B*Temp*isOn*kA,
B'=-A*B*Temp*isOn*kB,C'=A*B*Temp*isOn*kC,Temp'=A*B*Temp*
isOn*kT,t'=1&t<=T&A>=0&B>=0&C>=0&Temp>=0}](kA>0&kB>0&kT
>=0&A>=0&B>=0&%28kB*A<=kA*B&kA*Temp+kT*A<=kA*Tmax|kA*B
<=kB*A&kB*Temp+kT*B<=kB*Tmax%29)
Box
from
-1:  kA>0 Greater
-2:  kB>0 Greater
-3:  kC>0 Greater
-4:  kT>=0 GreaterEqual
-5:  T>0 Greater
-6:  kA>0 Greater
-7:  kB>0&kT>=0&A>=0&B>=0&(kB*A<=kA*B&kA*Temp+kT*A<=kA*
Tmax|kA*B<=kB*A&kB*Temp+kT*B<=kB*Tmax)
And
==> 1:  [{isOn:=0;++isOn:=1;}t:=0;{A'=-A*B*Temp*isOn*kA,

```

```

5964 B'=-A*B*Temp*isOn*kB,C'=A*B*Temp*isOn*kC,Temp'=A*B*Temp
5965 *isOn*kT,t'=1&t<=T&A>=0&B>=0&C>=0&Temp>=0}] (kA>0&kB>0&
5966 kT>=0&A>=0&B>=0&%28kB*A<=kA*B&kA*Temp+kT*A<=kA*Tmax|kA*
5967 B<=kB*A&kB*Temp+kT*B<=kB*Tmax%29)
5968 Box}))
5969
5970 in <( (unfold;auto;print("Init branch after auto.")),
5971 (unfold;print("Step branch after unfolding body.")),
5972 >>_<< )
5973 (Failed on branch (andL(-1);andL(-1);andL(-1);andL(-1);
5974 andL(-1);orL(-1);<( (andL(-1);QE("Z3"));
5975 print("Post branch case 1 closed (from kA*Temp + kT*A <=
5976 kA*Tmax) ."
5977 ), (andL(-1);QE("Z3"));
5978 print("Post branch case 2 closed (from kB*Temp + kT*B <=
5979 kB*Tmax) ."
5980 )) )))
5981
5982 in <( (unfold;auto;print("Init branch after auto.")),
5983 (unfold;print("Step branch after unfolding body.")),
5984 >>_<< )
5985 [Full trace elided]
5986 at edu.cmu.cs.ls.keymaerax.bellerophon.CoreLeftTactic.
5987 computeResult(BellerophonSyntax.scala:623)
5988 t... 40 more
5989
5990 [launcher] Restarting KeyMaera X with sufficient stack
5991 space
5992 /usr/lib/jvm/java-17-openjdk-amd64/bin/java -Xss20M -da
5993 -jar /home/.../keymaerax.jar
5994 -launch -prove /home/.../keymaerax_conjectures/kA0.kyx
5995 -tactic unfold;
5996 loop("kA>0 & kB>0 & kT>=0 & A>=0 & B>=0 & ( (kB*A <= kA*B &
5997 kA*Temp + kT*A <= kA*Tmax) | (kA*B <= kB*A & kB*Temp + kT*B
5998 <= kB*Tmax) )", 1); <(
5999 /* init: assumptions imply invariant */
6000 unfold; auto; print("Init branch after auto."
6001 ),
6002 /* step: unfold body, we will handle preservation next */
6003 unfold; print("
6004 Step branch after unfolding body."),
6005 /* post: invariant implies Temp<=Tmax */
6006 andL(-1); andL(-1); andL(-1); andL(-1); andL(-1);
6007 orL(-1); <(

```

```

andL(-1); QE("
Z3"); print("Post branch case 1 closed (from kA*Temp + kT*A
<= kA*Tmax)."),
andL(-1); QE("Z3"); print("Post branch case 2 closed (from
kB*Temp + kT*B <= kB*Tmax).")
)
)
)
-verbose -timeout 30
KeyMaera X Prover 5.1.2
Use option -help for usage and license information
Proving entries from 1 files
Proving /home/.../keymaerax_conjectures/kA0.kyx#Delphyne/kA0
...
unfold... unfold done (transformed goal(s), 44ms)
loop("kA>0&kB>0&kT>=0&A>=0&B>=0&
      (kB*A<=kA*B&kA*Temp+kT*A<=kA*Tmax|
      kA*B<=kB*A&kB*Temp+kT*B<=kB*Tmax) "
, 1)... loop done (added 2 goal(s), 60ms)
unfold... unfold done (no progress, 2ms)
auto... auto done (proved, 16ms)
print("Init branch after auto.")... INFO ===== Init branch
after auto. ===== ElidingProvable(Provable{   -1:  kA>0
Greater
-2:  kB>0 Greater
-3:  kC>0 Greater
-4:  kT>=0 GreaterEqual
-5:  T>0 Greater
-6:  A>=0 GreaterEqual
-7:  B>=0 GreaterEqual
-8:  C>=0 GreaterEqual
-9:  Temp>=0 GreaterEqual
-10:  kB*A<=kA*B&kA*Temp+kT*A<=kA*Tmax|
      kA*B<=kB*A&kB*Temp+kT*B
<=kB*Tmax
Or
==> 1:  kA>0&kB>0&kT>=0&A>=0&B>=0& (kB*A<=kA*B&kA*Temp+kT*A
<=kA*Tmax| kA*B<=kB*A&kB*Temp+kT*B<=kB*Tmax)
And proved}) =====
print done (proved, 4ms)
INFO ===== Step branch after unfolding body. =====
ElidingProvable(Provable{
-1:  kA>0&kB>0&kT>=0&A>=0&B>=0& (kB*A<=kA*B&kA*Temp+kT*A
<=kA*Tmax| kA*B<=kB*A&kB*Temp+kT*B<=kB*Tmax)

```

```

6052     And
6053     -2:  kA>0 Greater
6054     -3:  kB>0 Greater
6055     -4:  kC>0 Greater
6056     -5:  kT>=0
6057     GreaterEqual
6058     -6:  T>0 Greater
6059     ==> 1:  Temp<=Tmax LessEqual
6060     from
6061     -1:  kA>0
6062     Greater
6063     -2:  kB>0 Greater
6064     -3:  kC>0 Greater
6065     -4:  kT>=0 GreaterEqual
6066     -5:  T>0 Greater
6067     -6:  kA>0 Greater
6068     -7:  kB>0 Greater
6069     -8:  kT>=0
6070     GreaterEqual
6071     -9:  A>=0 GreaterEqual
6072     -10: B>=0 GreaterEqual
6073     -11: kB*A<=kA*B&kA*Temp+kT*A<=kA*Tmax |
6074           kA*B<=kB*A&kB*Temp+kT*B<=kB*Tmax
6075     Or
6076     ==> 1:  Temp<=Tmax LessEqual})) =====
6077     Done /home/.../keymaerax_conjectures/kA0.kyx#Delphyne/kA0
6078     (failed)
6079     FAILED Delphyne/kA0: tactic=user,tacticsize=-1,budget=30[s],
6080     duration=-1[ms],qe=-1[ms],rcf=-1,steps=-1
6081
6082
6083     Now please provide your summary.
6084     This ends the listing of the Verification Prompts.

```

6085 C.5 Synthesis Prompts

```

6086 Similarly to Appendix C.4, this section lists the synthesis prompts. Again prompts are listed in
6087 the order in which they are called during the synthesis process.
6088 1. AnalyzeGameLoop is the same as in Appendix C.4.
6089 2. PlanControlStrategy uses the analysis so far and the provided informal guideline to come upw
6090 ith a strategy that Angel should play.
6091 - role: system
6092 content: |

```


You are given a differential game logic (dGL) loop
and a desired postcondition that Angel (the controller)
aims to maintain.

Your task is to propose a succinct plan describing the
Angel's control strategy to guarantee the postcondition.
You just need to find one control strategy that works,
not necessarily the optimal or most general one. You may
be provided guidelines about what kind of control
solution to find. If so, the chose strategy should be in
accordance with those guidelines.

Background on dGL:
[Identical to the dGL syntax guide of AnalyzeGameLoop
followed
by the Bellerophon Tactic guide of GetTactic.]

You are given:

- The full loop.
- The postcondition that Angel aims to maintain.
- A prior analysis of Angel vs. Demon controls and control modes.
- Optionally, guidelines about what kind of control solution to find.

Goal:
Produce a strategy that maintains safety. If the control
involves a Demon controlled loop, carefully account for
infinite iterations.

Keep the strategy simple and minimalist. Avoid
unnecessarily long or involved control strategies.

It is OK if your strategy requires special initial
conditions or assumptions to work, such as starting at
specific equilibrium point or assuming the sign of some
constants. Just be clear about what these assumptions
are.

Requirements for the plan:

- State any initial conditions/assumptions the plan relies on, and what control mode it starts in. You must choose exactly one control mode to start in so as to make subsequent analysis easier. If the control

6137 mode requires specific initial conditions, this should
 6138 be part of your assumptions.
 6139 - After this, list Angel's strategy as a sequence of
 6140 steps, corresponding to control modes.
 6141 * Each step should start with an action word (e.g.,
 6142 "Choose", "Set", "Switch to"), possibly with a
 6143 conditional clause (e.g., "When X holds"). End
 6144 with a temporal indicator (e.g., "repeatedly",
 6145 "until Y", "forever").
 6146 * Each step should focus on Angel's actions. Mention
 6147 Demon only when necessary to motivate Angel's
 6148 tactics, or when he forces a mode change.
 6149 * Expect to transition between modes mainly by using
 6150 Angel's controllable actions and choices.
 6151 * Use clear English, no mathematical notation unless
 6152 necessary for clarity. This will later be
 6153 translated into formal constraints.
 6154 * Produce a numbered list of Angel's decisions.
 6155 * Be as concise as possible without superfluous
 6156 words. Aim for no more than ~120 words.
 6157 * Avoid hedging or verbose meta-commentary; write
 6158 only Angel's action steps.
 6159

6160 The input includes the game and a prior analysis
 6161 identifying Angel vs. Demon controls. Use that analysis
 6162 to ensure that the plan accounts for any possible
 6163 adversarial behavior.
 6164

6165 - Carefully consider the differential equation. There
 6166 can be coupling between variables that affects your
 6167 reasoning.
 6168 - Based on the dynamics, you may some times need to come
 6169 up with upper/lower bounds of quantities and use that
 6170 to derive sound preconditions. It's best to use
 6171 simplistic, polynomial bounds so that they verify more
 6172 easily.
 6173

6174 Output Structure:

6175 - Start with a one-line comment on what new region the
 6176 strategy is trying to cover.
 6177 - Then write section per the requirements listed above.
 6178 * Write a description of initial
 6179 conditions/assumptions, and mode.
 6180 * Then proceed with the plan steps.

You will be shown zero or more examples of how to answer, followed by a new question to answer.

- role: user

content: |-

Full Game:

```
{? A > 0 & B > 0 & kA > 0 & kB > 0 & kC > 0 & kT >= 0 &
T > 0;^@}; {subgame_c: {{{{isOn := 0;} ++ {isOn := 1;}}};
{t := 0;}}; {subgame_j: {A' = - A * B * Temp * isOn * kA,
B' = - A * B * Temp * isOn * kB, C' = A * B * Temp * isOn *
kC, Temp' = A * B * Temp * isOn * kT, t' = 1 & t <= T &
A >= 0 & B >= 0 & C >= 0 & Temp >= 0}
}}*^@}
```

Postcondition Angel aims to maintain after the loop:

Temp <= Tmax

Prior control analysis:

1) Angel-controlled actions

- None. The only potentially Angelic constructs (the choice `isOn := 0 ++ isOn := 1` and the ODE duration) are inside a loop that is wrapped by a single dual `^@`, so they are Demon-controlled.

2) Demon-controlled actions

- Initial test `{? A > 0 & B > 0 & kA > 0 & kB > 0 & kC > 0 & kT >= 0 & T > 0}^@` (subsequent play assumes these hold; otherwise Demon would have lost).
- Loop count `{subgame_c: {...}*}^@`: Demon chooses how many iterations (including zero).
- Inside each iteration (because the whole loop is under one `^@`):
- Branch choice `{{{isOn := 0;} ++ {isOn := 1;}}:` Demon chooses whether the reaction is Off or On for that iteration's flow.
- ODE duration in `subgame_j`: Demon controls how long to flow along

$$\begin{aligned} A' &= -A B \text{ Temp } \text{isOn } kA, \\ B' &= -A B \text{ Temp } \text{isOn } kB, \\ C' &= A B \text{ Temp } \text{isOn } kC, \\ \text{Temp}' &= A B \text{ Temp } \text{isOn } kT, \\ t' &= 1 \end{aligned}$$
 subject to the evolution domain $t \leq T, A \geq 0, B \geq 0, C \geq 0, \text{Temp} \geq 0$. He may stop anytime before hitting

6225 the domain boundary; he cannot exceed it.

6226 - Deterministic assignment $t := 0$ is not a choice (it

6227 always executes).

6228

6229 3) Control modes (qualitative behaviors and invariants)

6230 - OFF mode ($isOn = 0$, or equivalently any boundary where

6231 $A = 0$ or $B = 0$ or $Temp = 0$ makes the vector field zero):

6232 - $A' = B' = C' = Temp' = 0$, $t' = 1$. Chemical state

6233 $(A, B, C, Temp)$ is frozen; only the timer t increases until

6234 $t = T$.

6235 - Equilibrium for the chemical variables.

6236 - ON mode ($isOn = 1$, with $A, B, Temp > 0$):

6237 - Monotonicity: $A' = -k_A A B Temp \leq 0$, $B' = -k_B A B Temp$

6238 ≤ 0 , $C' = k_C A B Temp \geq 0$, $Temp' = k_T A B Temp \geq 0$

6239 ($Temp' = 0$ when $k_T = 0$). Thus A, B decrease; C increases;

6240 $Temp$ is nondecreasing.

6241 - Domain bounds: $A, B, C, Temp$ remain nonnegative; $0 \leq t \leq T$

6242 within each flow.

6243 - Polynomial invariants (valid in both ON and OFF, hence

6244 globally):

6245 - $k_B A - k_A B = \text{const.}$

6246 - $C + (k_C/k_A) A = \text{const}$ and $C + (k_C/k_B) B = \text{const.}$

6247 - $Temp + (k_T/k_A) A = \text{const}$ and $Temp + (k_T/k_B) B = \text{const}$

6248 (these degenerate correctly when $k_T = 0$).

6249 - Equilibria within ON: if $A = 0$ or $B = 0$ or $Temp = 0$ is

6250 reached, the system becomes stationary (same as OFF for

6251 the chemical variables).

6252 - Asymptotic/terminal behavior: Trajectories move along the

6253 invariant line $k_B A - k_A B = \text{const}$ toward the nonnegative

6254 axes; one reactant may be exhausted ($A=0$ or $B=0$) while the

6255 other remains positive. C accumulates according to the

6256 invariants; $Temp$ can only increase (if $k_T > 0$) or remain

6257 constant (if $k_T = 0$).

6258

6259 Special parameter submode:

6260 - Isothermal ON ($k_T = 0$): Same as ON but $Temp' = 0$, so $Temp$

6261 is constant; other invariants and monotonicity still hold.

6262

6263 4) Overall control pattern

6264 - Start with a Demon test ensuring $A > 0$, $B > 0$, k_A, k_B, k_C

6265 > 0 , $k_T \geq 0$, $T > 0$ (Angel may assume these).

6266 - Then repeat zero or more iterations (Demon loop):

6267 1) Demon chooses $isOn \in \{0, 1\}$.

6268 2) Deterministically reset $t := 0$.

3) Demon evolves the ODE for any duration up to the domain boundary (notably $t \leq T$), stopping at will.

- By sequencing iterations, Demon effectively applies a piecewise-constant ON/OFF control with segment durations at most T , steering the monotone reaction dynamics (consume A, B and produce C , increase Temp if $kT > 0$) or freezing them when OFF.

More information from previous proof attempts:
Find the control solution for the situation where the reactant runs out before temperature can become too large.

Provide the control strategy now.

3. The next query, which gets the LLM to guess a subvalue for a subgame, has three variants, one each for loops, nondeterministic assignments, and ODEs. We show the loop variant, Guess-LoopInvControl, but the others are similar.

- role: system
content: |

Your task is to guess a loop invariant for a game loop given the loop body and the postcondition.

A loop of the form $\{\alpha\}^* @$ executes the body α zero or more times (and we must account for every possibility). A loop invariant I is a property that:

1. Holds before the loop starts (precondition implies I)
2. Is preserved by each iteration of the loop body (if I holds before α , then I holds after α)
3. Combined with loop termination, implies the postcondition

A loop of the form $\{\alpha\}^*$ executes the body α zero or more times, and how many times is our choice. In this case, the invariant should be such if we run the loop enough (finite) times, the postcondition will be satisfied.

While choosing the precondition/invariant, you need to consider the essential properties that remain true throughout all possible iterations of the loop body, considering both angel and demon choices.

The loop invariant should be typically involves bounds, safety properties, or relationships between variables

Return your answer as a logical formula in KeYmaera X syntax:

- Use & for logical AND
- Use | for logical OR
- Use ! for logical NOT
- Use = for equality
- Use <, <=, >, >= for comparisons
- Use parentheses for grouping

Use only variables you see in the problem. For example, if the problem has 'x', use 'x', not 'x(0)'.

Provide up to 4 lines describing your reasoning. Then after a blank line, write a code block with the loop invariant in KeYmaera X syntax.

Some syntax instructions to keep in mind are:

- Do not add comments to your code blocks, as this may cause parsing issues.

Some information about dGL:

"@" is the dual symbol that flips who controls the action between Angel and Demon.

For example:

In game "x := *", Angel gets to choose what value is assigned.

The weakest precondition for such an assignment with postcondition P is the formula that holds *for some* value that could be assigned to x.

In game "{x := *;}^@", any real number can be assigned, per Demon's decision.

The weakest precondition for such an assignment with postcondition P is the formula that holds *for all* possible values that could be assigned to x.

In game "{{ x:= *;}^@}^@" the control flips twice, so Angel gets to choose the assignment to x.

- Carefully consider the differential equation. There can be coupling between variables that affects your reasoning.
- Based on the dynamics, you may some times need to come

up with upper/lower bounds of quantities and use that to derive sound preconditions. It's best to use simplistic, polynomial bounds so that they verify more easily.

- role: user

content: |

Loop body:

```
{isOn := 0; ++ isOn := 1;}^@; t := 0; {A' = - A * B *
Temp * isOn
* kA, B' = - A * B * Temp * isOn * kB, C' = A * B * Temp *
isOn * kC, Temp' = A *
B * Temp * isOn * kT, t' = 1 & t <= T & A >= 0 & B >= 0 &
C >= 0 & Temp >= 0}^@
```

Postcondition:

Temp <= Tmax

Overall game in which this loop appears (for context):

```
? A > 0 & B > 0 & kA > 0 & kB > 0 & kC > 0 & kT >= 0 &
T > 0;^@; {{isOn := 0; ++
isOn := 1;}; t := 0; {A' = - A * B * Temp * isOn * kA,
B' = - A * B * Temp * isOn
* kB, C' = A * B * Temp * isOn * kC, Temp' = A * B * Temp
* isOn * kT, t' = 1 & t
<= T & A >= 0 & B >= 0 & C >= 0 & Temp >= 0}}*^@
```

Here is an analysis of the game, the strategy that Angel is trying to play, and a log of things that you have already tried in the past.

Carefully consider this information.

Analysis of game:

1) Angel-controlled actions

- None. All controllable constructs in the body lie inside a single $^@$ (the outer loop dual), so their control is flipped to Demon. There are no remaining Angel decisions.

2) Demon-controlled actions

- Initial test $\{? A > 0 \& B > 0 \& kA > 0 \& kB > 0 \& kC > 0 \& kT \geq 0 \& T > 0\}^@$: Demon test. Angel may assume these hold initially; otherwise she already wins.

- Loop $\{subgame_c; subgame_j\}^@$: Demon loop. Demon decides how many iterations to execute (including

```

6400     zero).
6401 - Within each iteration:
6402 - Choice isOn := 0 ++ isOn := 1: Demon chooses whether
6403     the reaction is off or on during this iteration's flow
6404     (the ++ is inside the loop's ^@).
6405 - ODE duration in subgame_j: Demon controls how long to
6406     flow subject to the evolution domain ( $t \leq T$ ,
6407      $A, B, C, \text{Temp} \geq 0$ ). He may stop immediately or any time
6408     before hitting the boundary. - Deterministic
6409     assignments (no player control once chosen to execute):
6410 -  $t := 0$  each iteration (resets the local clock)
6411 - The chosen assignment to isOn is deterministic once the
6412     branch is selected.
6413
6414 3) Control modes (qualitative behavior)
6415 - OFF mode (isOn = 0):
6416 - Dynamics:  $A' = B' = C' = \text{Temp}' = 0$ ,  $t' = 1$ . State
6417     ( $A, B, C, \text{Temp}$ ) is constant while time advances to at most
6418      $T$ .
6419 - Equilibria: Entire state subspace is an equilibrium for
6420     the state variables.
6421
6422 - ON mode (isOn = 1):
6423 - Dynamics:  $A' = -k_A A B \text{Temp} \leq 0$ ,  $B' = -k_B A B \text{Temp} \leq 0$ ,
6424      $C' = k_C A B \text{Temp} \geq 0$ ,  $\text{Temp}' = k_T A B \text{Temp} \geq 0$ 
6425     (with  $k_T \geq 0$ ). If any of  $A=0$  or  $B=0$  or  $\text{Temp}=0$ , then
6426      $A'=B'=C'=\text{Temp}'=0$  (equilibrium).
6427 - Monotonicity (when
6428      $A, B, \text{Temp} > 0$ ):  $A$  and  $B$  strictly decrease;  $C$  and  $\text{Temp}$ 
6429     strictly increase ( $\text{Temp}$  is constant if  $k_T = 0$ ).
6430 - Global linear invariants (valid in both ON and OFF,
6431     since OFF has zero derivatives):
6432     -  $k_C A + k_A C = \text{const}$ 
6433     -  $k_C B + k_B C = \text{const}$ 
6434     -  $\text{Temp} - (k_T/k_C) C = \text{const}$ 
6435 - Bounds and asymptotes:
6436
6437 -  $A(t)$ ,  $B(t)$  remain  $\geq 0$  and decrease to 0 at most.
6438 - From invariants,  $C$  is bounded above by
6439      $C \leq \min(C_0 + (k_C/k_A) A_0, C_0 + (k_C/k_B) B_0)$ .
6440 -  $\text{Temp}$  follows  $\text{Temp} = \text{Temp}_0 + (k_T/k_C)(C - C_0)$ , hence
6441      $\text{Temp} \leq \min(\text{Temp}_0 + (k_T/k_A) A_0, \text{Temp}_0 + (k_T/k_B) B_0)$ .
6442 - Reaction effectively ceases upon reaching any
6443     equilibrium surface  $A=0$  or  $B=0$  or  $\text{Temp}=0$ ; otherwise

```


Demon may stop earlier at $t = T$. There are no oscillations; trajectories move toward these saturating bounds. 6444

- One-dimensional reduction: 6446

Using the invariants, C' becomes a cubic polynomial in C with roots at the limiting values where A or B hits zero (and a nonphysical root at negative C from $\text{Temp} - (kT/kC)C$ if $kT > 0$), confirming growth of C to a finite saturation. 6448 6449 6450 6451 6452 6453 6454

Special sub-modes: 6455

- Temperature-constant case ($kT = 0$): $\text{Temp}' = 0$; Temp is invariant. A, B decrease and C increases until $A=0$ or $B=0$ or Demon stops at $t=T$. 6456 6457 6458 6459

4) Overall control pattern 6460

- Initial Demon test enforces positive parameters and $T > 0$. 6461 6462

- The game proceeds through Demon-controlled iterations: 6463

1) Demon sets $\text{isOn} \in \{0,1\}$. 6464

2) Reset $t := 0$. 6465

3) Demon lets the ODE evolve for some duration up to T within the nonnegativity domain (or stops immediately). 6466 6467

- Demon also decides how many such iterations occur. 6468

- Angel has no direct control; Demon's only lever is to toggle the reaction ON/OFF and pick durations/number of pulses. The state evolves along the linear invariant manifolds, 6469 6470 6471 6472

with OFF holding the state and ON pushing it monotonically toward reagent depletion 6473 6474

(A or B) with bounded increases in C and Temp . 6475 6476

Angel control strategy: 6477

Covering the region where reactant depletion caps temperature growth before hitting $T_{\max}$. 6478 6479 6480

Initial conditions/assumptions and starting mode: 6481

- Assume Demon's test holds: $A > 0$, $B > 0$, $k_A, k_B, k_C > 0$, $kT \geq 0$, $T > 0$. 6482 6483

- Start in ON mode ($\text{isOn}=1$) for worst-case analysis of Temp growth. 6484 6485

- Safety margin: 6486

$\text{Temp}_0 + \min((kT/k_A) \cdot A_0, (kT/k_B) \cdot B_0) \leq T_{\max}$. 6487

```

6488     (Special
6489         case: if  $kT=0$ , it suffices that  $Temp0 \leq Tmax$ .)
6490
6491     Angel's strategy:
6492     1) Do nothing;
6493     permit any Demon choices of ON/OFF, durations, and
6494     iterations, forever. The invariants
6495     cap Temp by  $Temp0 + \min((kT/kA)A0, (kT/kB)B0)$ , and reagent
6496     depletion ( $A$  or  $B \rightarrow 0$ ) stops
6497     further increase, ensuring  $Temp \leq Tmax$  throughout.
6498
6499     Starting precondition computation.
6500
6501     What is a suitable loop invariant for this loop?
6502 4. GetNextActionCtrl is the query that decides what to do next: prove the current guess with the
6503 verification pipeline, or backtrack, and if so, to where?
6504 - role: system
6505   content: |-
6506     You are trying to assign the preconditions of a
6507     differential game logic game.
6508     You have been proposing preconditions using backwards
6509     reasoning where you first proposed something for subgames
6510     closest to the end and then worked backwards recursively.
6511     Now you have reached a cross-roads and must choose a next
6512     action.
6513     You are given the original program, a log of what you
6514     were doing last, and the options of actions available to
6515     you.
6516     Reply with a brief explanation, followed by a newline,
6517     followed by a code block containing only the word for the
6518     action.
6519
6520     ## Background information
6521
6522     Here is some information about dGL.
6523
6524     In differential game logic (dGL), there are two players:
6525     - Angel: The controllable player who makes choices we
6526       can control. Angel constructs include:
6527       - Angel choices: 'a ++ b' (we choose between a and b)
6528       - Angel loops: '{a}*' (we choose how many iterations)
6529       - Angel tests: '?P' (we can proceed if P holds,
6530         otherwise we immediately lose)

```

- Angel ODEs: $\{x' = f(x) \ \& \ Q\}$ (we control the duration) 6531

- Angel nondeterministic assignments: $x := *$ 6532
(we choose the value) 6533

6534

There is the dual operator $^@$ that switches Angel and 6535
Demon roles in a game construct. 6536

Duals can be nested arbitrarily, so carefully check 6537
nesting levels to determine which player controls each 6538
action. 6539

Being within an odd number of $^@$ operators indicates 6540
Demon control, while an even number (including zero) 6541
indicates Angel control. 6542

For example, in the game $\{(a ++ b)^@; (c ++ d)\}^@$, 6543
the choice between 'a' and 'b' is controlled by Angel, 6544
being within two $^@$ operators, while the choice 6545
between 'c' and 'd' is controlled by Demon. 6546

Using this notation, we go over the Demon constructs 6547
below, indicating that they are Demon-controlled by 6548
wrapping them in a single $^@$ operator. 6549

6550

- ****Demon****: The adversarial player who makes choices we 6551
cannot control. Demon has all the same constructs as 6552
Angel. 6553

- Demon choices: $\{a ++ b\}^@$ (adversary chooses between 6554
a and b) 6555

- Demon loops: $\{a^*\}^@$ (adversary chooses how many 6556
iterations) 6557

- Demon tests: $\{?P\}^@$ (adversary can proceed if P 6558
holds, and immediately loses otherwise. So Angel can 6559
assume in subsequent game play that P held at this 6560
point; otherwise she would already have won.) 6561

- Demon ODEs: $\{x' = f(x) \ \& \ Q\}^@$ (adversary controls the 6562
duration) 6563

- Demon nondeterministic assignments: $\{x := *\}^@$ 6564
(adversary chooses the value) 6565

6566

Additionally, the non-controlled operations are: 6567

- Assignment: $x := e$ (deterministic assignment like in 6568
regular programs) 6569

- Sequential composition: $a; b$ (first play a, then b) 6570

6571

Notes 6572

- ODEs can sometimes introduce unmatched single quotes, 6573
so account for that when parsing. 6574

6575 - In syntax such as { a++b; x:=* }*^@, the ``^@` applies to
6576 the entire loop including its body. In this example,
6577 Demon controls the choice within the loop between a and
6578 b, as well as the assignment to x, because both are
6579 inside one ``^@`.

6580 Some subgames are labeled with unique identifiers with
6581 the following syntax:
6582 <subgame_id>: <subgame>

6583

6584 These subgame ids will help make your answers more
6585 specific.

6586 A *precondition* for a given game is a logical formula
6587 that, when it is true before the game starts, there
6588 exists a strategy for Angel to ensure that the
6589 *postcondition* holds after the game ends, regardless of
6590 how Demon plays.

6591

6592 ## Input

6593

- 6594 1. The current subgame for which you were trying to
6595 propose preconditions.
- 6596 2. overall_game: The overall game in which the current
6597 subgame appears.
- 6598 3. Analysis of the game and the control strategy Angel is
6599 trying to execute that the chosen preconditions should
6600 support.

6601 This is followed by a log of previous attempts at
6602 identifying preconditions for subgames and verifying
6603 whether they hold by stepping through the game and
6604 checking if the preconditions compose to ensure the
6605 postcondition is satisfied. Examine this history to
6606 figure out what just happened. Did you propose a bad ODE
6607 precondition? Did the precondition you proposed end up
6608 not being compatible with an earlier chosen loop
6609 invariant? Did you come up with a good invariant but you
6610 think it can be improved upon?

- 6611 4. The options you can choose for your next action.

6612

6613 ## Possible Actions

6614

6615 The preconditions that you proposed most recently didn't
6616 check out.

6617 Either it was not provably a precondition of the current
6618 subgame, or if part of a loop, your provided

preconditions led to a precondition for the loop body
that was not implied by the loop invariant.

The error could lie in the most recent condition, or
earlier in the process where you may have made a choice
previously that backed you into a corner for this
subgame, or maybe the condition is correct but too
difficult for the automatic theorem prover to check. Now
you must decide till what point to backtrack, or whether
to try harder to prove the current subgame result using a
tactic with KeYmaera X.

You are given the original program, in which some points
in the game are labeled as subgame_id.

- 'backtrack-to:id': Backtrack to the subgame with the
given id and propose a new precondition there. If you
choose this option, in your reply, write
'backtrack-to:id' where id is the id of the subgame you
want to backtrack to. When backtracking, you can even
choose the current subgame to retry. Loops are
considered to occur "later" than their body, so if you
backtrack to a subgame inside a loop, you won't be
asked for a new loop invariant and the existing
invariant will be used.
- 'try-proof': Try to prove the current subgame using
KeYmaera X with the precondition you proposed most
recently. If you aren't sure why the most recent
precondition was bad, this could be a good option
because the manual proof attempt can give you feedback
on what's wrong with the precondition.

Output

Write a few lines of explanation, followed by a newline,
and then a code block containing the action you choose.
For example, a valid code block is:

```
```\ntry-proof\n```
```

I will now show you zero of more examples of how to  
respond, followed by the current case.

- role: user

```

6663 content: |-
6664 The current subgame is:
6665 subgame_p: {x' = a * x - b * x * y, y' = d * x * y - g *
6666 y}^@
6667
6668
6669 The overall game with labels is:
6670 {? x > 0 & y > 0 & a > 0 & b > 0 & d > 0 & g > 0 & x >=
6671 xmin & y >= ymin;}^@; {subgame_c: {{{{{{subgame_j:
6672 xadd := *;}^@
6673 }; {? xadd >= 0;}^@}}; {subgame_l: yadd := *;}^@
6674 }}; {? yadd >= 0;}^@}}; {x := x + xadd;}; {y := y +
6675 yadd;}}; {subgame_p: {x' = a * x - b * x * y, y' = d *
6676 x * y - g * y}
6677 }}*^@}
6678
6679 Here is an analysis of the game followed by logs of
6680 precondition computation and checking so far:
6681 Analysis of game:
6682 1) Angel-controlled actions
6683 - ODE duration (subgame_p): In each iteration, Angel
6684 controls how long to flow along $x' = a x - b x y$, $y' =$
6685 $d x y - g y$ (no evolution domain constraint, so any
6686 nonnegative duration, including 0).
6687
6688 2) Demon-controlled actions
6689 - Initial test: $\{? x > 0 \ \& \ y > 0 \ \& \ a > 0 \ \& \ b > 0 \ \& \ d > 0 \ \& \ g > 0 \ \& \ x \geq x_{\min} \ \& \ y \geq y_{\min}\}^@$. Angel may assume these
6690 hold initially; otherwise Demon would have already lost.
6691 - Looping: $\{\dots\}^@$. Demon chooses how many iterations to
6692 perform (finite but unbounded, including zero).
6693 - Nondeterministic increments with constraints:
6694 - subgame_j: $\{xadd := *\}^@$ followed by $\{? xadd \geq 0\}^@$:
6695 Demon chooses xadd and must satisfy $xadd \geq 0$.
6696 - subgame_l: $\{yadd := *\}^@$ followed by $\{? yadd \geq 0\}^@$:
6697 Demon chooses yadd and must satisfy $yadd \geq 0$.
6698 - Deterministic updates $x := x + xadd$; $y := y + yadd$ then
6699 apply each iteration.
6700
6701
6702 3) Control modes (qualitative dynamics)
6703 The hybrid iteration has two distinct phases per round:
6704
6705 A. Discrete jump (Demon): $x := x + xadd$, $y := y + yadd$
6706 with $xadd, yadd \geq 0$

```

- Monotone, upward-only resets of both coordinates. Demon can make  $x, y$  arbitrarily large in a single iteration.

- Effect on the continuous invariant (see below):  $H(x, y) = g \ln x - d x + a \ln y - b y$  has gradient

- $\Delta H / \Delta x = g/x - d$ , positive if  $x < g/d$  and negative if  $x > g/d$ .
- $\Delta H / \Delta y = a/y - b$ , positive if  $y < a/b$  and negative if  $y > a/b$ .

So increasing a coordinate raises  $H$  when that coordinate is below its nullcline threshold, and lowers  $H$  when above. Demon's nonnegative jumps can thus move the state to higher or lower  $H$ -level sets depending on where  $(x, y)$  is relative to  $(g/d, a/b)$ .

B. Continuous Lotka{Volterra flow (Angel-controlled duration):

- Nullclines:
  - $x' = 0$  when  $y = a/b$  (horizontal line).
  - $y' = 0$  when  $x = g/d$  (vertical line).
- Unique interior equilibrium:  $E^* = (g/d, a/b)$ .
- First integral (conserved along the ODE):  $H(x, y) = g \ln x - d x + a \ln y - b y = \text{constant}$ . Orbits are closed, encircling  $E^*$ ; no asymptotic convergence or divergence within the ODE phase.
- Sign structure by regions ( $x > 0, y > 0$  always preserved):
  - $y < a/b, x > g/d$ :  $x' > 0, y' > 0$  (both grow).
  - $y > a/b, x > g/d$ :  $x' < 0, y' > 0$ .
  - $y > a/b, x < g/d$ :  $x' < 0, y' < 0$  (both decay).
  - $y < a/b, x < g/d$ :  $x' > 0, y' < 0$ .

Angel can stop the flow at any point on the current  $H$ -level set (including immediately with 0 time).

- Simple bounds during any flow (using positivity and  $a, b, d, g > 0$ ):
  - $x' = x(a - b y) \leq a x \implies x(t) \leq x_0 e^{a t}$ .
  - $y' = y(d x - g) \geq -g y \implies y(t) \geq y_0 e^{-g t}$ .

Positivity is invariant:  $x(t), y(t)$  stay  $> 0$  for all flow times.

Special submodes within the ODE:

- Exact equilibrium: If  $(x, y) = (g/d, a/b)$ , then  $x' = y' = 0$  (rest).
- Oscillation: Otherwise, neutral cycles on  $H$ -level sets around  $E^*$ ; Angel chooses how far to traverse each cycle segment by choosing the duration.

6751  
6752 4) Overall control pattern

- 6753 - Start: Demon test establishes initial positivity and  
6754 parameter bounds.
- 6755 - Iteration (repeats a Demon-chosen number of times):
- 6756 1) Demon picks  $x_{add} \geq 0$  (subgame\_j) and  $y_{add} \geq 0$   
6757 (subgame\_l) and applies the discrete increases to  $x, y$ .
- 6758 2) Angel chooses a duration to flow along the  
6759 Lotka{Volterra ODE (subgame\_p), moving along the  
6760 current closed orbit (conserved  $H$ ).
- 6761 - Interaction:
- 6762 - Demon's lever: choose when to repeat and how large the  
6763 nonnegative jumps are, thereby reselecting the  $H$ -level  
6764 set before each flow.
- 6765 - Angel's lever: select the stopping point along the  
6766 chosen  $H$ -level set each iteration (including skipping  
6767 the flow by choosing zero duration).
- 6768 - Because Demon can pick unbounded  $x_{add}, y_{add}$  in any  
6769 iteration,  $x$  and/or  $y$  can be made arbitrarily large;  
6770 the ODE phase then oscillates around  $E^*$  on the newly  
6771 determined level set until Angel stops.

6772  
6773 Angel control strategy:

6774 Covering the "freeze the flow" region so Demon's  
6775 nonnegative jumps alone preserve safety.

6776  
6777 Initial conditions/assumptions and start mode:

- 6778 - Assume the initial test holds:  $x > 0$ ,  $y > 0$ ,  $a, b, d, g > 0$ , and  
6779  $x \geq x_{min}$ ,  $y \geq y_{min}$ .
- 6780 - Start and remain in Immediate-stop ODE mode (Angel always  
6781 ends the flow instantly).

6782  
6783 Angel's strategy:

- 6784 1) Choose zero ODE duration immediately whenever the  
6785 Lotka{Volterra flow is offered, ending the continuous  
6786 phase instantly, repeatedly.
- 6787 2) Repeat step 1 after every Demon discrete increment and  
6788 update, forever (or until Demon stops the loop).

6789  
6790 Rationale: Discrete steps only increase  $x$  and  $y$   
6791 ( $x_{add}, y_{add} \geq 0$ ). By never flowing,  $x$  and  $y$  never  
6792 decrease. Since the initial test ensures  $x \geq x_{min}$  and  
6793  $y \geq y_{min}$ , these inequalities remain true throughout all  
6794 iterations.



Starting precondition computation.	6795
Proposed the following loop invariant for loop	6796
{xadd := *;^@; ? xadd >= 0;^@; yadd := *;^@;	6797
? yadd >= 0;^@; x := x + xadd; y := y + yadd;	6798
{x' = a * x - b * x * y, y' = d * x * y - g * y}}*^@.	6799
x >= xmin & y >= ymin	6800
Attempting to check invariant by precondition computation	6801
on loop body.	6802
Proposed precondition for subgame {x' = a * x - b * x * y,	6803
y' = d * x * y - g * y}^@ and postcondition x >= xmin &	6804
y >= ymin.	6805
Precondition or invariant: x >= xmin & y >= ymin.	6806
Check succeeded: False.	6807
Feedback: The formula check failed, indicating it is	6808
wrong, or too complex for the current level of	6809
automation. Please revise your guess or try a more	6810
detailed proof tactic.	6811
	6812
	6813
The options you can choose are:	6814
['try-proof', 'backtrack-to:p', 'backtrack-to:c']	6815



## 6816

## 6817

6818

6819

6820

6821

6822

6823

6824

6825

6826

6827

6828

6829

6830

6831

6832

6833

6834

is already assumed to be an invariant, it is enough to prove that  $I$  is an invariant itself.  $I$  holds initially since  $I \rightarrow I$  is valid and it implies **safe** by definition of a controllable invariant. Preservation holds by the definition of  $G$ :

$$\begin{array}{c} \text{id} \quad \frac{\frac{*}{I, G_i \vdash G_i}}{I, G_i \vdash [\text{act}_i; \text{plant}] I} \\ \frac{[\cdot], [\cup], [?], \wedge R}{\vdash I \rightarrow [\cup_i (?G_i; \text{act}_i); \text{plant}] I} \end{array}$$

where the axioms  $[\cdot], [\cup], [?]$  are used to unpack and repack the games leading to one conjunct for each action (treated separately via  $\wedge R$ ). We also need to prove that an action is always available, which holds by virtue of  $I$  being a controllable invariant:

$$\frac{\vdash I \rightarrow \bigvee_i [\text{act}_i; \text{plant}] I}{\vdash I \rightarrow \bigvee_i G_i}$$

□ □

**Lemma 5.3.** *If  $(I, G)$  is a solution, then  $I' \equiv (\text{assum} \wedge I)$  is a controllable invariant. Moreover, we have  $(I, G) \sqsubseteq (I', \mathcal{G}(I'))$ .*

*Proof.* Consider a solution  $(I, G)$ . We prove that  $I' \equiv (\text{assum} \wedge I)$  is a controllable invariant. Per Def. 5.1,  $I'$  is an invariant for  $\text{prob}(I, G)$  and so  $\models I' \rightarrow \text{safe}$ . Also, we have the following derivation which repacks games via axioms  $[\cup], [?], [\cdot]$  using their equivalences:

$$\begin{array}{c} \frac{I' \vdash [\cup_i (?G_i; \text{act}_i); \text{plant}] I'}{I' \vdash \bigwedge_i [?G_i; \text{act}_i; \text{plant}] I'} \quad \text{M} \\ \frac{I' \vdash \bigwedge_i [?G_i; \text{act}_i; \text{plant}] I'}{I' \vdash [?G_i; \text{act}_i; \text{plant}] I'} \quad \rightarrow R, [?], [\cdot] \\ \frac{I' \vdash \bigvee_i G_i \quad \frac{I', \bigvee_i G_i \vdash \bigvee_i [\text{act}_i; \text{plant}] I'}{\vdash I' \vdash \bigvee_i [\text{act}_i; \text{plant}] I'} \quad \text{cut}} \end{array}$$

where the open premises are part of the definition of  $(I, G)$  being a solution according to Def. 5.1. Let us now prove that  $(I, G) \sqsubseteq (I', \mathcal{G}(I'))$ . Trivially, we have  $\models \text{assum} \rightarrow (I \rightarrow (\text{assum} \wedge I))$ . Let us now derive  $\models \text{assum} \wedge I \rightarrow \bigwedge_i (G_i \rightarrow \mathcal{G}(I')_i)$ :

$$\frac{\frac{\text{assum} \wedge I \vdash [(\cup_i (?G_i; \text{act}_i)); \text{plant}] (\text{assum} \wedge I)}{[\cup], [?], [\cdot]} \quad \frac{\text{assum} \wedge I \vdash \bigwedge_i (G_i \rightarrow [\text{act}_i; \text{plant}] (\text{assum} \wedge I))}{\text{assum} \wedge I \vdash \bigwedge_i (G_i \rightarrow \mathcal{G}(I')_i)}}$$

where the remaining premise is part of the definition of  $(I, G)$  being a solution. This concludes the proof. □

□

**Lemma 5.4.** *If  $I$  and  $I'$  are controllable invariants, then  $(I, \mathcal{G}(I)) \sqsubseteq (I', \mathcal{G}(I'))$  if and only if  $\models \text{assum} \rightarrow (I \rightarrow I')$ .*

*Proof.* Let us first assume that  $\text{assum} \models I \rightarrow I'$  and prove that  $(I, \mathcal{G}(I)) \sqsubseteq (I', \mathcal{G}(I'))$ . It remains to show either  $\text{assum} \models \neg(I' \rightarrow I)$  or  $(\text{assum} \wedge I) \rightarrow \bigwedge_i (G_i \rightarrow G'_i)$  is valid. We show that  $\text{assum} \models I \rightarrow (\mathcal{G}(I)_i \rightarrow \mathcal{G}(I')_i)$  for all  $i$ . To do so, we leverage the fact that  $\text{assum}$  is an invariant.

$$\frac{\frac{\frac{*}{\text{assum}, I \vdash I'}}{\boxed{\wedge, M}[\text{act}_i; \text{plant}] \text{assum}, [\text{act}_i; \text{plant}] I \vdash [\text{act}_i; \text{plant}] I'}{[\ast] \text{assum}, [\text{act}_i; \text{plant}] I \vdash [\text{act}_i; \text{plant}] I'}{\text{assum} \vdash I \rightarrow (\mathcal{G}(I)_i \rightarrow \mathcal{G}(I')_i)}$$

The reverse direction follows trivially from the definition of  $\sqsubseteq$ .  $\square$

**Theorem 5.1.**  $S^{\text{opt}}$  is an optimal solution (i.e. a maximum w.r.t.  $\sqsubseteq$ ) of Def. 5.1.

*Proof.* We have  $S^{\text{opt}} \equiv (I^{\text{opt}}, \mathcal{G}(I^{\text{opt}}))$ . From Lemma 5.1 and Lemma 5.2,  $S^{\text{opt}}$  is a solution. Let  $(I, G)$  be another solution. From Lemma 5.3, there exists a controllable invariant  $I'$  such that  $(I, G) \sqsubseteq (I', \mathcal{G}(I'))$ . Then, from Lemma 5.4 and from the optimality of  $I^{\text{opt}}$  (Lemma 5.1), we have  $(I', \mathcal{G}(I')) \sqsubseteq (I^{\text{opt}}, \mathcal{G}(I^{\text{opt}}))$ . By transitivity,  $(I, G) \sqsubseteq S^{\text{opt}}$ . This concludes the proof.  $\square$

**Lemma 5.5.** If  $\text{act}_i$  is permanent,  $\models [(\text{act}_i; \text{plant})^*] \text{safe} \leftrightarrow [\text{act}_i; \text{plant}_\infty] \text{safe}$ .

*Proof.* We first prove that  $(\text{act}_i; \text{plant})^n \equiv (\text{act}_i; \text{plant}^n)$  by induction on  $n \geq 1$ . The base case is trivial. Regarding the induction case, we have

$$\begin{aligned} (\text{act}_i; \text{plant})^{n+1} &\equiv (\text{act}_i; \text{plant}; (\text{act}_i; \text{plant})^n) \\ &\equiv (\text{act}_i; \text{plant}; \text{act}_i; \text{plant}^n) \\ &\equiv (\text{act}_i; \text{plant}; \text{act}_i; \text{plant}^n) \\ &\equiv (\text{act}_i; \text{plant}; \text{plant}^n) \\ &\equiv (\text{act}_i; \text{plant}^{n+1}). \end{aligned}$$

From this, we get  $(\text{act}_i; \text{plant})^* \equiv ?\text{true} \cup (\text{act}_i; \text{plant}^*)$  from the semantics of loops in dGL. Thus, we have  $\models [(\text{act}_i; \text{plant})^*] \text{safe} \leftrightarrow \text{safe} \wedge [\text{act}_i; \text{plant}_\infty] \text{safe}$  since  $t$  does not appear free in  $\text{safe}$ . From this, we prove our theorem by noting that  $\text{safe} \wedge [\text{act}_i; \text{plant}_\infty] \text{safe} \leftrightarrow [\text{act}_i; \text{plant}_\infty] \text{safe}$  since  $\text{act}_i$  cannot write any variable that appears in  $\text{safe}$ .  $\square$

**Theorem 5.2.**  $I^0$  is a controllable invariant.

*Proof.* Trivially, we have  $\models I^0 \rightarrow \text{safe}$ . More interestingly, let us prove that  $I^0 \rightarrow \bigvee_i [\alpha_i] I^0$  where  $\alpha_i \equiv (\text{act}_i; \text{plant})$ . The proof crucially leverages the permanence assumption via the

6887 **identity**  $\models I^0 \leftrightarrow \bigvee_{i \in P} [\alpha_i^*]$  **safe**.

	$*$
$[*]$	$[\alpha_i^*] \text{ safe} \vdash [\alpha_i] [\alpha_i^*] \text{ safe}$
$M[\cdot]$	$[\alpha_i^*] \text{ safe} \vdash [\alpha_i] (\bigvee_{j \in P} [\alpha_j^*] \text{ safe})$
	$[\alpha_i^*] \text{ safe} \vdash [\alpha_i] I^0$
$\vee L, M[\cdot]$	$\bigvee_{i \in P} [\alpha_i^*] \text{ safe} \vdash \bigvee_{i \in P} [\alpha_i] I^0$
	$I^0 \vdash \bigvee_{i \in P} [\alpha_i] I^0$
$WR$	$I^0 \vdash \bigvee_i [\alpha_i] I^0$

6889

6890 **Theorem 5.3.**  *$I^n$  is a controllable invariant for all  $n \geq 0$ .*

6891 *Proof.* We proceed by induction on  $n$ . The base case is covered by Theorem 5.2. Assume that  
6892  $I^n$  is a controllable invariant and prove that  $I^{n+1}$  is one also. Abbreviate  $\alpha_i \equiv (\text{act}_i; \text{plant})$ .  
6893 Without loss of generality, assume that all actions are permanent since non-permanent actions  
6894 play no role in computing  $I^n$ . The hard part is in proving that  $\models I^n \rightarrow \bigvee_i [\alpha_i] I^{n+1}$ .

$$\frac{\frac{I^n \vdash \vee_i [\alpha_i] I^n \quad \vdash I^n \rightarrow I^{n+1}}{\text{M}[\cdot] \quad I^n \vdash \vee_i [\alpha_i] I^{n+1}} \quad \frac{\text{[step]} \quad I^n \vdash \vee_i [\alpha_i] I^{n+1}}{\text{VL} \quad I^n \vee \text{[step]} \quad I^n \vdash \vee_i [\alpha_i] I^{n+1}}}{I^{n+1} \vdash \vee_i [\alpha_i] I^{n+1}}$$

6896 The first premise is a consequence of  $(I^n, G^n)$  being a solution (our induction hypothesis) and  
6897 the second one is a trivial consequence of the definition of  $I^{n+1}$ . We can now focus on proving  
6898 the last premise.

6899 To do so, it is useful to introduce the following predicate:

$$R(a, b) \equiv [\text{plant}_b] (\text{safe} \wedge (t \geq a \rightarrow I^n))$$

Intuitively,  $R(a, b)$  is true if following the dynamics leads to reaching  $I^n$  within time interval  $[a, b]$  while being safe the whole time. Using this predicate, we can reformulate  $[\text{step}] I^n$  as follows:

$$[\text{step}] I^n \equiv \vee_i \exists \theta \geq 0 [\text{act}_i] R(\theta, \theta + T). \quad (\text{D.2})$$

6905 In addition, Lemma 5.6 gives us the following key property of  $R$ :

$$\vdash c \leq b \wedge R(a, b) \rightarrow [\text{plant}_c] R(a - t, b - t). \quad (\text{D.3})$$

6907 We can now complete the proof using Eq. (D.2):

$$\frac{\frac{\frac{\Gamma \vdash [\alpha_i] (t \leq \theta \rightarrow I^{n+1}) \quad \Gamma \vdash [\alpha_i] (t \geq \theta \rightarrow I^{n+1})}{\theta \geq 0, [\mathbf{act}_i] R(\theta, \theta + T) \vdash [\alpha_i] I^{n+1}}}{\vee_{\mathbf{R}}, \vee_{\mathbf{L}} \vee_i \exists \theta \geq 0 [\mathbf{act}_i] R(\theta, \theta + T) \vdash \vee_i [\alpha_i] I^{n+1}}}{[\mathbf{step}] I^n \vdash \vee_i [\alpha_i] I^{n+1}}$$

where we abbreviate  $\Gamma \equiv \theta \geq 0, [\mathbf{act}_i] R(\theta, \theta + T)$ . In the case where  $t \geq \theta$  after a control cycle, the agent has reached  $I^n$  and therefore  $I^{n+1}$ :

$$\frac{\frac{*}{\Gamma \vdash [\mathbf{act}_i] R(\theta, \theta + T)}}{\Gamma \vdash [\mathbf{act}_i] [\mathbf{plant}_{\theta+T}] (\mathbf{safe} \wedge (t \geq \theta \rightarrow I^n))} \quad \frac{[\cdot]}{\Gamma \vdash [\mathbf{act}_i; \mathbf{plant}] (t \geq \theta \rightarrow I^{n+1})} \quad 6911$$

In the case where  $t \leq \theta$  after a control cycle, the agent must perform the same action again with a timeout of  $\theta - t$ .

$$\frac{\frac{*}{\Gamma \vdash [\mathbf{act}_i] R(\theta, \theta + T)}}{\Gamma \vdash [\mathbf{act}_i; \mathbf{plant}] R(\theta - t, \theta - t + T)} \quad \frac{\Gamma \vdash [\mathbf{act}_i; \mathbf{plant}] R(\theta - t, \theta - t + T)}{\Gamma \vdash [\mathbf{act}_i; \mathbf{plant}; \mathbf{act}_i] R(\theta - t, \theta - t + T)} \quad \frac{\Gamma \vdash [\alpha_i] (t \leq \theta \rightarrow [\mathbf{act}_i] R(\theta - t, \theta - t + T))}{\Gamma \vdash [\alpha_i] (t \leq \theta \rightarrow (\exists \rho \leq 0 [\mathbf{act}_i] R(\rho, \rho + T)))} \quad \frac{\Gamma \vdash [\alpha_i] (t \leq \theta \rightarrow (\exists \rho \leq 0 [\mathbf{act}_i] R(\rho, \rho + T)))}{\Gamma \vdash [\alpha_i] (t \leq \theta \rightarrow I^{n+1})} \quad 6914$$

This concludes the proof. □ 6915

**Lemma 5.6.** Consider a property of the form  $R(a, b) \equiv [\alpha_b] (S \wedge (t \geq a \rightarrow I))$  with  $\alpha_b \equiv (t := 0; \{x' = f(x), t' = 1 \ \& \ Q \wedge t \leq b\})$ . Then this formula is valid:

$$c \leq b \wedge R(a, b) \rightarrow [\alpha_c] R(a - t, b - t). \quad 6918$$

*Proof.* This follows from the semantics of **dl** since all involved differential equations are the same and  $t \leq c \leq b$  is the duration that passes during  $\alpha_c$ , and thus explaining the offset of  $-t$  on the time interval arguments of  $R(a, b)$ . □ 6921

**Lemma 5.7.** If  $I^n \leftrightarrow I^{n+1}$  is valid for some  $n \geq 0$ , then  $I^n \leftrightarrow I^m$  is valid for all  $m \geq n$  and  $I^n \leftrightarrow I^\omega$  is valid where  $I^\omega \equiv [\mathbf{step}^\times; \mathbf{forever}] \mathbf{safe}$ . 6922 6923

*Proof.* The first part is simply a case of a recursive sequence  $I^{n+1} \equiv F(I^n)$  reaching a fixpoint ( $F(I) \equiv I \vee [\mathbf{step}] I$ ). Let us then prove the  $\models I^n \leftrightarrow I^\omega$  equivalence, or rather the nontrivial direction  $\models I^\omega \rightarrow I^n$ . From  $\models I^n \leftrightarrow I^{n+1}$ , we get  $\models I^n \leftrightarrow I^n \vee [\mathbf{step}] I^n$  and so  $\models [\mathbf{step}] I^n \rightarrow I^n$ . In addition, by the monotonicity of  $(I^n)_n$ , we have  $\models I^0 \rightarrow I^n$ . The rest follows from the  $\mathbf{FP}^\times$  rule:

$$\frac{\frac{\frac{I^0 \vdash I^n \quad [\mathbf{step}] I^n \vdash I^n}{\vee L \quad I^0 \vee [\mathbf{step}] I^n \vdash I^n}}{\mathbf{FP}^\times \quad [\mathbf{step}^\times] I^0 \vdash I^n}}{\frac{I^\omega \vdash I^n}} \quad 6930$$

□ □ 6931

**Theorem 5.4.** Consider a finite number of discrete **dl** programs  $\alpha_i$  such that  $\models \langle \alpha_i \rangle$  true for all  $i$  and  $p \equiv \{x' = f(x) \ \& \ q \geq 0\}$ . Then, provided that  $\mathbf{best}_j((\alpha_i)_i, p, m)$  and  $\mathbf{best}_j((\alpha_i)_i, p, -q)$  (no other action stops earlier because of the domain constraint), we have:

$$\models \langle ((\cap \alpha_i); p)^* \rangle m \leq 0 \leftrightarrow \langle (\alpha_j; p)^* \rangle m \leq 0. \quad 6935$$

*Proof.* The nontrivial implication to prove is:

$$\models \langle (\alpha_j; p)^* \rangle m \leq 0 \rightarrow \langle ((\cap_i \alpha_i); p)^* \rangle m \leq 0.$$

We do so by proving:

$$\Gamma \models \langle (\alpha_j^{(1)}; p^{(1)})^* \rangle m^{(1)} \leq 0 \rightarrow \langle ((\cap_i \alpha_i^{(2)}); p^{(2)})^* \rangle m^{(2)} \leq 0, \quad (\text{D.4})$$

where  $\Gamma \equiv \bigwedge_{x \in V} (x^{(1)} = x^{(2)})$  and  $V \equiv \text{BV}(p) \cup \bigcup_i \text{BV}(\alpha_i)$ . To prove Eq. (D.4), we chain together three implications:

1.  $\Gamma \models \langle (\alpha_j^{(1)}; p^{(1)})^* \rangle m^{(1)} \leq 0 \rightarrow \langle (\alpha_j^{(1)}; (\cap_i \alpha_i^{(2)}); p^{(1,2)})^* \rangle m^{(1)} \leq 0$ :
  - (a) To prove the implication above, we consider a sequence of states  $s_1 \dots s_n$  such that  $s_1 \in \llbracket \Gamma \rrbracket$ ,  $s_n \in \llbracket m^{(1)} \leq 0 \rrbracket$  and  $(s_i, s_{i+1} \in \llbracket \alpha_j^{(1)}; p^{(1)} \rrbracket)$  for all  $i$ . We must prove that  $s_1 \in \langle (\alpha_j^{(1)}; (\cap_i \alpha_i^{(2)}); p^{(1,2)})^* \rangle m^{(1)} \leq 0$ .
  - (b) We say that two states  $s$  and  $s'$  are *1-equivalent* (written  $s \sim_{(1)} s'$ ) if they only differ on variables tagged with 2. Using this definition, it is enough to prove the following fact: for all  $i \leq n$  and any state  $s$  such that  $s \sim_{(1)} s_i$  and  $s \in \llbracket q^{(1)} \leq q^{(2)} \rrbracket$ , we have  $s \in \langle (\alpha_j^{(1)}; (\cap_i \alpha_i^{(2)}); p^{(1,2)})^* \rangle m^{(1)} \leq 0$ .
  - (c) We prove the fact above by descending induction on  $i$ . The base case for  $i = n$  follows from our assumption on  $s_n$ . The inductive case considers a state  $s$  such that  $s \sim_{(1)} s_{i-1}$  and  $s \in \llbracket q^{(1)} \leq q^{(2)} \rrbracket$ . By assumption,  $(s_{i-1}, s_i) \in \llbracket \alpha_j^{(1)}; p^{(1)} \rrbracket$ . Thus, there exists  $s'$  such that  $(s, s') \in \llbracket \alpha_j^{(1)}; (\cap_i \alpha_i^{(2)}); p^{(1,2)} \rrbracket$ ,  $s' \sim_{(1)} s_i$  and  $s' \in \llbracket q^{(1)} \leq q^{(2)} \rrbracket$ . Note that the assumption that  $\text{best}_j((\alpha_i)_i, p, -q)$  is critical in establishing the existence of  $s'$ , by ensuring that  $p^{(1,2)}$  can be run for at least as long as  $p^{(1)}$ . We conclude by using the induction hypothesis on  $s'$ .
2.  $\Gamma \models \langle \gamma \rangle m^{(1)} \leq 0 \rightarrow \langle \gamma \rangle m^{(2)} \leq 0$  with  $\gamma \equiv (\alpha_j^{(1)}; (\cap_i \alpha_i^{(2)}); p^{(1,2)})^*$ :
  - (a) The key fact we are using here is that for any game  $\beta$  and formulas  $P, Q$ , we have  $\models [\beta^{-d}] (P \rightarrow Q) \rightarrow \langle \beta \rangle P \rightarrow \langle \beta \rangle Q$  where  $\beta^{-d}$  is obtained from  $\beta$  by removing all applications of the dual operator. Intuitively, this is true since  $[\beta^{-d}] (P \rightarrow Q)$  ensures that  $P \rightarrow Q$  is true in every reachable game state, independently of both players' strategies.
  - (b) Per the theorem assumption, we have  $\Gamma \models [\gamma^{-d}] (m^{(1)} \geq m^{(2)})$ . Using the monotonicity rule M, we obtain  $\Gamma \models [\gamma^{-d}] (P \rightarrow Q)$  with  $P \equiv m^{(2)} \leq 0$  and  $Q \equiv m^{(1)} \leq 0$ . We can then conclude using the previous point.
3.  $\models \langle (\alpha_j^{(1)}; (\cap_i \alpha_i^{(2)}); p^{(1,2)})^* \rangle m^{(2)} \leq 0 \rightarrow \langle ((\cap_i \alpha_i^{(2)}); p^{(2)})^* \rangle m^{(2)} \leq 0$ :
  - (a) This last implication can be proved in a similar way than (1.), although the  $\text{best}_j((\alpha_i)_i, p, -q)$  assumption is not needed since we are removing ODE domain constraints instead of adding them.

By chaining everything, we get:

$$\Gamma \models \langle (\alpha_j^{(1)}; p^{(1)})^* \rangle m^{(1)} \leq 0 \rightarrow \langle ((\cap_i \alpha_i^{(2)}); p^{(2)})^* \rangle m^{(2)} \leq 0,$$

which closes the proof. □



**Lemma 6.1** (Game Suffixes are Operational Gameplay Suffixes). *For subgame  $b:\beta$  in overall game  $a:\alpha$ , let the game suffix be  $a:\alpha_{b:}$ . Let  $\sigma$  be some state. The states reachable from playing  $a:\alpha$  such that  $\sigma$  is reached at subgame  $b:\beta$  at some point during the gameplay are the same as the states reachable from playing  $a:\alpha_{b:}$  starting in  $\sigma$ . That is,*

$$\{\pi_{-\mathfrak{S}}(s) : p \cdot b \cdot s \in \mathcal{T}(a:\alpha, \sigma'), [p]_{\sigma'} = \sigma \text{ and } \pi_{-\mathfrak{S}}(s) \neq ()\} = \pi_{-\mathfrak{S}}(\mathcal{T}(a:\alpha_{b:}, \sigma)),$$

where  $()$  is the empty path.

*Proof.* We prove

$$\{\pi_{-\mathfrak{S}}(s) : p \cdot b \cdot s \in \mathcal{T}(a:\alpha, \sigma'), [p]_{\sigma'} = \sigma \text{ and } \pi_{-\mathfrak{S}}(s) \neq ()\} = \pi_{-\mathfrak{S}}(\mathcal{T}(a:\alpha_{b:}, \sigma))$$

by induction on the structure of  $a:\alpha$ , and case analysis.

- When  $a:\alpha$  is atomic and not controlled by Angel, i.e.,  $\alpha \in \{x := e, x := \otimes, ?Q, !Q, \{x' = f(x) \ \& \ Q\}^d\}$ , then the only subgame  $b$  in  $\text{subgames}(a:\alpha)$  is  $a:\alpha$  itself. Thus the only possibility is  $b = a$ .  $a:\alpha_{a:}$  is the game  $a:\alpha$ .  $\mathcal{T}(a:\alpha_{a:}, \sigma)$  is then  $\mathcal{T}(a:\alpha, \sigma)$ . The left-hand side,  $\pi_{-\mathfrak{S}}(\mathcal{T}(a:\alpha, \sigma))$ , is  $\mathcal{T}(\alpha, \sigma)$ , the label-free game tree of the original game (see Def. B.4). Per the definition of labeled game trees for any atomic subgame  $\alpha$  (Def. B.2),  $\mathcal{T}(a:\alpha, \sigma)$  is  $a \cdot t : t \in \mathcal{T}(\alpha, \sigma)$ . Thus, the right-hand side is also  $\mathcal{T}(\alpha, \sigma)$ .
- Extending the previous case to compositional subgames, when  $b = a$  and  $a:\alpha$  is not a loop, then a similar argument holds. The left-hand side is  $\pi_{-\mathfrak{S}}(\mathcal{T}(a:\alpha, \sigma))$ . The right-hand side is  $\{\pi_{-\mathfrak{S}}(s) : p \cdot a \cdot s \in \mathcal{T}(a:\alpha, \sigma'), \pi_{-\mathfrak{S}}(s) \neq () \text{ and } [p]_{\sigma'} = \sigma\}$ . Per the definition of (non-loop) labeled game trees, the only occurrence of label  $a$  is at the beginning of all plays and thus the right-hand side is equal to  $\{\pi_{-\mathfrak{S}}(s) : a \cdot s \in \mathcal{T}(a:\alpha, \sigma), \pi_{-\mathfrak{S}}(s) \neq ()\}$ . This is just  $\pi_{-\mathfrak{S}}(\mathcal{T}(a:\alpha, \sigma))$ .
- When  $a:\alpha$  is a loop, i.e.,  $a:(g:\gamma)^*$  or  $a:(g:\gamma)^\times$ , and  $b \neq a$ , then  $a:\alpha_{b:} = (g:\gamma_{b:}); a:\alpha$ . The left-hand side is then  $\pi_{-\mathfrak{S}}(\mathcal{T}(g:\gamma_{b:}; a:\alpha, \sigma))$ . Per the inductive hypothesis,  $\pi_{-\mathfrak{S}}(\mathcal{T}(g:\gamma_{b:}, \sigma))$  is equal to  $\{\pi_{-\mathfrak{S}}(s) : p \cdot b \cdot s \in \mathcal{T}(g:\gamma, \sigma') \text{ and } [p]_{\sigma'} = \sigma\}$ . Thus, per the definition of labeled game trees for sequential composition games, the left-hand side is  $\{\pi_{-\mathfrak{S}}(s) : p \cdot b \cdot s \in \mathcal{T}(g:\gamma, \sigma') \text{ and } [p]_{\sigma'} = \sigma\} \cup \{\pi_{-\mathfrak{S}}(s \cdot t) : p \cdot b \cdot s \in \text{leaf}(\mathcal{T}(g:\gamma, \sigma')), [p]_{\sigma'} = \sigma, [s]_\sigma = \sigma'' \text{ and } t \in \mathcal{T}(a:\alpha, \sigma'')\}$ , where  $\text{leaf}$  returns the set of leaf nodes in a tree as discussed in Appendix B.6.2. The right-hand side is  $\{\pi_{-\mathfrak{S}}(s) : p \cdot b \cdot s \in \mathcal{T}(a:\alpha, \sigma'), \pi_{-\mathfrak{S}}(s) \neq () \text{ and } [p]_{\sigma'} = \sigma\}$ . Per the definition of labeled game trees for loops, we can break this down by unrolling the loop:  $\{\pi_{-\mathfrak{S}}(s) : p \cdot b \cdot s \in \mathcal{T}(g:\gamma, \sigma') \text{ and } [p]_{\sigma'} = \sigma\} \cup \{\pi_{-\mathfrak{S}}(s \cdot t) : p \cdot b \cdot s \in \text{leaf}(\mathcal{T}(g:\gamma, \sigma')), [p]_{\sigma'} = \sigma, [s]_\sigma = \sigma'' \text{ and } t \in \mathcal{T}(a:\alpha, \sigma'')\}$ .
- Otherwise, when  $b = a$  and  $a:\alpha$  is a loop, i.e.,  $a:(g:\gamma)^*$  or  $a:(g:\gamma)^\times$ , then  $a:\alpha_{a:} = a:\alpha$ . The left-hand side is  $\pi_{-\mathfrak{S}}(\mathcal{T}(a:\alpha, \sigma))$ . The right-hand side is  $\{\pi_{-\mathfrak{S}}(s) : p \cdot a \cdot s \in \mathcal{T}(a:\alpha, \sigma'), \pi_{-\mathfrak{S}}(s) \neq () \text{ and } [p]_{\sigma'} = \sigma\}$ . By the choice of  $\sigma' = \sigma$ ,  $p = ()$  and the definition of labeled game trees, we can show that this is a superset of  $\pi_{-\mathfrak{S}}(\mathcal{T}(a:\alpha, \sigma))$ . Moreover, by the definition of labeled game trees

for loops, we know that any suffix  $s$  that follows label  $a$  must be generated by the loop fixed point function  $f$  from Def. B.2 for initial state  $\sigma$ . So,  $a \cdot s$  must be within the game tree  $\mathcal{T}(a:\alpha, \sigma)$ . Therefore, the right-hand side is also a subset of  $\pi_{-\mathcal{G}}(\mathcal{T}(a:\alpha, \sigma))$ . Thus, the left-hand side and right-hand side must be equal.

- When  $a:\alpha$  is a choice, i.e.,  $a:(g:\gamma \cup d:\delta)$  or  $a:(g:\gamma \cap d:\delta)$ , and  $b \neq a$ , then there are two cases.

1. If  $b \in \text{subgames}(g:\gamma)$ , then  $a:\alpha_{b:} = g:\gamma_{b:}$ .

The *left-hand side* is  $\pi_{-\mathcal{G}}(\mathcal{T}(g:\gamma_{b:}, \sigma))$ . Per the inductive hypothesis, this is  $\{\pi_{-\mathcal{G}}(s) : p \cdot b \cdot s \in \mathcal{T}(g:\gamma, \sigma'), \pi_{-\mathcal{G}}(s) \neq () \text{ and } \lfloor p \rfloor_{\sigma'} = \sigma\}$ .

The *right-hand side* is  $\{\pi_{-\mathcal{G}}(s) : p \cdot b \cdot s \in \mathcal{T}(a:\alpha, \sigma'), \pi_{-\mathcal{G}}(s) \neq () \text{ and } \lfloor p \rfloor_{\sigma'} = \sigma\}$ . Per the definition of labeled game trees for choices,  $\mathcal{T}(a:\alpha, \sigma')$  is the union of the game trees of  $g:\gamma$  and  $d:\delta$ . However,  $b$  does not appear in  $d:\delta$ . Thus the right-hand side can be simplified to  $\{\pi_{-\mathcal{G}}(s) : p \cdot b \cdot s \in \mathcal{T}(g:\gamma, \sigma'), \pi_{-\mathcal{G}}(s) \neq () \text{ and } \lfloor p \rfloor_{\sigma'} = \sigma\}$ .

2. If  $b \in \text{subgames}(d:\delta)$ , then  $a:\alpha_{b:} = d:\delta_{b:}$ , and a symmetric argument applies, with  $g:\gamma$  and  $d:\delta$  swapped.

- When  $a:\alpha$  is a sequential composition, i.e.,  $a:(g:\gamma ; d:\delta)$ , and  $b \neq a$ , then there are two cases.

1. If  $b \in \text{subgames}(g:\gamma)$ , then  $a:\alpha_{b:} = (g:\gamma_{b:}) ; d:\delta$ . The argument is similar to the loop case.

The *left-hand side* is  $\pi_{-\mathcal{G}}(\mathcal{T}((g:\gamma_{b:}) ; d:\delta, \sigma))$ .

As per the inductive hypothesis,  $\pi_{-\mathcal{G}}(\mathcal{T}(g:\gamma_{b:}, \sigma))$  is equal to  $\{\pi_{-\mathcal{G}}(s) : p \cdot b \cdot s \in \mathcal{T}(g:\gamma, \sigma'), \pi_{-\mathcal{G}}(s) \neq () \text{ and } \lfloor p \rfloor_{\sigma'} = \sigma\}$ .

Thus, per the definition of labeled game trees for sequential composition games, the left-hand side is  $\{\pi_{-\mathcal{G}}(s) : p \cdot b \cdot s \in \mathcal{T}(g:\gamma, \sigma'), \lfloor p \rfloor_{\sigma'} = \sigma, \pi_{-\mathcal{G}}(s) \neq ()\} \cup \{\pi_{-\mathcal{G}}(s \cdot t) : p \cdot b \cdot s \in \text{leaf}(\mathcal{T}(g:\gamma, \sigma')), t \in \mathcal{T}(d:\delta, \sigma''), \lfloor p \rfloor_{\sigma'} = \sigma, \pi_{-\mathcal{G}}(s \cdot t) \neq () \text{ and } \lfloor s \rfloor_{\sigma} = \sigma''\}$ .

The *right-hand side* is  $\{\pi_{-\mathcal{G}}(s) : p \cdot b \cdot s \in \mathcal{T}(a:\alpha, \sigma'), \pi_{-\mathcal{G}}(s) \neq () \text{ and } \lfloor p \rfloor_{\sigma'} = \sigma\}$ . Per the definition of labeled game trees for sequential composition, since  $b \notin \text{subgames}(d:\delta)$ , this is:  $\{\pi_{-\mathcal{G}}(s) : p \cdot b \cdot s \in \mathcal{T}(g:\gamma, \sigma'), \lfloor p \rfloor_{\sigma'} = \sigma, \pi_{-\mathcal{G}}(s) \neq ()\} \cup \{\pi_{-\mathcal{G}}(s \cdot t) : p \cdot b \cdot s \in \text{leaf}(\mathcal{T}(g:\gamma, \sigma')), t \in \mathcal{T}(d:\delta, \sigma''), \lfloor p \rfloor_{\sigma'} = \sigma, \pi_{-\mathcal{G}}(s \cdot t) \neq () \text{ and } \lfloor s \rfloor_{\sigma} = \sigma''\}$ .

2. If  $b \in \text{subgames}(d:\delta)$ , then  $a:\alpha_{b:} = d:\delta_{b:}$ .

The *left-hand side* is  $\pi_{-\mathcal{G}}(\mathcal{T}(d:\delta_{b:}, \sigma))$ .

Per the inductive hypothesis, this is  $\{\pi_{-\mathcal{G}}(s) : p \cdot b \cdot s \in \mathcal{T}(d:\delta, \sigma'), \pi_{-\mathcal{G}}(s) \neq () \text{ and } \lfloor p \rfloor_{\sigma'} = \sigma\}$ .

The *right-hand side* is  $\{\pi_{-\mathcal{G}}(s) : p \cdot b \cdot s \in \mathcal{T}(a:\alpha, \sigma'), \pi_{-\mathcal{G}}(s) \neq () \text{ and } \lfloor p \rfloor_{\sigma'} = \sigma\}$ .

Per the definition of labeled game trees for sequential composition, using the fact that  $b \notin \text{subgames}(g:\gamma)$ , this can be simplified to  $\{\pi_{-\mathcal{G}}(s) : p \cdot b \cdot s \in \mathcal{T}(d:\delta, \sigma') \text{ and } \lfloor p \rfloor_{\sigma'} = \sigma\}$ .

□

**Lemma 6.2** (Execution Prefixes are Original Gameplay Prefixes). *For subgame  $b:\beta$  in  $a:\alpha$ , for any initial state  $\sigma$ , the states reachable at  $b:\beta$  while playing  $a:\alpha$  are equal to those reachable by*

playing game prefix  $a:\alpha.b$ . That is, prefix set  $\{\pi_{-\ominus}(p) : p \cdot b \cdot s \in \mathcal{T}(a:\alpha, \sigma)\}$  is equal, under prefix closure, to prefix game tree  $\pi_{-\ominus}(\mathcal{T}(a:\alpha.b, \sigma))$ <sup>1</sup>. 7052  
7053

*Proof.* We show that 7054

$$\pi_{-\ominus}(\mathcal{T}(a:\alpha.b, \sigma)) = \{\pi_{-\ominus}(t) : p \cdot b \cdot s \in \mathcal{T}(a:\alpha, \sigma) \text{ and } p = t \cdot u\} \quad 7055$$

Similar to Lemma 6.1, we perform induction on the structure of  $a:\alpha$  and case analysis. 7056

- When  $b = a$  and  $a:\alpha$  is not a loop, then  $a:\alpha.a$  is the empty game skip. 7057  
 The *left-hand side*  $\pi_{-\ominus}(\mathcal{T}(a:\alpha.a, \sigma))$  is then  $\emptyset$ . 7058  
 Indeed, on the *right-hand side*, all plays in  $\mathcal{T}(a:\alpha, \sigma)$  are of the form  $a \cdot p$  where  $p$  does not contain  $a$ , so the condition of the theorem holds. 7059  
7060
- When  $a:\alpha$  is atomic and not controlled by Angel, i.e.,  $\alpha \in \{x := e, x := \otimes, ?Q, !Q, \{x' = f(x) \ \& \ Q\}^d\}$ , then the only subgame  $b$  in  $\text{subgames}(a:\alpha)$  is  $a:\alpha$  itself. Thus the only possibility is  $b = a$ , handled already by the first case. 7061  
7062  
7063
- When  $a:\alpha$  is a choice, i.e.,  $a:(g:\gamma \cup d:\delta)$  or  $a:(g:\gamma \cap d:\delta)$ , then we analyze cases depending on the value of  $b \neq a$ . 7064  
7065

1. If  $b$  is in  $g:\gamma$ , then  $a:\alpha.b = g:\gamma.b$ . 7066

The *left-hand side* is  $\pi_{-\ominus}(\mathcal{T}(g:\gamma.b, \sigma))$ . Per the inductive hypothesis, this consists of 7067

$$\{\pi_{-\ominus}(t) : p \cdot b \cdot s \in \mathcal{T}(g:\gamma, \sigma), p = t \cdot u, \pi_{-\ominus}(t) \neq ()\}. \quad 7068$$

The *right-hand side* is  $\{\pi_{-\ominus}(t) : p \cdot b \cdot s \in \mathcal{T}(a:\alpha, \sigma), p = t \cdot u, \pi_{-\ominus}(t) \neq ()\}$ . Per the definition of game trees for choices, since  $\mathcal{T}(a:\alpha, \sigma)$  is the union of the game trees for  $g:\gamma$  and  $d:\delta$ , and  $b$  does not appear in  $d:\delta$ , this simplifies to 7069  
7070  
7071

$$\{\pi_{-\ominus}(t) : p \cdot b \cdot s \in \mathcal{T}(g:\gamma, \sigma), p = t \cdot u, \pi_{-\ominus}(t) \neq ()\}. \quad 7072$$

2. If  $b$  is in  $d:\delta$ , then the argument is symmetric to the previous case, but  $g:\gamma$  swapped with  $d:\delta$ . 7073  
7074

- When  $a:\alpha$  is a sequential composition, i.e.,  $a:(g:\gamma ; d:\delta)$ , then we analyze cases depending on the value of  $b \neq a$ . 7075  
7076

1. If  $b$  is in  $g:\gamma$ , then  $a:\alpha.b = g:\gamma.b$ . 7077

The *left-hand side* is  $\pi_{-\ominus}(\mathcal{T}(g:\gamma.b, \sigma))$ . Per the inductive hypothesis, this is 7078

$$\{\pi_{-\ominus}(t) : p \cdot b \cdot s \in \mathcal{T}(g:\gamma, \sigma), p = t \cdot u, \pi_{-\ominus}(t) \neq ()\}. \quad 7079$$

The *right-hand side* is  $\{\pi_{-\ominus}(t) : p \cdot b \cdot s \in \mathcal{T}(a:\alpha, \sigma), p = t \cdot u, \pi_{-\ominus}(t) \neq ()\}$ . Since  $b \notin \text{subgames}(d:\delta)$ , using the definition of labeled game trees for sequential composition, this can be simplified to 7080  
7081  
7082

$$\{\pi_{-\ominus}(t) : p \cdot b \cdot s \in \mathcal{T}(g:\gamma, \sigma), p = t \cdot u, \pi_{-\ominus}(t) \neq ()\}. \quad 7083$$

<sup>1</sup>The prefix closure is required to recover tree structure. Unlike with the suffix construction, the prefix set builder collects only *parent* subgames of  $b:\beta$  and *not all predecessors*. Expanding the prefix closure results in the condition:  $\{\pi_{-\ominus}(t) : p \cdot b \cdot s \in \mathcal{T}(a:\alpha, \sigma) \text{ and } p = t \cdot u\} = \pi_{-\ominus}(\mathcal{T}(a:\alpha.b, \sigma))$

2. If  $b$  is in  $d\delta$ , then  $a\alpha.b = g\gamma ; d\delta.b$ .

The *left-hand side* is  $\pi_{-\mathfrak{S}}(\mathcal{T}(g\gamma ; d\delta.b, \sigma))$ .

As per the inductive hypothesis,  $\pi_{-\mathfrak{S}}(\mathcal{T}(d\delta.b, \sigma))$  is equal to  $\{\pi_{-\mathfrak{S}}(t) : p \cdot b \cdot s \in \mathcal{T}(d\delta, \sigma), p = t \cdot u, \pi_{-\mathfrak{S}}(t) \neq ()\}$ .

Thus, per the definition of labeled game trees for sequential composition games, the left-hand side is

$$\pi_{-\mathfrak{S}}(\mathcal{T}(g\gamma, \sigma)) \cup \{\pi_{-\mathfrak{S}}(q \cdot t) : q \in \text{leaf}(\mathcal{T}(g\gamma, \sigma)), [q]_{\sigma} = \sigma', p \cdot b \cdot s \in \mathcal{T}(d\delta, \sigma'), p = t \cdot u\} \blacksquare$$

where *leaf* indicates a leaf node of a game tree, as indicated in Appendix B.6.2.

The *right-hand side* is  $\{\pi_{-\mathfrak{S}}(t) : p \cdot b \cdot s \in \mathcal{T}(a\alpha, \sigma), p = t \cdot u, \pi_{-\mathfrak{S}}(t) \neq ()\}$ .

Per the definition of labeled game trees for sequential composition, since  $b \notin \text{subgames}(g\gamma)$ , this can be written as

$$\pi_{-\mathfrak{S}}(\mathcal{T}(g\gamma, \sigma)) \cup \{\pi_{-\mathfrak{S}}(q \cdot t) : q \in \text{leaf}(\mathcal{T}(g\gamma, \sigma)), [q]_{\sigma} = \sigma', p \cdot b \cdot s \in \mathcal{T}(d\delta, \sigma'), p = t \cdot u\}. \blacksquare$$

- When  $a\alpha$  is a loop, i.e.,  $a(g\gamma)^*$  or  $a(g\gamma)^{\times}$ , then we analyze cases depending on the value of  $b$ .

1. If  $b = a$ , then  $a\alpha.b = a\alpha$ .

The *left-hand side* is then  $\pi_{-\mathfrak{S}}(\mathcal{T}(a\alpha, \sigma))$ .

The *right-hand side* is  $\{\pi_{-\mathfrak{S}}(t) : p \cdot a \cdot s \in \mathcal{T}(a\alpha, \sigma), p = t \cdot u, \pi_{-\mathfrak{S}}(t) \neq ()\}$ . By the definition of labeled game trees (Def. B.2), all plays in  $\mathcal{T}(a\alpha, \sigma)$  end with  $a \cdot \mathfrak{s}^d$  or  $a \cdot \mathfrak{s}$ . Thus, from the elements where play  $s$  is  $\mathfrak{s}$  or  $\mathfrak{s}^d$ ,  $\{\pi_{-\mathfrak{S}}(t) : p \cdot a \cdot s \in \mathcal{T}(a\alpha, \sigma), p = t \cdot u, \pi_{-\mathfrak{S}}(t) \neq ()\}$  is a superset of  $\pi_{-\mathfrak{S}}(\mathcal{T}(a\alpha, \sigma))$ . Moreover, as labeled game trees are prefix-closed,  $\{\pi_{-\mathfrak{S}}(t) : p \cdot a \cdot s \in \mathcal{T}(a\alpha, \sigma), p = t \cdot u, \pi_{-\mathfrak{S}}(t) \neq ()\}$  is also a subset of  $\pi_{-\mathfrak{S}}(\mathcal{T}(a\alpha, \sigma))$ . Thus, the left-hand side and right-hand side must be equal.

2. If  $b \neq a$ , then  $a\alpha.b = (a\alpha ; g\gamma.b)$ .

The *left-hand side* is

$$\pi_{-\mathfrak{S}}(\mathcal{T}(a\alpha ; (g\gamma.b), \sigma)).$$

As per the inductive hypothesis,  $\pi_{-\mathfrak{S}}(\mathcal{T}(g\gamma.b, \sigma))$  is equal to  $\{\pi_{-\mathfrak{S}}(t) : p \cdot b \cdot s \in \mathcal{T}(g\gamma, \sigma), p = t \cdot u, \pi_{-\mathfrak{S}}(t) \neq ()\}$ . Thus, per the definition of labeled game trees for sequential composition games, the left-hand side is

$$\pi_{-\mathfrak{S}}(\mathcal{T}(a\alpha, \sigma)) \cup \{\pi_{-\mathfrak{S}}(q \cdot t) : q \in \text{leaf}(\mathcal{T}(a\alpha, \sigma)), [q]_{\sigma} = \sigma', p \cdot b \cdot s \in \mathcal{T}(g\gamma, \sigma'), p = t \cdot u\}. \blacksquare$$

The *right-hand side* is  $\{\pi_{-\mathfrak{S}}(t) : p \cdot b \cdot s \in \mathcal{T}(a\alpha, \sigma), p = t \cdot u, \pi_{-\mathfrak{S}}(t) \neq ()\}$ . This is a subset of  $\pi_{-\mathfrak{S}}(\mathcal{T}(a\alpha, \sigma))$ , and therefore the left-hand side, because game trees are prefix-closed. Next we show it is also a superset. Consider any play in the left-hand side set. For Angelic loops, it must be a prefix of some play of the form  $q \cdot p$  where  $q \in \text{leaf}(f^n(a \hat{\mathfrak{s}}, a \hat{\mathfrak{g}}))$  for some  $n$ ,  $[q]_{\sigma} = \sigma'$ , and  $p \cdot b \cdot s \in \mathcal{T}(g\gamma, \sigma')$  where  $f$  and  $\hat{\cdot}$  are from Def. B.2. Such a play and all its prefixes feature in the right-hand side set because they are in  $f^{n+1}(a \hat{\mathfrak{s}}, a \hat{\mathfrak{g}})$  which is in  $\mathcal{T}(a\alpha, \sigma)$ . For Demonic loops, the same argument applies but for  $f^n(a \hat{\mathfrak{s}}^d, a \hat{\mathfrak{g}}^d)$  as defined in Def. B.2.

□

**Lemma 6.3** (Existential projection correspondence). *For any Angelic subvalue map  $S$  for game  $a:\alpha$  and compatible winning condition  $\phi$ , in initial state  $\sigma \models \langle \alpha \rangle \phi$ , Angel has a winning strategy for the game  $a:\alpha \Downarrow S$  if and only if Angel can win  $\langle \alpha \rangle \phi$  while following the policy of  $S$ . That is,  $\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$  iff  $\mathcal{S}_{a:\alpha}(S)(\sigma) \neq \emptyset$ .*

*Proof.* We use structural induction on the game  $a:\alpha$ . The cases for  $a:\alpha$  are as follows.

- When  $\alpha$  is atomic and not controlled by Angel, that is,  $\alpha \in \{x := e, x := \otimes, ?Q, !Q, \{x' = f(x) \ \& \ Q\}^d\}$ , then:  
 $\langle a:\alpha \Downarrow S \rangle \phi$  is  
 $\langle \alpha \rangle \phi$  because  $a:\alpha \Downarrow S = \alpha$ ,  
which holds because by assumption  $\sigma \models \langle \alpha \rangle \phi$ .  
The corresponding strategy sets are all nonempty and always contain exactly one strategy per Def. B.7 as Angel has no decisions to make and the Angelic strategy is just the full game tree.
- If  $a:\alpha$  is a non-deterministic Angel assignment  $a:x := *$ , then  $\mathcal{S}_{a:\alpha}(S)(\sigma)$  has the strategy set  $\{\{(x := e) \mid \sigma(x \mapsto e) \models S(\text{end})\}\}$ .  
 $\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$  iff  
 $\sigma \models \langle x := *; ?S(\text{end}) \rangle \phi$  because  $a:\alpha \Downarrow S$  is  $x := *; ?S(\text{end})$   
iff  $\sigma \models \langle x := * \rangle (S(\text{end}) \wedge \phi)$  applying axiom  $\langle ? \rangle$   
iff  $\sigma \models \langle x := * \rangle (S(\text{end}))$   $S(\text{end}) \wedge \phi \leftrightarrow S(\text{end})$  as  $\phi$  is a compatible condition  
iff  $\exists e \sigma(x \mapsto e) \models S(\text{end})$  by semantics of dGL  
which is exactly the condition for which the strategy set is nonempty.
- If  $a:\alpha$  is an Angelic ODE  $a:\{x' = f(x) \ \& \ Q\}$ , then  $\mathcal{S}_{a:\alpha}(S)(\sigma)$  has the strategy set  $\{\{(x' = f(x) \ \& \ Q @ t) : t \geq 0, \forall s \in [0, t], \varphi(s) \models S(b) \text{ and } \varphi(t) \models S(\text{succ}(b, a:\alpha)) \text{ where } \varphi : [0, t] \rightarrow \mathcal{S} \text{ is differentiable, } \varphi(0) = \sigma, \text{ and } \forall s \in [0, t], \varphi(s) \models x' = f(x) \wedge Q\}\}$ .  
 $\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$  iff  
 $\sigma \models \langle \{x' = f(x) \ \& \ Q\}; ?S(\text{end}) \rangle \phi$  because  $a:\alpha \Downarrow S$  is  $\{x' = f(x) \ \& \ Q\}; ?S(\text{end})$ ,  
iff  $\sigma \models \langle \{x' = f(x) \ \& \ Q\} \rangle (S(\text{end}) \wedge \phi)$  applying axiom  $\langle ? \rangle$   
iff  $\sigma \models \langle \{x' = f(x) \ \& \ Q\} \rangle (S(\text{end}))$   $S(\text{end}) \wedge \phi \leftrightarrow S(\text{end})$  as  $\phi$  is a compatible winning condition  
Which per the semantics of dGL is exactly when the strategy set is nonempty.
- If  $a:\alpha$  is an Angelic choice  $a:(g:\gamma \cup d:\delta)$ , then  $\mathcal{S}_{a:\alpha}(S)(\sigma)$  there are four possibilities.
  1. If  $\sigma \not\models S(g)$  and  $\sigma \not\models S(d)$ , then the strategy set is empty.  
Accordingly, Angel cannot win the existential projection game:  $\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$  iff  
 $\sigma \models \langle (?S(g); g:\gamma) \cup (?S(d); d:\delta) \rangle \phi$  because  $a:\alpha \Downarrow S$  is  $(?S(g); g:\gamma) \cup (?S(d); d:\delta)$   
iff  $\sigma \models \langle (?S(g); g:\gamma \Downarrow S) \rangle \phi \vee \langle (?S(d); d:\delta \Downarrow S) \rangle \phi$  by axiom  $\langle \cup \rangle$   
iff  $\sigma \models (S(g) \wedge \langle g:\gamma \Downarrow S \rangle \phi) \vee (S(d) \wedge \langle d:\delta \Downarrow S \rangle \phi)$  by axiom  $\langle ? \rangle, \langle ; \rangle$   
Which is false by assumption  $\sigma \not\models S(g)$  and  $\sigma \not\models S(d)$ .
  2. If  $\sigma \models S(g)$  and  $\sigma \not\models S(d)$ , then the strategy set is  $\{\{t\} : t \in \mathcal{S}_{g:\gamma}(S)(\sigma)\}$ , which is nonempty when  $\mathcal{S}_{g:\gamma}(S)(\sigma)$  is nonempty. By the inductive hypothesis,  $\mathcal{S}_{g:\gamma}(S)(\sigma)$  is nonempty when  $\sigma \models \langle g:\gamma \Downarrow S \rangle \phi$ .

$\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$  iff  
 $\sigma \models \langle ((?S(g); g:\gamma) \cup (?S(d); d:\delta)) \rangle \phi$  because  $a:\alpha \Downarrow S$  is  $(?S(g); g:\gamma) \cup (?S(d); d:\delta)$   
iff  $\sigma \models \langle (?S(g); g:\gamma \Downarrow S) \rangle \phi \vee \langle (?S(d); d:\delta \Downarrow S) \rangle \phi$  by axiom  $\langle \cup \rangle$   
iff  $\sigma \models \langle g:\gamma \Downarrow S \rangle \phi \vee \langle S(d) \wedge \langle d:\delta \Downarrow S \rangle \phi \rangle$   $\sigma \models S(g)$ , applying axioms  $\langle ? \rangle$ ,  $\langle ; \rangle$   
iff  $\sigma \models \langle g:\gamma \Downarrow S \rangle \phi$   $\sigma \not\models S(g)$  by assumption  
which is exactly the condition for which the strategy set is nonempty.

3. If  $\sigma \models S(d)$  and  $\sigma \not\models S(g)$ , the argument is symmetric but with the roles of  $\gamma$  and  $\delta$  swapped.

4. If  $\sigma \models S(g)$  and  $\sigma \models S(d)$ , then the strategy set is  $\{\{l \hat{t}\} : t \in \mathcal{S}_{g:\gamma}(S)(\sigma)\} \cup \{\{r \hat{t}\} : t \in \mathcal{S}_{d:\delta}(S)(\sigma)\}$ . This set is nonempty when either  $\mathcal{S}_{g:\gamma}(S)(\sigma)$  or  $\mathcal{S}_{d:\delta}(S)(\sigma)$  is nonempty. By the inductive hypothesis, this is when  $\sigma \models \langle g:\gamma \Downarrow S \rangle \phi$  or  $\sigma \models \langle d:\delta \Downarrow S \rangle \phi$ .

$\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$  iff  
 $\sigma \models \langle ((?S(g); g:\gamma) \cup (?S(d); d:\delta)) \rangle \phi$  because  $a:\alpha \Downarrow S$  is  $(?S(g); g:\gamma) \cup (?S(d); d:\delta)$   
iff  $\sigma \models \langle (?S(g); g:\gamma \Downarrow S) \rangle \phi \vee \langle (?S(d); d:\delta \Downarrow S) \rangle \phi$  by axiom  $\langle \cup \rangle$   
iff  $\sigma \models \langle g:\gamma \Downarrow S \rangle \phi \vee \langle S(d) \wedge \langle d:\delta \Downarrow S \rangle \phi \rangle$   $\sigma \models S(g)$ ,  $\sigma \models S(d)$  which is exactly the condition  
for which the strategy set is nonempty.

- If  $a:\alpha$  is an Angelic loop  $a:(g:\gamma)^*$ . By the inductive hypothesis,  $\mathcal{S}_{g:\gamma}(S)(\sigma)$  is nonempty when  $\sigma \models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(a)) \rangle S(a)$ .

$\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$  iff  
 $\sigma \models \langle a:\alpha \Downarrow S \rangle S(\text{end})$   $\phi$  is a compatible winning condition

We show that if there is a strategy in the subvalue map strategy set then there is a strategy to win  $\langle (?S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^* \rangle (S(\text{end}))$  from  $\sigma$  and vice versa by induction over iterations.

First we build up a translation function  $i_Z$  inductively builds up translations for sequences (elements) in strategy set  $Z$ .

It maps elements of a strategy for  $\langle (?S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^* \rangle (S(\text{end}))$  to elements of a strategy in the subvalue map strategy set.

For example,  $i_{\{g\}}(g) = g$ .  $i_Z$  is overloaded to also apply to strategies by applying to every element of the strategy.

$i_{f(Z)} \equiv \{s \hat{ } (?S(\text{end}))\} \mapsto \{s\}; \quad \{g\} \mapsto \{g\};$   
for  $t \cdot g \in \text{leaf}(Z)$ ,  $(t \hat{ } g \hat{ } (?S(g)) \hat{ } u_t) \mapsto (i_Z(t \hat{ } g) \hat{ } h_{g:\gamma}(u_t))$   
 $(t \hat{ } g \hat{ } (?S(g)) \hat{ } u_t \hat{ } g) \mapsto (i_Z(t \hat{ } g) \hat{ } h_{g:\gamma}(u_t) \hat{ } g)$   
 $(t \hat{ } g \hat{ } (?S(g)) \hat{ } u_t \hat{ } s \hat{ } (?S(\text{end}))) \mapsto (i_Z(t \hat{ } g) \hat{ } h_{g:\gamma}(u_t) \hat{ } s)$   
where  $u_t$  is a winning strategy for game  $\langle g:\gamma \Downarrow S(\text{end} \mapsto S(a)) \rangle S(a)$  and  
corresponding translation function  $h_{g:\gamma}(u_t)$  for strategies of game  
 $\langle g:\gamma \Downarrow S(\text{end} \mapsto S(a)) \rangle S(a)$  to strategies induced by subvalue map  
 $S(\text{end} \mapsto S(a))$ .

Above, where for readability, the map implicitly respects prefix closure over *caret*, i.e., if

$a \hat{ } b \mapsto c \hat{ } d$  then  $a \mapsto c$  and  $a \cdot b \mapsto c \cdot d$ . The exception is for the extra  $?S(g)$  and  $?S(\text{end})$  actions which have no corresponding mapped action. The mapping function  $h_{g:\gamma}(u_t)$  can be recursively defined to map a strategy winning the existential projection of a game onto a subvalue map to a strategy from the strategy set generated by that subvalue map where both strategies result in the same final state, though we do not show the full construction here.

Now we argue that the translated functions are amongst the subvalue map's induced strategies.

- As base case, if  $\{\mathfrak{s}, \mathfrak{s} \cdot (?S(\text{end}))\}$  wins the projection game from initial state  $\sigma$  then  $\{\mathfrak{s}\}$  is in  $\mathcal{S}_{a:\alpha}(S)(\sigma)$ . Otherwise if  $\mathfrak{g}$  is the first step in the projection game's strategy, then  $\sigma \models S(g)$  of the Angelic test that follows,  $\{\mathfrak{g}\}$  can start strategies in  $\mathcal{S}_{a:\alpha}(S)(\sigma)$ .
- For the recursive case, suppose that  $Z \cup (\bigcup_{t \cdot \mathfrak{g} \in \text{leaf}(Z)} t \hat{ } \mathfrak{g} \hat{ } (?S(g)) \hat{ } u_t \hat{ } \mathfrak{s} \hat{ } (?S(\text{end})))$  is a strategy that wins for the projection game. Then  $(i_Z(t \hat{ } \mathfrak{g}) \hat{ } h_{g:\gamma}(u_t) \hat{ } \mathfrak{s})$  should be in  $\mathcal{S}_{a:\alpha}(S)(\sigma)$  for the original game.  $i_Z(t \hat{ } \mathfrak{g})$  can begin a strategy in  $\mathcal{S}_{a:\alpha}(S)(\sigma)$  because of the inductive hypothesis of the inner induction on iterations.  $h_{g:\gamma}(u_t)$  is some strategy for game  $\langle g:\gamma \Downarrow S(\text{end} \mapsto S(a)) \rangle S(a)$  which must exist because of the inductive hypothesis from the outer proof.  $\mathfrak{s}$  appears in  $\mathcal{P}_{a:\alpha}(S)(\lfloor \sigma \rfloor_{t \hat{ } \mathfrak{g} \hat{ } h_{g:\gamma}(u_t) \hat{ } \mathfrak{s}})$ , appearing as a final action during strategy generation because in the original projection game, because in the existential projection it was in a state where the test  $(?S(\text{end}))$  is satisfied.
- Otherwise, if  $(Z \cup \bigcup_{t \cdot \mathfrak{g} \in \text{leaf}(Z)} t \hat{ } \mathfrak{g} \hat{ } (?S(g)) \hat{ } u_t)$  or  $(Z \cup \bigcup_{t \cdot \mathfrak{g} \in \text{leaf}(Z)} t \hat{ } \mathfrak{g} \hat{ } (?S(g)) \hat{ } u_t)$  appear in a winning strategy of the projection game and do not yet lose, then their translations can generate strategies in  $\mathcal{S}_{a:\alpha}(S)(\sigma)$  as well because of a similar argument.

This establishes that if there is a way to win the projection game, then the subvalue map strategy set is nonempty.

For the other direction, a similar translation function can be written by inverting the construction above. A similar argument can be made to show that the translated strategies are winning for the projection game.

- If  $a:\alpha$  is a Demonic loop  $a:(g:\gamma)^*$ , we use an approach similar to the Angelic loop case. By the inductive hypothesis,  $\mathcal{S}_{g:\gamma}(S)(\sigma)$  is nonempty when  $\sigma \models \langle g:\gamma \Downarrow S \rangle S(a)$ .  
 $\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$  iff  
 $\sigma \models \langle a:\alpha \Downarrow S \rangle S(\text{end}) \quad \phi \text{ is a compatible winning condition}$

We show that if there is a strategy in the set then there is a strategy to win

$\sigma \langle (?S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^* \rangle (S(\text{end}))$  and vice versa by induction over iterations.

We again build up a translation function  $i_Z$ . It inductively builds up translations for sequences (elements) in strategy set  $Z$ . It is similar to the previous case but with dual actions and without the added Angelic tests.

$$i_{f(Z)} \equiv \{\mathfrak{s}^d\} \mapsto \{\mathfrak{s}^d\}; \quad \{\mathfrak{g}^d\} \mapsto \{\mathfrak{g}^d\};$$

$$\text{for } t \cdot \mathfrak{g}^d \in \text{leaf}(Z), (t \hat{ } \mathfrak{g}^d \hat{ } u_t) \mapsto (i_Z(t \hat{ } \mathfrak{g}^d) \hat{ } h_{g:\gamma}(u_t))$$

$(t \hat{g}^d \wedge u_t \hat{g}^d) \mapsto (i_Z(t \hat{g}^d) \wedge h_{g:\gamma}(u_t) \wedge (?S(a)) \wedge g^d)$   
 $(t \hat{g}^d \wedge u_t \hat{s}^d \wedge (?S(\text{end}))) \mapsto (i_Z(t \hat{g}^d) \wedge h_{g:\gamma}(u_t) \wedge s^d)$   
 where  $u_t$  is a winning strategy for game  $\langle g:\gamma \exists S(\text{end} \mapsto S(a)) \rangle S(a)$  and  
 corresponding  $h_{g:\gamma}(u_t)$  for game  $\langle \gamma \rangle S(a)$  must exist per inductive hypothesis.

Above, where for readability, the map implicitly respects prefix closure over *caret*. A full strategy is translated by translating each sequence.

The argument that the translated function is in the strategy set is also similar to the previous case. As with the previous case, an inverse translation function shows the opposite direction.

- If  $a:\alpha$  is a Demonic choice  $a:(g:\gamma \cap d:\delta)$ , then  $\mathcal{S}_{a:\alpha}(S)(\sigma)$  is  $\{t_l \hat{u} t_r : t_l \in \mathcal{S}_{g:\gamma}(S)(g, \sigma) \text{ and } t_r \in \mathcal{S}_{d:\delta}(S)(d, \sigma)\}$ , which is nonempty when both  $\mathcal{S}_{g:\gamma}(S)(\sigma)$  and  $\mathcal{S}_{d:\delta}(S)(\sigma)$  are nonempty. By the inductive hypothesis, this is when  $\sigma \models \langle g:\gamma \exists S \rangle \phi$  and  $\sigma \models \langle d:\delta \exists S \rangle \phi$ .  
 $\sigma \models \langle a:\alpha \exists S \rangle \phi$  iff  
 $\sigma \models \langle (g:\gamma \exists S) \cap (d:\delta \exists S) \rangle \phi$  by definition of  $a:\alpha \exists S$   
 iff  $\sigma \models \langle g:\gamma \exists S \rangle \phi \wedge \langle d:\delta \exists S \rangle \phi$  by axiom  $\langle \cap \rangle$  which is exactly the condition for which the strategy set is nonempty.
- If  $a:\alpha$  is a sequence  $a:(g:\gamma; d:\delta)$ , then  $\mathcal{S}_{a:\alpha}(S)(\sigma)$  is  $\{t \cup \bigcup_{v \in \text{leaf}(t)} (v \hat{u}) : t \in \mathcal{S}_{g:\gamma}(S(\text{end} \mapsto S(d)))(g, \sigma) \text{ and } u \in \mathcal{S}_{d:\delta}(S)(d)([v]_\sigma)\}$ , which is nonempty when both  $\mathcal{S}_{g:\gamma}(S)(\sigma)$  is nonempty and for some choice of  $v$ ,  $\mathcal{S}_{d:\delta}(S)([v]_\sigma)$  is nonempty.  
 By the inductive hypothesis, this is when  $\sigma \models \langle g:\gamma \exists S(\text{end} \mapsto S(d)) \rangle S(d)$  and for some  $v \in \mathcal{S}_{g:\gamma}(S)(\sigma)$ ,  $[v]_\sigma \models \langle d:\delta \exists S \rangle \phi$ .  
 $\sigma \models \langle a:\alpha \exists S \rangle \phi$  iff  
 $\sigma \models \langle (g:\gamma \exists S(\text{end} \mapsto S(d))) ; (d:\delta \exists S) \rangle \phi$  by definition of  $a:\alpha \exists S$   
 $\sigma \models \langle (g:\gamma \exists S(\text{end} \mapsto S(d))) \rangle \langle (d:\delta \exists S) \rangle \phi$  by axiom  $\langle ; \rangle$   
 which matches the condition for which the strategy set is nonempty.

□

**Lemma 6.4** (Universal projection correspondence). *For any Angelic subvalue map  $S$  for game  $a:\alpha$ , in initial state  $\sigma \models \langle \alpha \rangle a$ , for compatible winning condition  $\phi$ , Angel has a winning strategy for the game  $a:\alpha \Downarrow S$  if and only if all ways to pursue the policy of  $S$  for game  $a:\alpha$  that complete the game end in Angel's winning region. That is,  $\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$  iff all elements of set  $\mathcal{S}_{a:\alpha}(S)(\sigma)$  are Angel winning strategies.*

*Proof.* We show the proof by induction over the structure of  $a:\alpha$ .

- If  $a:\alpha$  is atomic, i.e.,  $\alpha \in \{x := e, x := \otimes, ?Q, !Q, \{x' = f(x) \ \& \ Q\}^d\}$ , then  $a:\alpha \Downarrow S$  is  $\alpha$ .  
 $\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$  iff  $\sigma \models \langle \alpha \rangle \phi$  by definition of  $a:\alpha \Downarrow S$ . This is true because  $\sigma \models \langle \alpha \rangle \phi$ .  
 The strategy set consists of a single strategy with no Angelic decisions, that is just the entire game tree. This strategy is a winning strategy because we know there is one winning strategy because by assumption  $\sigma \models \langle \alpha \rangle \phi$ , and this strategy is the only candidate.



- If  $a:\alpha$  is a non-deterministic Angel assignment  $a:x := *$ , then  $\mathcal{S}_{a:\alpha}(S)(\sigma)$  has the strategy set  $\{\{(x := e) \mid \sigma(x \mapsto e) \models S(\text{end})\}\}$ , where all strategies are winning strategies when  $\sigma(x \mapsto e) \models \phi \mid \sigma(x \mapsto e) \models S(\text{end})$ , which is always, since  $\phi$  is a compatible winning condition so  $S(\text{end}) \rightarrow \phi$ .  
 $\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$  iff  
 $\sigma \models \langle x := \otimes; !S(\text{end}) \rangle \phi$  by definition of  $a:\alpha \Downarrow S$   
iff  $\sigma \models \langle x := \otimes \rangle (S(\text{end}) \rightarrow \phi)$  by axioms  $\langle ; \rangle, [?]$   
iff  $\sigma \models \langle x := \otimes \rangle \top \quad S(\text{end}) \rightarrow \phi$  because  $\phi$  is a compatible winning condition  
But this always holds by the semantics of dGL.
- If  $a:\alpha$  is an Angelic ODE  $a:\{x' = f(x) \ \& \ Q\}$ , then  $\mathcal{S}_{a:\alpha}(S)(\sigma)$  has the strategy set  $\{\{(x' = f(x) \ \& \ Q @ t) : t \geq 0, \forall s \in [0, t], \varphi(s) \models S(b) \text{ and } \varphi(t) \models S(\text{succ}(b, a:\alpha)) \text{ where } \varphi : [0, t] \rightarrow \mathcal{S} \text{ is differentiable, } \varphi(0) = \sigma, \text{ and } \forall s \in [0, t], \varphi(s) \models x' = f(x) \wedge Q\}\}$ . All strategies are always winning strategies by a similar argument to for the previous case, since  $S(\text{end}) \rightarrow \phi$ .  
 $\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$  iff  
 $\sigma \models \langle \{x' = f(x) \ \& \ Q\}^d; !S(\text{end}) \rangle \phi$  by definition of  $a:\alpha \Downarrow S$   
iff  $\sigma \models \langle \{x' = f(x) \ \& \ Q\}^d \rangle (S(\text{end}) \rightarrow \phi)$  by axioms  $\langle ; \rangle, [?]$   
iff  $\sigma \models \langle \{x' = f(x) \ \& \ Q\}^d \rangle \top \quad S(\text{end}) \rightarrow \phi$   
But this always holds by the semantics of dGL.
- If  $a:\alpha$  is an Angelic choice  $a:(g:\gamma \cup d:\delta)$ , then  
 $\langle a:\alpha \Downarrow S \rangle \phi$  iff  
 $\langle (!S(g); g:\gamma \Downarrow S) \cap (!S(d); d:\delta \Downarrow S) \rangle \phi$  by definition of  $a:\alpha \Downarrow S$   
iff  $\sigma \models \langle !S(g); g:\gamma \Downarrow S \rangle \phi \wedge \langle !S(d); d:\delta \Downarrow S \rangle \phi$  by axiom  $\langle \cap \rangle$   
iff  $\sigma \models (S(g) \rightarrow \langle g:\gamma \Downarrow S \rangle \phi) \wedge (S(d) \rightarrow \langle d:\delta \Downarrow S \rangle \phi)$  by axiom  $[?], \langle ; \rangle$

There are four cases:

- $\sigma \not\models S(g)$  and  $\sigma \not\models S(d)$ : then the Angel wins the projection game because of the vacuous premises of the implications, and the strategy set has only winning strategies vacuously since it has no strategies.
- $\sigma \models S(g)$  and  $\sigma \not\models S(d)$ : then the Angel wins the projection game iff  
 $\sigma \models (S(g) \rightarrow \langle g:\gamma \Downarrow S \rangle \phi)$  second conjunct true as  $\sigma \not\models S(d)$   
iff  $\sigma \models \langle g:\gamma \Downarrow S \rangle \phi$  as  $\sigma \models S(g)$

By the inductive hypothesis, this is exactly when the strategy set  $\mathcal{S}_{g:\gamma}(S)(\sigma)$  has only winning strategies for Angel in game  $\langle \gamma \rangle \phi$ .

The strategy set for the overall game is  $\{\{!t\} : t \in \mathcal{S}_{g:\gamma}(S)(g, \sigma)\}$  which consists of winning strategies exactly when  $\mathcal{S}_{g:\gamma}(S)(\sigma)$  has only winning strategies for Angel in game  $\langle \gamma \rangle \phi$ .

- $\sigma \not\models S(g)$  and  $\sigma \models S(d)$ , the argument is symmetric to the previous case, with the roles of  $g$  and  $d$  swapped.
- $\sigma \models S(g)$  and  $\sigma \models S(d)$ : then the Angel wins the projection game iff  
 $\sigma \models \langle g:\gamma \Downarrow S \rangle \phi \wedge \langle d:\delta \Downarrow S \rangle \phi$  as  $\sigma \models S(g)$  and  $\sigma \models S(d)$

By the inductive hypothesis, this is exactly when both strategy sets  $\mathcal{S}_{g:\gamma}(S)(\sigma)$  and

$\mathcal{S}_{d\delta}(S)(\sigma)$  have only winning strategies for Angel in games  $\langle \gamma \rangle \phi$  and  $\langle \delta \rangle \phi$  respectively.

The strategy set for the overall game is  $\{\{l^{\wedge}t\} : t \in \mathcal{S}_{g\gamma}(S)(g, \sigma)\} \cup \{\{r^{\wedge}t\} : t \in \mathcal{S}_{d\delta}(S)(d, \sigma)\}$ .

This consists of winning strategies exactly when both  $\mathcal{S}_{g\gamma}(S)(\sigma)$  and  $\mathcal{S}_{d\delta}(S)(\sigma)$  have only winning strategies for Angel in games  $\langle \gamma \rangle \phi$  and  $\langle \delta \rangle \phi$  respectively.

- If  $a:\alpha$  is an Angelic loop  $a:(g:\gamma)^*$ :

$\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$  iff

$\sigma \models \langle (!S(g); (g:\gamma \Downarrow (S(\text{end} \mapsto S(a))))^{\times}; !S(\text{end})) \rangle \phi$  by definition of  $a:\alpha \Downarrow S$

iff  $\sigma \models \langle (!S(g); (g:\gamma \Downarrow (S(\text{end} \mapsto S(a))))^{\times} \rangle S(\text{end}) \rightarrow \phi$  by axioms  $\langle ; \rangle$ ,  $[?]$

iff  $\sigma \models \langle (!S(g); (g:\gamma \Downarrow (S(\text{end} \mapsto S(a))))^{\times} \rangle \top \quad S(\text{end}) \rightarrow \phi$

But this always holds applying the loop rule with invariant  $\langle \alpha \rangle \phi$  and using the inductive hypothesis on the subgame  $\langle \gamma \rangle S(\text{end} \mapsto S(a))$  and Theorem 6.1.

The strategy set also consists of only winning strategies. Consider any leaf  $t$  in any strategy tree in the set. If  $t$  ends with  $\mathfrak{s}$  then by construction (Def. B.7),  $[t]_{\sigma} \models S(\text{end})$  and  $t$  is a winning play for Angel. Otherwise, if  $t$  does not end in  $\mathfrak{s}$ , then it must have ended while playing loop body  $\gamma$ . Let  $t = u \cdot v$  where  $v$  plays the final (incomplete) iteration of the loop body.  $v$  ends with Demon getting stuck because it is a winning strategy for  $\gamma$  by the inductive hypothesis in combination with Theorem 6.1. Thus,  $t$  is also a winning strategy for Angel in the larger game.

- If  $a:\alpha$  is a Demonic loop  $a:(g:\gamma)^{\times}$ , then

$\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$  iff

$\sigma \models \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^{\times} \rangle \phi$

This always holds: we can apply the loop invariant rule with invariant  $\langle \alpha \rangle \phi$  and use Theorem 6.1 on the subgame  $\langle \gamma \rangle S(\text{end} \mapsto S(a))$ .

The argument that the strategy set has only winning strategies is similar to the previous case. Consider any leaf in any strategy tree in the set. If it ends with  $\mathfrak{s}$ , it is a winning play for Angel because  $[t]_{\sigma} \models S(\text{end})$ . Otherwise, if it does not end in  $\mathfrak{s}$ , then it must have ended while playing loop body  $\gamma$ . It must end with Demon getting stuck because by the inductive hypothesis in combination with Theorem 6.1, and so must be a winning strategy for Angel in the larger game.

- If  $a:\alpha$  is a Demonic choice  $a:(g:\gamma \cap d:\delta)$ , then  $\mathcal{S}_{a\alpha}(S)(\sigma)$  has the strategy set  $\{l^d \wedge t_l \cup r^d \wedge t_r : t_l \in \mathcal{S}_{g\gamma}(S)(g, \sigma) \text{ and } t_r \in \mathcal{S}_{d\delta}(S)(d, \sigma)\}$ , which contains only winning strategies exactly when  $\mathcal{S}_{g\gamma}(S)(\sigma)$  and  $\mathcal{S}_{d\delta}(S)(\sigma)$  have only winning strategies for Angel in games  $\langle \gamma \rangle \phi$  and  $\langle \delta \rangle \phi$  respectively.

$\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$  iff

$\sigma \models \langle (g:\gamma \Downarrow S) \cap (d:\delta \Downarrow S) \rangle \phi$  by definition of  $a:\alpha \Downarrow S$

iff  $\sigma \models \langle g:\gamma \Downarrow S \rangle \phi \wedge \langle d:\delta \Downarrow S \rangle \phi$  by axiom  $\langle \cap \rangle$

which by the inductive hypothesis applied for  $g:\gamma$  and  $d:\delta$  is exactly when the overall strategy set has only winning Angel strategies.

- If  $a:\alpha$  is sequential composition  $a:(g:\gamma; d:\delta)$ , then  $\mathcal{S}_{a\alpha}(S)(\sigma)$  has the strategy set  $\{t \cup \bigcup_{v \in \text{leaf}(t)} (v^{\wedge}u) : t \in \mathcal{S}_{g\gamma}(S(\text{end} \mapsto S(d)))(g, \sigma) \text{ and } u \in \mathcal{S}_{d\delta}(S)(d)([v]_{\sigma})\}$ . These are always winning strategies:  $\mathcal{S}_{g\gamma}(S(\text{end} \mapsto S(d)))(g, \sigma)$  and  $\mathcal{S}_{d\delta}(S)(d)([v]_{\sigma})$  have

only winning strategies using the inductive hypothesis along with Theorem 6.1. All plays  
that get stuck win for Angel, and all plays that run till the end result in a state  $\sigma' \in S(\text{end})$ ,  
and so win for Angel because  $S(\text{end}) \rightarrow \phi$ .  
 $\sigma \models \langle a:\alpha \Downarrow S \rangle \phi$  always holds by Theorem 6.1.

□ 7373

**Theorem 6.1** (Subvalue Map stays in Winning Region). *Suppose  $S$  is an Angelic subvalue map  
for game  $a:\alpha$  compatible with winning condition  $\phi$ , i.e., for every subgame  $b \in \text{subgames}(a:\alpha)$ ,  
 $\models S(b) \rightarrow \langle a:\alpha_b \rangle S(\text{end})$  and  $\models S(\text{end}) \rightarrow \phi$ . Upon starting in any state  $\sigma \in \llbracket S(a) \rrbracket$ , and  
reaching subgame  $b$  by following any strategy induced by the subvalue map  $S$ , there exists a win-  
ning strategy for Angel to win the remainder of the game. That is,  $S(a) \models \langle (a:\alpha \Downarrow S)_b \rangle \langle a:\alpha_b \rangle \phi$ .*

*Proof.* We perform induction on the structure of  $a:\alpha$ .

- When  $a:\alpha$  is atomic and not controlled by Angel, i.e.,  $\alpha \in \{x := e, x := \otimes, ?Q, !Q, \{x' = f(x) \ \& \ Q\}^d\}$ , then The only subgame  $b$  in  $\text{subgames}(a:\alpha)$  is  $a:\alpha$  itself. Thus the only  
possibility is  $b = a$ .  $a:\alpha_a$  is the empty game **skip**, and  $a:\alpha_a$  is the game  $a:\alpha$ , and  $a:\alpha \Downarrow S$   
is just  $a:\alpha$ . Thus,  $\langle a:\alpha_a \Downarrow S \rangle \langle a:\alpha_a \rangle \phi$  is equivalent to  $\langle \text{skip} \rangle \langle \alpha \rangle \phi$ . We need to show that  
 $S(a) \models \langle a:\alpha \rangle \phi$  which holds because by definition of an Angelic subvalue map,  $\models S(a) \rightarrow$   
 $\langle \alpha \rangle \phi$ .
- When  $a:\alpha$  is an Angelic nondeterministic assignment, i.e.,  $a:x := *$ , then again the only  
subgame  $b$  in  $\text{subgames}(a:\alpha)$  is  $a:\alpha$  itself. Thus the only possibility is  $b = a$ .  $a:\alpha_a$  is  
the empty game **skip**, and  $a:\alpha \Downarrow S$  is just  $a:\alpha ; ?S(\text{end})$ , and  $a:\alpha ; ?S(\text{end})_a$  is the game  
 $a:\alpha ; ?S(\text{end})$ . Thus,  $\langle a:\alpha_a \Downarrow S \rangle \langle a:\alpha_a \rangle \phi$  is equivalent to  $\langle \text{skip} \rangle \langle \alpha ; ?S(\text{end}) \rangle \phi$ . We need  
to show that  $S(a) \models \langle \alpha \rangle (?S(\text{end}) \wedge \phi)$ . Since  $\phi$  is a compatible winning condition, this is  
 $S(a) \models \langle \alpha \rangle ?S(\text{end})$  which holds by the definition of an inductive Angelic subvalue map.
- When  $a:\alpha$  is an Angelic ODE, i.e.,  $a:\{x' = f(x) \ \& \ Q\}$ , then the argument is similar to the  
previous case. As before the only possibility is  $b = a$ , and we need to show that  $S(a) \models$   
 $\langle \alpha \rangle ?S(\text{end}) \wedge \phi$ . Since  $\phi$  is a compatible winning condition, this is  $S(a) \models \langle a:\alpha \rangle ?S(\text{end})$   
which holds by the definition of an inductive Angelic subvalue map.
- When  $a:\alpha$  is an Angelic choice,  $a:(g:\gamma \cup d:\delta)$ , then we analyze cases depending on the value  
of  $b$ .
  1. If  $b = a$ , then  $a:\alpha_a \Downarrow S_b = \text{skip}$ , and  $a:\alpha_b = a:\alpha$ . Thus,  $\langle a:\alpha_b \Downarrow S \rangle \langle a:\alpha_b \rangle \phi$  is  
equivalent to  $\langle a:\alpha \rangle \phi$ .  $S(a) \models \langle a:\alpha \rangle \phi$  This can be proved by applying the axioms  $\rightarrow R$   
along with  $S(a) \rightarrow \langle a:\alpha \rangle \phi$  from the definition of an Angelic subvalue map.
  2. Otherwise, if  $b \in \text{subgames}(g:\gamma)$ , then  $a:\alpha_a \Downarrow S_b = !S(g) ; g:\gamma \Downarrow S_b$  and  $a:\alpha_b =$   
 $g:\gamma_b$ . We can show that  $S(a) \models \langle !S(g) ; g:\gamma \Downarrow S_b \rangle \langle g:\gamma_b \rangle \phi$  by first applying axioms  
 $\langle ; \rangle$ ,  $[?]$ ,  $[!]$ ,  $\langle^d \rangle$ ,  $\rightarrow R$  to get the goal  $S(a), S(g) \models \langle g:\gamma \Downarrow S_b \rangle \langle g:\gamma_b \rangle \phi$  Notice that  $S$  is  
also an inductive Angelic subvalue map for  $g:\gamma$  and winning condition  $\phi$ . Thus by the  
inductive hypothesis,  $S(g) \models \langle g:\gamma_b \Downarrow S \rangle \langle g:\gamma_b \rangle \phi$  completing the proof.
  3. Otherwise, if  $b \in \text{subgames}(d:\delta)$ , then the proof proceeds similarly to the previous  
case, but with  $d:\delta$  replacing  $g:\gamma$ .
- When  $a:\alpha$  is an Angelic loop, i.e.,  $a:(g:\gamma)^*$ , then we analyze cases depending on the value  
of  $b$ .

1. If  $b = a$ , then  $a:\alpha_{b:} = a:\alpha$  and  $a:\alpha \Downarrow S_{:b} = (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times; !S(\text{end})$ .

Thus,

$$\langle a:\alpha \Downarrow S_{:b} \rangle \langle a:\alpha_{b:} \rangle \phi$$

is by definition

$$\langle (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times; !S(\text{end}) \rangle \langle a:\alpha \rangle \phi.$$

By the inductive hypothesis,  $S(g) \models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(a)) \rangle S(a)$ . Then applying loop with invariant  $S(a)$  we have

$$S(a) \models \langle (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times \rangle S(a).$$

By the definition of an Angelic subvalue map,  $\models S(a) \rightarrow \langle a:\alpha \rangle \phi$ . Thus by M,  $\rightarrow R$ , and  $\langle ? \rangle$ , we prove the desired goal

$$S(a) \models \langle (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times; !S(\text{end}) \rangle \langle a:\alpha \rangle \phi.$$

2. Otherwise, if  $b \in \text{subgames}(g:\gamma)$ , then

$$a:\alpha \Downarrow S_{:b} = (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times; !S(g); (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{:b},$$

$$a:\alpha_{b:} = g:\gamma_{b:}; a:\alpha.$$

Want to show:

$$S(a) \models \langle (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times; !S(g); (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{:b} \rangle \langle (g:\gamma)_{b:}; a:\alpha \rangle \phi. \quad \blacksquare$$

First, by Lemma B.5, we have  $S(a) \models \langle (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times; !S(\text{end}) \rangle \top$ , so by axioms  $[?]$  and  $\langle ; \rangle$ ,  $S(a) \models \langle (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times \rangle \top$ .

Next, by the inductive hypothesis and applications of axioms  $\langle ; \rangle$ ,  $\langle ? \rangle$  and  $\rightarrow R$ , we have

$$\top \models \langle !S(g); (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{:b} \rangle \langle g:\gamma_{b:} \rangle S(a)$$

Finally, since  $S$  is an Angelic subvalue map, we have  $S(a) \models \langle a:\alpha \rangle \phi$ . Thus by M and  $\langle ; \rangle$  we can conclude the desired result.

- When  $a:\alpha$  is a Demon loop, i.e.,  $a:(g:\gamma)^\times$ , then we analyze cases depending on the value of  $b$ .

1. If  $b = a$ , then  $(a:\alpha \Downarrow S)_{:b} = a:\alpha \Downarrow S$  and  $(a:\alpha)_{b:} = a:\alpha$ .

Thus we must show that  $S(a) \models \langle a:\alpha \Downarrow S \rangle \langle a:\alpha \rangle \phi$ .

$S(a) \models \langle a:\alpha \Downarrow S \rangle S(a)$  because  $S(a)$  is a loop invariant of  $(g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times$  by Lemma B.5.

$S(a) \models \langle a:\alpha \rangle \phi$  by definition of a subvalue map. Thus by M, we can conclude the desired result.

2. Otherwise, if  $b \in \text{subgames}(g:\gamma)$ , then:

$$(a:\alpha \Downarrow S)_{:b} = a:\alpha \Downarrow S; (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{:b},$$

$$(a:\alpha)_{b:} = (g:\gamma)_{b:}; a:\alpha.$$

Thus we must show that

$$S(a) \models \langle a:\alpha \Downarrow S; (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{:b} \rangle \langle (g:\gamma)_{b:}; a:\alpha \rangle \phi$$

First, we have  $S(a) \models \langle a:\alpha \Downarrow S \rangle S(a)$  because  $S(a)$  is a loop invariant of  $(g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times$  by Lemma B.5.

Next, by the inductive hypothesis, we have

$$S(a) \models \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{:b} ; (g:\gamma)_{b:} \rangle S(a).$$

By definition of a subvalue map, we have  $S(a) \models \langle a:\alpha \rangle \phi$ .

Thus by M and  $\langle ; \rangle$ , we can conclude the desired result.

- When  $a:\alpha$  is sequential composition, i.e.,  $a:(g:\gamma ; d:\delta)$ , then we analyze cases depending on the value of  $b$ .

1. If  $b = a$ , then  $(a:\alpha \Downarrow S)_{:b} = \text{skip}$  and  $a:\alpha_{b:} = a:\alpha$ . Thus we must show that  $S(a) \models \langle \text{skip} \rangle \langle a:\alpha \rangle \phi$ , i.e.,  $S(a) \models \langle a:\alpha \rangle \phi$ , which hold by definition of a subvalue map.
2. Otherwise, if  $b \in \text{subgames}(g:\gamma)$ , then  $(a:\alpha \Downarrow S)_{:b} = (g:\gamma \Downarrow S(\text{end} \mapsto S(d)))_{:b}$  and  $a:\alpha_{b:} = (g:\gamma)_{b:} ; d:\delta$ .

Thus we must show that  $S(a) \models \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(d)))_{:b} \rangle \langle (g:\gamma)_{b:} ; d:\delta \rangle \phi$ .

This is  $S(a) \models \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(d)))_{:b} \rangle \langle (g:\gamma)_{b:} \rangle \langle d:\delta \rangle \phi$  applying  $\langle ; \rangle$ .

This is  $S(a) \models \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(d)))_{:b} ; (g:\gamma)_{b:} \rangle \langle d:\delta \rangle \phi$  again applying  $\langle ; \rangle$ .

By the inductive hypothesis,  $S(a) \models \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(d)))_{:b} \rangle S(d)$  since  $S(a) \rightarrow S(g)$  in an inductive subvalue map.

By definition of an Angelic subvalue map,  $S(d) \models \langle d:\delta \rangle \phi$ .

By rule M, we can conclude the desired result.

3. Otherwise, if  $b \in \text{subgames}(d:\delta)$ , then a similar argument applies.

$a:\alpha \Downarrow S_{:b} = \{g:\gamma \Downarrow S(\text{end} \mapsto S(d)) ; (d:\delta \Downarrow S)_{:b}\}$ , and  $a:\alpha_{b:} = d:\delta_{b:}$ .

Thus we must show that  $S(a) \models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)) ; (d:\delta \Downarrow S)_{:b} \rangle \langle d:\delta_{b:} \rangle \phi$ .

By the  $\langle ; \rangle$  axiom, this is  $S(a) \models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)) \rangle \langle (d:\delta \Downarrow S)_{:b} ; (d:\delta)_{b:} \rangle \phi$ .

By Lemma B.5, and since in an inductive subvalue map,  $\models S(a) \rightarrow S(g)$ ,  $S(a) \models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(d)) \rangle S(d)$ .

By the inductive hypothesis,  $S(d) \models \langle (d:\delta \Downarrow S)_{:b} ; (d:\delta)_{b:} \rangle \phi$ .

By rule M, we can conclude the desired result.

□

**Theorem 6.2** (Inductive subvalue map ensures winning actions). *Suppose  $S$  is an inductive Angelic subvalue map for game  $a:\alpha$  and compatible with winning condition  $\phi$ , i.e.,  $a:\alpha \models S$  and  $S(\text{end}) \models \phi$ . Upon starting in any state  $\sigma \models S(a)$ , and reaching subgame  $b$  by following any strategy induced by the subvalue map  $S$ , there is a way for Angel to win by continuing to follow  $S$ . That is,  $S(a) \models \langle (a:\alpha \Downarrow S)_{:b} \rangle \langle (a:\alpha \Downarrow S)_{b:} \rangle \phi$ .*

*Proof.* We perform induction on the structure of  $a:\alpha$ .

- When  $a:\alpha$  is atomic and not controlled by Angel, i.e.,  $\alpha \in \{x := e, x := \otimes, ?Q, !Q, \{x' = f(x) \ \& \ Q\}^d\}$ , then The only subgame  $b$  in  $\text{subgames}(a:\alpha)$  is  $a:\alpha$  itself. Thus the only possibility is  $b = a$ .  $a:\alpha_{:a}$  is the empty game  $\text{skip}$ , and  $a:\alpha_{a:}$  is the game  $a:\alpha$ , and  $a:\alpha \Downarrow S$  is just  $a:\alpha$ . Thus,  $\langle (a:\alpha \Downarrow S)_{:a} \rangle \langle (a:\alpha \Downarrow S)_{a:} \rangle \phi$  is equivalent to  $\langle \text{skip} \rangle \langle a:\alpha \rangle \phi$ . We need to show that  $S(a) \models \langle a:\alpha \Downarrow S \rangle \phi$  which holds because by definition of an inductive Angelic subvalue map along with the fact that  $S(\text{end}) \rightarrow \phi$ ,  $S(a) \rightarrow \langle a:\alpha \rangle \phi$ .

- When  $a:\alpha$  is an Angelic nondeterministic assignment, i.e.,  $a:x := *$ , then again the only subgame  $b$  in  $\text{subgames}(a:\alpha)$  is  $a:\alpha$  itself. Thus the only possibility is  $b = a$ .  $a:\alpha:a$  is the empty game **skip**, and  $a:\alpha \forall S$  is just  $a:\alpha ; ?S(\text{end})$ , and  $a:\alpha ; ?S(\text{end})_a$  is the game  $a:\alpha ; ?S(\text{end})$ . Thus,  $\langle a:\alpha:a \forall S \rangle \langle a:\alpha:a \exists S \rangle \phi$  is equivalent to  $\langle \text{skip} \rangle \langle \alpha ; ?S(\text{end}) \rangle \phi$ . We need to show that  $S(a) \models \langle \alpha \rangle (?S(\text{end}) \wedge \phi)$ . Since  $\phi$  is a compatible winning condition, this is  $S(a) \models \langle \alpha \rangle ?S(\text{end})$  which holds by the definition of an inductive Angelic subvalue map.
- When  $a:\alpha$  is an Angelic ODE, i.e.,  $a:\{x' = f(x) \ \& \ Q\}$ , then the argument is similar to the previous case. As before the only possibility is  $b = a$ , and we need to show that  $S(a) \models \langle \alpha \rangle ?S(\text{end}) \wedge \phi$ . Since  $\phi$  is a compatible winning condition, this is  $S(a) \models \langle a:\alpha \exists S \rangle ?S(\text{end})$  which holds by the definition of an inductive Angelic subvalue map.
- When  $a:\alpha$  is an Angelic choice, i.e.,  $a:(g:\gamma \cup d:\delta)$ , then we analyze cases depending on the value of  $b$ .
  1. If  $b = a$ , then  $(a:\alpha \forall S)_b = \text{skip}$  and  $a:\alpha \exists S_b = a:\alpha \exists S$ . Thus we must show that  $S(a) \models \langle \text{skip} \rangle \langle a:\alpha \exists S \rangle \phi$ , i.e.,  $S(a) \models \langle a:\alpha \exists S \rangle \phi$ , which holds by Lemma B.6.
  2. Otherwise, if  $b \in \text{subgames}(g:\gamma)$ , then  $(a:\alpha \forall S)_b = !S(g) ; g:\gamma \forall S_b$  and  $a:\alpha \exists S_b = g:\gamma \exists S_b$ . Thus we must show that  $S(a) \models \langle !S(g) ; g:\gamma \forall S_b \rangle \langle g:\gamma \exists S_b \rangle \phi$ . This is  $S(a) \wedge S(g) \models \langle g:\gamma \forall S_b \rangle \langle g:\gamma \exists S_b \rangle \phi$  using axioms  $\langle ; \rangle$ ,  $[?]$ . This holds by the inductive hypothesis.
  3. Otherwise, if  $b \in \text{subgames}(d:\delta)$ , then the proof proceeds similarly to the previous case, but with  $d:\delta$  replacing  $g:\gamma$ .
- When  $a:\alpha$  is a Demonic choice, i.e.,  $a:(g:\gamma \cap d:\delta)$ , then the argument is similar to Angelic choice, except that we get assumptions  $S(g)$  or  $S(d)$  not from the execution prefix but from  $S(a)$  because in an inductive subvalue map,  $\models S(a) \rightarrow S(g) \wedge S(d)$ . We analyze cases depending on the value of  $b$ .
  1. If  $b = a$ , then  $(a:\alpha \forall S)_b = \text{skip}$  and  $a:\alpha \exists S_b = a:\alpha \exists S$ . Thus we must show that  $S(a) \models \langle \text{skip} \rangle \langle a:\alpha \exists S \rangle \phi$ , i.e.,  $S(a) \models \langle a:\alpha \exists S \rangle \phi$ , which holds by Lemma B.6.
  2. Otherwise, if  $b \in \text{subgames}(g:\gamma)$ , then  $(a:\alpha \forall S)_b = g:\gamma \forall S_b$  and  $a:\alpha \exists S_b = g:\gamma \exists S_b$ . Thus we must show that  $S(a) \models \langle g:\gamma \forall S_b \rangle \langle g:\gamma \exists S_b \rangle \phi$ . This holds by the inductive hypothesis since as discussed,  $S(a) \rightarrow S(g)$ .
  3. Otherwise, if  $b \in \text{subgames}(d:\delta)$ , then the proof proceeds similarly to the previous case, but with  $d:\delta$  replacing  $g:\gamma$ .
- When  $a:\alpha$  is sequential composition, i.e.,  $a:(g:\gamma ; d:\delta)$ , then we analyze cases depending on the value of  $b$ .
  1. If  $b = a$ , then  $(a:\alpha \forall S)_b = \text{skip}$  and  $a:\alpha \exists S_b = a:\alpha \exists S$ . Thus we must show that  $S(a) \models \langle \text{skip} \rangle \langle a:\alpha \exists S \rangle \phi$ , i.e.,  $S(a) \models \langle a:\alpha \exists S \rangle \phi$ , which holds by Lemma B.6.
  2. Otherwise, if  $b \in \text{subgames}(g:\gamma)$ , then  $(a:\alpha \forall S)_b = (g:\gamma \forall S(\text{end} \mapsto S(d)))_b$  and  $a:\alpha \exists S_b = (g:\gamma \exists S(\text{end} \mapsto S(d)))_b ; d:\delta \exists S$ . Thus we must show that:  
 $S(a) \models \langle g:\gamma \forall S(\text{end} \mapsto S(d)) \rangle_b \langle (g:\gamma \exists S(\text{end} \mapsto S(d)))_b ; d:\delta \exists S \rangle \phi$ .  
 This is  $S(a) \models \langle (g:\gamma \forall S(\text{end} \mapsto S(d)))_b \rangle \langle (g:\gamma \exists S(\text{end} \mapsto S(d)))_b \rangle \langle d:\delta \exists S \rangle \phi$  applying  $\langle ; \rangle$ .

- This is  $S(a) \models \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(d)))_{:b}; (g:\gamma \Downarrow S(\text{end} \mapsto S(d)))_{:b} \rangle \langle d:\delta \Downarrow S \rangle \phi$  again applying  $\langle ; \rangle$ .  
 By the inductive hypothesis,  $S(a) \models \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(d)))_{:b} \rangle S(d)$  since  $S(a) \rightarrow S(g)$  in an inductive subvalue map.  
 By Lemma B.6,  $S(d) \models \langle d:\delta \Downarrow S \rangle \phi$ .  
 By rule M, we can conclude the desired result.
3. Otherwise, if  $b \in \text{subgames}(d:\delta)$ , then a similar argument applies.  
 $a:\alpha \Downarrow S_{:b} = \{g:\gamma \Downarrow S(\text{end} \mapsto S(d)); (d:\delta \Downarrow S)_{:b}\}$ , and  $a:\alpha \Downarrow S_{b:} = d:\delta \Downarrow S_{b:}$ .  
 Thus we must show that  $S(a) \models \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(d)))_{:b}; (d:\delta \Downarrow S)_{:b} \rangle \langle d:\delta \Downarrow S_{b:} \rangle \phi$ .  
 By the  $\langle ; \rangle$  axiom, this is  $S(a) \models \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(d)))_{:b}; (d:\delta \Downarrow S)_{:b} \rangle \langle (d:\delta \Downarrow S)_{b:} \rangle \phi$ .  
 By Lemma B.7, and since in an inductive subvalue map,  $\models S(a) \rightarrow S(g)$ ,  $S(a) \models \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(d)))_{:b} \rangle S(d)$ .  
 By the inductive hypothesis,  $S(d) \models \langle (d:\delta \Downarrow S)_{:b}; (d:\delta \Downarrow S)_{b:} \rangle \phi$ .  
 By rule M, we can conclude the desired result.
- When  $a:\alpha$  is an Angelic loop, i.e.,  $a:(g:\gamma)^*$ , then we analyze cases depending on the value of  $b$ .
    1. If  $b = a$  then  $(a:\alpha \Downarrow S)_{:b} = (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times$  (with the terminal test missing compared to  $a:\alpha \Downarrow S$ ) and  $a:\alpha \Downarrow S_{b:} = a:\alpha \Downarrow S$ .  
 We need to show that  $S(a) \models \langle (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times \rangle \langle a:\alpha \Downarrow S \rangle \phi$ .  
 We first show that  $S(a) \models \langle (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times \rangle S(a)$ . This follows from applying the loop rule with invariant  $S(a)$  and then using the fact that per Lemma B.7,  $S(g) \models \langle g:\gamma \Downarrow S(\text{end} \mapsto S(a)) \rangle S(a)$ .  
 Then by Lemma B.6,  $S(a) \models \langle a:\alpha \Downarrow S \rangle \phi$ . Thus by M, we can conclude the desired result.
    2. Otherwise, if  $b \neq a$  then it is in  $\text{subgames}(g:\gamma)$ .  
 $(a:\alpha \Downarrow S)_{:b} = (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{:b}^\times; !S(g); (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{:b}$ .  
 $a:\alpha \Downarrow S_{b:} = (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{b:}; a:\alpha \Downarrow S$ .  
 Want to show:  

$$S(a) \models \langle (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{:b}^\times; !S(g); (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{:b} \rangle \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{b:}; a:\alpha \Downarrow S \rangle \phi$$
  
 First, by Lemma B.7, we have  $S(a) \models \langle (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{:b}^\times; !S(\text{end}) \rangle \top$ , so by axioms  $[?]$  and  $\langle ; \rangle$ ,  $S(a) \models \langle (!S(g); g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{:b}^\times \rangle \top$ .  
 Next, by the inductive hypothesis and applications of axioms  $\langle ; \rangle$ ,  $\langle ? \rangle$  and  $\rightarrow R$ , we have  

$$\top \models \langle !S(g); (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{:b} \rangle \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{b:} \rangle S(a)$$
  
 Finally, by Lemma B.6, we have  $S(a) \models \langle a:\alpha \Downarrow S \rangle \phi$ . Thus by M and  $\langle ; \rangle$  we can conclude the desired result.
  - When  $a:\alpha$  is a Demonic loop, i.e.,  $a:(g:\gamma)^\times$ , then we analyze cases depending on the value of  $b$ .

7567 1. If  $b = a$ , then  $(a:\alpha \Downarrow S)_{:b} = a:\alpha \Downarrow S$  and  $(a:\alpha \Downarrow S)_{b:} = a:\alpha \Downarrow S$ .  
 7568 Thus we must show that  $S(a) \models \langle a:\alpha \Downarrow S \rangle \langle a:\alpha \Downarrow S \rangle \phi$ .  
 7569  $S(a) \models \langle a:\alpha \Downarrow S \rangle S(a)$  because  $S(a)$  is a loop invariant of  $(g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times$   
 7570 by Lemma B.7.  
 7571  $S(a) \models \langle a:\alpha \Downarrow S \rangle \phi$  by Lemma B.6. Thus by M, we can conclude the desired result.  
 7572 2. Otherwise, if  $b \in \text{subgames}(g:\gamma)$ , then:  
 7573  $(a:\alpha \Downarrow S)_{:b} = a:\alpha \Downarrow S ; (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{:b}$ ,  
 7574  $(a:\alpha \Downarrow S)_{b:} = (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{b:} ; a:\alpha \Downarrow S$ .  
 7575 Thus we must show that

$$7576 S(a) \models \langle a:\alpha \Downarrow S ; (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{:b} \rangle \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{b:} ; a:\alpha \Downarrow S \rangle \phi$$

7577 First, we have  $S(a) \models \langle a:\alpha \Downarrow S \rangle S(a)$  because  $S(a)$  is a loop invariant of  
 7578  $(g:\gamma \Downarrow S(\text{end} \mapsto S(a)))^\times$  by Lemma B.7.  
 7579 Next, by the inductive hypothesis, we have  $S(a) \models \langle (g:\gamma \Downarrow S(\text{end} \mapsto S(a)))_{:b} \rangle S(a)$ .  
 7580 By Lemma B.6, we have  $S(a) \models \langle a:\alpha \Downarrow S \rangle \phi$ .  
 7581 Thus by M and  $\langle ; \rangle$ , we can conclude the desired result.

7582 □

7583 **Theorem 6.3** (Inductive subvalue maps are subvalue maps). *For dGL game  $a:\alpha$ , if  $S$  is an induc-*  
 7584 *tive Angelic subvalue map ( $a:\alpha \models S$ ), then it is also an Angelic subvalue map for  $a:\alpha$ , i.e., for*  
 7585 *every subgame  $b:\beta$  in  $\text{subgames}(a:\alpha)$ ,  $\models S(b) \rightarrow \langle a:\alpha_{b:} \rangle S(\text{end})$ . Dually, if  $S$  is an inductive*  
 7586 *Demonic subvalue map for  $a:\alpha$ , then  $S$  is a Demonic subvalue map for  $a:\alpha$ .*

7587 *Proof.* The proof uses structural induction. When  $a:\alpha$  is atomic, i.e.  $\alpha \in \{x := e, x := *,$   
 7588  $?Q, !Q, \{x' = f(x) \ \& \ Q\}, \{x' = f(x) \ \& \ Q\}^d\}$ , according to Def. 6.8,  $a:\alpha \models S$  exactly when  
 7589  $\models S(a) \rightarrow \langle \alpha \rangle S(\text{end})$ . Since  $a:\alpha_{a:} = a:\alpha$  and  $a$  is the only subgame in  $\text{subgames}(a:\alpha)$ , we can  
 7590 conclude that  $S$  is a subvalue map. For the recursive cases, if  $a:\alpha$  has the structure:

- 7591 1.  $g:\gamma \cup d:\delta$ , for any subgame  $b$  in  $\text{subgames}(g:\gamma)$ ,  $a:\alpha_{b:} = g:\gamma_{b:}$ . Since per the inductive  
 7592 hypothesis,  $\models S(b) \rightarrow \langle g:\gamma_{b:} \rangle S(\text{end})$ , we can conclude that  $\models S(b) \rightarrow \langle a:\alpha_{b:} \rangle S(\text{end})$ .  
 7593 The same argument applies to all subgames in  $\text{subgames}(d:\delta)$ . It remains to show that  
 7594 for the overall game  $a$ ,  $\models S(a) \rightarrow \langle a:\alpha_{a:} \rangle S(\text{end})$ . By the inductive hypothesis, because  
 7595  $g:\gamma \models S$ , so  $\models S(g) \rightarrow \langle \gamma \rangle S(\text{end})$  and because  $d:\delta \models S$ , so  $\models S(d) \rightarrow \langle \delta \rangle S(\text{end})$ . Using  
 7596 the disjunction of implications,  $\models S(g) \vee S(d) \rightarrow \langle \gamma \rangle S(\text{end}) \vee \langle \delta \rangle S(\text{end})$ . But according  
 7597 to dGL axiom  $\langle \cup \rangle$ ,  $\models \langle \gamma \rangle S(\text{end}) \vee \langle \delta \rangle S(\text{end}) \leftrightarrow \langle \gamma \cup \delta \rangle S(\text{end})$ , so  $\models S(g) \vee S(d) \rightarrow$   
 7598  $\langle \gamma \cup \delta \rangle S(\text{end})$ . Because  $S$  is valid,  $\models S(a) \rightarrow S(g) \vee S(d)$ . By transitivity of implication,  
 7599  $\models S(a) \rightarrow \langle \gamma \cup \delta \rangle S(\text{end})$ , proving the desired result.
- 7600 2.  $g:\gamma \cap d:\delta$ , for any subgame  $b$  in  $\text{subgames}(g:\gamma)$ ,  $a:\alpha_{b:} = g:\gamma_{b:}$ . Since per the inductive  
 7601 hypothesis,  $\models S(b) \rightarrow \langle g:\gamma_{b:} \rangle S(\text{end})$ , we can conclude that  $\models S(b) \rightarrow \langle a:\alpha_{b:} \rangle S(\text{end})$ .  
 7602 The same argument applies to all subgames in  $\text{subgames}(d:\delta)$ . It remains to show that  
 7603 for the overall game  $a$ ,  $\models S(a) \rightarrow \langle a:\alpha_{a:} \rangle S(\text{end})$ . By the inductive hypothesis, because  
 7604  $g:\gamma \models S$ , so  $\models S(g) \rightarrow \langle \gamma \rangle S(\text{end})$  and because  $d:\delta \models S$ , so  $\models S(d) \rightarrow \langle \delta \rangle S(\text{end})$ . Using  
 7605 the conjunction of implications,  $\models S(g) \wedge S(d) \rightarrow \langle \gamma \rangle S(\text{end}) \wedge \langle \delta \rangle S(\text{end})$ . But according  
 7606 to dGL axiom  $\langle \cap \rangle$ ,  $\models \langle \gamma \rangle S(\text{end}) \wedge \langle \delta \rangle S(\text{end}) \leftrightarrow \langle \gamma \cap \delta \rangle S(\text{end})$ , so  $\models S(g) \wedge S(d) \rightarrow$



- $\langle \gamma \cap \delta \rangle S(\text{end})$ . Because  $S$  is valid,  $\models S(a) \rightarrow S(g) \wedge S(d)$ . By transitivity of implication,  $\models S(a) \rightarrow \langle \gamma \cap \delta \rangle S(\text{end})$ , proving the desired result. 7607  
7608
3.  $g:\gamma; d:\delta$ , for any subgame  $b$  in  $\text{subgames}(d:\delta)$ ,  $d:\delta_b = a:\alpha_b$ . Since per the inductive hypothesis,  $\models S(b) \rightarrow \langle d:\delta_b \rangle S(\text{end})$ , we can conclude that  $\models S(b) \rightarrow \langle a:\alpha_b \rangle S(\text{end})$ . 7609  
7610
- Consider now any subgame  $b$  in  $\text{subgames}(g:\gamma)$ . Because  $g:\gamma \models S$ ,  $\models S(b) \rightarrow \langle g:\gamma_b \rangle S(d)$ . 7611
- Now, because  $d:\delta \models S$ ,  $\models S(d) \rightarrow \langle \delta \rangle S(\text{end})$ . According to the dGL monotonicity rule M, 7612
- this means that  $\models S(b) \rightarrow \langle g:\gamma_b \rangle \langle \delta \rangle S(\text{end})$ . According to the dGL axiom  $\langle ; \rangle$ ,  $\models S(b) \rightarrow$  7613
- $\langle g:\gamma_b; \delta \rangle S(\text{end})$ . Since  $a:\alpha_b = g:\gamma_b; d:\delta$ , we can conclude that  $\models S(b) \rightarrow \langle a:\alpha_b \rangle S(\text{end})$ . 7614
- It remains to show that for the overall game  $a$ ,  $\models S(a) \rightarrow \langle a:\alpha_a \rangle S(\text{end})$ . By the in- 7615
- ductive hypothesis, because  $g:\gamma \models S$ , so  $\models S(g) \rightarrow \langle \gamma \rangle S(\delta)$  and because  $d:\delta \models S$ , 7616
- so  $\models S(d) \rightarrow \langle \delta \rangle S(\text{end})$ . According to the dGL monotonicity rule M, this means 7617
- that  $\models S(g) \rightarrow \langle \gamma \rangle \langle \delta \rangle S(\text{end})$ . According to the dGL axiom  $\langle ; \rangle$ ,  $\models \langle \gamma \rangle \langle \delta \rangle S(\text{end}) \leftrightarrow$  7618
- $\langle \gamma; \delta \rangle S(\text{end})$ . Thus,  $\models S(g) \rightarrow \langle \gamma; \delta \rangle S(\text{end})$ . Because  $S$  is valid,  $\models S(a) \rightarrow S(g)$ . By 7619
- transitivity of implication,  $\models S(a) \rightarrow \langle \gamma; \delta \rangle S(\text{end})$ , proving the desired result. 7620
4.  $g:\gamma^*$ , then because  $a:(g:\gamma)^* \models S$ , then immediately as the first validity condition  $\models S(a) \rightarrow$  7621
- $\langle a:\alpha \Downarrow S \rangle S(\text{end})$ . Per Lemma B.2,  $\models \langle a:\alpha \Downarrow S \rangle S(\text{end}) \rightarrow \langle \alpha \rangle S(\text{end})$ . By the transitivity of 7622
- implication,  $\models S(a) \rightarrow \langle \alpha \rangle S(\text{end})$ . 7623
5.  $g:\gamma^\times$ , then because  $a:(g:\gamma^\times) \models S$ ,  $\models S(a) \rightarrow S(g)$  and  $\models S(a) \rightarrow S(\text{end})$ . Additionally, 7624
- by the inductive hypothesis, because  $g:\gamma \models S$ ,  $\models S(g) \rightarrow \langle \gamma \rangle S(a)$ . By the transitivity of 7625
- implication,  $\models S(a) \rightarrow \langle \gamma \rangle S(a)$ . Applying the ind rule of dGL, this permits us to infer 7626
- $\models S(a) \rightarrow \langle \gamma^\times \rangle S(a)$ . Recall that because  $a:(g:\gamma^\times) \models S$ ,  $\models S(a) \rightarrow S(\text{end})$ . By the 7627
- monotonicity rule M,  $\models S(a) \rightarrow \langle \gamma^\times \rangle S(\text{end})$ , proving that  $\models S(a) \rightarrow a:\alpha_a$ . 7628
- It remains to show that for any subgame  $b$  in  $\text{subgames}(g:\gamma)$ ,  $\models S(b) \rightarrow \langle a:\alpha_b \rangle S(\text{end})$ . 7629
- Because  $g:\gamma \models S$ , by the inductive hypothesis  $\models S(b) \rightarrow \langle g:\gamma_b \rangle S(a)$ . Since  $\models S(a) \rightarrow$  7630
- $\langle \gamma^\times \rangle S(\text{end})$ , by the monotonicity rule M,  $\models S(b) \rightarrow \langle g:\gamma_b \rangle \langle \gamma^\times \rangle S(\text{end})$ . By dGL ax- 7631
- iom  $\langle ; \rangle$ ,  $\models S(b) \rightarrow \langle g:\gamma_b; \gamma^\times \rangle S(\text{end})$ . But,  $a:\alpha_b = g:\gamma_b; (\gamma^\times)$ . Thus,  $\models S(b) \rightarrow$  7632
- $\langle a:\alpha_b \rangle S(\text{end})$ . 7633

□ 7634

**Corollary D.1** (Inductive subvalue map correctness). *For dGL game  $a:\alpha$  and inductive Angelic subvalue map  $S$ , if  $a:\alpha \models S$ , then  $\models S(a) \rightarrow \langle \alpha \rangle S(\text{end})$ . Dually, if  $a:\alpha \sqsubseteq S$ , then  $\models S(a) \rightarrow$*  7635  
7636  
7637

*Proof.* Corollary of Theorem 6.3, which shows that inductive subvalue maps are subvalue maps. 7638

Def. 6.2 says that the winning subregion for every subgame in the subvalue map implies the 7639

winning region for starting at that subgame and running the rest of the game. For overall game 7640

$a$ , this means  $\models S(a) \rightarrow \langle \alpha \rangle S(\text{end})$ . 7641

**Theorem 6.4** (Sound solving).  $S := \Diamond\text{map}(a:\alpha, \phi)$  is an inductive Angelic subvalue map for 7642

game  $a:\alpha$  ( $a:\alpha \models S$ ) with all subvalues in  $\mathcal{P}_{\mathbb{R}}$ , compatible with Angel winning condition  $\phi \in \mathcal{P}_{\mathbb{R}}$ . 7643

Dually,  $S := \Box\text{map}(a:\alpha, \phi)$  is an inductive Demonic subvalue map with all subvalues in  $\mathcal{P}_{\mathbb{R}}$ , 7644

compatible with Demon winning condition  $\phi$ . 7645

*Proof.* The proof is by structural induction on  $a:\alpha$ . We show the proof for  $\Diamond\text{map}$ , the proof for 7646

$\Box\text{map}$  is symmetric.  $S(\text{end})$  is compatible with  $\phi$  because  $S(\text{end})$  is set to  $\phi$ . In the case where 7647

$a:\alpha$  is atomic, i.e., when  $\alpha \in \{x := e, x := *, ?f, !f, \{x' = f(x) \& Q\}, \{x' = f(x) \& Q\}^d\}$ , the result is immediate. According to Def. 6.8,  $a:\alpha \models S$  exactly when  $\models S(a) \rightarrow \langle \alpha \rangle \phi$ . But per the definition of  $\Diamond \text{map}$ ,  $S(a)$  is  $\text{simpl}(\langle \alpha \rangle \phi)$  which by definition implies  $\langle \alpha \rangle \phi$ .

In the compositional cases, we use structural induction. If  $a:\alpha$  has the structure:

- $a:(g:\gamma \cup d:\delta)$ , then let  $S_1 := \Diamond \text{map}(g:\gamma, \phi)$  and  $S_2 := \Diamond \text{map}(d:\delta, \phi)$ .  $S$  is the disjoint union of  $S_1$  and  $S_2$  extended by mapping  $a$  to  $S_1(g) \vee S_2(d)$ . We must show that  $\models S(a) \rightarrow S(g) \vee S(d)$  and  $g:\gamma \models S$  and  $d:\delta \models S$ . The first implication is immediate as  $S(g) = S_1(g)$  and  $S(d) = S_2(d)$  by the construction of  $S$ . The second and third formulas hold by the inductive hypothesis.
- $a:(g:\gamma \cap d:\delta)$ , then let  $S_1 := \Diamond \text{map}(g:\gamma, \phi)$  and  $S_2 := \Diamond \text{map}(d:\delta, \phi)$ .  $S$  is the disjoint union of  $S_1$  and  $S_2$  extended by mapping  $a$  to  $S_1(g) \wedge S_2(d)$ . We must show that  $\models S(a) \rightarrow S(g) \wedge S(d)$  and  $g:\gamma \models S$  and  $d:\delta \models S$ . The first implication is immediate as  $S(g) = S_1(g)$  and  $S(d) = S_2(d)$  by the construction of  $S$ . The second and third formulas hold by the inductive hypothesis.
- $a:(g:\gamma ; d:\delta)$ , then let  $S_1 := \Diamond \text{map}(d:\delta, \phi)$  and  $S_2 := \Diamond \text{map}(g:\gamma, S_1(d))$ .  $S$  is the disjoint union of  $S_1$  and  $S_2$  extended by mapping  $a$  to  $S_2(g)$ . We must show that  $\models S(a) \rightarrow S(g)$  and  $g:\gamma \models S$  and  $d:\delta \models S$ . The first implication is immediate as  $S(g) = S_2(g)$  by the construction of  $S$ . The second and third formulas hold by the inductive hypothesis.
- $a:(g:\gamma)^*$ , then let  $S' := \Diamond \text{map}(g:\gamma, \text{Inv} \vee \phi)$ .  $S$  is constructed by extending  $S'$  by mapping  $a$  to  $\text{Inv}$ . Additionally, per the side condition  $\models \text{Inv} \rightarrow \langle a:\alpha \exists S' \rangle \phi$ . We must show  $S(a) \rightarrow \langle a:\alpha \exists S' \rangle \phi$  and  $g:\gamma \models S$ . The first implication follows from the side condition along with the fact that  $S(g) = S'(g)$  by the construction of  $S$ . The second formula holds by the inductive hypothesis as it applies to  $S'$ .
- $a:(g:\gamma)^\times$ , then let  $S' := \Diamond \text{map}(g:\gamma, \text{Inv})$ .  $S$  is constructed by extending  $S'$  by mapping  $a$  to  $\text{Inv}$ . Additionally, per the side condition  $\models \text{Inv} \rightarrow S(g) \wedge \phi$ . We must show that  $S(a) \rightarrow \phi \wedge S(g)$ , and  $g:\gamma \models S$ . The first implication follows from the side condition along with the fact that  $S(g) = S'(g)$  by the construction of  $S$ . The second formula holds by the inductive hypothesis as it applies to  $S'$ .

□

**Theorem 6.5** (Maximal inductive subvalue map). *Amongst the inductive Angelic subvalue maps for game  $a:\alpha$  compatible with winning condition  $\phi$ , the model predictive Angelic map given by Def. 6.9 is maximal under the ordering of Def. 6.10. That is, for all  $S$  such that  $a:\alpha \models S$ , model predictive Angelic subvalue map  $S'$  satisfies  $S' \sqsupseteq S$ . Dually, the model predictive Demonic subvalue map (Def. 6.9) is maximal amongst inductive Demonic subvalue maps per the ordering of Def. 6.10.*

*Proof.* Lemma B.4 shows that the MPC solution is valid. Next we show that it is maximal. We first show that  $S' \succsim S$  for all  $S$  such that  $a:\alpha \models S$  under the weaker ordering of Def. B.1. A symmetric proof applies when  $a:\alpha \models S$  is replaced by  $a:\alpha \sqsubseteq S$ . The proof uses structural induction.

For **end**, if  $a:\alpha \models S$  then  $\models S(\text{end}) \rightarrow \phi$ . But for the MPC solution,  $S'(\text{end})$  is  $\phi$ , so  $\models S(a) \rightarrow S'(a)$ .

When  $a:\alpha$  is atomic, i.e.  $\alpha \in \{x := e, x := *, ?f, !f, \{x' = f(x) \ \& \ Q\}, \{x' = f(x) \ \& \ Q\}^d\}$ , the result is immediate: according to Def. 6.8, if  $a:\alpha \models S$  then  $\models S(a) \rightarrow \langle \alpha \rangle \phi$ . But per the definition of the MPC solution,  $S'(a)$  is  $\langle \alpha \rangle \phi$ . Thus,  $\models S(a) \rightarrow S'(a)$ .

For the recursive cases, if  $a:\alpha$  has the structure:

1.  $g:\gamma \cup d:\delta$ , then observe that restrictions  $S'|_{\text{subgames}(\gamma)}$  and  $S'|_{\text{subgames}(\delta)}$  of  $S'$  are the MPC solutions for Angel win condition  $\phi$  for games  $g:\gamma$  and  $d:\delta$  respectively. By the inductive hypothesis, for all subgames  $b \in \text{subgames}(g:\gamma)$ ,  $S(b) \rightarrow S'(b)$ . Similarly, for all subgames  $b \in \text{subgames}(d:\delta)$ ,  $S(b) \rightarrow S'(b)$ . The only remaining subgame in  $\text{subgames}(\alpha)$  is  $a$ . Per the inductive hypothesis,  $\models S(g) \rightarrow S'(g)$  and  $\models S(d) \rightarrow S'(d)$ . Taking the disjunction of these implications, we get  $\models S(g) \vee S(d) \rightarrow S'(g) \vee S'(d)$ . Because  $S$  is valid,  $\models S(a) \rightarrow S(g) \vee S(d)$ . By transitivity of implication,  $\models S(a) \rightarrow S'(g) \vee S'(d)$ . By definition,  $S'(a)$  is  $S'(g) \vee S'(d)$ . Thus we can conclude that  $\models S(a) \rightarrow S'(a)$ .
2.  $g:\gamma \cap d:\delta$ , then observe that restrictions  $S'|_{\text{subgames}(\gamma)}$  and  $S'|_{\text{subgames}(\delta)}$  of  $S'$  are the MPC solutions for Angel win condition  $\phi$  for games  $g:\gamma$  and  $d:\delta$  respectively. By the inductive hypothesis, for all nodes  $b \in \text{subgames}(g:\gamma)$ ,  $S(b) \rightarrow S'(b)$ . Similarly, for all subgames  $b \in \text{subgames}(d:\delta)$ ,  $S(b) \rightarrow S'(b)$ . The only remaining subgame in  $\text{subgames}(\alpha)$  is  $a$ . Per the inductive hypothesis,  $\models S(g) \rightarrow S'(g)$  and  $\models S(d) \rightarrow S'(d)$ . Taking the conjunction of these implications, we get  $\models S(g) \wedge S(d) \rightarrow S'(g) \wedge S'(d)$ . Because  $S$  is valid,  $\models S(a) \rightarrow S(g) \wedge S(d)$ . By transitivity of implication,  $\models S(a) \rightarrow S'(g) \wedge S'(d)$ . By definition,  $S'(a)$  is  $S'(g) \wedge S'(d)$ . Thus we can conclude that  $\models S(a) \rightarrow S'(a)$ .
3.  $g:\gamma; d:\delta$ , then observe that  $S'|_{\text{subgames}(\delta)}$  is the MPC solutions for Angel win condition  $\phi$  for game  $d:\delta$ . By the inductive hypothesis, for all subgames  $b \in \text{subgames}(d:\delta)$ ,  $\models S(b) \rightarrow S'(b)$ . Now, let  $S''$  be the MPC solution for Angel win condition  $S(d)$  for game  $g:\gamma$ . By the inductive hypothesis, for all subgames  $b \in \text{subgames}(g:\gamma)$ ,  $\models S(b) \rightarrow S''(b)$ . By definition, for any subgame  $b \in \text{subgames}(g:\gamma)$ ,  $S''(b)$  is  $\langle g:\gamma_b \rangle S(d)$  while  $S'(b)$  is  $\langle g:\gamma_b \rangle \phi$ . As argued already, by the inductive hypothesis applied to game  $\delta$ ,  $\models S(d) \rightarrow S'(d)$ . By the dGL monotonicity rule M,  $\models \langle g:\gamma_b \rangle S(d) \rightarrow \langle g:\gamma_b \rangle S'(d)$ . Thus,  $\models S''(b) \rightarrow S'(b)$ . By the transitivity of implication,  $\models S(b) \rightarrow S'(b)$ . We have shown the correct implication for all the nodes in  $\text{subgames}(g:\gamma)$  and  $\text{subgames}(d:\delta)$ . The only subgame remaining is  $a$ . Because  $a:\alpha \models S$ ,  $\models S(a) \rightarrow S(g)$ . We have shown that  $\models S(g) \rightarrow S'(g)$ . By the definition of the MPC solution,  $S'(a)$  is  $S'(g)$ . Thus,  $\models S(a) \rightarrow S'(a)$ .
4.  $g:\gamma^*$ , then because  $a:(\gamma^*) \models S$ ,  $\models S(a) \rightarrow \langle a:\alpha \downarrow S \rangle \phi$ . Further, by Lemma B.2,  $\models S(a) \rightarrow \langle \gamma^* \rangle \phi$ . But by the definition of the MPC solution,  $S'(a)$  is  $\langle \gamma^* \rangle \phi$ . By transitivity of implication,  $\models S(a) \rightarrow S'(a)$ . The subgames that remain besides  $a$  all belong to  $\gamma$ . Consider  $S''$ , the MPC solution for Angel win condition  $S(a)$  for game  $g:\gamma$ . By the inductive hypothesis, for every subgame  $b$  in  $\gamma$ ,  $\models S(b) \rightarrow S''(b)$ . Now, we argue that  $\models S''(b) \rightarrow S'(b)$ , so that by transitivity of implication,  $\models S(b) \rightarrow S'(b)$  completing the proof. Per the definition of the MPC solution,  $S''(b)$  is  $\langle g:\gamma_b \rangle S(a)$  while  $S'(b)$  is  $\langle g:\gamma_b \rangle \phi$ . By the monotonicity rule M, since  $\models S(a) \rightarrow \phi$ , we have  $\models \langle g:\gamma_b \rangle S(a) \rightarrow \langle g:\gamma_b \rangle \phi$ . Thus,  $\models S''(b) \rightarrow S'(b)$ .
5.  $g:\gamma^\times$ , then because  $a:(\gamma^\times) \models S$ , per Theorem 6.3,  $\models S(a) \rightarrow \langle \gamma^\times \rangle \phi$ . By the definition of the MPC solution,  $S'(a)$  is  $\langle \gamma^\times \rangle \phi$ . So,  $\models S(a) \rightarrow S'(a)$ . The nodes that remain besides  $a$  all belong to  $\text{subgames}(g:\gamma)$ . Consider  $S''$ , the MPC solution for Angel win condition  $S(a)$  for game  $g:\gamma$ . By the inductive hypothesis, for every subgame  $b$  in  $\gamma$ ,  $\models$

7732  $S(b) \rightarrow S''(b)$ . Now, we argue that  $\models S''(b) \rightarrow S'(b)$ , so that by transitivity of implication,  
 7733  $\models S(b) \rightarrow S'(b)$ , completing the proof. Per the definition of the MPC solution,  $S''(b)$  is  
 7734  $\langle g:\gamma_b \rangle S(a)$  while  $S'(b)$  is  $\langle g:\gamma_b \rangle S'(a)$ . Since  $\models S(a) \rightarrow S'(a)$ , by the dGL monotonicity  
 7735 rule M,  $\models \langle g:\gamma_b \rangle S(a) \rightarrow \langle g:\gamma_b \rangle S'(a)$ . Thus,  $\models S''(b) \rightarrow S'(b)$ .

7736 We have shown that  $S' \succeq S$  for all  $S$  such that  $a:\alpha \models S$  under the weaker ordering of Def. B.1.  
 7737 From Lemma B.1, it also follows that  $S' \sqsupseteq S$  under the stronger ordering of Def. 6.10. Thus,  $S'$   
 7738 is maximal.  $\square$

7739 **Theorem D.1** (Subvalue maps are sound monitors). *For dGL game  $a:\alpha$  and subvalue map  $S$ , if*  
 7740  *$a:\alpha \models S$ , then  $\models S(a) \rightarrow \langle a:\alpha \rangle S\phi$ . Dually, if  $a:\alpha \sqmodels S$  then  $\models S(a) \rightarrow [a:\alpha] S\phi$ .*

7741 *Proof.* The proof proceeds by induction on the structure of the game  $a:\alpha$ . We show this for the  
 7742  $a:\alpha \setminus S$  case; the dual case is analogous. If  $a:\alpha$  has structure:

- 7743 • atomic and Angel does not take any decision, i.e.,  $\alpha \in \{x := e, x := \otimes, ?Q, !Q, \{x' =$   
 7744  $f(x) \ \& \ Q\}^d\}$ , then  $\langle a:\alpha \setminus S \rangle \phi$  is  $\langle \alpha \rangle \phi$ . Because  $a:\alpha \models S$ , we know that  $\models S(a) \rightarrow \langle \alpha \rangle \phi$ .
- 7745 •  $x := *$ , then  $\langle a:\alpha \setminus S \rangle \phi$  is  $\langle ? \langle \alpha \rangle \phi ; x := \otimes ; !\phi \rangle \phi$ . By the  $\langle ; \rangle$ ,  $[?]$  and  $\langle ? \rangle$  axioms, this is  
 7746 equivalent to  $\langle \alpha \rangle \phi \wedge \langle x := \otimes \rangle (\phi \rightarrow \phi)$ . Because  $a:\alpha \models S$ , we know that  $\models S(a) \rightarrow \langle \alpha \rangle \phi$ .  
 7747 By  $\langle := \rangle$ ,  $\langle x := \otimes \rangle \top$  is  $\forall x \top$  which is  $\top$ .
- 7748 •  $\{x' = f(x) \ \& \ Q\}$ , then  $\langle a:\alpha \setminus S \rangle \phi$  is  $\langle ? \langle \alpha \rangle \phi ; \{x' = f(x) \ \& \ Q\}^d ; !\phi \rangle \phi$ . By the  $\langle := \rangle$ ,  
 7749  $\langle ? \rangle$  and  $[?]$  axioms, this is equivalent to  $\langle \alpha \rangle \phi \wedge \langle \{x' = f(x) \ \& \ Q\}^d \rangle (\phi \rightarrow \phi)$ . Because  
 7750  $a:\alpha \models S$ , we know that  $\models S(a) \rightarrow \langle \alpha \rangle \phi$ . By  $[?]$  and  $\langle ^d \rangle$ ,  $\langle \{x' = f(x) \ \& \ Q\}^d \rangle \top$  is  
 7751  $\forall t \geq 0 [x := y(t)] \top$  where  $y'(t) = f(y)$ . By G,  $[x := y(t)] \top$  is  $\top$ , so  $\forall t \geq 0 \top$  is  $\top$ .
- 7752 •  $a:(g:\gamma \cup d:\delta)$  then  $\langle a:\alpha \setminus S \rangle \phi$  is  $\langle ?S(g) \vee S(d) ; ((g:\gamma \setminus S) \cap (d:\delta \setminus S)) \rangle \phi$ . By the  $\langle ? \rangle$ ,  $\langle \cup \rangle$ ,  $[?]$  and  
 7753  $\langle ; \rangle$  axioms, this is equivalent to  $(S(g) \vee S(d)) \wedge (S(g) \rightarrow \langle g:\gamma \setminus S \rangle \phi) \wedge (S(d) \rightarrow \langle d:\delta \setminus S \rangle \phi)$ .  
 7754 Because  $a:\alpha \models S$ , we know that  $\models S(a) \rightarrow S(g) \vee S(d)$ , and further,  $g:\gamma \models S$  and  
 7755  $d:\delta \models S$ . By the inductive hypothesis,  $\models S(g) \rightarrow \langle g:\gamma \setminus S \rangle \phi$  and  $\models S(d) \rightarrow \langle d:\delta \setminus S \rangle \phi$ .  
 7756 Thus,  $\models S(a) \rightarrow \langle a:\alpha \setminus S \rangle \phi$ .
- 7757 •  $a:(g:\gamma \cap d:\delta)$  then  $\langle a:\alpha \setminus S \rangle \phi$  is  $\langle ((g:\gamma \setminus S) \cap (d:\delta \setminus S)) \rangle \phi$ . We want to show  $S(a) \rightarrow \langle (g:\gamma \setminus S) \cap$   
 7758  $(d:\delta \setminus S) \rangle \phi$ . By the inductive hypothesis,  $\models S(g) \rightarrow \langle g:\gamma \setminus S \rangle \phi$  and  $\models S(d) \rightarrow \langle d:\delta \setminus S \rangle \phi$ .  
 7759 By  $\langle \cup \rangle$  and M, then,  $\models (S(g) \wedge S(d)) \rightarrow \langle (g:\gamma \setminus S) \cap (d:\delta \setminus S) \rangle \phi$ . By the definition of a  
 7760 valid solution,  $S(a) \rightarrow S(g) \wedge S(d)$ . By transitivity of implication,  $\models S(a) \rightarrow \langle (g:\gamma \setminus S) \cap$   
 7761  $(d:\delta \setminus S) \rangle \top$ .
- 7762 •  $a:(g:\gamma ; d:\delta)$  then  $\langle a:\alpha \setminus S \rangle \phi$  is  $\langle (g:\gamma \setminus S(\text{end} \mapsto S(d))) ; (d:\delta \setminus S) \rangle \phi$ . After applying the  $\langle ; \rangle$   
 7763 axiom, we want to show  $S(a) \rightarrow \langle g:\gamma \setminus S(\text{end} \mapsto S(d)) \rangle \langle d:\delta \setminus S \rangle \phi$ . By the inductive hy-  
 7764 pothesis,  $\models S(g) \rightarrow \langle g:\gamma \setminus S(\text{end} \mapsto S(d)) \rangle S(d)$  and  $\models S(d) \rightarrow \langle d:\delta \setminus S \rangle \phi$ . Since  $\models$   
 7765  $S(d) \rightarrow \langle d:\delta \setminus S \rangle \phi$ , by M, it remains to show that  $\models S(a) \rightarrow \langle g:\gamma \setminus S(\text{end} \mapsto S(d)) \rangle S(d)$ .  
 7766 Since  $a:\alpha \models S$ , we know that  $\models S(a) \rightarrow S(g)$ . Thus, by transitivity of implication,  
 7767  $\models S(a) \rightarrow \langle g:\gamma \setminus S(\text{end} \mapsto S(d)) \rangle S(d)$ , completing the proof.
- 7768 •  $a:(g:\gamma)^*$  then  $\langle a:\alpha \setminus S \rangle \phi$  is  $\langle ?(S(a)) ; (!S(g) ; g:\gamma \setminus S(\text{end} \mapsto S(a)) ; ?S(a))^\times ; !S(\text{end}) \rangle \phi$ .  
 7769 By the  $\langle ? \rangle$ ,  $[?]$  and  $\langle ; \rangle$  axioms, this is equivalent to  
 7770  $(S(a)) \wedge \langle (!S(g) ; g:\gamma \setminus S(\text{end} \mapsto S(a)) ; ?S(a))^\times \rangle (\phi \rightarrow \phi)$ .  
 7771 Thus, we want to show  $\models S(a) \rightarrow (S(a)) \wedge \langle (!S(g) ; g:\gamma \setminus S(\text{end} \mapsto S(a)) ; ?S(a))^\times \rangle \top$ .

We want to show that  $S(a) \rightarrow \langle (!S(g); g:\gamma \backslash S(\text{end} \mapsto S(a)); ?S(a))^\times \rangle \top$  holds. We prove this using the loop rule. 7772

$\top$  serves as an invariant of this loop game since it holds initially ( $\models \top$ ), implies the postcondition ( $\models \top \rightarrow \top$ ), and is inductive ( $\langle (!S(g); g:\gamma \backslash S(\text{end} \mapsto S(a)); ?S(a)) \rangle \top$ ) as argued next. Since  $a:\alpha \models S$ , so  $g:\gamma \models S(\text{end} \mapsto S(a))$ , By the inductive hypothesis,  $\models S(g) \rightarrow \langle g:\gamma \backslash S(\text{end} \mapsto S(a)) \rangle (S(a))$ . Applying the axioms  $[?]$ ,  $\langle ; \rangle$  and  $\langle ? \rangle$ , this is equivalent to 7773

$\models \langle (!S(g); g:\gamma \backslash S(\text{end} \mapsto S(a)); ?S(a)) \rangle \top$  7774  
 showing that  $\top$  is an inductive invariant. 7775

- $a:(g:\gamma)^\times$  then  $\langle a:\alpha \backslash S \rangle \phi$  is  $\langle (g:\gamma \backslash S)^\times \rangle \phi$ . 7776

We want to show  $\models S(a) \rightarrow \langle (g:\gamma \backslash S(\text{end} \mapsto S(a)))^\times \rangle \phi$ . 7777

Since  $a:\alpha \models S$ , it must be the case that  $\models S(a) \rightarrow (S(g) \wedge \phi)$  and  $g:\gamma \models S$ . By the inductive hypothesis,  $\models S(g) \rightarrow \langle g:\gamma \backslash S(\text{end} \mapsto S(a)) \rangle S(a)$ . We apply the loop rule to prove the conclusion  $\models S(a) \rightarrow \langle (g:\gamma \backslash S)^\times \rangle \phi$ .  $S(a)$  serves as an invariant of this loop. It holds initially. It is inductive per the inductive hypothesis (using weakening and  $\models S(a) \rightarrow S(g)$ ), and implies the postcondition ( $\models S(a) \rightarrow \phi$ ). 7778

□ 7779

**Theorem A.1** (Correctness of reduce). *For any loop-free dGL formula  $F$  and assumptions  $A \in \mathcal{P}_{\mathbb{R}}$  the function `odereduce` either sets `exact=true` and the formula  $A \rightarrow (\text{reduce}(F, A) \leftrightarrow F)$  is valid, or else it sets `exact=false` and the formula  $A \rightarrow (\text{reduce}(F, A) \rightarrow F)$  is valid.* 7780

*Proof.* Follows from dGL axioms being defined in terms of the decidable fragment of  $\text{FOL}_{\mathbb{R}}$ , quantifier elimination being decidable, and the properties of `odereduce` (Def. A.1). 7781

□ 7782



# Bibliography

7794

- [1] Alessandro Abate, Iury Bessa, Lucas C. Cordeiro, Cristina David, Pascal Kesseli, Daniel Kroening, and Elizabeth Polgreen. Automated formal synthesis of provably safe digital controllers for continuous plants. *Acta Informatica*, 57(1-2):223–244, 2020. doi: 10.1007/s00236-019-00359-1. 1.3, 5.5
- [2] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, and U. Topcu. Safe reinforcement learning via shielding. *Proceedings of the Aaai Conference on Artificial Intelligence*, 32, 2018. doi: 10.1609/aaai.v32i1.11797. 1.3, 5.5
- [3] Matthias Althoff. Guaranteed state estimation in CORA 2021. In Goran Frehse and Matthias Althoff, editors, *8th International Workshop on Applied Verification of Continuous and Hybrid Systems (ARCH21), Brussels, Belgium, July 9, 2021*, volume 80 of *EPiC Series in Computing*, pages 161–175. EasyChair, 2021. doi: 10.29007/7M2K. 4.1
- [4] Aaron D. Ames, Samuel Coogan, Magnus Egerstedt, Gennaro Notomista, Koushil Sreenath, and Paulo Tabuada. Control barrier functions: Theory and applications. In *17th European Control Conference, ECC 2019, Naples, Italy, June 25-28, 2019*, pages 3420–3431. IEEE, 2019. doi: 10.23919/ECC.2019.8796030. 1.3, 6.11
- [5] M. Antoniotti and B. Mishra. Discrete event models+temporal logic=supervisory controller: automatic synthesis of locomotion controllers. In *Proceedings of 1995 IEEE International Conference on Robotics and Automation*, volume 2, pages 1441–1446 vol.2, 1995. doi: 10.1109/ROBOT.1995.525480. 1.3, 5.5
- [6] Zvi Artstein. Stabilization with relaxed controls. *Nonlinear Analysis: Theory, Methods & Applications*, 7(11):1163–1173, 1983. ISSN 0362-546X. 1.3, 6.11
- [7] Robert Atkey, Bruno Gavranović, Neil Ghani, Clemens Kupke, Jérémy Ledent, and Fredrik Nordvall Forsberg. Compositional Game Theory, Compositionally. In *3rd international Applied Category Theory Conference*, volume 333, pages 198 – 214, Online, United States, July 2020. doi: 10.4204/eptcs.333.14. URL <https://hal.science/hal-04470659>. 6.11
- [8] Robert J. Aumann. Backward induction and common knowledge of rationality. *Games and Economic Behavior*, 8(1):6–19, 1995. ISSN 0899-8256. doi: 10.1016/S0899-8256(05)80015-6. 6.7
- [9] G. Basile and Giovanni Marro. Controlled and conditioned invariant subspaces in linear system theory. *Journal of Optimization Theory and Applications*, 3:306–315, 05 1969. doi: 10.1007/BF00931370. 5.3.1

- [10] Richard Bellman. *Dynamic Programming*. Dover Publications, 1957. ISBN 9780486428093. 1.1, 1.2, 6.3, 6.11
- [11] Richard E. Bellman and Stuart E Dreyfus. *Applied Dynamic Programming*. Princeton University Press, Princeton, 1962. ISBN 9781400874651. doi: 10.1515/9781400874651. 6.3
- [12] Calin Belta, Boyan Yordanov, and Ebru Aydin Gol. *Formal Methods for Discrete-Time Dynamical Systems*. Springer Cham, 2017. ISBN 978-3-319-50763-7. 1.3, 5.5
- [13] Massimo Benerecetti and Marco Faella. Automatic synthesis of switching controllers for linear hybrid systems: Reachability control. *ACM Trans. Embed. Comput. Syst.*, 16(4), May 2017. ISSN 1539-9087. doi: 10.1145/3047500. URL <https://doi.org/10.1145/3047500>. 1.3, 6.10, 6.11, 1
- [14] Ulrich Berger, Phillip James, Andrew Lawrence, Markus Roggenbach, and Monika Seisenberger. Verification of the European Rail Traffic Management System in Real-Time Maude. *Science of Computer Programming*, 154:61–88, 2018. doi: <https://doi.org/10.1016/j.scico.2017.10.011>. 3.2
- [15] Julien Bernet, David Janin, and Igor Walukiewicz. Permissive strategies: from parity games to safety games. *RAIRO Theor. Informatics Appl.*, 36(3):261–275, 2002. doi: 10.1051/ITA:2002013. URL <https://doi.org/10.1051/ita:2002013>. 6.11
- [16] Rastislav Bodik, Satish Chandra, Joel Galenson, Doug Kimelman, Nicholas Tung, Shaon Barman, and Casey Rodarmor. Programming with angelic nondeterminism. In *Proceedings of the 37th Annual ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages*, POPL ’10, pages 339–352, New York, NY, USA, 2010. Association for Computing Machinery. ISBN 9781605584799. doi: 10.1145/1706299.1706339. URL <https://doi.org/10.1145/1706299.1706339>. 1.3, 6.11
- [17] Sergiy Bogomolov, Marcelo Forets, Goran Frehse, Kostiantyn Potomkin, and Christian Schilling. Juliareach: a toolbox for set-based reachability. In Necmiye Ozay and Pavithra Prabhakar, editors, *Proceedings of the 22nd ACM International Conference on Hybrid Systems: Computation and Control, HSCC 2019, Montreal, QC, Canada, April 16-18, 2019*, pages 39–44. ACM, 2019. doi: 10.1145/3302504.3311804. 4.1
- [18] Brandon Bohrer, Yong Kiam Tan, Stefan Mitsch, Magnus O. Myreen, and André Platzer. VeriPhy: Verified controller executables from verified cyber-physical system models. In *PLDI*, pages 617–630, 2018. doi: 10.1145/3192366.3192406. 3.9
- [19] Rose Bohrer. Chemical case studies in keymaera x. In *Formal Methods for Industrial Critical Systems: 27th International Conference, FMICS 2022, Warsaw, Poland, September 14-15, 2022, Proceedings*, pages 103–120, Berlin, Heidelberg, 2022. Springer-Verlag. ISBN 978-3-031-15007-4. doi: 10.1007/978-3-031-15008-1\_8. C.3
- [20] Rose Bohrer and André Platzer. Constructive hybrid games. In *IJCAR*, pages 454–473, Berlin, Heidelberg, 2020. Springer. ISBN 978-3-030-51073-2. doi: 10.1007/978-3-030-51074-9\_26. 1.3, 6.9, 6.11



- [21] Andrea Bonacchi and Alessandro Fantechi. On the validation of an interlocking system by model-checking. pages 94–108, 09 2014. doi: 10.1007/978-3-319-10702-87. 3.2
- [22] Michele Boreale. Complete algorithms for algebraic strongest postconditions and weakest preconditions in polynomial ODE’s. In A Min Tjoa, Ladjel Bellatreche, Stefan Biffl, Jan van Leeuwen, and Jiri Wiedermann, editors, *SOFSEM 2018: Theory and Practice of Computer Science - 44th International Conference on Current Trends in Theory and Practice of Computer Science, Krems, Austria, January 29 - February 2, 2018, Proceedings*, volume 10706 of *LNCS*, pages 442–455. Springer, 2018. 5.3.4
- [23] Joe Brosseau, Bill Moore Ede, Shad Pate, RB Wiley, and Joe Drapa. Development of an operationally efficient PTC braking enforcement algorithm for freight trains. Technical Report DOT/FRA/ORD-13/34, Federal Railroad Administration, 2013. 3.1, 3.1, 3.3, 3.4, 3.8, 3.8
- [24] Joseph Brosseau and Bill Moore Ede. Development of an adaptive predictive braking enforcement algorithm. Technical Report FRA/DOT/ORD-9/13, Federal Railroad Administration, 2009. 1.2, 3.1, 3.3, 3.3, 3.4, 3.7, 3.8, 3.8, 3.8.1, 3.2a, 3.2b, 3.4a, 3.4c
- [25] Janusz A. Brzozowski. Derivatives of regular expressions. *J. ACM*, 11(4):481–494, October 1964. ISSN 0004-5411. doi: 10.1145/321239.321249. 1.1, 1, 2
- [26] Lei Bu, You Li, Linzhang Wang, and Xuandong Li. Bach: Bounded reachability checker for linear hybrid automata. In *Proceedings of the 2008 International Conference on Formal Methods in Computer-Aided Design, FMCAD ’08*. IEEE Press, 2008. ISBN 9781424427352. 4.1
- [27] Christof Budnik, Marco Gario, Georgi Markov, and Zhu Wang. Guided test case generation through AI enabled output space exploration. pages 53–56, 05 2018. doi: 10.1145/3194733.3194740. 3.2
- [28] Gustavo Carvalho, Ana Cavalcanti, and Augusto Sampaio. Modelling timed reactive systems from natural-language requirements. *Formal Aspects of Computing*, 28(4):725–765, 2016. doi: 10.1007/s00165-016-0387-x. 8.3
- [29] Bob F Caviness and Jeremy R Johnson. *Quantifier elimination and cylindrical algebraic decomposition*. Springer Science & Business Media, 2012. 5.3.7
- [30] Yongchao Chen, Rujul Gandhi, Yang Zhang, and Chuchu Fan. NL2TL: Transforming natural languages to temporal logics using large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 15880–15903, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.985. URL <https://aclanthology.org/2023.emnlp-main.985/>. 8, 8.3
- [31] Alessandro Cimatti, Raffaele Corvino, Armando Lazzaro, Iman Narasamdya, Tiziana Rizzo, Marco Roveri, Angela Sanseviero, and Andrei Tchaltsev. Formal verification and validation of ERTMS industrial railway train spacing system. In *CAV*, pages 378–393, 2012. doi: 10.1007/978-3-642-31424-729. 3.2

- [32] Mariano Di Claudio, Alessandro Fantechi, Giacomo Martelli, Simone Menabeni, and Paolo Nesi. Model-based development of an automatic train operation component for communication based train control. In *ITSC*, pages 1015–1020, 2014. doi: 10.1109/ITSC.2014.6957821. 3.2
- [33] Max H. Cohen and Calin Belta. Approximate optimal control for safety-critical systems with control barrier functions. In *2020 59th IEEE Conference on Decision and Control (CDC)*, pages 2062–2067, 2020. doi: 10.1109/CDC42340.2020.9303896. 1.3, 6.11
- [34] Charles R. Cutler and Brian L. Ramaker. Dynamic matrix control—a computer control algorithm. *IEEE Transactions on Automatic Control*, 17:72, 1979. doi: 10.1109/JACC.1980.4232009. 4
- [35] J. Daafouz, P. Riedinger, and C. Iung. Stability analysis and control synthesis for switched systems: a switched Lyapunov function approach. *IEEE Transactions on Automatic Control*, 47(11):1883–1887, 2002. doi: 10.1109/TAC.2002.804474. 1.3, 6.11
- [36] Hongkai Dai, Benoit Landry, Marco Pavone, and Russ Tedrake. Counter-example guided synthesis of neural network lyapunov functions for piecewise linear systems. *2020 59th IEEE Conference on Decision and Control (CDC)*, pages 1274–1281, 2020. 1.3, 5.5
- [37] Jeremy Dasher and Kenny Morrison. Evaluation of ptc braking enforcement algorithms for passenger and commuter trains. Technical report, Federal Railroad Administration, 2020. 3.3, 3.5.1
- [38] James H. Davenport and Joos Heintz. Real quantifier elimination is doubly exponential. *J. Symb. Comput.*, 5(1/2):29–35, 1988. 3.5.1, 1, 5.3.7, 3, 9
- [39] Leonardo de Moura and Nikolaj Bjørner. Z3: An efficient SMT solver. In C. R. Ramakrishnan and Jakob Rehof, editors, *Tools and Algorithms for the Construction and Analysis of Systems*, pages 337–340, Berlin, Heidelberg, 2008. Springer Berlin Heidelberg. 2.1, 4.3.4, 5.4, 7.1.1, 8.2
- [40] Edsger W. Dijkstra. Guarded commands, nondeterminacy and formal derivation of programs. *Commun. ACM*, 18(8):453–457, August 1975. ISSN 0001-0782. doi: 10.1145/360933.360975. 1.1, 6.3, 6.7, 5, 7.1.1
- [41] Catalin Dima, Mariem Hammami, Youssef Oualhadj, and Régine Laleau. Deciding the Synthesis Problem for Hybrid Games Through Bisimulation. In Kazuhiro Ogata, Dominique Mery, Meng Sun, and Shaoying Liu, editors, *Formal Methods and Software Engineering*, pages 181–198, Singapore, 2024. Springer Nature Singapore. ISBN 978-981-96-0617-7. doi: 10.1007/978-981-96-0617-7\_11. 1.3, 6.11
- [42] Mengge Du, Yuntian Chen, Zhongzheng Wang, Longfeng Nie, and Dongxiao Zhang. Large language models for automatic equation discovery of nonlinear dynamics. *Physics of Fluids*, 36(9), 2024. 8.3
- [43] Jony Eckert, Ícaro Teodoro, Luis da Silva Teixeira, Thiago Martins, Paulo Kurka, and Auteliano Santos. A fast simulation approach to assess draft gear loads for heavy haul trains during braking. *Mechanics Based Design of Structures and Machines*, 2021. doi:

- [44] Didier Essamé and Daniel Dollé. B in large-scale projects: The canarsie line cbtc experience. pages 252–254, 12 2006. ISBN 978-3-540-68760-3. doi: 10.1007/11955757\_21. 3.2 7948 7949 7950
- [45] Alessandro Fantechi. Connected or autonomous trains? In *RSSRail*, pages 3–19, 2019. doi: 10.1007/978-3-030-18744-6\_1. 3.2 7951 7952
- [46] Alessio Ferrari, Maurice H. ter Beek, Franco Mazzanti, D. Basile, Alessandro Fantechi, Stefania Gnesi, Andrea Piattino, and Daniele Trentini. Survey on formal methods and tools in railways: The astrail approach. In *RSSRail*, 2019. 3.2 7953 7954 7955
- [47] J. Fisac, A. Akametalu, M. Zeilinger, S. Kaynama, J. Gillula, and C. Tomlin. A general safety framework for learning-based control in uncertain robotic systems. *Ieee Transactions on Automatic Control*, 64:2737–2752, 2019. doi: 10.1109/tac.2018.2876389. 1.3, 5.5 7956 7957 7958 7959
- [48] Robert W. Floyd. Assigning meanings to programs. In Timothy R. Colburn, James H. Fetzer, and Terry L. Rankin, editors, *Program Verification - Fundamental Issues in Computer Science*, volume 14 of *Studies in Cognitive Systems*, pages 65–81. Springer Netherlands, 1993. doi: 10.1007/978-94-011-1793-7\_4. URL [https://doi.org/10.1007/978-94-011-1793-7\\_4](https://doi.org/10.1007/978-94-011-1793-7_4). 1 7960 7961 7962 7963 7964
- [49] Nathan Fulton and André Platzer. Safe reinforcement learning via formal methods: Toward safe control through proof and learning. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence and Thirtieth Innovative Applications of Artificial Intelligence Conference and Eighth AAAI Symposium on Educational Advances in Artificial Intelligence*, AAAI’18/IAAI’18/EAAI’18. AAAI Press, 2018. ISBN 978-1-57735-800-8. 1, 1.3, 9 7965 7966 7967 7968 7969 7970
- [50] Nathan Fulton and André Platzer. Verifiably safe off-model reinforcement learning. In Tomáš Vojnar and Lijun Zhang, editors, *TACAS*, pages 413–430, Cham, 2019. Springer International Publishing. doi: 10.1007/978-3-030-17462-0\_28. 10 7971 7972 7973
- [51] Nathan Fulton, Stefan Mitsch, Jan-David Quesel, Marcus Völpl, and André Platzer. KeYmaera X: An axiomatic tactical theorem prover for hybrid systems. In *CADE*, pages 527–538, 2015. doi: 10.1007/978-3-319-21401-6\_36. 3.1, 4.1, 4.2, 5.4, 8.2 7974 7975 7976
- [52] Nathan Fulton, Stefan Mitsch, Brandon Bohrer, and André Platzer. Bellerophon: Tactical theorem proving for hybrid systems. In Mauricio Ayala-Rincón and César A. Muñoz, editors, *ITP*, volume 10499 of *LNCIS*, pages 207–224. Springer, 2017. ISBN 978-3-319-66106-3. doi: 10.1007/978-3-319-66107-0\_14. 4.3.1, 5.1, A.1, A.3.3, 13 7977 7978 7979 7980
- [53] Carlos E. García, David M. Prett, and Manfred Morari. Model predictive control: Theory and practice—a survey. *Automatica*, 25(3):335–348, 1989. ISSN 0005-1098. doi: 10.1016/0005-1098(89)90002-2. 4 7981 7982 7983
- [54] Luca Geretti, Julien Alexandre Dit Sandretto, Matthias Althoff, Luis Benet, Pieter Collins, Marcelo Forets, Stefan Mitsch, Ismail Patel, Maximilian Perschl, Christian Schilling, and Joris Tillet. Arch-comp25 category report: Continuous and hybrid systems with nonlinear 7984 7985 7986

- dynamics. In Goran Frehse and Matthias Althoff, editors, *Proceedings of 12th Int. Workshop on Applied Verification for Continuous and Hybrid Systems*, volume 108 of *EPiC Series in Computing*, pages 39–70. EasyChair, 2025. doi: 10.29007/7br2. 4.3.4, C.3
- [55] Susan L. Gerhart, Dan Craigen, and Ted Ralston. Case study: Paris metro signaling system. *IEEE Software*, 11:32–28, 1994. 3.2
- [56] Neil Ghani, Jules Hedges, Viktor Winschel, and Philipp Zahn. Compositional game theory. In *Proceedings of the 33rd Annual ACM/IEEE Symposium on Logic in Computer Science*, LICS ’18, page 472–481, New York, NY, USA, 2018. Association for Computing Machinery. ISBN 9781450355834. doi: 10.1145/3209108.3209165. URL <https://doi.org/10.1145/3209108.3209165>. 6.11
- [57] Khalil Ghorbal and André Platzer. Characterizing algebraic invariants by differential radical invariants. In Erika Ábrahám and Klaus Havelund, editors, *TACAS*, volume 8413 of *LNCSS*, pages 279–294. Springer, 2014. ISBN 978-3-642-54861-1. doi: 10.1007/978-3-642-54862-8\_19. 4.3.4
- [58] Khalil Ghorbal, Andrew Sogokon, and André Platzer. Invariance of conjunctions of polynomial equalities for algebraic differential equations. In Markus Müller-Olm and Helmut Seidl, editors, *SAS*, volume 8723 of *LNCSS*, pages 151–167. Springer, 2014. ISBN 978-3-319-10935-0. doi: 10.1007/978-3-319-10936-7\_10. 4.1
- [59] Bijoy K. Ghosh. Controlled invariant and feedback controlled invariant subspaces in the design of a generalized dynamical system. In *1985 24th IEEE Conference on Decision and Control*, pages 872–873, 1985. doi: 10.1109/CDC.1985.268620. 5.3.1
- [60] Ichiro Hasuo, Clovis Eberhart, James Haydon, Jérémy Dubut, Rose Bohrer, Tsutomu Kobayashi, Sasinee Pruekprasert, Xiao-Yi Zhang, Erik André Pallas, Akihisa Yamada, Kohei Suenaga, Fuyuki Ishikawa, Kenji Kamijo, Yoshiyuki Shinya, and Takamasa Suetomi. Goal-aware rss for complex scenarios via program logic. *IEEE Transactions on Intelligent Vehicles*, 8(4):3040–3072, 2023. doi: 10.1109/TIV.2022.3169762. 1.3, 6.11
- [61] William Walter Hay. *Railroad engineering*. Wiley, New York, 2nd ed. edition, 1982. ISBN 0471364002. 3.3, 3.6.1, 4
- [62] Thomas A. Henzinger, Benjamin Horowitz, and Rupak Majumdar. Rectangular hybrid games. In Jos C. M. Baeten and Sjouke Mauw, editors, *CONCUR’99 Concurrency Theory*, pages 320–335, Berlin, Heidelberg, 1999. Springer. ISBN 978-3-540-48320-5. doi: 10.1007/3-540-48320-9\_23. 1.3, 6.11
- [63] Thomas A. Henzinger, Benjamin Horowitz, and Rupak Majumdar. Rectangular hybrid games. In *Proceedings of the 10th International Conference on Concurrency Theory*, CONCUR ’99, page 320–335, Berlin, Heidelberg, 1999. Springer-Verlag. ISBN 3540664254. 8.1
- [64] C. A. R. Hoare. An axiomatic basis for computer programming. *Commun. ACM*, 12(10):576–580, 1969. doi: 10.1145/363235.363259. URL <https://doi.org/10.1145/363235.363259>. 1

- [65] Xuhan Huang, Qingning Shen, Yan Hu, Anningzhe Gao, and Benyou Wang. LLMs for mathematical modeling: Towards bridging the gap between natural and mathematical languages. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 2678–2710, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-195-7. URL <https://aclanthology.org/2025.findings-naacl.146/>. 8.3
- [66] I. E. Irodov. *Problems in General Physics*. Mir Publishers, 1988. 8, 8.1, 8.1, 8.4
- [67] Aditi Kabra, Stefan Mitsch, and Andre Platzer. Verified Train Controllers for the Federal Railroad Administration Train Kinematics Model: Balancing Competing Brake and Track Forces (Models and Proofs). 8 2022. doi: 10.1184/R1/19542610.v1. URL [https://kilthub.cmu.edu/articles/software/Verified\\_Train\\_Controllers\\_for\\_the\\_Federal\\_Railroad\\_Administration\\_Train\\_Kinematics\\_Model\\_Balancing\\_Competing\\_Brake\\_and\\_Track\\_Forces\\_Models\\_and\\_Proofs\\_/19542610.1.3](https://kilthub.cmu.edu/articles/software/Verified_Train_Controllers_for_the_Federal_Railroad_Administration_Train_Kinematics_Model_Balancing_Competing_Brake_and_Track_Forces_Models_and_Proofs_/19542610.1.3)
- [68] Aditi Kabra, Stefan Mitsch, and André Platzer. Verified train controllers for the federal railroad administration train kinematics model: Balancing competing brake and track forces. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 41(11):4409–4420, 2022. doi: 10.1109/TCAD.2022.3197690. 1.2, 3, 9, 9.1, C.3
- [69] Aditi Kabra, Jonathan Laurent, Stefan Mitsch, and André Platzer. CESAR: Control envelope synthesis via angelic refinements. In Laura Kovacs and Bernd Finkbeiner, editors, *TACAS, LNCS*. Springer, 2024. 1.2, 1.3, 6.7, 6.7, 6.7.1, 6.10, 6.10, 6.3b, 9, 9, B.4.1, B.4.4
- [70] Aditi Kabra, Jonathan Laurent, Stefan Mitsch, and Andre Platzer. Control Envelope Synthesis via Angelic Refinements (CESAR): Artifact, 1 2024. URL [https://figshare.com/articles/software/Control\\_Envelope\\_Synthesis\\_via\\_Angelic\\_Refinements\\_CESAR\\_Artifact/24922896.5.4](https://figshare.com/articles/software/Control_Envelope_Synthesis_via_Angelic_Refinements_CESAR_Artifact/24922896.5.4)
- [71] Aditi Kabra, Jonathan Laurent, Sagar Bharadwaj, Ruben Martins, Stefan Mitsch, and André Platzer. Can large language models autoformalize kinematics? In Ahmed Irfan and Daniela Kaufmann, editors, *FMCAD*, pages 78–83. TU Wien Academic Press, 2025. doi: 10.34727/2025/isbn.978-3-85448-084-6\_13. 1.3, 9.1
- [72] Aditi Kabra, Jonathan Laurent, Sagar Bharadwaj, Ruben Martins, Stefan Mitsch, and Andre Platzer. Artifact for Can Large Language Models Autoformalize Kinematics? 2025. [https://kilthub.cmu.edu/articles/software/Artifact\\_for\\_Can\\_Large\\_Language\\_Models\\_Autoformalize\\_Kinematics\\_/28934195](https://kilthub.cmu.edu/articles/software/Artifact_for_Can_Large_Language_Models_Autoformalize_Kinematics_/28934195). DOI: 10.1184/R1/28934195. 8.1, 8.2, 8.2, 8.4
- [73] Aditi Kabra, Jonathan Laurent, Stefan Mitsch, and André Platzer. Hybrid game control envelope synthesis, 2025. arXiv:2508.05997. 1.3, 9
- [74] Aditi Kabra, Jonathan Laurent, Ruben Martins, Stefan Mitsch, and André Platzer. Llm-powered automatic theorem proving and synthesis for hybrid systems and game, 2026. 4.2
- [75] Aditi Kabra, Jonathan Laurent, Stefan Mitsch, and Andre Platzer. Artifact for Hybrid

Game Control Envelope Synthesis. 2 2026. doi: 10.1184/R1/30310993. 6.12

- [76] Eduard Kamburjan and Reiner Hähnle. Deductive verification of railway operations. In *RSSRail*, pages 131–147, 2017. ISBN 978-3-319-68499-4. 3.2
- [77] Shyam Lal Karra, Kim Guldstrand Larsen, Florian Lorber, and Jiří Srba. Safe and time-optimal control for railway games. In *RSSRail*, pages 106–122, 2019. ISBN 978-3-030-18744-6. 3.2
- [78] Donald E. Knuth. *The Art of Computer Programming*. Addison Wesley Longman Publishing Co., Inc., USA, 1997. ISBN 0201896834. 5.4
- [79] Donald E. Knuth and Ronald W. Moore. An analysis of alpha-beta pruning. *Artificial Intelligence*, 6(4):293–326, 1975. ISSN 0004-3702. doi: 10.1016/0004-3702(75)90019-3. 9
- [80] Mykel J Kochenderfer, Jessica E Holland, and James P Chryssanthacopoulos. Next generation airborne collision avoidance system. *Lincoln Laboratory Journal*, 19(1):17–33, 2012. 1.3, 5.5
- [81] Yanni Kouskoulas, David W. Renshaw, André Platzer, and Peter Kazanzides. Certifying the safe design of a virtual fixture control algorithm for a surgical robot. In Calin Belta and Franjo Ivancic, editors, *Hybrid Systems: Computation and Control, HSCC, Philadelphia, PA, USA*, pages 263–272. ACM, 2013. doi: 10.1145/2461328.2461369. 6.3a, 6.10, B.4.1, 19
- [82] Atanu Kundu, Sarthak Das, and Rajarshi Ray. Sat-reach: A bounded model checker for affine hybrid systems. *ACM Trans. Embed. Comput. Syst.*, 22(2), January 2023. ISSN 1539-9087. doi: 10.1145/3567425. 4.1
- [83] Martin Lara, Rosario López, Iván Pérez, and Juan F. San-Juan. Exploring the long-term dynamics of perturbed keplerian motion in high degree potential fields. *Communications in Nonlinear Science and Numerical Simulation*, 82:105053, 2020. ISSN 1007-5704. doi: <https://doi.org/10.1016/j.cnsns.2019.105053>. URL <https://www.sciencedirect.com/science/article/pii/S1007570419303727>. 5.4
- [84] Andrea Lattuada, Travis Hance, Chanhee Cho, Matthias Brun, Isitha Subasinghe, Yi Zhou, Jon Howell, Bryan Parno, and Chris Hawblitzel. Verus: Verifying rust programs using linear ghost types. *Proc. ACM Program. Lang.*, 7(OOPSLA1):286–315, 2023. doi: 10.1145/3586037. URL <https://doi.org/10.1145/3586037>. 4.2
- [85] Jonathan Laurent and André Platzer. Oracular programming: A modular foundation for building llm-enabled software. *arXiv preprint arXiv:2502.05310*, 2025. 4.3.1
- [86] Jonathan Laurent and André Platzer. Oracular programming: A modular foundation for building llm-enabled software, 2025. URL <https://arxiv.org/abs/2502.05310>. 8.2
- [87] K. Rustan M. Leino. Dafny: An automatic program verifier for functional correctness. In Edmund M. Clarke and Andrei Voronkov, editors, *Logic for Programming, Artificial Intelligence, and Reasoning - 16th International Conference, LPAR-16, Dakar, Senegal, April 25-May 1, 2010, Revised Selected Papers*, volume 6355 of *Lecture Notes in Computer*

- Science*, pages 348–370. Springer, 2010. doi: 10.1007/978-3-642-17511-4\_20. URL [https://doi.org/10.1007/978-3-642-17511-4\\_20](https://doi.org/10.1007/978-3-642-17511-4_20). 8108  
8109
- [88] Zenan Li, Yifan Wu, Zhaoyu Li, Xinming Wei, Fan Yang, Xian Zhang, and Xiaoxing Ma. Autoformalize mathematical statements by symbolic equivalence and semantic consistency. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 53598–53625. Curran Associates, Inc., 2024. URL [https://proceedings.neurips.cc/paper\\_files/paper/2024/file/6034a661584af6c28fd97a6f23e56c0a-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2024/file/6034a661584af6c28fd97a6f23e56c0a-Paper-Conference.pdf). 8, 8.3 8110  
8111  
8112  
8113  
8114  
8115  
8116
- [89] Samuel J. Ling, Jeff Sanny, and William Moebs. *University Physics: Volume 1*. OpenStax, Houston, TX, September 19 2016. ISBN 978-1506698175. URL <https://openstax.org/books/university-physics-volume-1>. 8, 8.1, 8.4 8117  
8118  
8119
- [90] Siyuan Liu, Ashutosh Trivedi, Xiang Yin, and Majid Zamani. Secure-by-construction synthesis of cyber-physical systems. *Annual Reviews in Control*, 53:30–50, 2022. ISSN 1367-5788. doi: <https://doi.org/10.1016/j.arcontrol.2022.03.004>. 1.3, 5.5 8120  
8121  
8122  
8123
- [91] Sarah M. Loos and André Platzer. Safe intersections: At the crossing of hybrid systems and verification. In Kyongsu Yi, editor, *ITSC*, pages 1181–1186, 2011. doi: 10.1109/ITSC.2011.6083138. 1.2 8124  
8125  
8126
- [92] Sarah M. Loos and André Platzer. Differential refinement logic. In Martin Grohe, Eric Koskinen, and Natarajan Shankar, editors, *LICS*, pages 505–514. ACM, 2016. doi: 10.1145/2933575.2934555. 3.1 8127  
8128  
8129
- [93] Sarah M. Loos, André Platzer, and Ligia Nistor. Adaptive cruise control: Hybrid, distributed, and now formally verified. In Michael Butler and Wolfram Schulte, editors, *FM*, volume 6664 of *LNCS*, pages 42–56. Springer, 2011. doi: 10.1007/978-3-642-21437-0\_6. 6.3a, 6.10, B.4.1 8130  
8131  
8132  
8133
- [94] Konstantinos Mamouras. Synthesis of strategies and the hoare logic of angelic nontermination. In Andrew Pitts, editor, *Foundations of Software Science and Computation Structures*, pages 25–40, Berlin, Heidelberg, 2015. Springer. ISBN 978-3-662-46678-0. doi: 10.1007/978-3-662-46678-0\_2. 1.3, 6.11 8134  
8135  
8136  
8137
- [95] MIT OpenCourseWare. 8.01SC Classical Mechanics Problem Set 1. [https://ocw.mit.edu/courses/8-01sc-classical-mechanics-fall-2016/resources/mit8\\_01f16\\_pset1\\_new/](https://ocw.mit.edu/courses/8-01sc-classical-mechanics-fall-2016/resources/mit8_01f16_pset1_new/), 2016. Fall 2016, undergraduate level. 8, 8.1, 8.4 8138  
8139  
8140  
8141
- [96] Ian M. Mitchell and Jeremy A. Templeton. A toolbox of hamilton-jacobi solvers for analysis of nondeterministic continuous and hybrid systems. In Manfred Morari and Lothar Thiele, editors, *Hybrid Systems: Computation and Control*, pages 480–494, Berlin, Heidelberg, 2005. Springer. ISBN 978-3-540-31954-2. doi: 10.1007/978-3-540-31954-2\_31. 1.3, 6.11 8142  
8143  
8144  
8145  
8146
- [97] Stefan Mitsch and André Platzer. ModelPlex: Verified runtime validation of verified cyber-physical system models. In Borzoo Bonakdarpour and Scott A. Smolka, edi- 8147  
8148

- tors, *RV*, volume 8734 of *LNCS*, pages 199–214. Springer, 2014. doi: 10.1007/978-3-319-11164-3\_17. 1, 1.3
- [98] Stefan Mitsch and André Platzer. Modelplex: verified runtime validation of verified cyber-physical system models. *Formal Methods Syst. Des.*, 49(1-2):33–74, 2016. doi: 10.1007/s10703-016-0241-z. 3.1, 3.8, 6.9, 8, 9.1
- [99] Stefan Mitsch, Sarah M. Loos, and André Platzer. Towards formal verification of freeway traffic control. In Chenyang Lu, editor, *ICCPs*, pages 171–180. IEEE, 2012. doi: 10.1109/ICCPs.2012.25. 1.2
- [100] Stefan Mitsch, Marco Gario, Christof J. Budnik, Michael Golm, and André Platzer. Formal verification of train control with air pressure brakes. In *RSSRail*, pages 173–191, 2017. doi: 10.1007/978-3-319-68499-4\_12. 1.2, 3.1, 3.2, 3.5.1, C.3
- [101] Stefan Mitsch, Huanhuan Sheng, Bohua Zhan, Shuling Wang, Simon Foster, and Jonathan Julian Huerta Y Munive. Arch-comp23 category report: Hybrid systems theorem proving. In Goran Frehse and Matthias Althoff, editors, *Proceedings of 10th International Workshop on Applied Verification of Continuous and Hybrid Systems (ARCH23)*, volume 96 of *EPiC Series in Computing*, pages 170–188. EasyChair, 2023. doi: 10.29007/57g4. URL /publications/paper/TtcQ. 4.2
- [102] Stefan Mitsch, Ismail Patel, Hari Hara Sudhan Kannan, Xiangyu Jin, Bohua Zhan, and Shuling Wang. Arch-comp25 category report: Hybrid systems theorem proving. In Goran Frehse and Matthias Althoff, editors, *Proceedings of 12th Int. Workshop on Applied Verification for Continuous and Hybrid Systems*, volume 108 of *EPiC Series in Computing*, pages 152–168. EasyChair, 2025. doi: 10.29007/pdlw. URL /publications/paper/ZlZZ. 4.2
- [103] Thomas Moor and Jennifer M. Davoren. Robust controller synthesis for hybrid systems using modal logic. In Maria Domenica Di Benedetto and Alberto L. Sangiovanni-Vincentelli, editors, *HSCC*, volume 2034 of *LNCS*, pages 433–446. Springer, 2001. 6.1
- [104] Carroll Morgan. *Programming from specifications (2nd ed.)*. Prentice Hall International (UK) Ltd., GBR, 1994. ISBN 0131232746. 1.3, 6.11
- [105] Andreas Müller, Stefan Mitsch, and André Platzer. Verified traffic networks: Component-based verification of cyber-physical flow systems. In *ITSC*, pages 757–764, 2015. doi: 10.1109/ITSC.2015.128. 1.2
- [106] Daniel Neider and Rajarshi Roy. *What Is Formal Verification Without Specifications? A Survey on Mining LTL Specifications*, pages 109–125. Springer Nature Switzerland, Cham, 2025. ISBN 978-3-031-75778-5. doi: 10.1007/978-3-031-75778-5\_6. URL [https://doi.org/10.1007/978-3-031-75778-5\\_6](https://doi.org/10.1007/978-3-031-75778-5_6). 8.3
- [107] A. Nerode and A. Yakhnis. Modelling hybrid systems as games. In *Decision and Control, 1992., Proceedings of the 31st IEEE Conference on*, pages 2947–2952 vol.3, 1992. doi: 10.1109/CDC.1992.371272. 1.3, 5.5, 6.1, 9
- [108] Bruno Oliveira, Gustavo Carvalho, Mohammad Reza Mousavi, and Augusto Sampaio. Simulation of hybrid systems from natural-language requirements. In *2017 13th IEEE*



- Conference on Automation Science and Engineering (CASE)*, pages 1320–1325, 2017. doi: 10.1109/COASE.2017.8256284. 8.3
- [109] Jiayi Pan, Glen Chou, and Dmitry Berenson. Data-efficient learning of natural language to linear temporal logic translators for robot task specification. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pages 11554–11561, 2023. doi: 10.1109/ICRA48891.2023.10161125. 8, 8.3
- [110] Rohit Parikh. Propositional game logic. In *24th Annual Symposium on Foundations of Computer Science (sfcs 1983)*, pages 195–200, 1983. doi: 10.1109/SFCS.1983.47. 6.11
- [111] Jainta Paul, Stefan Mitsch, and Luis Garcia. Hytwin: Hybrid program semantics for digital twin-based security interventions in industrial control systems. In *NASA Formal Methods: 17th International Symposium, NFM 2025, Williamsburg, VA, USA, June 11–13, 2025, Proceedings*, page 294–312, Berlin, Heidelberg, 2025. Springer-Verlag. ISBN 978-3-031-93705-7. doi: 10.1007/978-3-031-93706-4\_17. 1.3
- [112] Lawrence C. Paulson and Jasmin Christian Blanchette. Three years of experience with sledgehammer, a practical link between automatic and interactive theorem provers. In Geoff Sutcliffe, Stephan Schulz, and Eugenia Ternovska, editors, *The 8th International Workshop on the Implementation of Logics, IWIL 2010, Yogyakarta, Indonesia, October 9, 2011*, volume 2 of *EPiC Series in Computing*, pages 1–11. EasyChair, 2010. doi: 10.29007/36DT. 4.2
- [113] Marc Pauly. A modal logic for coalitional power in games. *Journal of Logic and Computation*, 12(1):149–166, 02 2002. ISSN 0955-792X. doi: 10.1093/logcom/12.1.149. URL <https://doi.org/10.1093/logcom/12.1.149>. 6.11
- [114] Christian Pek and Matthias Althoff. Fail-safe motion planning for online verification of autonomous vehicles using convex optimization. *IEEE Transactions on Robotics*, 37(3): 798–814, 2020. 5.3.5
- [115] André Platzer. Differential dynamic logic for hybrid systems. *J. Autom. Reas.*, 41(2): 143–189, 2008. ISSN 0168-7433. doi: 10.1007/s10817-008-9103-8. 1.2, 2.1, 3.1, 3.2
- [116] André Platzer. *Differential Dynamic Logics: Automated Theorem Proving for Hybrid Systems*. PhD thesis, Department of Computing Science, University of Oldenburg, 2008. 2, 2.3, 2.2
- [117] André Platzer. *Logical Analysis of Hybrid Systems: Proving Theorems for Complex Dynamics*. Springer, Heidelberg, 2010. doi: 10.1007/978-3-642-14509-4. 2.1
- [118] André Platzer. Stochastic differential dynamic logic for stochastic hybrid programs. In *CADE*, volume 6803 of *LNCS*. Springer, 2011. doi: 10.1007/978-3-642-22438-6\_34. 9.1
- [119] André Platzer. Logics of dynamical systems. In *LICS*, pages 13–24. IEEE, 2012. ISBN 978-1-4673-2263-8. doi: 10.1109/LICS.2012.13. 4.1
- [120] André Platzer. Differential game logic. *ACM Trans. Comput. Log.*, 17(1):1:1–1:51, 2015.

- ISSN 1529-3785. doi: 10.1145/2817824. 1.1, 1.2, 2, 2.2, 2.3, 2.3, 4.2, 6.1, 6.2, 1, 6.4.2, 3, 6.4.2, 6.7, 6.11, 8, 8.1, 9, 2, B.6.1, B.6.2
- [121] André Platzer. A complete uniform substitution calculus for differential dynamic logic. *J. Autom. Reas.*, 59(2):219–265, 2017. doi: 10.1007/s10817-016-9385-1. 2.1, 2.1, 3.1, 3.1, 3.2, 3.4, 3.4, 3.7
- [122] André Platzer. *Logical Foundations of Cyber-Physical Systems*. Springer, 2018. doi: 10.1007/978-3-319-63588-0. 2.1
- [123] André Platzer. *Logical Foundations of Cyber-Physical Systems*. Springer, Cham, 2018. ISBN 978-3-319-63587-3. doi: 10.1007/978-3-319-63588-0. 2.1, 2.2, 2.2, 1, 5.1
- [124] André Platzer. Intersymbolic AI: Interlinking Symbolic AI and Subsymbolic AI. In Tiziana Margaria and Bernhard Steffen, editors, *ISoLA 2024*, volume 15222 of *LNCS*, pages 162–180. Springer, 2024. doi: 10.1007/978-3-031-75387-9\_11. 9.1
- [125] André Platzer and Edmund M. Clarke. Computing differential invariants of hybrid systems as fixedpoints. *Form. Methods Syst. Des.*, 35(1):98–120, 2009. doi: 10.1007/s10703-009-0079-8. 6.7, 6.7.1
- [126] André Platzer and Jan-David Quesel. European train control system: A case study in formal verification. In *Formal Methods and Software Engineering, 11th International Conference on Formal Engineering Methods, ICFEM 2009, Rio de Janeiro, Brazil, December 9-12, 2009. Proceedings*, pages 246–265, 2009. doi: 10.1007/978-3-642-10373-5\_13. 1.2, 3.1, 3.2, 3.5.1, 3.5.2, 7, 5.2, 5.3.6, 5.1, 5.5, A.1, A.3.1, B.4.1
- [127] André Platzer and Yong Kiam Tan. Differential equation axiomatization: The impressive power of differential ghosts. In Anuj Dawar and Erich Grädel, editors, *LICS*, pages 819–828, New York, 2018. ACM. ISBN 978-1-4503-5583-4. doi: 10.1145/3209108.3209147. 1
- [128] André Platzer and Yong Kiam Tan. Differential equation invariance axiomatization. *J. ACM*, 67(1):6:1–6:66, 2020. doi: 10.1145/3380825. 1
- [129] André Platzer and Yong Kiam Tan. Differential equation invariance axiomatization. *Journal of the ACM (JACM)*, 67(1):1–66, 2020. 1.3, 5.3.4
- [130] Stanislas Polu and Ilya Sutskever. Generative language modeling for automated theorem proving. *ArXiv*, abs/2009.03393, 2020. 4.1, 4.2
- [131] Stephen Prajna and Ali Jadbabaie. Safety verification of hybrid systems using barrier certificates. In Rajeev Alur and George J. Pappas, editors, *Hybrid Systems: Computation and Control*, pages 477–492, Berlin, Heidelberg, 2004. Springer. ISBN 978-3-540-24743-2. doi: 10.1007/978-3-540-24743-2\_32. 1.3, 6.11
- [132] Marian Qian and Stefan Mitsch. Reward shaping from hybrid systems models in reinforcement learning. In Kristin Yvonne Rozier and Swarat Chaudhuri, editors, *NFM*, volume 13903 of *LNCS*. Springer, 2023. 1, 9
- [133] Jianxing Qin, Alexander Du, Danfeng Zhang, Matthew Lentz, and Danyang Zhuo. Can

- large language models verify system software? a case study using fscq as a benchmark. In *Proceedings of the 2025 Workshop on Hot Topics in Operating Systems, HotOS '25*, page 34–41, New York, NY, USA, 2025. Association for Computing Machinery. ISBN 9798400714757. doi: 10.1145/3713082.3730382. 4.1, 4.2
- [134] Hadi Ravanbakhsh and Sriram Sankaranarayanan. Robust controller synthesis of switched systems using counterexample guided framework. In *2016 International Conference on Embedded Software, EMSOFT 2016, Pittsburgh, Pennsylvania, USA, October 1-7, 2016*, pages 8:1–8:10, 2016. doi: 10.1145/2968478.2968485. 1.3, 5.5
- [135] Rajarshi Ray, Amit Gurung, Binayak Das, Ezio Bartocci, Sergiy Bogomolov, and Radu Grosu. Xspeed: Accelerating reachability analysis on multi-core processors. In Nir Piterman, editor, *Hardware and Software: Verification and Testing - 11th International Haifa Verification Conference, HVC 2015, Haifa, Israel, November 17-19, 2015, Proceedings*, volume 9434 of *Lecture Notes in Computer Science*, pages 3–18. Springer, 2015. doi: 10.1007/978-3-319-26287-1\_1. 4.1
- [136] Jacques Richalet, André Rault, JL Testud, and J Papon. Model predictive heuristic control. *Automatica*, 14(5):413–428, 1978. doi: 10.1016/0005-1098(78)90001-8. 4
- [137] Daniel Richardson. Some undecidable problems involving elementary functions of a real variable. *J. Symb. Log.*, 33(4):514–520, 1968. 3.5.1
- [138] Kristin Yvonne Rozier. Specification: The biggest bottleneck in formal methods and autonomy. In Sandrine Blazy and Marsha Chechik, editors, *Verified Software. Theories, Tools, and Experiments*, pages 8–26, Cham, 2016. Springer International Publishing. ISBN 978-3-319-48869-1. 8
- [139] D. Seto, B. Krogh, L. Sha, and A. Chutinan. The simplex architecture for safe online control system upgrades. In *Proceedings of the 1998 American Control Conference. ACC (IEEE Cat. No.98CH36207)*, volume 6, pages 3504–3508 vol.6, 1998. doi: 10.1109/ACC.1998.703255. 1.3
- [140] Claude E. Shannon. Programming a computer for playing chess. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 41(314):256–275, 1950. doi: 10.1080/14786445008521796. 1.3, 6.11
- [141] Huanhuan Sheng, Alexander Bentkamp, and Bohua Zhan. Hhlpy: Practical verification of hybrid systems using hoare logic. In Marsha Chechik, Joost-Pieter Katoen, and Martin Leucker, editors, *Formal Methods - 25th International Symposium, FM 2023, Lübeck, Germany, March 6-10, 2023, Proceedings*, volume 14000 of *Lecture Notes in Computer Science*, pages 160–178. Springer, 2023. doi: 10.1007/978-3-031-27481-7\_11. URL [https://doi.org/10.1007/978-3-031-27481-7\\_11](https://doi.org/10.1007/978-3-031-27481-7_11). 4.2
- [142] J. Tanner Slagel, Mariano M. Moscato, Lauren M. White, César A. Muñoz, Swee Balachandran, and Aaron Dutle. Embedding differential dynamic logic in PVS. In Temur Kutsia, Daniel Ventura, David Monniaux, and José F. Morales, editors, *Proceedings 18th International Workshop on Logical and Semantic Frameworks, with Applications and 10th Workshop on Horn Clauses for Verification and Synthesis, LSFA/HCVS 2023, and 10th Workshop on Horn Clauses for Verification and Synthesis Rome, Italy & Paris, France*,

1-2 July, 2023 & 23rd April 2023, volume 402 of *EPTCS*, pages 43–62, 2023. doi: 10.4204/EPTCS.402.7. URL <https://doi.org/10.4204/EPTCS.402.7>. 4.2

[143] Andrew Sogokon, Stefan Mitsch, Yong Kiam Tan, Katherine Cordwell, and André Platzer. Pegasus: Sound continuous invariant generation. *Form. Methods Syst. Des.*, 58(1):5–41, 2022. ISSN 0925-9856. doi: 10.1007/s10703-020-00355-z. Special issue for selected papers from FM’19. 4.1, 4.3.3, 5.3.4, 5.3.7, 6.7, 6.7.1, 6.10, 8.5, A.1

[144] Armando Solar-Lezama. Program sketching. *STTT*, 15(5-6):475–495, 2013. doi: 10.1007/s10009-012-0249-7. 1.3, 5.5

[145] Eduardo D. Sontag. A Lyapunov-like characterization of asymptotic controllability. *SIAM Journal on Control and Optimization*, 21(3):462–471, 1983. doi: 10.1137/0321028. 1.3, 6.11

[146] R. S. Sutton and A. Barto. Reinforcement learning: an introduction. *IEEE Transactions on Neural Networks*, 9:1054–1054, 1998. doi: 10.1109/tnn.1998.712192. 1.3, 6.1, 6.3, 6.11

[147] Richard S. Sutton. Learning to predict by the methods of temporal differences. *Machine Learning*, 3:9–44, 1988. doi: 10.1007/BF00115009. 6.3, 6.11

[148] C. Szepesvári and M. Littman. A unified analysis of value-function-based reinforcement-learning algorithms. *Neural Computation*, 11:2017–2060, 1999. doi: 10.1162/089976699300016070. 6.3, 6.11

[149] Paulo Tabuada. *Verification and Control of Hybrid Systems: A Symbolic Approach*. Springer, Berlin, 2009. doi: 10.1007/978-1-4419-0224-5. 1.3, 5.5

[150] Yong Kiam Tan and André Platzer. Switched systems as hybrid programs\*\*this research was sponsored by the afosr under grant number fa9550-16-1-0288. the first author was also supported by a \*star, singapore. *IFAC-PapersOnLine*, 54(5):247–252, 2021. ISSN 2405-8963. doi: <https://doi.org/10.1016/j.ifacol.2021.08.506>. URL <https://www.sciencedirect.com/science/article/pii/S2405896321012817>. 7th IFAC Conference on Analysis and Design of Hybrid Systems ADHS 2021. 8.1

[151] Alfred Tarski. A decision method for elementary algebra and geometry. In Bob F. Cavi-ness and Jeremy R. Johnson, editors, *Quantifier Elimination and Cylindrical Algebraic Decomposition*, pages 24–84, Vienna, 1998. Springer Vienna. ISBN 978-3-7091-9459-1. 5.3.7, 6.3

[152] Ícaro P. Teodoro, Jony J. Eckert, Pedro F. Lopes, Thiago S. Martins, and Auteliano A. Santos. Parallel simulation of railway pneumatic brake using openMP. *International Journal of Rail Transportation*, 8(2):180–194, 04 2020. doi: 10.1080/23248378.2019.1660239. 3.2

[153] Samuel Teuber, Stefan Mitsch, and André Platzer. Provably safe neural network controllers via differential dynamic logic. In *NeurIPS*. Curran Associates, Inc., 2024. doi: 10.5555/3737916.3737967. 1, 9

- [154] Claire J. Tomlin, John Lygeros, and Shankar Sastry. A game theoretic approach to controller design for hybrid systems. *Proc. IEEE*, 88(7):949–970, 2000. doi: 10.1109/5.871303. 1.3, 5.5, 6.1, 8.1
- [155] Alan C. van Hulst, Michel A. Reniers, and Wan J. Fokkink. Maximally permissive controlled system synthesis for modal logic. In Giuseppe F. Italiano, Tiziana Margaria-Steffen, Jaroslav Pokorný, Jean-Jacques Quisquater, and Roger Wattenhofer, editors, *SOFSEM 2015: Theory and Practice of Computer Science*, pages 230–241, Berlin, Heidelberg, 2015. Springer. ISBN 978-3-662-46078-8. doi: 10.1007/978-3-662-46078-8\_19. 1.3, 6.11
- [156] Sumanth Varambally, Thomas Voice, Yanchao Sun, Zhifeng Chen, Rose Yu, and Ke Ye. Hilbert: Recursively building formal proofs with informal reasoning. In *NeurIPS Workshop*, 2025. URL <https://arxiv.org/abs/2509.22819>. 4.1, 4.2
- [157] John von Neumann. Zur Theorie der Gesellschaftsspiele. *Mathematische Annalen*, 100: 295–320, 1928. doi: 10.1007/BF01448847. 1.3, 6.11
- [158] Linh Hong Vu, Anne E. Haxthausen, and Jan Peleska. Formal modelling and verification of interlocking systems featuring sequential release. *Science of Computer Programming*, 133:91–115, 2017. Formal Techniques for Safety-Critical Systems. 3.2
- [159] Martine Wahl. *Survey of railway embedded network solutions - Towards the use of Industrial Ethernet technologies*. In: *Les Collections de l’Inrets, Synthèse S61, 104 pages*. 2010. 3.2
- [160] Shuling Wang, Naijun Zhan, and Liang Zou. An improved HHL prover: An interactive theorem prover for hybrid systems. In Michael J. Butler, Sylvain Conchon, and Fatiha Zaïdi, editors, *Formal Methods and Software Engineering - 17th International Conference on Formal Engineering Methods, ICFEM 2015, Paris, France, November 3-5, 2015, Proceedings*, volume 9407 of *Lecture Notes in Computer Science*, pages 382–399. Springer, 2015. doi: 10.1007/978-3-319-25423-4\_25. 4.1, 4.2
- [161] Volker Weispfenning. The complexity of linear problems in fields. *J. Symb. Comput.*, 5 (1-2):3–27, 1988. 3.5.1, 5.3.7
- [162] Max Willsey, Chandrakana Nandi, Yisu Remy Wang, Oliver Flatt, Zachary Tatlock, and Pavel Panchekha. Egg: Fast and extensible equality saturation. *Proc. ACM Program. Lang.*, 5(POPL), jan 2021. doi: 10.1145/3434304. URL <https://doi.org/10.1145/3434304>. 5.4
- [163] Yuhuai Wu, Albert Q. Jiang, Wenda Li, Markus N. Rabe, Charles Staats, Mateja Jamnik, and Christian Szegedy. Autoformalization with large language models. In *Proceedings of the 36th International Conference on Neural Information Processing Systems, NIPS ’22*, Red Hook, NY, USA, 2022. Curran Associates Inc. ISBN 9781713871088. 8, 8.3
- [164] Jonathan Julián Huerta y Munive, Simon Foster, Mario Gleirscher, Georg Struth, Christian Pardillo Laursen, and Thomas Hickman. Isavodes: Interactive verification of cyber-physical systems at scale. *J. Autom. Reason.*, 68(4):21, 2024. doi: 10.1007/s10817-024-09709-2. URL <https://doi.org/10.1007/s10817-024-09709-2>. 4.2

- 8391 [165] Shuo Yang, Xiang Yin, Shaoyuan Li, and Majid Zamani. Secure-by-construction optimal  
8392 path planning for linear temporal logic tasks. In *2020 59th IEEE Conference on Deci-*  
8393 *sion and Control (CDC)*, pages 4460–4466, 2020. doi: 10.1109/CDC42340.2020.  
8394 9304153. 1.3, 5.5
- 8395 [166] Chaochen Zhou, Ji Wang, and Anders P. Ravn. A formal description of hybrid systems.  
8396 In Rajeev Alur, Thomas A. Henzinger, and Eduardo D. Sontag, editors, *Hybrid Systems*  
8397 *III: Verification and Control, Proceedings of the DIMACS/SYCON Workshop on Verifi-*  
8398 *cation and Control of Hybrid Systems, October 22-25, 1995, Rutgers University, New*  
8399 *Brunswick, NJ, USA*, volume 1066 of *Lecture Notes in Computer Science*, pages 511–  
8400 530. Springer, 1995. doi: 10.1007/BFB0020972. 4.1
- 8401 [167] Shufang Zhu and Giuseppe De Giacomo. Synthesis of maximally permissive strategies for  
8402 ltl specifications. In Lud De Raedt, editor, *Proceedings of the Thirty-First International*  
8403 *Joint Conference on Artificial Intelligence, IJCAI-22*, pages 2783–2789. International  
8404 Joint Conferences on Artificial Intelligence Organization, 7 2022. doi: 10.24963/  
8405 ijcai.2022/386. URL <https://doi.org/10.24963/ijcai.2022/386>.  
8406 1.3, 6.11
- 8407 [168] Liang Zou, Jidong Lv, Shuling Wang, Naijun Zhan, Tao Tang, Lei Yuan, and Yu Liu.  
8408 Verifying chinese train control system under a combined scenario by theorem proving. In  
8409 *Verified Software: Theories, Tools, Experiments*, pages 262–280, 2014. 3.1, 3.2